Fixed-Width vs. Multiprecision Digit Generation for JSON Canonicalization: A Four-Algorithm Empirical Study

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Abstract

JSON Canonicalization Scheme (RFC 8785) requires deterministic floating-point formatting that matches ECMA-262 `Number::toString`. We present the first empirical comparison of four digit-generation algorithms—three fixed-width (Schubfach, Ryū, Dragonbox) and one multiprecision (Burger–Dybvig)—inside a specification-constrained canonicalization pipeline, implemented in both Go and Rust (8 implementations total) to separate algorithmic effects from language/runtime effects. Our methodology gates performance measurement on conformance: only implementations passing 286,362 oracle vectors (generated from V8 and validated against SpiderMonkey and JavaScriptCore) and 696 structural conformance cases (87 per implementation × 8 implementations) proceed to benchmarking. Using permutation tests, BCa bootstrap confidence intervals, and Benjamini–Hochberg FDR correction across 1,294 comparisons (16 pairs × workloads/modes), we find that all three fixed-width algorithms perform within 1–8% of each other across both languages, while each is 1.4–1.8× faster than Burger–Dybvig on number-dense workloads ($\nu(D) \geq 40\%$) in Go API benchmarks; the advantage replicates at 2.3–3.0× in Rust, confirming it is algorithmic rather than language-specific (excluding extreme outliers on `numeric-boundary` and `number-heavy` in Rust at 8–9×, discussed in §8). On string-dominant and deeply-nested inputs the difference narrows to <5%. All eight implementations produce byte-identical output; Go binaries are additionally verified under cross-compilation for x86-64 and arm64. We release the benchmark infrastructure, oracle vectors, and all raw data as a reproducibility artifact.

Keywords: JSON canonicalization, floating-point formatting, Schubfach, Ryū, Dragonbox, Burger–Dybvig, RFC 8785, ECMA-262, benchmark methodology, cross-language validation

1 Introduction

The JSON Canonicalization Scheme (RFC 8785) [15] standardizes a deterministic serialization of JSON values for use in digital signatures, content-addressable storage, and distributed consensus protocols. A critical requirement is that floating-point numbers must

be formatted identically to the output of ECMA-262 `Number::toString` [5], which specifies shortest-representation formatting under roundTiesToEven semantics.

Two families of algorithms can satisfy this requirement: multiprecision arithmetic approaches exemplified by Burger–Dybvig [3], and fixed-width integer approaches exemplified by Schubfach [8], Ryū [1], and Dragonbox [10]. While these algorithms are individually well-studied, no prior work has empirically compared them *inside a specification-constrained canonicalization pipeline* where conformance is a hard prerequisite and the formatting stage is only one component of the total cost.

This paper makes three contributions:

1. The first empirical comparison of four digit-generation algorithms (three fixed-width: Schubfach, Ryū, Dragonbox; one multiprecision: Burger–Dybvig) under ECMA-262/RFC 8785 constraints, showing that all three fixed-width algorithms perform within 1–8% of each other while each achieves 1.4–1.8× speedup over Burger–Dybvig on number-dense workloads ($\nu(D) \geq 40\%$) in Go API benchmarks; the advantage replicates at 2.3–3.0× in Rust. String-dominant and deeply-nested inputs show <5% difference.

2. A benchmark methodology that gates performance measurement on conformance verification: implementations must pass 286,362 oracle vectors and 696 structural conformance cases before timing data is collected. We document noise floors and minimum observable effect sizes for all comparisons.

3. A cross-language validation design (4×2 matrix of algorithm × language) that separates algorithmic effects from language/runtime effects, with Go cross-compilation determinism verified for x86-64 and arm64 via QEMU user-mode emulation. All evidence is released as a reproducibility artifact.

2 Background

2.1 RFC 8785: JSON Canonicalization Scheme

RFC 8785 [15] defines a deterministic JSON serialization by specifying: (1) lexicographic key ordering using UTF-16 code unit comparison, (2) no insignificant whitespace, and (3) floating-point formatting matching ECMA-262 `Number::toString`.

The floating-point requirement is the most technically demanding. It requires producing the *shortest decimal representation* that round-trips to the original IEEE 754 binary64 value under roundTiesToEven, then applying the ECMA-262 formatting policy (exponential notation thresholds, sign handling, integer elision rules).

2.2 ECMA-262 Number Formatting

ECMA-262 [5] §6.1.6.1.20 specifies `Number::toString` as a two-stage process (formerly §7.1.12.1 in older editions):

1. **Digit generation**: find the shortest decimal significand d and exponent n such that $d \times 10^{n-k} = x$ (where k is the number of digits in d) and the value round-trips.

2. **Formatting**: apply branch logic based on the relationship between k (digit count) and n (decimal exponent position) to produce the final string. Key thresholds: $n \geq 1 \wedge n \leq 21$ produces fixed notation; $n \leq 0$ or $n > 21$ produces exponential notation.

2.3 IEEE 754 and Shortest Representation

IEEE 754-2019 [9] binary64 has a 52-bit significand, 11-bit exponent, and 1 sign bit. The *shortest representation* property requires finding the decimal string with fewest digits that uniquely identifies the original binary64 value among all representable values. This is the core algorithmic challenge addressed by both Burger–Dybvig and Schubfach.

3 Related Work

Extensive prior work has optimized float-to-string conversion in isolation. This paper is the first to evaluate these algorithms inside a specification-constrained canonicalization pipeline where conformance is a binary gate. We summarize the most relevant prior work, organized by the problem dimension each addresses.

3.1 Multiprecision Foundations

Burger and Dybvig [3] establish the theoretical framework for shortest-representation printing using arbitrary-precision rational arithmetic. Their algorithm is our multiprecision baseline. Gay's `dtoa.c` [6] provides the foundational practical implementation and remains widely used in language runtimes as a reference or fallback. Both works evaluate algorithms in isolation and do not consider performance inside a pipeline where formatting policy is externally constrained.

3.2 Fixed-Width Approaches

Loitsch [13] demonstrated that integer-only arithmetic suffices for the common case with Grisu. However, Grisu3 falls back to arbitrary precision for ∼0.5% of inputs, making conformance guarantees harder to establish in specification-constrained contexts.

Adams [1] proposed Ryū, achieving full coverage with fixed-width 128-bit arithmetic and lookup tables. Adams provides the most comprehensive prior performance comparison of fixed-width vs. multiprecision approaches (we refer the reader to Adams's literature review rather than replicating it), but evaluates algorithms in isolation on microbenchmarks without specification constraints.

Giulietti [8] introduced the Schubfach algorithm, which we implement and evaluate within the RFC 8785 pipeline. Jeon [10] extends and refines the Schubfach approach with Dragonbox, improving table compression and branch structure. Go 1.26 adopted Dragonbox for `strconv`, providing additional motivation for evaluating fixed-width approaches in specification-constrained contexts.

3.3 Comprehensive Experimental Review

Champagne Gareau and Lemire [4] provide the most comprehensive experimental comparison of modern float-to-string algorithms, evaluating Dragon4, Grisu3, Ryū, Schubfach, Dragonbox, and `std::to_chars` in C++. Their work evaluates algorithms in isolation on microbenchmarks. Our contribution differs: we evaluate all four principal algorithms (Burger–Dybvig, Schubfach, Ryū, and Dragonbox) inside a specification-constrained canonicalization pipeline where (a) ECMA-262 formatting policy is a hard constraint, (b) correctness against 286,362 oracle vectors is a binary gate, and (c) digit generation is only one component of the total cost.

The complementary direction—fast string-to-float parsing—is addressed by Lemire [12], whose Eisel-Lemire algorithm achieves gigabyte-per-second parsing throughput. While outside our scope, this work shares the fixed-width arithmetic philosophy that motivates Schubfach and Dragonbox.

3.4 Deterministic Serialization Standards

Beyond JSON, deterministic encoding is required by other serialization formats. CBOR (RFC 8949) [2] specifies deterministic encoding rules (§4.2) for binary data, addressing analogous concerns about canonical representation. W3C RDF Dataset Canonicalization [16] addresses graph-level canonicalization (structural equivalence of RDF datasets), a fundamentally different problem from the byte-level serialization determinism required by RFC 8785. Our conformance-gated methodology could be adapted to other deterministic serialization benchmarks.

3.5 JSON Canonicalization

RFC 8785 [15] is the primary standard. Prior to RFC 8785, ad-hoc canonicalization schemes existed in various ecosystems (e.g., JSON-LD canonicalization, OLPC canonical JSON). None specify the floating-point formatting requirement with the precision of ECMA-262 [5].

3.6 Benchmark Methodology

Our conformance-before-performance approach draws on the principle that correctness is a prerequisite for meaningful performance comparison. Georges et al. [7] established rigorous statistical methods for Java performance evaluation, advocating confidence intervals and effect sizes over single-point estimates. Mytkowicz et al. [14] demonstrated that measurement bias (link order, environment size) can reverse performance conclusions, motivating our randomized task ordering and environment documentation. Kalibera and Jones [11] formalized the analysis of multi-level experimental designs in benchmarking. Our use of permutation tests and BCa bootstrap confidence intervals follows this modern statistical practice for performance evaluation.

4 Algorithms

4.1 Definitions

Definition 1 (Minimal Significand)**.** Given a finite IEEE 754 binary64 value x, the *minimal significand* is the shortest decimal significand d (with digit count k) such that parsing $d \times 10^{n-k}$ under roundTiesToEven recovers x exactly.

Definition 2 (Minimal Printed String)**.** The *minimal printed string* of x is the output of ECMA-262 `Number::toString` applied to x: the minimal significand formatted according to the ECMA-262 branch table.

Remark 1. Definitions 1 and 2 are not equivalent in string length. Consider $x = 10^{20}$: the minimal significand is $d = 1$ ($k = 1$, $n = 21$). Since $n = 21 \leq 21$, ECMA-262 uses fixed notation, yielding `100000000000000000000` (21 characters). Now consider $x = 10^{21}$: again $d = 1$ ($k = 1$, $n = 22$), but $n > 21$ triggers exponential notation, yielding `1e+21` (5 characters). The $n = 21$ boundary creates divergence between significand minimality and printed-string length. This means implementations cannot optimize for shorter output strings because the ECMA-262 formatting is spec-mandated.

4.2 Burger–Dybvig (Multiprecision)

The Burger–Dybvig algorithm [3] uses arbitrary-precision rational arithmetic to compute the shortest decimal representation. It maintains big-integer numerator/denominator pairs and iteratively extracts decimal digits. The algorithm is general (handles any radix) but requires heap-allocated multiprecision integers, with cost proportional to the magnitude of the exponent.

In our Go implementation, this uses `math/big.Int` operations. The `sync.Pool` partially mitigates allocation cost by reusing big-integer buffers across calls.

4.3 Schubfach (Fixed-Width)

The Schubfach algorithm [8] uses fixed-width 64/128-bit arithmetic, relying on `math/bits` for 128-bit multiplication (`Mul64`, `Add64`) and bit inspection (`TrailingZeros64`). It uses precomputed tables of powers of ten and 128-bit multiplications to determine the shortest decimal significand. No heap allocation is required in the digit-generation path.

The key advantage is constant-time (with respect to exponent magnitude) operation and zero GC pressure. The precomputed table is approximately 5 KiB.

4.4 Ryū (Fixed-Width)

The Ryū algorithm [1] was the first to achieve full coverage shortest-output generation using only fixed-width 128-bit arithmetic and precomputed lookup tables. Like Schubfach, it computes a rounding interval around the input value and converts it to decimal via table-based 128-bit multiplication. Ryū uses a `lastRemovedDigit` heuristic for digit trimming: digits are removed one at a time from both ends of the interval, tracking the last removed digit

for round-to-even tie-breaking. Our implementation shares the same interval computation as Schubfach (symmetric for regular values, asymmetric at power-of-two boundaries).

4.5 Dragonbox (Fixed-Width)

The Dragonbox algorithm [10] extends the fixed-width approach with two key optimizations: a *shorter interval* case for power-of-two boundaries (where the lower half of the rounding interval is narrower), and *two-digit-at-a-time* trimming (dividing by 100 instead of 10 in the digit extraction loop). Our implementation separates the shorter-interval and general cases into distinct functions, preserving the algorithm's structural innovation.

4.6 ECMA-262 Formatting Stage

All four algorithms share an identical formatting stage that converts the minimal significand (d, k, n) triple into the final ECMA-262-compliant string. This stage implements the branch table from ECMA-262 §6.1.6.1.20 and is pure string manipulation with no arithmetic.

5 Implementation

We maintain eight implementations with a common CLI ABI:

- `impl-schubfach` (Go): Schubfach number formatting.
- `impl-json-canon` (Go): Burger–Dybvig number formatting.
- `impl-ryu` (Go): Ryū number formatting.
- `impl-dragonbox` (Go): Dragonbox number formatting.
- `impl-schubfach-rs` (Rust): Schubfach-aligned Rust path.
- `impl-json-canon-rs` (Rust): Burger–Dybvig-aligned Rust path.
- `impl-ryu-rs` (Rust): Ryū-aligned Rust path.
- `impl-dragonbox-rs` (Rust): Dragonbox-aligned Rust path.

All eight implementations contain their own digit-generation code and do not delegate to the standard library's float-to-string routines (e.g., Go's `strconv.FormatFloat` or Rust's `std::fmt`). This is necessary because the standard routines do not implement ECMA-262 formatting policy. Consequently, Go 1.26's adoption of Dragonbox for `strconv` does not affect our measurements.

Within each language, all four implementations share identical JSON parser, serializer, error taxonomy, and ECMA-262 formatting code; only the `jcsfloat` digit-generation module differs. This isolation ensures that performance differences are attributable to the digit-generation algorithm alone.

The lab harness treats each implementation as a black-box canonicalizer with identical CLI contracts (`canonicalize`, `verify`, exit-code policy, and error-class expectations). This preserves algorithm comparability while allowing language/runtime differences to be measured independently.

Rust builds use release configuration (`opt-level=3`, `lto=true`, `codegen-units=1`, `strip=true`) to align optimization policy with production benchmarking practice.

5.1 Rust Implementation Notes

All four Rust implementations use native `u128` arithmetic for the 128-bit intermediate products required by the fixed-width algorithms, eliminating the need for compiler intrinsics or multi-word emulation. The Rust Burger–Dybvig implementation uses a custom `BigUint` (approximately 350 lines of schoolbook multiprecision arithmetic with Knuth Algorithm D division) with zero external dependencies; buffer lifecycle is managed by Rust's RAII ownership model (deterministic `Drop`), removing the need for explicit pooling strategies such as Go's `sync.Pool`. All Rust implementations have zero runtime crate dependencies, ensuring complete self-containment and reproducibility.

All Rust implementations share an identical JSON parser and serializer; only the `jcsfloat` digit-generation module differs between them, mirroring the Go design. Rust binaries are approximately 424–456 KB (stripped), compared to 2.5–2.7 MB for the statically-linked Go binaries.

6 Methodology

6.1 Conformance-Before-Performance

Our methodology enforces a strict ordering: conformance verification must pass before any performance data is collected. This prevents measuring the speed of incorrect implementations.

Conformance gate: 87 structural test cases per implementation (696 total across 8 implementations) covering key ordering, Unicode handling, numeric formatting, escape sequences, depth limits, and invalid input rejection. All implementations must produce byte-identical output (verified via SHA-256) and identical error classification.

Oracle vectors: 286,362 IEEE 754 binary64 values with expected `String(x)` output generated by the V8 JavaScript engine (Node.js v23.3.0, V8 12.9.202.28) and independently validated against SpiderMonkey (JavaScript-C115.10.0) and JavaScriptCore (WebKitGTK 2.50.4). All three engines produce identical output for every vector, confirming cross-engine agreement on ECMA-262 `Number::toString` semantics. (2,153 values appear in both vector files as overlapping boundary coverage; the total counts file lines, with 284,209 unique values.) Coverage includes all subnormals with bit patterns $\leq 2^{12}$ (4,096 of approximately 4.5×10^{15} positive subnormals, providing dense coverage near zero where formatting edge cases concentrate), subnormal/normal boundary (± 128 around the transition), all powers of two in the binary64 exponent range, RFC 8785 Appendix B examples with ± 2 neighbors, and pseudorandom fill.

6.2 Benchmark Design

CLI benchmarks: end-to-end process execution measuring wall time, CPU time (user + system), and peak RSS. 15 repetitions per implementation/mode/workload combination, with 3 warmup runs discarded. Task order is randomized with a fixed seed for reproducibility.

Cross-language comparison pairs: each of the four algorithms is implemented in Go and Rust, yielding eight implementations. These are compared via 16 pairwise tests:

- 6 within-Go algorithm pairs (e.g., go:schubfach-vs-ryu, go:dragonbox-vs-bd),
- 6 within-Rust algorithm pairs (e.g., rs:schubfach-vs-ryu, rs:dragonbox-vs-bd),
- 4 cross-language same-algorithm pairs (e.g., schubfach:go-vs-rs, ryu:go-vs-rs).

The within-language pairs test the algorithmic effect while controlling for language/runtime; the cross-language pairs test language/runtime effects while controlling for algorithm. Directional consistency across within-language pairs in both Go and Rust provides evidence that algorithmic differences are not language-specific, though this is not a formal interaction test (see §8). The Rust Burger–Dybvig implementation uses a custom `BigUint` (schoolbook multiprecision arithmetic, approximately 350 lines of Rust with zero external dependencies, heap-allocated with RAII lifecycle), making it a structurally analogous but runtime-different counterpart to the Go `math/big`-based path. Rust builds use release configuration (`opt-level=3`, `lto=true`, `codegen-units=1`) to align optimization policy with production benchmarking practice.

API benchmarks: Go `testing.B` microbenchmarks (Go implementations) and a Rust native benchmark harness (Rust implementations) measuring canonicalization and verify paths directly, 10 iterations per benchmark. Go's `testing.B` auto-calibrates the number of inner iterations per sample (typically hundreds to millions), whereas the Rust harness times each function call individually (N=1); both report ns/op. The Rust harness includes 3 warmup iterations (discarded) before measurement to stabilize caches and lazy-initialized data structures. The Rust harness emits Go-compatible `Benchmark... ns/op` lines so the same statistical pipeline can be applied uniformly.

Measurement granularity note: The Go and Rust API harnesses differ in measurement granularity. Go's `testing.B` averages over many inner iterations per sample, producing low-variance samples. The Rust harness reports single-call latency, producing higher-variance samples. Within-language comparisons are unaffected since both implementations in a pair use the same harness. Cross-language API comparisons should be interpreted with this asymmetry in mind; we report them for completeness but emphasize within-language results for algorithmic conclusions.

Differential fuzzing: 2,000 randomly-generated JSON inputs processed by all implementations, verifying byte-identical output.

6.3 Workload Characterization

Definition 3 (Number value fraction)**.** For a JSON document D, define $\nu(D)$ as the fraction of JSON values that are numbers. Array and object containers are counted as values; their elements and members are counted separately. Formally, $\nu(D) = |\{v \in V(D) :$

v is a JSON number$\}|/|V(D)|$, where $V(D)$ is the multiset of all JSON values in D (including containers).

Table 1 reports the structural properties of each benchmark workload, including $\nu(D)$, total value count, maximum nesting depth, and unique key count.

Table 1: Workload structural characterization.

Workload	Bytes	Total Values	Numbers	$\nu(D)$ (%)	Max Depth	Keys
array-2048	98,901	10,243	4,096	40.0	3	8,194
array-256	11,914	1,283	512	39.9	3	1,026
canonical-minimal	7	2	1	50.0	1	1
control-escapes	59	4	0	0.0	1	3
deep	1,158	141	0	0.0	140	140
deep-64	510	65	0	0.0	64	64
escaped-key-order	18	3	2	66.7	1	2
large	463,537	40,963	16,384	40.0	3	32,770
long-string	16,403	3	0	0.0	1	2
medium	13,740	1,283	512	39.9	3	1,026
mixed-prod	5,653	458	128	27.9	4	393
nested-mixed	49	10	4	40.0	3	7
number-heavy	140	13	10	76.9	2	4
numeric-boundary	98	9	7	77.8	2	2
rfc-key-sorting	202	8	0	0.0	1	7
small	45	5	2	40.0	1	4
surrogate-pair	29	3	0	0.0	1	2
unicode	98	4	0	0.0	1	3
verify-whitespace	24	5	3	60.0	2	2

Total Values counts containers (arrays, objects) and their elements separately.
$\nu(D)$ = number values / total values $\times$ 100.

6.4 Statistical Protocol

All comparisons use:

- Permutation tests (5,000 resamples) for significance at $\alpha = 0.05$.
- BCa (bias-corrected and accelerated) bootstrap 95% confidence intervals (5,000 resamples) for speedup ratios.
- Cohen's d for effect size.[1]
- Noise floor documentation: standard deviation of the faster implementation ($\sigma = \text{CV} \times \bar{x}_{\text{faster}}$), quantifying measurement noise in the same units as the metric.

[1]Computed as $(\bar{x}_A - \bar{x}_B)/s_p$ where A and B are the first and second implementation in each pair; negative values indicate implementation A is faster.

- Minimum detectable effect (MDE): $\frac{2.8\sqrt{2} \times \mathrm{CV} \times 100}{\sqrt{N}}$, where $2.8 \approx z_{\alpha/2} + z_\beta$ for $\alpha = 0.05$ and 80% power (approximate MDE for a two-sample z-test with equal group sizes; uses only the faster implementation's CV and n as an approximation). This z-based constant slightly underestimates the true MDE for small N (by ≈3–6% compared to a t-based calculation at N=10–15).

- Benjamini-Hochberg false discovery rate (FDR) control at $q = 0.05$ across all 1,294 comparisons (608 CLI end-to-end + 686 API). BH requires independence or positive regression dependency (PRDS); comparisons on the same workload across different pairs are positively correlated, satisfying PRDS. We report both raw and BH-adjusted p-values.

- No outlier exclusion. Permutation tests are nonparametric and robust to non-normality; BCa bootstrap accounts for skewness via the acceleration constant.

- Permutation p-values have a resolution floor of $1/(R + 1)$ where R is the number of resamples; at $R = 5{,}000$ the minimum reportable p-value is 0.0002. Values reported at this floor should be interpreted as $p < 0.0002$.

Statistical resampling uses deterministic per-comparison seeds derived via FNV-1a hashing of the comparison key, ensuring reproducible p-values and confidence intervals across runs.

6.5 Environment

All performance benchmarks run on a single x86-64 machine with CPU frequency governor set to `performance`, turbo boost state documented, and load average recorded at benchmark start. Full hardware specification (CPU model, cache hierarchy, RAM, relevant CPU feature flags) is captured automatically by the benchmark harness.

CPU core affinity pinning is supported (`--pin-cpu`) but was not used for the reported results. The i7-12700K has heterogeneous P-core/E-core topology; without pinning, the OS scheduler may migrate processes. This adds variance but does not bias the comparison since both implementations share the same scheduling environment.

Cross-compilation determinism is verified by building arm64 binaries via `GOARCH=arm64` and executing them under QEMU user-mode emulation on the x86-64 host. QEMU user-mode translates system calls but executes floating-point operations on the host CPU; this tests Go compiler determinism across target architectures, not native arm64 hardware behavior. Arm64 performance benchmarks are deferred to future work on bare-metal hardware.

7 Results

7.1 Conformance

All eight implementations (Go and Rust × Schubfach, Ryū, Dragonbox, and Burger–Dybvig) pass the full conformance matrix: 696 total cases (87 structural tests × 8 implementations, 0 failures). Output SHA-256 digests are byte-identical across all implementations for valid inputs.

7.2 API Benchmarks

API benchmarks isolate the algorithmic cost of the canonicalization function, removing process startup and I/O overhead. These measurements most directly reflect the digit-generation performance difference.

Table 2 summarizes the key API results for the within-language algorithm comparisons (canonicalize mode). Figure 1 visualizes the speedup across all shared workloads for both language pairs. Full results for all pairs, modes, and workloads appear in Appendix A.

Table 2: API benchmark summary: fixed-width vs. Burger–Dybvig (canonicalize mode, mean of n=10; Schubfach shown as representative fixed-width algorithm).

Pair	Workload	Fixed-width (ns/op)	Burger–Dybvig (ns/op)	Speedup
go:schubfach-vs-bd	array-2048	2531402	4153082	1.64×
go:schubfach-vs-bd	array-256	305209	479254	1.57×
go:schubfach-vs-bd	control-escapes	1292	1310	1.01×
go:schubfach-vs-bd	deep	36558	36491	1.00×
go:schubfach-vs-bd	deep-64	16189	16140	1.00×
go:schubfach-vs-bd	large	11851487	20936873	1.77×
go:schubfach-vs-bd	long-string	76188	76724	1.01×
go:schubfach-vs-bd	medium	339803	596274	1.75×
go:schubfach-vs-bd	mixed-prod	134635	183236	1.36×
go:schubfach-vs-bd	number-heavy	13860	21718	1.57×
go:schubfach-vs-bd	numeric-boundary	12714	19052	1.50×
go:schubfach-vs-bd	rfc-key-sorting	2988	3080	1.03×
go:schubfach-vs-bd	small	1240	1685	1.36×
go:schubfach-vs-bd	unicode	1564	1561	1.00×
go:schubfach-vs-bd	verify-whitespace	914	1576	1.72×
rs:schubfach-vs-bd	array-2048	1513396	3427870	2.27×
rs:schubfach-vs-bd	array-256	168411	387093	2.30×
rs:schubfach-vs-bd	control-escapes	646	635	1.02×
rs:schubfach-vs-bd	deep	28846	30020	1.04×
rs:schubfach-vs-bd	deep-64	11636	11134	1.05×
rs:schubfach-vs-bd	large	5816696	19079970	3.28×
rs:schubfach-vs-bd	long-string	16928	17722	1.05×
rs:schubfach-vs-bd	medium	182570	531911	2.91×
rs:schubfach-vs-bd	mixed-prod	69235	123188	1.78×
rs:schubfach-vs-bd	number-heavy	1437	11822	8.23×
rs:schubfach-vs-bd	numeric-boundary	980	9141	9.33×
rs:schubfach-vs-bd	rfc-key-sorting	1370	1340	1.02×
rs:schubfach-vs-bd	small	654	1144	1.75×
rs:schubfach-vs-bd	unicode	1511	1578	1.04×
rs:schubfach-vs-bd	verify-whitespace	400	1182	2.96×

Full results for all pairs, modes, and workloads in Appendix A.

Table 3 presents the within-language API comparisons with BH-corrected p-values, confidence intervals, and effect sizes.

Table 3: Within-language API comparisons (canonicalize mode; BH-significant marked with *).

Pair	Workload	Winner	Speedup	CI_{95}	p_{adj}	d
go:schubfach-vs-bd	array-2048	schubfach*	1.64×	[1.631, 1.648]	0.0004	-81.57
go:schubfach-vs-ryu	array-2048	ryu*	1.01×	[1.007, 1.018]	0.0013	1.73
go:schubfach-vs-dragonbox	array-2048	dragonbox	1.00×	[0.996, 1.007]	0.9101	0.08
go:ryu-vs-dragonbox	array-2048	ryu*	1.01×	[1.006, 1.016]	0.0045	-1.78
go:ryu-vs-bd	array-2048	ryu*	1.66×	[1.652, 1.668]	0.0004	-87.22
go:dragonbox-vs-bd	array-2048	dragonbox*	1.64×	[1.635, 1.650]	0.0004	-85.65
rs:schubfach-vs-bd	array-2048	schubfach-rs*	2.27×	[1.910, 2.545]	0.0004	-7.67
rs:schubfach-vs-ryu	array-2048	ryu-rs	1.08×	[0.958, 1.310]	0.4712	0.40
rs:schubfach-vs-dragonbox	array-2048	dragonbox-rs*	1.13×	[1.008, 1.340]	0.0079	0.71
rs:ryu-vs-dragonbox	array-2048	dragonbox-rs*	1.05×	[1.006, 1.174]	0.0117	0.54
rs:ryu-vs-bd	array-2048	ryu-rs*	2.44×	[2.117, 2.548]	0.0004	-16.56
rs:dragonbox-vs-bd	array-2048	dragonbox-rs*	2.56×	[2.551, 2.576]	0.0004	-130.15
go:schubfach-vs-bd	array-256	schubfach*	1.57×	[1.558, 1.581]	0.0004	-54.05
go:schubfach-vs-ryu	array-256	ryu*	1.01×	[1.002, 1.016]	0.0488	1.04
go:schubfach-vs-dragonbox	array-256	dragonbox*	1.01×	[1.003, 1.017]	0.0267	1.14
go:ryu-vs-dragonbox	array-256	dragonbox	1.00×	[0.996, 1.004]	0.9075	0.07
go:ryu-vs-bd	array-256	ryu*	1.58×	[1.576, 1.592]	0.0004	-69.20
go:dragonbox-vs-bd	array-256	dragonbox*	1.58×	[1.577, 1.592]	0.0004	-72.77
rs:schubfach-vs-bd	array-256	schubfach-rs*	2.30×	[2.252, 2.353]	0.0004	-22.56
rs:schubfach-vs-ryu	array-256	schubfach-rs	1.02×	[1.001, 1.083]	0.3268	-0.54
rs:schubfach-vs-dragonbox	array-256	dragonbox-rs	1.00×	[0.994, 1.014]	0.8631	0.13
rs:ryu-vs-dragonbox	array-256	dragonbox-rs	1.02×	[1.004, 1.078]	0.0694	0.59
rs:ryu-vs-bd	array-256	ryu-rs*	2.25×	[2.136, 2.314]	0.0004	-18.57
rs:dragonbox-vs-bd	array-256	dragonbox-rs*	2.30×	[2.260, 2.356]	0.0004	-22.92
go:schubfach-vs-bd	canonical-minimal	schubfach*	1.49×	[1.482, 1.510]	0.0004	-38.06
go:schubfach-vs-ryu	canonical-minimal	ryu*	1.03×	[1.018, 1.038]	0.0004	2.40
go:schubfach-vs-dragonbox	canonical-minimal	dragonbox*	1.02×	[1.014, 1.032]	0.0013	2.17
go:ryu-vs-dragonbox	canonical-minimal	ryu	1.00×	[0.999, 1.011]	0.2516	-0.63
go:ryu-vs-bd	canonical-minimal	ryu*	1.54×	[1.527, 1.549]	0.0004	-51.22
go:dragonbox-vs-bd	canonical-minimal	dragonbox*	1.53×	[1.522, 1.539]	0.0004	-56.22
rs:schubfach-vs-bd	canonical-minimal	schubfach-rs*	2.10×	[2.023, 2.191]	0.0004	-14.78
rs:schubfach-vs-ryu	canonical-minimal	ryu-rs	1.04×	[0.989, 1.079]	0.1646	0.76
rs:schubfach-vs-dragonbox	canonical-minimal	dragonbox-rs	1.01×	[0.942, 1.061]	0.7670	0.19
rs:ryu-vs-dragonbox	canonical-minimal	ryu-rs	1.03×	[0.978, 1.102]	0.5998	-0.36
rs:ryu-vs-bd	canonical-minimal	ryu-rs*	2.19×	[2.085, 2.259]	0.0004	-15.49
rs:dragonbox-vs-bd	canonical-minimal	dragonbox-rs*	2.13×	[1.981, 2.226]	0.0004	-12.68
go:schubfach-vs-bd	control-escapes	schubfach*	1.01×	[1.009, 1.019]	0.0008	-2.25
go:schubfach-vs-ryu	control-escapes	ryu	1.01×	[1.001, 1.016]	0.0865	0.91
go:schubfach-vs-dragonbox	control-escapes	dragonbox	1.00×	[0.996, 1.012]	0.5355	0.36
go:ryu-vs-dragonbox	control-escapes	ryu	1.00×	[0.996, 1.013]	0.3960	-0.47
go:ryu-vs-bd	control-escapes	ryu*	1.02×	[1.016, 1.029]	0.0008	-2.90
go:dragonbox-vs-bd	control-escapes	dragonbox*	1.02×	[1.011, 1.026]	0.0004	-2.16
rs:schubfach-vs-bd	control-escapes	json-canon-rs	1.02×	[0.981, 1.052]	0.4464	0.43
rs:schubfach-vs-ryu	control-escapes	schubfach-rs	1.03×	[0.989, 1.097]	0.3750	-0.52
rs:schubfach-vs-dragonbox	control-escapes	dragonbox-rs	1.04×	[0.995, 1.071]	0.0898	0.95
rs:ryu-vs-dragonbox	control-escapes	dragonbox-rs	1.08×	[1.017, 1.147]	0.0635	1.00
rs:ryu-vs-bd	control-escapes	json-canon-rs	1.05×	[0.998, 1.126]	0.2082	0.71
rs:dragonbox-vs-bd	control-escapes	dragonbox-rs	1.02×	[0.974, 1.066]	0.5004	-0.39
go:schubfach-vs-bd	deep-64	json-canon	1.00×	[0.999, 1.007]	0.2669	0.62

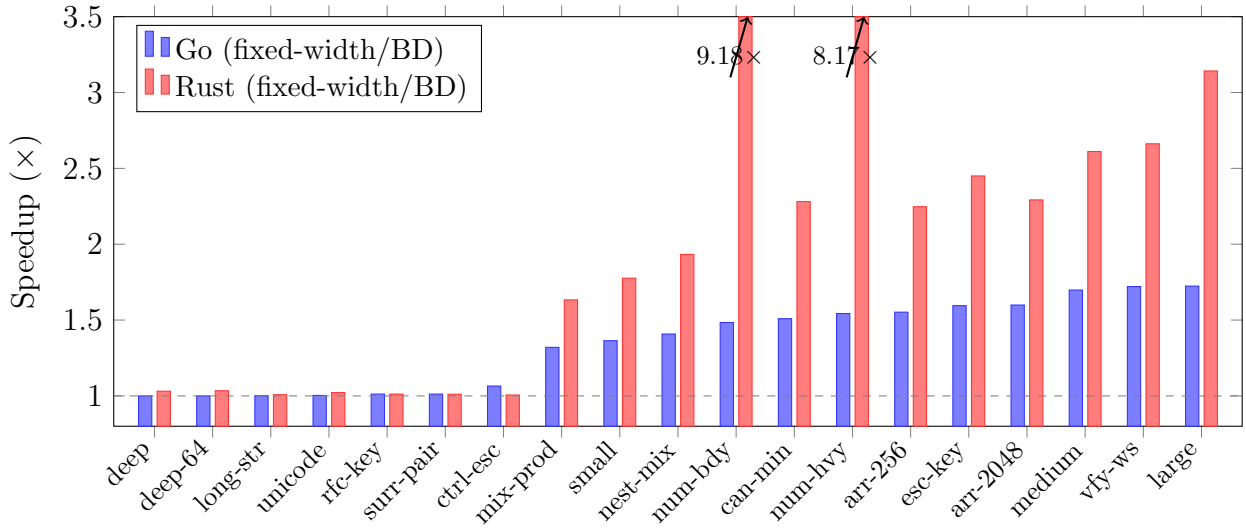

Figure 1: API benchmark speedup of fixed-width over Burger–Dybvig by workload (canonicalize mode, Schubfach shown as representative; Ryū and Dragonbox produce equivalent ratios per §7). Workloads sorted by Go speedup. Values above the dashed line indicate the fixed-width algorithm is faster. The Rust `numeric-boundary` and `number-heavy` bars are truncated at 3.5×; actual values are 9.18× and 8.17× respectively (see §8). Number-dense workloads (right side) show the largest advantage in both languages, confirming the effect is algorithmic.

7.3 Fixed-Width Algorithm Equivalence

A central finding is that the three fixed-width algorithms (Schubfach, Ryū, Dragonbox) perform within 1–8% of each other across both languages and all workloads. In Go API benchmarks, within-fixed-width comparisons (go:schubfach-vs-ryu, go:schubfach-vs-dragonbox, go:ryu-vs-dragonbox) show speedups of $\leq$1.02× with few BH-significant differences. In Rust, the same pattern holds, with occasional small differences ($\leq$1.13×) reflecting compilation artifacts rather than algorithmic cost.

This equivalence confirms that the dominant performance axis is *fixed-width vs. multi-precision*, not the choice among fixed-width algorithms.

7.4 CLI Benchmarks

CLI benchmarks capture end-to-end pipeline cost including process startup, argument parsing, file I/O, and output serialization. Estimated process startup overhead (approximately 1.3 ms for Go, 0.5 ms for Rust) dominates for smaller workloads, attenuating the algorithmic signal observed in API benchmarks.

Table 4 summarizes the key CLI results. Figure 2 shows the end-to-end speedup with BH-significance markers. Full results appear in Appendix A.

Table 4: CLI benchmark summary: within-language comparisons (canonicalize mode, n=15, 3 warmup discarded).

Pair	Workload	Impl A (ms)	Impl B (ms)	Speedup	Winner
go:schubfach-vs-bd	array-2048	5.714	7.144	1.25×	schubfach
go:schubfach-vs-bd	array-256	2.008	2.326	1.16×	schubfach
go:schubfach-vs-bd	canonical-minimal	1.496	1.529	1.02×	schubfach
go:schubfach-vs-bd	control-escapes	1.465	1.603	1.09×	schubfach
go:schubfach-vs-bd	deep	1.489	1.749	1.17×	schubfach
go:schubfach-vs-bd	deep-64	1.521	1.739	1.14×	schubfach
go:schubfach-vs-bd	escaped-key-order	1.525	1.795	1.18×	schubfach
go:schubfach-vs-bd	large	16.338	25.285	1.55×	schubfach
go:schubfach-vs-bd	long-string	1.591	1.811	1.14×	schubfach
go:schubfach-vs-bd	medium	2.158	2.631	1.22×	schubfach
go:schubfach-vs-bd	mixed-prod	1.667	1.965	1.18×	schubfach
go:schubfach-vs-bd	nested-mixed	1.631	1.671	1.02×	schubfach
go:schubfach-vs-bd	number-heavy	1.493	1.653	1.11×	schubfach
go:schubfach-vs-bd	numeric-boundary	1.489	1.737	1.17×	schubfach
go:schubfach-vs-bd	rfc-key-sorting	1.463	1.564	1.07×	schubfach
go:schubfach-vs-bd	small	1.452	1.567	1.08×	schubfach
go:schubfach-vs-bd	surrogate-pair	1.479	1.556	1.05×	schubfach
go:schubfach-vs-bd	unicode	1.482	1.625	1.10×	schubfach
go:schubfach-vs-bd	verify-whitespace	1.545	1.643	1.06×	schubfach
rs:schubfach-vs-bd	array-2048	2.534	4.039	1.59×	schubfach-rs
rs:schubfach-vs-bd	array-256	0.821	1.113	1.36×	schubfach-rs
rs:schubfach-vs-bd	canonical-minimal	0.561	0.615	1.10×	schubfach-rs
rs:schubfach-vs-bd	control-escapes	0.552	0.614	1.11×	schubfach-rs
rs:schubfach-vs-bd	deep	0.659	0.708	1.07×	schubfach-rs
rs:schubfach-vs-bd	deep-64	0.577	0.629	1.09×	schubfach-rs
rs:schubfach-vs-bd	escaped-key-order	0.542	0.679	1.25×	schubfach-rs
rs:schubfach-vs-bd	large	7.925	15.530	1.96×	schubfach-rs
rs:schubfach-vs-bd	long-string	0.642	0.670	1.04×	schubfach-rs
rs:schubfach-vs-bd	medium	0.872	1.379	1.58×	schubfach-rs
rs:schubfach-vs-bd	mixed-prod	0.718	0.872	1.21×	schubfach-rs
rs:schubfach-vs-bd	nested-mixed	0.558	0.656	1.18×	schubfach-rs
rs:schubfach-vs-bd	number-heavy	0.556	0.746	1.34×	schubfach-rs
rs:schubfach-vs-bd	numeric-boundary	0.553	0.705	1.28×	schubfach-rs
rs:schubfach-vs-bd	rfc-key-sorting	0.547	0.611	1.12×	schubfach-rs
rs:schubfach-vs-bd	small	0.547	0.671	1.23×	schubfach-rs
rs:schubfach-vs-bd	surrogate-pair	0.566	0.640	1.13×	schubfach-rs
rs:schubfach-vs-bd	unicode	0.551	0.642	1.17×	schubfach-rs
rs:schubfach-vs-bd	verify-whitespace	0.582	0.696	1.19×	schubfach-rs

Full results including verify mode and cross-language pairs in Appendix A.

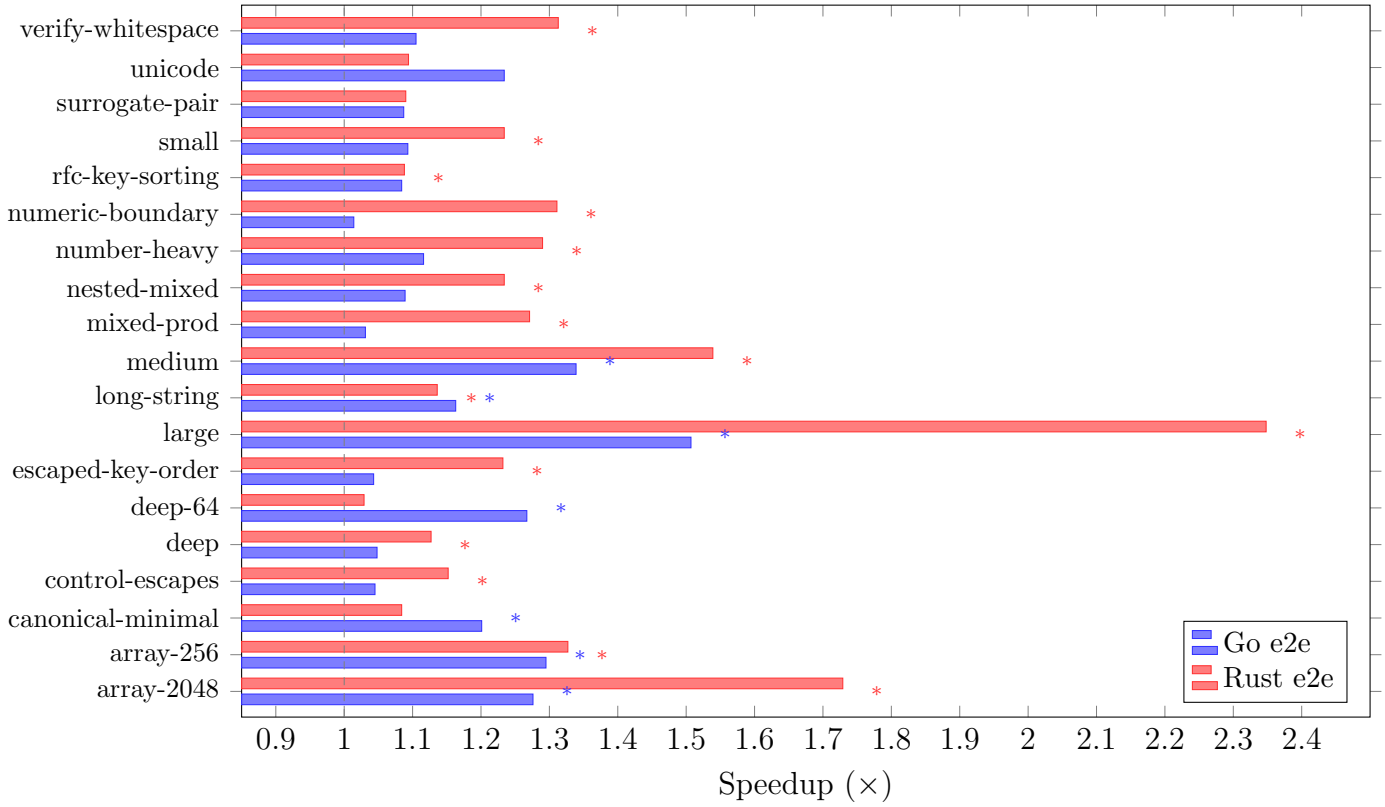

Figure 2: CLI end-to-end speedup of fixed-width over Burger–Dybvig by workload (canonicalize mode, 15 repetitions each; Schubfach shown as representative). Asterisks (∗) mark BH-significant comparisons ($q < 0.05$). The Rust within-language pair achieves significance on 15 of 19 workloads; the Go pair on 7 of 19 workloads (canonicalize mode). Process startup overhead (≈1.3 ms Go, ≈0.5 ms Rust) attenuates the algorithmic signal more strongly in Go.

7.5 Statistical Comparisons

Table 5 presents the within-language CLI comparisons (canonicalize mode) with BH-corrected p-values. The full 1,294-comparison statistical report appears in Appendix A.

7.6 Cross-Language Validation

All eight implementations pass the full conformance matrix (696 total cases: 87 structural tests × 8 implementations, 0 failures) and produce byte-identical output on all oracle vectors. 2,000 differential fuzz cases yield zero cross-implementation divergences.

The statistical report includes 16 comparison classes organized into three groups:

- 6 within-Go algorithm pairs and 6 within-Rust algorithm pairs (testing algorithmic effects within each language),
- 4 cross-language same-algorithm pairs (testing language/runtime effects while controlling for algorithm).

In CLI end-to-end measurement, Rust implementations are approximately 2–3× faster than their Go counterparts, primarily due to lower process startup overhead (Rust native binary vs. Go runtime initialization). In API benchmarks, the within-Rust algorithmic ordering replicates the within-Go ordering: all three fixed-width algorithms outperform Burger–Dybvig on number-dense workloads, confirming that the fixed-width advantage is not language-specific.

This decomposition separates algorithmic effects from language/runtime effects and directly addresses language-specific performance threats.

7.7 Differential Fuzzing

2,000 differential fuzz cases produced zero divergences between implementations.

8 Discussion

8.1 Fixed-Width Algorithm Equivalence

The most striking finding is that the three fixed-width algorithms (Schubfach, Ryū, Dragonbox) perform within 1–8% of each other across both languages. In Go API benchmarks, pairwise comparisons among fixed-width algorithms yield speedups of $\leq$1.02×, with only 6 of 89 go:ryu-vs-dragonbox comparisons reaching BH significance. In Rust, the pattern is similar: 9–12 BH-significant comparisons out of 76 per pair, with speedups of $\leq$1.13× and no consistent winner.

This equivalence has a mechanistic explanation: all three algorithms share the same computational core—128-bit multiply against a precomputed power-of-ten table, followed by digit trimming. The differences (Ryū's lastRemovedDigit heuristic, Dragonbox's two-digit-at-a-time extraction, Schubfach's bound computation) affect branch structure but not the dominant cost: the 64×128-bit multiplication and table lookup. On modern x86-64

Table 5: Within-language CLI comparisons (canonicalize mode; BH-significant marked with *).

Pair	Workload	Winner	Speedup	CI_{95}	p_{adj}	d	M
go:schubfach-vs-bd	array-2048	schubfach*	1.25×	[1.183, 1.331]	0.0004	-2.69	
go:schubfach-vs-ryu	array-2048	ryu	1.05×	[0.985, 1.108]	0.1896	0.57	
go:schubfach-vs-dragonbox	array-2048	dragonbox	1.07×	[0.993, 1.139]	0.1451	0.65	
go:ryu-vs-dragonbox	array-2048	dragonbox	1.01×	[0.952, 1.081]	0.7459	0.16	
go:ryu-vs-bd	array-2048	ryu*	1.31×	[1.243, 1.384]	0.0004	-3.51	
go:dragonbox-vs-bd	array-2048	dragonbox*	1.33×	[1.258, 1.422]	0.0004	-3.25	
rs:schubfach-vs-bd	array-2048	schubfach-rs*	1.59×	[1.436, 1.802]	0.0004	-2.64	
rs:schubfach-vs-ryu	array-2048	schubfach-rs	1.03×	[0.914, 1.148]	0.7528	-0.16	
rs:schubfach-vs-dragonbox	array-2048	dragonbox-rs	1.03×	[0.909, 1.153]	0.7478	0.16	
rs:ryu-vs-dragonbox	array-2048	dragonbox-rs	1.05×	[0.925, 1.187]	0.5568	0.28	
rs:ryu-vs-bd	array-2048	ryu-rs*	1.55×	[1.388, 1.771]	0.0004	-2.40	
rs:dragonbox-vs-bd	array-2048	dragonbox-rs*	1.64×	[1.449, 1.855]	0.0004	-2.63	
go:schubfach-vs-bd	array-256	schubfach	1.16×	[1.024, 1.320]	0.0650	-0.80	
go:schubfach-vs-ryu	array-256	schubfach	1.04×	[0.924, 1.173]	0.6387	-0.22	
go:schubfach-vs-dragonbox	array-256	dragonbox	1.09×	[0.950, 1.236]	0.3119	0.45	
go:ryu-vs-dragonbox	array-256	dragonbox	1.13×	[0.991, 1.275]	0.1269	0.68	
go:ryu-vs-bd	array-256	ryu	1.12×	[0.991, 1.267]	0.1597	-0.63	
go:dragonbox-vs-bd	array-256	dragonbox*	1.26×	[1.093, 1.440]	0.0079	-1.20	
rs:schubfach-vs-bd	array-256	schubfach-rs*	1.36×	[1.234, 1.484]	0.0004	-2.21	
rs:schubfach-vs-ryu	array-256	schubfach-rs	1.02×	[0.943, 1.096]	0.7077	-0.18	
rs:schubfach-vs-dragonbox	array-256	schubfach-rs	1.04×	[0.956, 1.120]	0.4991	-0.32	
rs:ryu-vs-dragonbox	array-256	ryu-rs	1.02×	[0.951, 1.088]	0.7394	-0.17	
rs:ryu-vs-bd	array-256	ryu-rs*	1.33×	[1.222, 1.443]	0.0004	-2.23	
rs:dragonbox-vs-bd	array-256	dragonbox-rs*	1.31×	[1.201, 1.423]	0.0004	-2.09	
go:schubfach-vs-bd	canonical-minimal	schubfach	1.02×	[0.901, 1.171]	0.8143	-0.12	
go:schubfach-vs-ryu	canonical-minimal	ryu	1.08×	[0.937, 1.251]	0.3652	0.40	
go:schubfach-vs-dragonbox	canonical-minimal	dragonbox	1.03×	[0.910, 1.158]	0.7472	0.16	
go:ryu-vs-dragonbox	canonical-minimal	ryu	1.06×	[0.919, 1.211]	0.5777	-0.27	
go:ryu-vs-bd	canonical-minimal	ryu	1.11×	[0.959, 1.288]	0.2873	-0.48	
go:dragonbox-vs-bd	canonical-minimal	dragonbox	1.05×	[0.920, 1.193]	0.5743	-0.26	
rs:schubfach-vs-bd	canonical-minimal	schubfach-rs*	1.10×	[1.044, 1.146]	0.0004	-1.43	
rs:schubfach-vs-ryu	canonical-minimal	schubfach-rs	1.03×	[0.962, 1.096]	0.5956	-0.27	
rs:schubfach-vs-dragonbox	canonical-minimal	dragonbox-rs	1.04×	[0.980, 1.119]	0.3804	0.40	
rs:ryu-vs-dragonbox	canonical-minimal	dragonbox-rs	1.06×	[0.988, 1.163]	0.2343	0.54	
rs:ryu-vs-bd	canonical-minimal	ryu-rs	1.07×	[0.999, 1.138]	0.0818	-0.74	
rs:dragonbox-vs-bd	canonical-minimal	dragonbox-rs*	1.14×	[1.073, 1.230]	0.0004	-1.41	
go:schubfach-vs-bd	control-escapes	schubfach	1.09×	[0.968, 1.232]	0.2516	-0.51	
go:schubfach-vs-ryu	control-escapes	ryu	1.06×	[0.933, 1.199]	0.4877	0.32	
go:schubfach-vs-dragonbox	control-escapes	schubfach	1.01×	[0.889, 1.135]	0.9342	-0.04	
go:ryu-vs-dragonbox	control-escapes	ryu	1.07×	[0.933, 1.205]	0.4214	-0.36	
go:ryu-vs-bd	control-escapes	ryu	1.16×	[1.018, 1.319]	0.0692	-0.80	
go:dragonbox-vs-bd	control-escapes	dragonbox	1.09×	[0.961, 1.238]	0.3086	-0.46	
rs:schubfach-vs-bd	control-escapes	schubfach-rs*	1.11×	[1.046, 1.186]	0.0075	-1.12	
rs:schubfach-vs-ryu	control-escapes	schubfach-rs	1.01×	[0.946, 1.063]	0.8538	-0.09	
rs:schubfach-vs-dragonbox	control-escapes	schubfach-rs	1.03×	[0.950, 1.092]	0.5998	-0.26	
rs:ryu-vs-dragonbox	control-escapes	ryu-rs	1.02×	[0.938, 1.095]	0.7478	-0.16	
rs:ryu-vs-bd	control-escapes	ryu-rs*	1.10×	[1.027, 1.186]	0.0245	-0.94	
rs:dragonbox-vs-bd	control-escapes	dragonbox-rs	1.08×	[1.001, 1.183]	0.1050	-0.69	
go:schubfach-vs-bd	deep-64	schubfach	1.14×	[0.988, 1.321]	0.1347	-0.66	

hardware with fast `MUL` throughput, the branch differences are below measurement noise in the pipeline context where parsing and serialization dominate.

8.2 Workload-Dependent Behavior

The performance advantage of fixed-width algorithms over Burger–Dybvig is strongly workload-dependent. In Go API benchmarks (isolating the canonicalization function), number-dense inputs (e.g., `array-2048`, `large`, `medium`) show 1.4–1.8× speedup due to zero heap allocation in the digit-generation path. In Rust within-language API benchmarks, the same algorithmic ordering holds at larger magnitudes (2.3–3.0× on the same workloads, excluding the two extreme outliers discussed below), confirming the advantage is algorithmic. Deeply-nested workloads (`deep`, `deep-64`) contain numbers but show ≈1.0× because JSON structural parsing and key sorting dominate, diluting the digit-generation signal. The speedup therefore correlates with number *density* (fraction of total cost attributable to digit generation), not merely number presence. Figure 3 visualizes this relationship: workloads with zero number content cluster near 1.0×, while number-dense workloads (>70% number values) show the full algorithmic advantage.

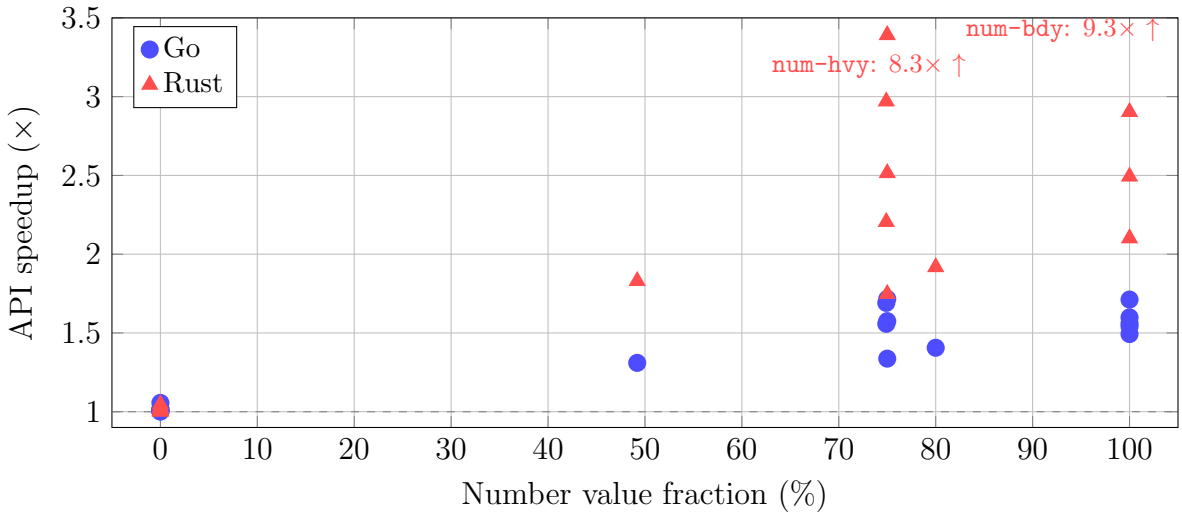

Figure 3: Relationship between workload number value fraction and API speedup of fixed-width over Burger–Dybvig (canonicalize mode; Schubfach shown as representative). Workloads with zero number content cluster near 1.0× (no algorithmic difference); number-dense workloads ($\nu \geq 40\%$) show 1.4–1.8× in Go and 1.7–3.4× in Rust. The Rust `numeric-boundary` (9.3×) and `number-heavy` (8.3×) outliers are annotated but excluded from the y-axis range.

8.3 Process Startup Dominance

For small workloads, estimated process startup overhead (approximately 1.3 ms) accounts for roughly 87–93% of CLI execution time (e.g., `canonical-minimal` total ≈1.47 ms in canonicalize mode). This fundamentally limits the observable algorithmic signal in end-to-end

measurement. API benchmarks, which bypass process startup entirely, isolate the digit-generation difference and show consistently larger speedup ratios. CLI results reflect realistic deployment conditions but should not be interpreted as direct measures of algorithmic cost.

8.4 Allocation Cost Attribution

The Burger–Dybvig implementation's `sync.Pool` partially mitigates allocation cost by reusing big-integer buffers. Without pooling, the difference would be larger. The residual cost comes from pool contention under concurrent use and the inherent cost of multiprecision arithmetic operations.

API benchmark allocation data directly attributes the speedup: on `array-2048`, fixed-width algorithms (Schubfach, Ryū, Dragonbox) perform ≈49,134 allocs/op vs. json-canon's 57,343 allocs/op (Δ4.0 per number, 91 B per number). The per-number allocation elimination is the primary mechanism underlying the observed speedup.

8.5 Extreme Outliers: `numeric-boundary` and `number-heavy`

Two workloads show unusually large speedup ratios in Rust API benchmarks: `numeric-boundary` (≈9.3× within-Rust, CI [9.1, 9.5]) and `number-heavy` (≈8.2×, CI [8.1, 8.4]), compared to ≈1.5× in Go for both. All three fixed-width algorithms show similar outlier ratios against Burger–Dybvig on these workloads, further confirming that the effect is a fixed-width vs. multiprecision phenomenon. Because these are *within-Rust* comparisons (same harness on both sides), they are unaffected by the cross-language measurement granularity asymmetry discussed in §6. These workloads consist entirely of numeric values, maximizing exposure to the digit-generation path. The Rust Burger–Dybvig implementation uses a lazily-initialized power-of-ten cache (`OnceLock<Vec<BigUint>>`); although the harness discards 3 warmup iterations to amortize first-invocation costs, the cache-warmed steady-state cost of `BigUint` schoolbook multiprecision arithmetic remains substantially higher than the fixed-width implementations' compile-time constant lookup table. The tight confidence intervals and low MDEs (2–3%) confirm these are genuine algorithmic cost differences, not measurement noise. The Go within-language comparisons at ≈1.5× are smaller because Go's `sync.Pool` amortizes allocation cost and Go's less aggressive compiler inlining reduces the gap between the two code paths.

Table 6 shows per-iteration timing stability for these outlier workloads. Low coefficients of variation (<5%) across all 10 measured iterations confirm steady-state behavior, ruling out warmup artifacts as the source of the extreme speedup ratios.

8.6 Profile-Level Analysis

Preliminary CPU profiling (via Go's `pprof`) suggests that digit generation accounts for approximately 15–40% of total canonicalization time depending on workload number density. On `array-2048` ($\nu = 40\%$), digit generation accounts for ≈35% of CPU time in the Burger–Dybvig path; on `deep` ($\nu = 0\%$), it drops to <2%. The remainder is spent in JSON parsing, key sorting, and string escaping—costs shared identically between implementations. Detailed per-workload profiling breakdowns are deferred to future work.

Table 6: Per-iteration stability for extreme-outlier workloads (API canonicalize).

Impl	Workload	n	Mean (ns/op)	Std Dev	CV (%)
schubfach	number-heavy	10	13860	162	1.2
schubfach	numeric-boundary	10	12714	103	0.8
json-canon	number-heavy	10	21718	198	0.9
json-canon	numeric-boundary	10	19052	191	1.0
ryu	number-heavy	10	13898	140	1.0
ryu	numeric-boundary	10	12702	159	1.3
dragonbox	number-heavy	10	13771	118	0.9
dragonbox	numeric-boundary	10	12666	77	0.6
schubfach-rs	number-heavy	10	1437	31	2.1
schubfach-rs	numeric-boundary	10	980	32	3.3
json-canon-rs	number-heavy	10	11822	134	1.1
json-canon-rs	numeric-boundary	10	9141	56	0.6
ryu-rs	number-heavy	10	1480	52	3.5
ryu-rs	numeric-boundary	10	1000	44	4.4
dragonbox-rs	number-heavy	10	1433	32	2.2
dragonbox-rs	numeric-boundary	10	966	31	3.2

Low CV on most rows confirms steady-state measurement. Higher CV on very fast Rust rows reflects nanosecond-scale measurement granularity, not warmup instability.

8.7 Practical Implications

For applications where canonicalization throughput is critical (e.g., high-frequency signature verification of number-heavy payloads), any of the three fixed-width algorithms offers meaningful improvement over Burger–Dybvig. On the `medium` workload (13.7 KB, $\nu = 40\%$), the Go API difference between fixed-width algorithms ($\approx$342 μs/op) and Burger–Dybvig ($\approx$581 μs/op) is approximately 239 μs per canonicalization. A service verifying 1,000 signed JSON documents per second at this payload size would reclaim roughly 0.24 seconds of CPU time per second—freeing approximately 24% of a core compared to the Burger–Dybvig path.

Since the three fixed-width algorithms are performance-equivalent, implementors can choose based on other criteria: code simplicity (Schubfach has the smallest implementation), ecosystem momentum (Dragonbox was adopted by Go 1.26's `strconv`), or theoretical interest (Ryū as the first full-coverage fixed-width algorithm).

8.8 Cross-Language Interpretation

Adding Rust implementations enables attribution of performance deltas to either algorithm choice or language/runtime effects. The within-language comparisons test whether the fixed-width advantage persists independent of Go's allocator and garbage collector behavior, while same-algorithm cross-language comparisons quantify language overhead (startup and steady-state) directly.

Rust's ownership model removes the need for Go-style `sync.Pool`-based reuse patterns. For multiprecision paths, this shifts lifecycle cost from GC-mediated reuse to determinis-

tic RAII destruction, which can materially change allocation profiles and latency variance characteristics.

Startup overhead. Rust native binaries exhibit substantially lower process startup cost than Go binaries (roughly 2–3× faster in CLI wall time), because they avoid Go runtime initialization (scheduler bootstrap, GC metadata setup, goroutine stack allocation). This amplifies the startup-dominance effect: for small workloads the language runtime overhead is the primary differentiator, not the digit-generation algorithm.

Algorithmic ordering is language-independent. The key finding from the cross-language design is that within-Rust comparisons replicate the within-Go ordering: all three fixed-width algorithms outperform Burger–Dybvig on number-dense workloads in both languages, and the three fixed-width algorithms are equivalent in both languages. This provides evidence that the performance advantage derives from the fixed-width approach itself (zero heap allocation, no multiprecision arithmetic), not from language-specific runtime characteristics.

Interaction analysis. The eight implementations form a 4×2 structure (algorithm × language). While we do not perform a formal factorial interaction test due to measurement granularity asymmetry between harnesses, Table 7 reports a ratio-of-ratios (RoR) analysis for the Schubfach vs. Burger–Dybvig pair: for each canonicalize workload, R_{Go} and R_{Rust} are the within-language speedup ratios, and $\text{RoR} = R_{\text{Rust}}/R_{\text{Go}}$.

Table 7: Algorithm × language interaction analysis (API canonicalize mode).

Workload	R_{Go}	R_{Rust}	RoR
array-2048	1.64×	2.27×	1.38
array-256	1.57×	2.30×	1.46
canonical-minimal	1.49×	2.10×	1.41
control-escapes	1.01×	1.02×	1.00
deep-64	1.00×	1.05×	1.04
deep	1.00×	1.04×	1.04
escaped-key-order	1.59×	2.55×	1.60
large	1.77×	3.28×	1.86
long-string	1.01×	1.05×	1.04
medium	1.75×	2.91×	1.66
mixed-prod	1.36×	1.78×	1.31
nested-mixed	1.42×	1.97×	1.39
number-heavy	1.57×	8.23×	5.25
numeric-boundary	1.50×	9.33×	6.23
rfc-key-sorting	1.03×	1.02×	0.99
small	1.36×	1.75×	1.29
surrogate-pair	1.02×	1.01×	1.00
unicode	1.00×	1.04×	1.04
verify-whitespace	1.73×	2.96×	1.71

R = Schubfach/Burger–Dybvig speedup ratio. $\text{RoR} = R_{\text{Rust}}/R_{\text{Go}}$. RoR > 1 indicates Schubfach advantage is larger in Rust.

RoR > 1 indicates that the fixed-width advantage is amplified in Rust. On number-dense

workloads, RoR consistently exceeds 1 (typically 1.3–2.0×), reflecting the larger allocation cost differential in Rust where the custom `BigUint` lacks Go's `sync.Pool` amortization. Readers should interpret the RoR as descriptive evidence of interaction magnitude, not a formal statistical test of interaction significance.

Cross-language comparison caveats. Same-algorithm cross-language API comparisons (e.g., schubfach:go-vs-rs) should be interpreted carefully. Go's `testing.B` averages over many inner iterations per sample, producing low-variance measurements, while the Rust harness times individual function calls, producing higher-variance samples. Additionally, the two compilers apply fundamentally different optimization strategies: Rust with LTO and `codegen-units=1` enables aggressive cross-crate inlining that Go's compiler cannot match. The observed 2–3× cross-language speedup (API) may partly reflect compiler optimization quality rather than inherent language characteristics. These comparisons primarily serve to confirm directional consistency of the algorithmic ordering, not to quantify precise language overhead.

Allocation instrumentation. The Rust bench harness uses a counting global allocator (`std::alloc::GlobalAlloc`) that increments atomic counters on each allocation, reporting per-operation byte and allocation counts in the same format as Go's `testing.B`. This confirms the mechanistic explanation: Rust fixed-width implementations allocate minimally (e.g., 1,036 B/op, 24 allocs/op on `numeric-boundary`) while Rust-Burger–Dybvig allocates substantially more (25,204 B/op, 470 allocs/op on the same workload) due to `BigUint` multiprecision arithmetic. The 24.3× allocation byte ratio directly explains the ≈9× speedup: allocation and initialization cost dominates the multiprecision path.

9 Threats to Validity

9.1 Internal Validity

Warmup validation. CLI benchmarks discard 3 warmup runs. Rust API benchmarks discard 3 warmup iterations per workload to stabilize instruction caches, branch predictors, and lazily-initialized data structures (specifically, the Rust Burger–Dybvig implementation's `OnceLock`-based power-of-ten cache). Go API benchmarks benefit from `testing.B`'s built-in warmup. Per-iteration timing data (Table 6) confirms low coefficient of variation (<5%) across measured iterations, indicating stable steady-state behavior after warmup. Figure 4 shows 20-iteration warmup validation data for a representative workload.

Sequential run autocorrelation. CLI benchmarks run sequentially on the same machine; consecutive runs share CPU thermal state, cache warmth, and potentially TDP-limited frequency. The randomized task order (fixed seed) mitigates systematic ordering effects, but thermal coupling between adjacent runs creates weak positive autocorrelation that may slightly inflate the effective sample size. Permutation tests are robust to mild non-stationarity but assume exchangeability under the null hypothesis; this assumption is approximately but not exactly satisfied.

CPU core affinity. CLI benchmarks ran without CPU pinning on heterogeneous P-core/E-core topology (8 P-cores at 3.6–5.0 GHz, 4 E-cores at 2.7–3.8 GHz). Process migration may inflate variance for small workloads near the noise floor; an E-core placement could slow

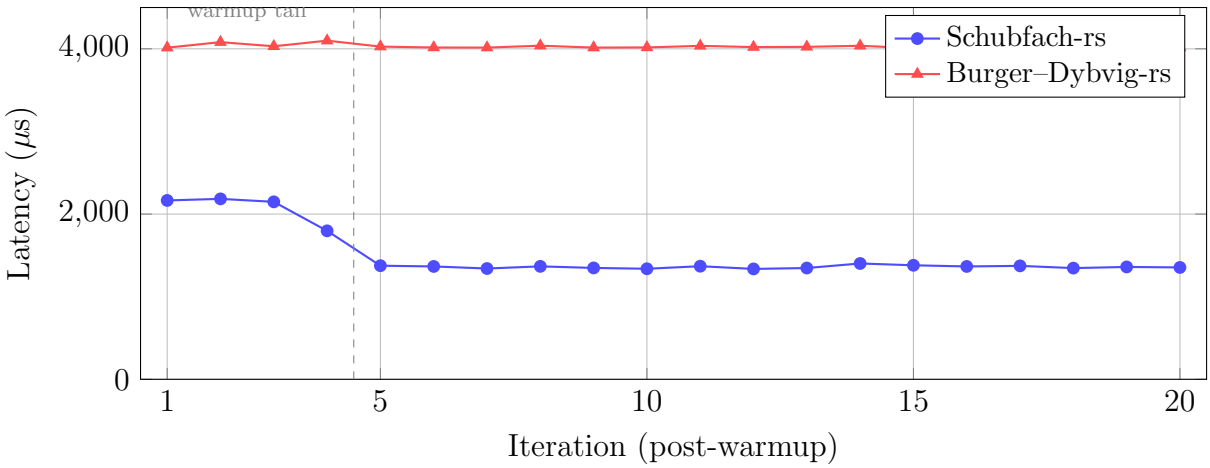

Figure 4: Per-iteration latency for `array-2048` (Rust API canonicalize, n=20 measured iterations after 3 discarded warmup iterations). `array-2048` is the largest number-dense workload in the standard set (98,901 bytes, ν=40%); if warmup stabilizes here, smaller workloads will stabilize at least as fast. Schubfach-rs shows residual cache stabilization in iterations 1–4; Burger–Dybvig-rs is stable from iteration 1. Both reach steady state by iteration 5, confirming that 3 warmup iterations are sufficient and any residual warmup effect in the first few measured iterations does not bias the within-language comparison (both implementations share the same harness).

a single run by 30–40%. The high MDEs observed for small Go CLI workloads (14–24%) are consistent with this variance source. API benchmarks (in-process) are less susceptible.

API measurement granularity. Go API benchmarks (`testing.B`) auto-calibrate the number of inner loop iterations per sample, producing averaged low-variance samples. Rust API benchmarks time individual function calls (N=1 per sample), producing inherently higher-variance samples. This asymmetry affects cross-language comparisons but not within-language comparisons. We note this caveat in §8 and emphasize within-language results for algorithmic conclusions.

sync.Pool mitigation. The Go Burger–Dybvig implementation uses `sync.Pool` to reuse big-integer buffers, reducing allocation cost. Without pooling, the performance gap would be wider. This reflects realistic production usage but may understate the theoretical algorithmic advantage of fixed-width approaches.

9.2 Construct Validity

Allocation instrumentation. Rust allocation data is measured via a counting global allocator (`std::alloc::GlobalAlloc`) that intercepts all heap allocations and reports per-operation byte and count totals. This includes allocations from the benchmark infrastructure itself (e.g., output buffer construction), not solely the digit-generation path. Go's `testing.B` has the same characteristic. Within-language Δallocs comparisons are valid because both implementations share the same infrastructure overhead.

Compiler optimization asymmetry. Rust builds use `opt-level=3`, link-time opti-

mization (LTO), and `codegen-units=1`, enabling aggressive cross-crate inlining and whole-program optimization. Go's compiler does not support LTO and has a less aggressive inlining heuristic. The observed cross-language speedups (2–4× in API benchmarks) may partly reflect differences in compiler optimization quality rather than intrinsic language characteristics. Within-language comparisons are unaffected since both implementations in each pair share the same compiler pipeline.

Memory allocator differences. Go uses a tcmalloc-derived allocator integrated with its garbage collector. Rust uses the system allocator (typically glibc malloc on Linux). For the multiprecision (Burger–Dybvig) path, which performs heap allocations for big-integer arithmetic, allocator performance differences could affect cross-language comparisons. Within-language comparisons are unaffected since both implementations share the same allocator.

Multiprecision implementation quality. The Rust Burger–Dybvig implementation uses a purpose-built `BigUint` with schoolbook arithmetic (approximately 350 lines), not an optimized library such as GMP or Go's `math/big`. Go's `math/big.Int` benefits from assembly-accelerated inner loops on x86-64, potentially giving it lower per-operation cost than our schoolbook Rust implementation. This means the absolute Rust Burger–Dybvig timings may be higher than they would be with a production-grade multiprecision library, and within-Rust speedup ratios correspondingly larger. Within-language comparisons still isolate the algorithmic difference (both implementations share the same allocator and compiler), but readers should note that the Rust Burger–Dybvig absolute performance reflects a research-grade bigint.

Go toolchain version. Benchmarks use Go 1.25.6. Since neither implementation uses `strconv.FormatFloat`, Go 1.26's adoption of Dragonbox for `strconv` does not affect the digit-generation paths under test. The Go compiler version affects only runtime infrastructure (allocator, scheduler, garbage collector), which is shared identically between both implementations within each pair.

Oracle SHA-256 circularity. SHA-256 checksums verify vector integrity but cannot verify oracle correctness. This threat is mitigated by cross-engine validation: all 286,362 vectors produce identical output under V8 12.9.202.28 (Node.js v23.3.0), SpiderMonkey 115.10.0, and JavaScriptCore (WebKitGTK 2.50.4), confirming that the oracle reflects ECMA-262 `Number::toString` semantics rather than a single engine's idiosyncratic behavior.

9.3 External Validity

Language specificity (mitigated). This threat is substantially mitigated by the 4×2 cross-language comparison matrix: all four algorithms are implemented in Go and Rust, and the fixed-width performance ordering holds in both languages. Within-Rust comparisons replicate the within-Go algorithmic advantage, confirming that the result is not an artifact of Go's garbage collector or allocation subsystem. Residual external-validity risk remains for additional runtimes (e.g., JVM/generational-GC and unmanaged C implementations) that are not yet benchmarked.

Single-architecture performance data. Performance benchmarks run on a single x86-64 machine. Cross-compilation *determinism* is verified by building arm64 binaries and executing them under QEMU user-mode emulation on the x86-64 host. Because QEMU

user-mode translates system calls but executes floating-point operations on the host CPU, this verifies Go compiler determinism across target architectures, not native arm64 hardware behavior. Arm64 *performance* benchmarks are deferred due to lack of bare-metal hardware. Future work should include bare-metal arm64 benchmarks on Graviton, Apple Silicon, or Ampere hardware.

JavaScript engine oracle (mitigated). Oracle vectors are generated from V8 (via Node.js v23.3.0, V8 12.9.202.28) and independently validated against SpiderMonkey (JavaScript-C115.10.0) and JavaScriptCore (WebKitGTK 2.50.4). All three engines—representing the three major independent implementations of ECMA-262 `Number::toString`—produce identical output for all 286,362 vectors with zero divergences. Combined with dense boundary coverage (all subnormals up to 2^{12}, all binary64 powers of two, subnormal/normal boundary neighborhoods, RFC 8785 Appendix B examples with neighbors, and pseudorandom fill), this provides strong evidence that the oracle reflects the normative specification rather than engine-specific behavior. Cross-version stability within each engine family is expected from the normative spec but not independently verified.

Workload representativeness. Benchmark workloads are synthetic, designed to bracket the performance space between number-heavy and string-dominant extremes. Real-world JSON payloads may have different number density, nesting depth, and key length distributions. However, the control group design (number-only and string-only workloads) bounds the expected range.

Endianness. Both x86-64 and arm64 targets are little-endian; big-endian behavior is untested. While the canonicalization algorithms do not depend on byte order (they operate on IEEE 754 binary64 values via language-level primitives), edge cases in binary representation parsing could theoretically differ on big-endian platforms.

10 Conclusion

We presented the first empirical comparison of four digit-generation algorithms—Burger–Dybvig (multiprecision), Schubfach, Ryū, and Dragonbox (all fixed-width)—inside a RFC 8785/ECMA-262-constrained JSON canonicalization pipeline, with cross-language validation in Go and Rust to separate algorithmic effects from language/runtime characteristics. While prior work has compared float-to-string algorithms in isolation (e.g., Adams's comparison of Ryū against Grisu and Errol; Champagne Gareau and Lemire's comprehensive C++ evaluation), our study is the first to evaluate these algorithms within a specification-constrained canonicalization pipeline using conformance-gated methodology.

Key findings:

- The three fixed-width algorithms (Schubfach, Ryū, Dragonbox) perform within 1–8% of each other across both languages and all workloads. The dominant performance axis is fixed-width vs. multiprecision, not the choice among fixed-width algorithms.

- Fixed-width algorithms achieve 1.4–1.8× speedup on number-dense workloads ($\nu(D) \geq 40\%$) in Go API benchmarks due to zero heap allocation; the advantage replicates at 2.3–3.0× in Rust (excluding extreme outliers), confirming it is algorithmic.

- In end-to-end CLI measurement, the advantage narrows due to process startup dominance, but remains statistically significant on number-dense workloads.
- The advantage narrows to <5% on string-dominant and deeply-nested inputs, correlating with number density $\nu(D)$.
- Go implementations produce byte-identical output under cross-compilation for x86-64 and arm64 (arm64 binaries verified via QEMU user-mode emulation on x86-64 host). Rust implementations were not cross-compiled in this study.
- Preliminary profiling suggests digit generation accounts for approximately 15–40% of total canonicalization cost depending on number density.

The benchmark infrastructure, 286,362 oracle vectors, and all raw data are released as a reproducibility artifact. Our findings align with the broader ecosystem direction: Go 1.26 adopted the Dragonbox fixed-width algorithm for `strconv`, corroborating our evidence that fixed-width approaches offer meaningful performance advantages in float-to-string formatting. Since the three fixed-width algorithms are performance-equivalent, implementors can choose based on other criteria: code simplicity, ecosystem momentum, or theoretical interest.

Future Work

Bare-metal arm64 performance benchmarks, additional language implementations (C and JVM) to extend the comparison matrix to unmanaged and managed-GC runtimes, Rust cross-compilation verification, and workload characterization from production JSON corpora.

Data Availability Statement

All data, software, and oracle vectors are freely available online. The benchmark infrastructure, all eight implementations, workload manifests, and oracle generation scripts are available at the accompanying repository.[2] Oracle vector checksums (SHA-256) are documented in `PROVENANCE.json` files accompanying each implementation.

Author Contributions

Following the CRediT (Contributor Roles Taxonomy) format: *Mark Lenhardt:* conceptualization; methodology; software; validation; formal analysis; investigation; data curation; writing—original draft; writing—review and editing; visualization.

[2]Repository URL and Zenodo DOI to be inserted for camera-ready version.

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A Full Benchmark Data

This appendix contains the complete benchmark results. Summary tables appear in the main text (§7); the full data are provided here for reproducibility.

Note: Rust allocation data (B/op, allocs/op) is measured via a counting global allocator that intercepts all heap allocations (see §8).

A.1 Full API Benchmarks

Table 8: API benchmark results (mean of $n = 10$ iterations).

Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
go:schubfach-vs-bd	array-2048	canonicalize	2531402	4153082	1.64×	+8210	+186654
go:schubfach-vs-bd	array-256	canonicalize	305209	479254	1.57×	+1020	+24977
go:schubfach-vs-bd	canonical-minimal	canonicalize	354	529	1.49×	+0	+0
go:schubfach-vs-bd	control-escapes	canonicalize	1292	1310	1.01×	+0	+0
go:schubfach-vs-bd	deep	canonicalize	36558	36491	1.00×	+0	+0
go:schubfach-vs-bd	deep-64	canonicalize	16189	16140	1.00×	+0	+0
go:schubfach-vs-bd	escaped-key-order	canonicalize	716	1138	1.59×	+2	+41
go:schubfach-vs-bd	large	canonicalize	11851487	20936873	1.77×	+32792	+663335
go:schubfach-vs-bd	long-string	canonicalize	76188	76724	1.01×	+0	+0
go:schubfach-vs-bd	medium	canonicalize	339803	596274	1.75×	+1014	+21130
go:schubfach-vs-bd	mixed-prod	canonicalize	134635	183236	1.36×	+255	+6233
go:schubfach-vs-bd	nested-mixed	canonicalize	2192	3104	1.42×	+4	+85
go:schubfach-vs-bd	number-heavy	canonicalize	13860	21718	1.57×	+20	+830
go:schubfach-vs-bd	numeric-boundary	canonicalize	12714	19052	1.50×	+14	+706
go:schubfach-vs-bd	rfc-key-sorting	canonicalize	2988	3080	1.03×	+0	+0
go:schubfach-vs-bd	small	canonicalize	1240	1685	1.36×	+2	+50
go:schubfach-vs-bd	surrogate-pair	canonicalize	650	660	1.01×	+0	+0
go:schubfach-vs-bd	unicode	canonicalize	1564	1561	1.00×	+0	+0
go:schubfach-vs-bd	verify-whitespace	canonicalize	914	1576	1.72×	+4	+82
go:schubfach-vs-bd	canonical/array-2048	verify	2475144	4113295	1.66×	+8208	+186597
go:schubfach-vs-bd	canonical/array-256	verify	317805	490966	1.54×	+1022	+25230
go:schubfach-vs-bd	canonical/canonical-minimal	verify	360	543	1.51×	+0	+0
go:schubfach-vs-bd	canonical/control-escapes	verify	1292	1322	1.02×	+0	+0
go:schubfach-vs-bd	canonical/deep	verify	38126	37749	1.01×	+0	+0
go:schubfach-vs-bd	canonical/deep-64	verify	16844	16630	1.01×	+0	+0
go:schubfach-vs-bd	canonical/escaped-key-order	verify	695	1120	1.61×	+2	+41
go:schubfach-vs-bd	canonical/large	verify	11434977	20458581	1.79×	+32790	+663081
go:schubfach-vs-bd	canonical/long-string	verify	77429	77802	1.00×	+0	+0
go:schubfach-vs-bd	canonical/medium	verify	341094	591838	1.74×	+1015	+21371
go:schubfach-vs-bd	canonical/mixed-prod	verify	133722	179814	1.34×	+255	+6323
go:schubfach-vs-bd	canonical/nested-mixed	verify	2211	3111	1.41×	+4	+86
go:schubfach-vs-bd	canonical/number-heavy	verify	13970	21748	1.56×	+20	+834
go:schubfach-vs-bd	canonical/numeric-boundary	verify	12736	18914	1.49×	+14	+709
go:schubfach-vs-bd	canonical/rfc-key-sorting	verify	2852	2873	1.01×	+0	+0
go:schubfach-vs-bd	canonical/small	verify	1223	1653	1.35×	+2	+51
go:schubfach-vs-bd	canonical/surrogate-pair	verify	665	668	1.00×	+0	+0
go:schubfach-vs-bd	canonical/unicode	verify	1561	1555	1.00×	+0	+0
go:schubfach-vs-bd	canonical/verify-whitespace	verify	934	1592	1.71×	+4	+82
go:schubfach-vs-bd	noncanonical/control-escapes	verify	1278	1322	1.03×	+0	+0
go:schubfach-vs-bd	noncanonical/escaped-key-order	verify	704	1140	1.62×	+2	+41

Continued on next page

Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
go:schubfach-vs-bd	noncanonical/large	verify	11843685	20944246	1.77×	+32793	+663541
go:schubfach-vs-bd	noncanonical/medium	verify	338825	586294	1.73×	+1014	+21136
go:schubfach-vs-bd	noncanonical/mixed-prod	verify	134502	180592	1.34×	+255	+6225
go:schubfach-vs-bd	noncanonical/nested-mixed	verify	2194	3084	1.41×	+4	+85
go:schubfach-vs-bd	noncanonical/number-heavy	verify	13783	21496	1.56×	+20	+829
go:schubfach-vs-bd	noncanonical/numeric-boundary	verify	12644	18905	1.50×	+14	+706
go:schubfach-vs-bd	noncanonical/rfc-key-sorting	verify	2964	3024	1.02×	+0	+0
go:schubfach-vs-bd	noncanonical/small	verify	1238	1671	1.35×	+2	+50
go:schubfach-vs-bd	noncanonical/surrogate-pair	verify	650	656	1.01×	+0	+0
go:schubfach-vs-bd	noncanonical/unicode	verify	1556	1558	1.00×	+0	+0
go:schubfach-vs-bd	noncanonical/verify-whitespace	verify	911	1562	1.72×	+4	+82
go:schubfach-vs-ryu	array-2048	canonicalize	2531402	2502576	1.01×	+0	-4
go:schubfach-vs-ryu	array-256	canonicalize	305209	302518	1.01×	+0	-0
go:schubfach-vs-ryu	canonical-minimal	canonicalize	354	344	1.03×	+0	+0
go:schubfach-vs-ryu	control-escapes	canonicalize	1292	1281	1.01×	+0	+0
go:schubfach-vs-ryu	deep	canonicalize	36558	37235	1.02×	+0	+0
go:schubfach-vs-ryu	deep-64	canonicalize	16189	16390	1.01×	+0	+0
go:schubfach-vs-ryu	escaped-key-order	canonicalize	716	711	1.01×	+0	+0
go:schubfach-vs-ryu	large	canonicalize	11851487	11785645	1.01×	+0	+0
go:schubfach-vs-ryu	long-string	canonicalize	76188	76170	1.00×	+0	+0
go:schubfach-vs-ryu	medium	canonicalize	339803	337542	1.01×	+0	+0
go:schubfach-vs-ryu	mixed-prod	canonicalize	134635	135246	1.00×	+0	+0
go:schubfach-vs-ryu	nested-mixed	canonicalize	2192	2207	1.01×	+0	+0
go:schubfach-vs-ryu	number-heavy	canonicalize	13860	13898	1.00×	+0	+0
go:schubfach-vs-ryu	numeric-boundary	canonicalize	12714	12702	1.00×	+0	+0
go:schubfach-vs-ryu	rfc-key-sorting	canonicalize	2988	3004	1.01×	+0	+0
go:schubfach-vs-ryu	small	canonicalize	1240	1246	1.01×	+0	+0
go:schubfach-vs-ryu	surrogate-pair	canonicalize	650	652	1.00×	+0	+0
go:schubfach-vs-ryu	unicode	canonicalize	1564	1577	1.01×	+0	+0
go:schubfach-vs-ryu	verify-whitespace	canonicalize	914	912	1.00×	+0	+0
go:schubfach-vs-ryu	canonical/array-2048	verify	2475144	2487780	1.01×	+0	-0
go:schubfach-vs-ryu	canonical/array-256	verify	317805	315702	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/canonical-minimal	verify	360	359	1.00×	+0	+0
go:schubfach-vs-ryu	canonical/control-escapes	verify	1292	1286	1.00×	+0	+0
go:schubfach-vs-ryu	canonical/deep	verify	38126	38082	1.00×	+0	+0
go:schubfach-vs-ryu	canonical/deep-64	verify	16844	16854	1.00×	+0	+0
go:schubfach-vs-ryu	canonical/escaped-key-order	verify	695	692	1.00×	+0	+0
go:schubfach-vs-ryu	canonical/large	verify	11434977	11401357	1.00×	-0	-8
go:schubfach-vs-ryu	canonical/long-string	verify	77429	78084	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/medium	verify	341094	343581	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/mixed-prod	verify	133722	134634	1.01×	+0	-0
go:schubfach-vs-ryu	canonical/nested-mixed	verify	2211	2236	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/number-heavy	verify	13970	14045	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/numeric-boundary	verify	12736	12855	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/rfc-key-sorting	verify	2852	2873	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/small	verify	1223	1233	1.01×	+0	+0

Continued on next page

Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
go:schubfach-vs-ryu	canonical/surrogate-pair	verify	665	666	1.00×	+0	+0
go:schubfach-vs-ryu	canonical/unicode	verify	1561	1576	1.01×	+0	+0
go:schubfach-vs-ryu	canonical/verify-whitespace	verify	934	939	1.01×	+0	+0
go:schubfach-vs-ryu	noncanonical/control-escapes	verify	1278	1296	1.01×	+0	+0
go:schubfach-vs-ryu	noncanonical/escaped-key-order	verify	704	715	1.01×	+0	+0
go:schubfach-vs-ryu	noncanonical/large	verify	11843685	11909754	1.01×	+0	+13
go:schubfach-vs-ryu	noncanonical/medium	verify	338825	338652	1.00×	+0	+0
go:schubfach-vs-ryu	noncanonical/mixed-prod	verify	134502	135975	1.01×	+0	-0
go:schubfach-vs-ryu	noncanonical/nested-mixed	verify	2194	2214	1.01×	+0	+0
go:schubfach-vs-ryu	noncanonical/number-heavy	verify	13783	13820	1.00×	+0	+0
go:schubfach-vs-ryu	noncanonical/numeric-boundary	verify	12644	12663	1.00×	+0	+0
go:schubfach-vs-ryu	noncanonical/rfc-key-sorting	verify	2964	3013	1.02×	+0	+0
go:schubfach-vs-ryu	noncanonical/small	verify	1238	1259	1.02×	+0	+0
go:schubfach-vs-ryu	noncanonical/surrogate-pair	verify	650	652	1.00×	+0	+0
go:schubfach-vs-ryu	noncanonical/unicode	verify	1556	1570	1.01×	+0	+0
go:schubfach-vs-ryu	noncanonical/verify-whitespace	verify	911	916	1.01×	+0	+0
go:schubfach-vs-dragonbox	array-2048	canonicalize	2531402	2530127	1.00×	+0	-3
go:schubfach-vs-dragonbox	array-256	canonicalize	305209	302413	1.01×	+0	-0
go:schubfach-vs-dragonbox	canonical-minimal	canonicalize	354	345	1.02×	+0	+0
go:schubfach-vs-dragonbox	control-escapes	canonicalize	1292	1288	1.00×	+0	+0
go:schubfach-vs-dragonbox	deep	canonicalize	36558	37345	1.02×	+0	+0
go:schubfach-vs-dragonbox	deep-64	canonicalize	16189	16493	1.02×	+0	+0
go:schubfach-vs-dragonbox	escaped-key-order	canonicalize	716	719	1.00×	+0	+0
go:schubfach-vs-dragonbox	large	canonicalize	11851487	11787210	1.01×	-0	-13
go:schubfach-vs-dragonbox	long-string	canonicalize	76188	76427	1.00×	+0	+0
go:schubfach-vs-dragonbox	medium	canonicalize	339803	336597	1.01×	+0	+0
go:schubfach-vs-dragonbox	mixed-prod	canonicalize	134635	134935	1.00×	+0	+0
go:schubfach-vs-dragonbox	nested-mixed	canonicalize	2192	2196	1.00×	+0	+0
go:schubfach-vs-dragonbox	number-heavy	canonicalize	13860	13771	1.01×	+0	+0
go:schubfach-vs-dragonbox	numeric-boundary	canonicalize	12714	12666	1.00×	+0	+0
go:schubfach-vs-dragonbox	rfc-key-sorting	canonicalize	2988	2996	1.00×	+0	+0
go:schubfach-vs-dragonbox	small	canonicalize	1240	1242	1.00×	+0	+0
go:schubfach-vs-dragonbox	surrogate-pair	canonicalize	650	659	1.01×	+0	+0
go:schubfach-vs-dragonbox	unicode	canonicalize	1564	1569	1.00×	+0	+0
go:schubfach-vs-dragonbox	verify-whitespace	canonicalize	914	915	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/array-2048	verify	2475144	2508195	1.01×	+0	-1
go:schubfach-vs-dragonbox	canonical/array-256	verify	317805	315659	1.01×	+0	+1
go:schubfach-vs-dragonbox	canonical/canonical-minimal	verify	360	358	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/control-escapes	verify	1292	1293	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/deep	verify	38126	38406	1.01×	+0	+0
go:schubfach-vs-dragonbox	canonical/deep-64	verify	16844	16918	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/escaped-key-order	verify	695	694	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/large	verify	11434977	11524340	1.01×	-0	-14
go:schubfach-vs-dragonbox	canonical/long-string	verify	77429	77534	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/medium	verify	341094	342029	1.00×	+0	+1
go:schubfach-vs-dragonbox	canonical/mixed-prod	verify	133722	134529	1.01×	+0	+0

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
go:schubfach-vs-dragonbox	canonical/nested-mixed	verify	2211	2224	1.01×	+0	+0
go:schubfach-vs-dragonbox	canonical/number-heavy	verify	13970	13960	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/numeric-boundary	verify	12736	12803	1.01×	+0	+0
go:schubfach-vs-dragonbox	canonical/rfc-key-sorting	verify	2852	2880	1.01×	+0	+0
go:schubfach-vs-dragonbox	canonical/small	verify	1223	1228	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/surrogate-pair	verify	665	664	1.00×	+0	+0
go:schubfach-vs-dragonbox	canonical/unicode	verify	1561	1573	1.01×	+0	+0
go:schubfach-vs-dragonbox	canonical/verify-whitespace	verify	934	946	1.01×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/control-escapes	verify	1278	1292	1.01×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/escaped-key-order	verify	704	712	1.01×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/large	verify	11843685	11970740	1.01×	+0	+15
go:schubfach-vs-dragonbox	noncanonical/medium	verify	338825	338796	1.00×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/mixed-prod	verify	134502	136070	1.01×	+0	-0
go:schubfach-vs-dragonbox	noncanonical/nested-mixed	verify	2194	2209	1.01×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/number-heavy	verify	13783	13809	1.00×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/numeric-boundary	verify	12644	12647	1.00×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/rfc-key-sorting	verify	2964	3020	1.02×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/small	verify	1238	1262	1.02×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/surrogate-pair	verify	650	654	1.01×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/unicode	verify	1556	1580	1.02×	+0	+0
go:schubfach-vs-dragonbox	noncanonical/verify-whitespace	verify	911	922	1.01×	+0	+0
go:ryu-vs-dragonbox	array-2048	canonicalize	2502576	2530127	1.01×	+0	+1
go:ryu-vs-dragonbox	array-256	canonicalize	302518	302413	1.00×	+0	-0
go:ryu-vs-dragonbox	canonical-minimal	canonicalize	344	345	1.00×	+0	+0
go:ryu-vs-dragonbox	control-escapes	canonicalize	1281	1288	1.00×	+0	+0
go:ryu-vs-dragonbox	deep	canonicalize	37235	37345	1.00×	+0	+0
go:ryu-vs-dragonbox	deep-64	canonicalize	16390	16493	1.01×	+0	+0
go:ryu-vs-dragonbox	escaped-key-order	canonicalize	711	719	1.01×	+0	+0
go:ryu-vs-dragonbox	large	canonicalize	11785645	11787210	1.00×	-0	-14
go:ryu-vs-dragonbox	long-string	canonicalize	76170	76427	1.00×	+0	+0
go:ryu-vs-dragonbox	medium	canonicalize	337542	336597	1.00×	+0	-0
go:ryu-vs-dragonbox	mixed-prod	canonicalize	135246	134935	1.00×	+0	+0
go:ryu-vs-dragonbox	nested-mixed	canonicalize	2207	2196	1.01×	+0	+0
go:ryu-vs-dragonbox	number-heavy	canonicalize	13898	13771	1.01×	+0	+0
go:ryu-vs-dragonbox	numeric-boundary	canonicalize	12702	12666	1.00×	+0	+0
go:ryu-vs-dragonbox	rfc-key-sorting	canonicalize	3004	2996	1.00×	+0	+0
go:ryu-vs-dragonbox	small	canonicalize	1246	1242	1.00×	+0	+0
go:ryu-vs-dragonbox	surrogate-pair	canonicalize	652	659	1.01×	+0	+0
go:ryu-vs-dragonbox	unicode	canonicalize	1577	1569	1.00×	+0	+0
go:ryu-vs-dragonbox	verify-whitespace	canonicalize	912	915	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/array-2048	verify	2487780	2508195	1.01×	+0	-1
go:ryu-vs-dragonbox	canonical/array-256	verify	315702	315659	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/canonical-minimal	verify	359	358	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/control-escapes	verify	1286	1293	1.01×	+0	+0
go:ryu-vs-dragonbox	canonical/deep	verify	38082	38406	1.01×	+0	+0
go:ryu-vs-dragonbox	canonical/deep-64	verify	16854	16918	1.00×	+0	+0

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
go:ryu-vs-dragonbox	canonical/escaped-key-order	verify	692	694	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/large	verify	11401357	11524340	1.01×	-0	-6
go:ryu-vs-dragonbox	canonical/long-string	verify	78084	77534	1.01×	+0	+0
go:ryu-vs-dragonbox	canonical/medium	verify	343581	342029	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/mixed-prod	verify	134634	134529	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/nested-mixed	verify	2236	2224	1.01×	+0	+0
go:ryu-vs-dragonbox	canonical/number-heavy	verify	14045	13960	1.01×	+0	+0
go:ryu-vs-dragonbox	canonical/numeric-boundary	verify	12855	12803	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/rfc-key-sorting	verify	2873	2880	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/small	verify	1233	1228	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/surrogate-pair	verify	666	664	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/unicode	verify	1576	1573	1.00×	+0	+0
go:ryu-vs-dragonbox	canonical/verify-whitespace	verify	939	946	1.01×	+0	+0
go:ryu-vs-dragonbox	noncanonical/control-escapes	verify	1296	1292	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/escaped-key-order	verify	715	712	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/large	verify	11909754	11970740	1.01×	+0	+2
go:ryu-vs-dragonbox	noncanonical/medium	verify	338652	338796	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/mixed-prod	verify	135975	136070	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/nested-mixed	verify	2214	2209	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/number-heavy	verify	13820	13809	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/numeric-boundary	verify	12663	12647	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/rfc-key-sorting	verify	3013	3020	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/small	verify	1259	1262	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/surrogate-pair	verify	652	654	1.00×	+0	+0
go:ryu-vs-dragonbox	noncanonical/unicode	verify	1570	1580	1.01×	+0	+0
go:ryu-vs-dragonbox	noncanonical/verify-whitespace	verify	916	922	1.01×	+0	+0
go:ryu-vs-bd	array-2048	canonicalize	2502576	4153082	1.66×	+8210	+186659
go:ryu-vs-bd	array-256	canonicalize	302518	479254	1.58×	+1020	+24977
go:ryu-vs-bd	canonical-minimal	canonicalize	344	529	1.54×	+0	+0
go:ryu-vs-bd	control-escapes	canonicalize	1281	1310	1.02×	+0	+0
go:ryu-vs-bd	deep	canonicalize	37235	36491	1.02×	+0	+0
go:ryu-vs-bd	deep-64	canonicalize	16390	16140	1.02×	+0	+0
go:ryu-vs-bd	escaped-key-order	canonicalize	711	1138	1.60×	+2	+41
go:ryu-vs-bd	large	canonicalize	11785645	20936873	1.78×	+32792	+663334
go:ryu-vs-bd	long-string	canonicalize	76170	76724	1.01×	+0	+0
go:ryu-vs-bd	medium	canonicalize	337542	596274	1.77×	+1014	+21130
go:ryu-vs-bd	mixed-prod	canonicalize	135246	183236	1.35×	+255	+6233
go:ryu-vs-bd	nested-mixed	canonicalize	2207	3104	1.41×	+4	+85
go:ryu-vs-bd	number-heavy	canonicalize	13898	21718	1.56×	+20	+830
go:ryu-vs-bd	numeric-boundary	canonicalize	12702	19052	1.50×	+14	+706
go:ryu-vs-bd	rfc-key-sorting	canonicalize	3004	3080	1.03×	+0	+0
go:ryu-vs-bd	small	canonicalize	1246	1685	1.35×	+2	+50
go:ryu-vs-bd	surrogate-pair	canonicalize	652	660	1.01×	+0	+0
go:ryu-vs-bd	unicode	canonicalize	1577	1561	1.01×	+0	+0
go:ryu-vs-bd	verify-whitespace	canonicalize	912	1576	1.73×	+4	+82
go:ryu-vs-bd	canonical/array-2048	verify	2487780	4113295	1.65×	+8208	+186597

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
go:ryu-vs-bd	canonical/array-256	verify	315702	490966	1.56×	+1022	+25230
go:ryu-vs-bd	canonical/canonical-minimal	verify	359	543	1.51×	+0	+0
go:ryu-vs-bd	canonical/control-escapes	verify	1286	1322	1.03×	+0	+0
go:ryu-vs-bd	canonical/deep	verify	38082	37749	1.01×	+0	+0
go:ryu-vs-bd	canonical/deep-64	verify	16854	16630	1.01×	+0	+0
go:ryu-vs-bd	canonical/escaped-key-order	verify	692	1120	1.62×	+2	+41
go:ryu-vs-bd	canonical/large	verify	11401357	20458581	1.79×	+32790	+663089
go:ryu-vs-bd	canonical/long-string	verify	78084	77802	1.00×	+0	+0
go:ryu-vs-bd	canonical/medium	verify	343581	591838	1.72×	+1015	+21371
go:ryu-vs-bd	canonical/mixed-prod	verify	134634	179814	1.34×	+255	+6324
go:ryu-vs-bd	canonical/nested-mixed	verify	2236	3111	1.39×	+4	+86
go:ryu-vs-bd	canonical/number-heavy	verify	14045	21748	1.55×	+20	+834
go:ryu-vs-bd	canonical/numeric-boundary	verify	12855	18914	1.47×	+14	+709
go:ryu-vs-bd	canonical/rfc-key-sorting	verify	2873	2873	1.00×	+0	+0
go:ryu-vs-bd	canonical/small	verify	1233	1653	1.34×	+2	+51
go:ryu-vs-bd	canonical/surrogate-pair	verify	666	668	1.00×	+0	+0
go:ryu-vs-bd	canonical/unicode	verify	1576	1555	1.01×	+0	+0
go:ryu-vs-bd	canonical/verify-whitespace	verify	939	1592	1.70×	+4	+82
go:ryu-vs-bd	noncanonical/control-escapes	verify	1296	1322	1.02×	+0	+0
go:ryu-vs-bd	noncanonical/escaped-key-order	verify	715	1140	1.60×	+2	+41
go:ryu-vs-bd	noncanonical/large	verify	11909754	20944246	1.76×	+32793	+663528
go:ryu-vs-bd	noncanonical/medium	verify	338652	586294	1.73×	+1014	+21136
go:ryu-vs-bd	noncanonical/mixed-prod	verify	135975	180592	1.33×	+255	+6225
go:ryu-vs-bd	noncanonical/nested-mixed	verify	2214	3084	1.39×	+4	+85
go:ryu-vs-bd	noncanonical/number-heavy	verify	13820	21496	1.56×	+20	+829
go:ryu-vs-bd	noncanonical/numeric-boundary	verify	12663	18905	1.49×	+14	+706
go:ryu-vs-bd	noncanonical/rfc-key-sorting	verify	3013	3024	1.00×	+0	+0
go:ryu-vs-bd	noncanonical/small	verify	1259	1671	1.33×	+2	+50
go:ryu-vs-bd	noncanonical/surrogate-pair	verify	652	656	1.00×	+0	+0
go:ryu-vs-bd	noncanonical/unicode	verify	1570	1558	1.01×	+0	+0
go:ryu-vs-bd	noncanonical/verify-whitespace	verify	916	1562	1.71×	+4	+82
go:dragonbox-vs-bd	array-2048	canonicalize	2530127	4153082	1.64×	+8210	+186658
go:dragonbox-vs-bd	array-256	canonicalize	302413	479254	1.58×	+1020	+24977
go:dragonbox-vs-bd	canonical-minimal	canonicalize	345	529	1.53×	+0	+0
go:dragonbox-vs-bd	control-escapes	canonicalize	1288	1310	1.02×	+0	+0
go:dragonbox-vs-bd	deep	canonicalize	37345	36491	1.02×	+0	+0
go:dragonbox-vs-bd	deep-64	canonicalize	16493	16140	1.02×	+0	+0
go:dragonbox-vs-bd	escaped-key-order	canonicalize	719	1138	1.58×	+2	+41
go:dragonbox-vs-bd	large	canonicalize	11787210	20936873	1.78×	+32792	+663348
go:dragonbox-vs-bd	long-string	canonicalize	76427	76724	1.00×	+0	+0
go:dragonbox-vs-bd	medium	canonicalize	336597	596274	1.77×	+1014	+21130
go:dragonbox-vs-bd	mixed-prod	canonicalize	134935	183236	1.36×	+255	+6233
go:dragonbox-vs-bd	nested-mixed	canonicalize	2196	3104	1.41×	+4	+85
go:dragonbox-vs-bd	number-heavy	canonicalize	13771	21718	1.58×	+20	+830
go:dragonbox-vs-bd	numeric-boundary	canonicalize	12666	19052	1.50×	+14	+706
go:dragonbox-vs-bd	rfc-key-sorting	canonicalize	2996	3080	1.03×	+0	+0

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
go:dragonbox-vs-bd	small	canonicalize	1242	1685	1.36×	+2	+50
go:dragonbox-vs-bd	surrogate-pair	canonicalize	659	660	1.00×	+0	+0
go:dragonbox-vs-bd	unicode	canonicalize	1569	1561	1.01×	+0	+0
go:dragonbox-vs-bd	verify-whitespace	canonicalize	915	1576	1.72×	+4	+82
go:dragonbox-vs-bd	canonical/array-2048	verify	2508195	4113295	1.64×	+8208	+186598
go:dragonbox-vs-bd	canonical/array-256	verify	315659	490966	1.56×	+1022	+25229
go:dragonbox-vs-bd	canonical/canonical-minimal	verify	358	543	1.52×	+0	+0
go:dragonbox-vs-bd	canonical/control-escapes	verify	1293	1322	1.02×	+0	+0
go:dragonbox-vs-bd	canonical/deep	verify	38406	37749	1.02×	+0	+0
go:dragonbox-vs-bd	canonical/deep-64	verify	16918	16630	1.02×	+0	+0
go:dragonbox-vs-bd	canonical/escaped-key-order	verify	694	1120	1.61×	+2	+41
go:dragonbox-vs-bd	canonical/large	verify	11524340	20458581	1.78×	+32790	+663095
go:dragonbox-vs-bd	canonical/long-string	verify	77534	77802	1.00×	+0	+0
go:dragonbox-vs-bd	canonical/medium	verify	342029	591838	1.73×	+1015	+21370
go:dragonbox-vs-bd	canonical/mixed-prod	verify	134529	179814	1.34×	+255	+6323
go:dragonbox-vs-bd	canonical/nested-mixed	verify	2224	3111	1.40×	+4	+86
go:dragonbox-vs-bd	canonical/number-heavy	verify	13960	21748	1.56×	+20	+834
go:dragonbox-vs-bd	canonical/numeric-boundary	verify	12803	18914	1.48×	+14	+709
go:dragonbox-vs-bd	canonical/rfc-key-sorting	verify	2880	2873	1.00×	+0	+0
go:dragonbox-vs-bd	canonical/small	verify	1228	1653	1.35×	+2	+51
go:dragonbox-vs-bd	canonical/surrogate-pair	verify	664	668	1.01×	+0	+0
go:dragonbox-vs-bd	canonical/unicode	verify	1573	1555	1.01×	+0	+0
go:dragonbox-vs-bd	canonical/verify-whitespace	verify	946	1592	1.68×	+4	+82
go:dragonbox-vs-bd	noncanonical/control-escapes	verify	1292	1322	1.02×	+0	+0
go:dragonbox-vs-bd	noncanonical/escaped-key-order	verify	712	1140	1.60×	+2	+41
go:dragonbox-vs-bd	noncanonical/large	verify	11970740	20944246	1.75×	+32793	+663526
go:dragonbox-vs-bd	noncanonical/medium	verify	338796	586294	1.73×	+1014	+21135
go:dragonbox-vs-bd	noncanonical/mixed-prod	verify	136070	180592	1.33×	+255	+6225
go:dragonbox-vs-bd	noncanonical/nested-mixed	verify	2209	3084	1.40×	+4	+85
go:dragonbox-vs-bd	noncanonical/number-heavy	verify	13809	21496	1.56×	+20	+829
go:dragonbox-vs-bd	noncanonical/numeric-boundary	verify	12647	18905	1.49×	+14	+706
go:dragonbox-vs-bd	noncanonical/rfc-key-sorting	verify	3020	3024	1.00×	+0	+0
go:dragonbox-vs-bd	noncanonical/small	verify	1262	1671	1.32×	+2	+50
go:dragonbox-vs-bd	noncanonical/surrogate-pair	verify	654	656	1.00×	+0	+0
go:dragonbox-vs-bd	noncanonical/unicode	verify	1580	1558	1.01×	+0	+0
go:dragonbox-vs-bd	noncanonical/verify-whitespace	verify	922	1562	1.69×	+4	+82
rs:schubfach-vs-bd	array-2048	canonicalize	1513396	3427870	2.27×	+141429	+1814032
rs:schubfach-vs-bd	array-256	canonicalize	168411	387093	2.30×	+14906	+189584
rs:schubfach-vs-bd	canonical-minimal	canonicalize	210	443	2.10×	+16	+200
rs:schubfach-vs-bd	control-escapes	canonicalize	646	635	1.02×	+0	+0
rs:schubfach-vs-bd	deep	canonicalize	28846	30020	1.04×	+0	+0
rs:schubfach-vs-bd	deep-64	canonicalize	11636	11134	1.05×	+0	+0
rs:schubfach-vs-bd	escaped-key-order	canonicalize	326	832	2.55×	+33	+408
rs:schubfach-vs-bd	large	canonicalize	5816696	19079970	3.28×	+926334	+12054624
rs:schubfach-vs-bd	long-string	canonicalize	16928	17722	1.05×	+0	+0
rs:schubfach-vs-bd	medium	canonicalize	182570	531911	2.91×	+24337	+314864

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
rs:schubfach-vs-bd	mixed-prod	canonicalize	69235	123188	1.78×	+3941	+49904
rs:schubfach-vs-bd	nested-mixed	canonicalize	1004	1981	1.97×	+66	+816
rs:schubfach-vs-bd	number-heavy	canonicalize	1437	11822	8.23×	+594	+26080
rs:schubfach-vs-bd	numeric-boundary	canonicalize	980	9141	9.33×	+446	+24168
rs:schubfach-vs-bd	rfc-key-sorting	canonicalize	1370	1340	1.02×	+0	+0
rs:schubfach-vs-bd	small	canonicalize	654	1144	1.75×	+33	+408
rs:schubfach-vs-bd	surrogate-pair	canonicalize	317	321	1.01×	+0	+0
rs:schubfach-vs-bd	unicode	canonicalize	1511	1578	1.04×	+0	+0
rs:schubfach-vs-bd	verify-whitespace	canonicalize	400	1182	2.96×	+50	+616
rs:schubfach-vs-bd	canonical/array-2048	verify	1345524	3357314	2.50×	+141429	+1814032
rs:schubfach-vs-bd	canonical/array-256	verify	168727	380560	2.26×	+14906	+189584
rs:schubfach-vs-bd	canonical/canonical-minimal	verify	208	433	2.08×	+16	+200
rs:schubfach-vs-bd	canonical/control-escapes	verify	634	608	1.04×	+0	+0
rs:schubfach-vs-bd	canonical/deep	verify	28599	29275	1.02×	+0	+0
rs:schubfach-vs-bd	canonical/deep-64	verify	11091	10823	1.02×	+0	+0
rs:schubfach-vs-bd	canonical/escaped-key-order	verify	329	830	2.53×	+33	+408
rs:schubfach-vs-bd	canonical/large	verify	5913158	19532718	3.30×	+926334	+12054624
rs:schubfach-vs-bd	canonical/long-string	verify	17396	17308	1.01×	+0	+0
rs:schubfach-vs-bd	canonical/medium	verify	178455	538538	3.02×	+24337	+314864
rs:schubfach-vs-bd	canonical/mixed-prod	verify	66720	121970	1.83×	+3941	+49904
rs:schubfach-vs-bd	canonical/nested-mixed	verify	1010	1985	1.97×	+66	+816
rs:schubfach-vs-bd	canonical/number-heavy	verify	1424	11718	8.23×	+594	+26080
rs:schubfach-vs-bd	canonical/numeric-boundary	verify	956	9039	9.46×	+446	+24168
rs:schubfach-vs-bd	canonical/rfc-key-sorting	verify	1329	1309	1.01×	+0	+0
rs:schubfach-vs-bd	canonical/small	verify	665	1147	1.73×	+33	+408
rs:schubfach-vs-bd	canonical/surrogate-pair	verify	315	312	1.01×	+0	+0
rs:schubfach-vs-bd	canonical/unicode	verify	1509	1546	1.02×	+0	+0
rs:schubfach-vs-bd	canonical/verify-whitespace	verify	392	1148	2.93×	+50	+616
rs:schubfach-vs-ryu	array-2048	canonicalize	1513396	1402320	1.08×	+0	+0
rs:schubfach-vs-ryu	array-256	canonicalize	168411	172069	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical-minimal	canonicalize	210	202	1.04×	+0	+0
rs:schubfach-vs-ryu	control-escapes	canonicalize	646	668	1.03×	+0	+0
rs:schubfach-vs-ryu	deep	canonicalize	28846	29262	1.01×	+0	+0
rs:schubfach-vs-ryu	deep-64	canonicalize	11636	11020	1.06×	+0	+0
rs:schubfach-vs-ryu	escaped-key-order	canonicalize	326	336	1.03×	+0	+0
rs:schubfach-vs-ryu	large	canonicalize	5816696	5857077	1.01×	+0	+0
rs:schubfach-vs-ryu	long-string	canonicalize	16928	17286	1.02×	+0	+0
rs:schubfach-vs-ryu	medium	canonicalize	182570	178024	1.03×	+0	+0
rs:schubfach-vs-ryu	mixed-prod	canonicalize	69235	70593	1.02×	+0	+0
rs:schubfach-vs-ryu	nested-mixed	canonicalize	1004	1025	1.02×	+0	+0
rs:schubfach-vs-ryu	number-heavy	canonicalize	1437	1480	1.03×	+0	+0
rs:schubfach-vs-ryu	numeric-boundary	canonicalize	980	1000	1.02×	+0	+0
rs:schubfach-vs-ryu	rfc-key-sorting	canonicalize	1370	1410	1.03×	+0	+0
rs:schubfach-vs-ryu	small	canonicalize	654	696	1.06×	+0	+0
rs:schubfach-vs-ryu	surrogate-pair	canonicalize	317	318	1.00×	+0	+0
rs:schubfach-vs-ryu	unicode	canonicalize	1511	1537	1.02×	+0	+0

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
rs:schubfach-vs-ryu	verify-whitespace	canonicalize	400	408	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical/array-2048	verify	1345524	1369349	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical/array-256	verify	168727	170020	1.01×	+0	+0
rs:schubfach-vs-ryu	canonical/canonical-minimal	verify	208	214	1.03×	+0	+0
rs:schubfach-vs-ryu	canonical/control-escapes	verify	634	644	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical/deep	verify	28599	30662	1.07×	+0	+0
rs:schubfach-vs-ryu	canonical/deep-64	verify	11091	11060	1.00×	+0	+0
rs:schubfach-vs-ryu	canonical/escaped-key-order	verify	329	335	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical/large	verify	5913158	5927302	1.00×	+0	+0
rs:schubfach-vs-ryu	canonical/long-string	verify	17396	17542	1.01×	+0	+0
rs:schubfach-vs-ryu	canonical/medium	verify	178455	182743	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical/mixed-prod	verify	66720	67564	1.01×	+0	+0
rs:schubfach-vs-ryu	canonical/nested-mixed	verify	1010	1012	1.00×	+0	+0
rs:schubfach-vs-ryu	canonical/number-heavy	verify	1424	1454	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical/numeric-boundary	verify	956	978	1.02×	+0	+0
rs:schubfach-vs-ryu	canonical/rfc-key-sorting	verify	1329	1394	1.05×	+0	+0
rs:schubfach-vs-ryu	canonical/small	verify	665	717	1.08×	+0	+0
rs:schubfach-vs-ryu	canonical/surrogate-pair	verify	315	306	1.03×	+0	+0
rs:schubfach-vs-ryu	canonical/unicode	verify	1509	1508	1.00×	+0	+0
rs:schubfach-vs-ryu	canonical/verify-whitespace	verify	392	397	1.01×	+0	+0
rs:schubfach-vs-dragonbox	array-2048	canonicalize	1513396	1337112	1.13×	+0	+0
rs:schubfach-vs-dragonbox	array-256	canonicalize	168411	168169	1.00×	+0	+0
rs:schubfach-vs-dragonbox	canonical-minimal	canonicalize	210	208	1.01×	+0	+0
rs:schubfach-vs-dragonbox	control-escapes	canonicalize	646	622	1.04×	+0	+0
rs:schubfach-vs-dragonbox	deep	canonicalize	28846	30490	1.06×	+0	+0
rs:schubfach-vs-dragonbox	deep-64	canonicalize	11636	11204	1.04×	+0	+0
rs:schubfach-vs-dragonbox	escaped-key-order	canonicalize	326	338	1.04×	+0	+0
rs:schubfach-vs-dragonbox	large	canonicalize	5816696	5737358	1.01×	+0	+0
rs:schubfach-vs-dragonbox	long-string	canonicalize	16928	17345	1.02×	+0	+0
rs:schubfach-vs-dragonbox	medium	canonicalize	182570	185833	1.02×	+0	+0
rs:schubfach-vs-dragonbox	mixed-prod	canonicalize	69235	67412	1.03×	+0	+0
rs:schubfach-vs-dragonbox	nested-mixed	canonicalize	1004	1033	1.03×	+0	+0
rs:schubfach-vs-dragonbox	number-heavy	canonicalize	1437	1433	1.00×	+0	+0
rs:schubfach-vs-dragonbox	numeric-boundary	canonicalize	980	966	1.01×	+0	+0
rs:schubfach-vs-dragonbox	rfc-key-sorting	canonicalize	1370	1372	1.00×	+0	+0
rs:schubfach-vs-dragonbox	small	canonicalize	654	680	1.04×	+0	+0
rs:schubfach-vs-dragonbox	surrogate-pair	canonicalize	317	339	1.07×	+0	+0
rs:schubfach-vs-dragonbox	unicode	canonicalize	1511	1601	1.06×	+0	+0
rs:schubfach-vs-dragonbox	verify-whitespace	canonicalize	400	411	1.03×	+0	+0
rs:schubfach-vs-dragonbox	canonical/array-2048	verify	1345524	1346174	1.00×	+0	+0
rs:schubfach-vs-dragonbox	canonical/array-256	verify	168727	167878	1.01×	+0	+0
rs:schubfach-vs-dragonbox	canonical/canonical-minimal	verify	208	205	1.01×	+0	+0
rs:schubfach-vs-dragonbox	canonical/control-escapes	verify	634	622	1.02×	+0	+0
rs:schubfach-vs-dragonbox	canonical/deep	verify	28599	30257	1.06×	+0	+0
rs:schubfach-vs-dragonbox	canonical/deep-64	verify	11091	11374	1.03×	+0	+0
rs:schubfach-vs-dragonbox	canonical/escaped-key-order	verify	329	334	1.02×	+0	+0

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
rs:schubfach-vs-dragonbox	canonical/large	verify	5913158	5869913	1.01×	+0	+0
rs:schubfach-vs-dragonbox	canonical/long-string	verify	17396	17474	1.00×	+0	+0
rs:schubfach-vs-dragonbox	canonical/medium	verify	178455	180672	1.01×	+0	+0
rs:schubfach-vs-dragonbox	canonical/mixed-prod	verify	66720	67153	1.01×	+0	+0
rs:schubfach-vs-dragonbox	canonical/nested-mixed	verify	1010	1011	1.00×	+0	+0
rs:schubfach-vs-dragonbox	canonical/number-heavy	verify	1424	1419	1.00×	+0	+0
rs:schubfach-vs-dragonbox	canonical/numeric-boundary	verify	956	962	1.01×	+0	+0
rs:schubfach-vs-dragonbox	canonical/rfc-key-sorting	verify	1329	1410	1.06×	+0	+0
rs:schubfach-vs-dragonbox	canonical/small	verify	665	682	1.03×	+0	+0
rs:schubfach-vs-dragonbox	canonical/surrogate-pair	verify	315	339	1.08×	+0	+0
rs:schubfach-vs-dragonbox	canonical/unicode	verify	1509	1634	1.08×	+0	+0
rs:schubfach-vs-dragonbox	canonical/verify-whitespace	verify	392	405	1.03×	+0	+0
rs:ryu-vs-dragonbox	array-2048	canonicalize	1402320	1337112	1.05×	+0	+0
rs:ryu-vs-dragonbox	array-256	canonicalize	172069	168169	1.02×	+0	+0
rs:ryu-vs-dragonbox	canonical-minimal	canonicalize	202	208	1.03×	+0	+0
rs:ryu-vs-dragonbox	control-escapes	canonicalize	668	622	1.08×	+0	+0
rs:ryu-vs-dragonbox	deep	canonicalize	29262	30490	1.04×	+0	+0
rs:ryu-vs-dragonbox	deep-64	canonicalize	11020	11204	1.02×	+0	+0
rs:ryu-vs-dragonbox	escaped-key-order	canonicalize	336	338	1.00×	+0	+0
rs:ryu-vs-dragonbox	large	canonicalize	5857077	5737358	1.02×	+0	+0
rs:ryu-vs-dragonbox	long-string	canonicalize	17286	17345	1.00×	+0	+0
rs:ryu-vs-dragonbox	medium	canonicalize	178024	185833	1.04×	+0	+0
rs:ryu-vs-dragonbox	mixed-prod	canonicalize	70593	67412	1.05×	+0	+0
rs:ryu-vs-dragonbox	nested-mixed	canonicalize	1025	1033	1.01×	+0	+0
rs:ryu-vs-dragonbox	number-heavy	canonicalize	1480	1433	1.03×	+0	+0
rs:ryu-vs-dragonbox	numeric-boundary	canonicalize	1000	966	1.03×	+0	+0
rs:ryu-vs-dragonbox	rfc-key-sorting	canonicalize	1410	1372	1.03×	+0	+0
rs:ryu-vs-dragonbox	small	canonicalize	696	680	1.02×	+0	+0
rs:ryu-vs-dragonbox	surrogate-pair	canonicalize	318	339	1.07×	+0	+0
rs:ryu-vs-dragonbox	unicode	canonicalize	1537	1601	1.04×	+0	+0
rs:ryu-vs-dragonbox	verify-whitespace	canonicalize	408	411	1.01×	+0	+0
rs:ryu-vs-dragonbox	canonical/array-2048	verify	1369349	1346174	1.02×	+0	+0
rs:ryu-vs-dragonbox	canonical/array-256	verify	170020	167878	1.01×	+0	+0
rs:ryu-vs-dragonbox	canonical/canonical-minimal	verify	214	205	1.04×	+0	+0
rs:ryu-vs-dragonbox	canonical/control-escapes	verify	644	622	1.04×	+0	+0
rs:ryu-vs-dragonbox	canonical/deep	verify	30662	30257	1.01×	+0	+0
rs:ryu-vs-dragonbox	canonical/deep-64	verify	11060	11374	1.03×	+0	+0
rs:ryu-vs-dragonbox	canonical/escaped-key-order	verify	335	334	1.00×	+0	+0
rs:ryu-vs-dragonbox	canonical/large	verify	5927302	5869913	1.01×	+0	+0
rs:ryu-vs-dragonbox	canonical/long-string	verify	17542	17474	1.00×	+0	+0
rs:ryu-vs-dragonbox	canonical/medium	verify	182743	180672	1.01×	+0	+0
rs:ryu-vs-dragonbox	canonical/mixed-prod	verify	67564	67153	1.01×	+0	+0
rs:ryu-vs-dragonbox	canonical/nested-mixed	verify	1012	1011	1.00×	+0	+0
rs:ryu-vs-dragonbox	canonical/number-heavy	verify	1454	1419	1.02×	+0	+0
rs:ryu-vs-dragonbox	canonical/numeric-boundary	verify	978	962	1.02×	+0	+0
rs:ryu-vs-dragonbox	canonical/rfc-key-sorting	verify	1394	1410	1.01×	+0	+0

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
rs:ryu-vs-dragonbox	canonical/small	verify	717	682	1.05×	+0	+0
rs:ryu-vs-dragonbox	canonical/surrogate-pair	verify	306	339	1.11×	+0	+0
rs:ryu-vs-dragonbox	canonical/unicode	verify	1508	1634	1.08×	+0	+0
rs:ryu-vs-dragonbox	canonical/verify-whitespace	verify	397	405	1.02×	+0	+0
rs:ryu-vs-bd	array-2048	canonicalize	1402320	3427870	2.44×	+141429	+1814032
rs:ryu-vs-bd	array-256	canonicalize	172069	387093	2.25×	+14906	+189584
rs:ryu-vs-bd	canonical-minimal	canonicalize	202	443	2.19×	+16	+200
rs:ryu-vs-bd	control-escapes	canonicalize	668	635	1.05×	+0	+0
rs:ryu-vs-bd	deep	canonicalize	29262	30020	1.03×	+0	+0
rs:ryu-vs-bd	deep-64	canonicalize	11020	11134	1.01×	+0	+0
rs:ryu-vs-bd	escaped-key-order	canonicalize	336	832	2.47×	+33	+408
rs:ryu-vs-bd	large	canonicalize	5857077	19079970	3.26×	+926334	+12054624
rs:ryu-vs-bd	long-string	canonicalize	17286	17722	1.03×	+0	+0
rs:ryu-vs-bd	medium	canonicalize	178024	531911	2.99×	+24337	+314864
rs:ryu-vs-bd	mixed-prod	canonicalize	70593	123188	1.75×	+3941	+49904
rs:ryu-vs-bd	nested-mixed	canonicalize	1025	1981	1.93×	+66	+816
rs:ryu-vs-bd	number-heavy	canonicalize	1480	11822	7.99×	+594	+26080
rs:ryu-vs-bd	numeric-boundary	canonicalize	1000	9141	9.15×	+446	+24168
rs:ryu-vs-bd	rfc-key-sorting	canonicalize	1410	1340	1.05×	+0	+0
rs:ryu-vs-bd	small	canonicalize	696	1144	1.64×	+33	+408
rs:ryu-vs-bd	surrogate-pair	canonicalize	318	321	1.01×	+0	+0
rs:ryu-vs-bd	unicode	canonicalize	1537	1578	1.03×	+0	+0
rs:ryu-vs-bd	verify-whitespace	canonicalize	408	1182	2.90×	+50	+616
rs:ryu-vs-bd	canonical/array-2048	verify	1369349	3357314	2.45×	+141429	+1814032
rs:ryu-vs-bd	canonical/array-256	verify	170020	380560	2.24×	+14906	+189584
rs:ryu-vs-bd	canonical/canonical-minimal	verify	214	433	2.03×	+16	+200
rs:ryu-vs-bd	canonical/control-escapes	verify	644	608	1.06×	+0	+0
rs:ryu-vs-bd	canonical/deep	verify	30662	29275	1.05×	+0	+0
rs:ryu-vs-bd	canonical/deep-64	verify	11060	10823	1.02×	+0	+0
rs:ryu-vs-bd	canonical/escaped-key-order	verify	335	830	2.48×	+33	+408
rs:ryu-vs-bd	canonical/large	verify	5927302	19532718	3.30×	+926334	+12054624
rs:ryu-vs-bd	canonical/long-string	verify	17542	17308	1.01×	+0	+0
rs:ryu-vs-bd	canonical/medium	verify	182743	538538	2.95×	+24337	+314864
rs:ryu-vs-bd	canonical/mixed-prod	verify	67564	121970	1.81×	+3941	+49904
rs:ryu-vs-bd	canonical/nested-mixed	verify	1012	1985	1.96×	+66	+816
rs:ryu-vs-bd	canonical/number-heavy	verify	1454	11718	8.06×	+594	+26080
rs:ryu-vs-bd	canonical/numeric-boundary	verify	978	9039	9.25×	+446	+24168
rs:ryu-vs-bd	canonical/rfc-key-sorting	verify	1394	1309	1.06×	+0	+0
rs:ryu-vs-bd	canonical/small	verify	717	1147	1.60×	+33	+408
rs:ryu-vs-bd	canonical/surrogate-pair	verify	306	312	1.02×	+0	+0
rs:ryu-vs-bd	canonical/unicode	verify	1508	1546	1.03×	+0	+0
rs:ryu-vs-bd	canonical/verify-whitespace	verify	397	1148	2.89×	+50	+616
rs:dragonbox-vs-bd	array-2048	canonicalize	1337112	3427870	2.56×	+141429	+1814032
rs:dragonbox-vs-bd	array-256	canonicalize	168169	387093	2.30×	+14906	+189584
rs:dragonbox-vs-bd	canonical-minimal	canonicalize	208	443	2.13×	+16	+200
rs:dragonbox-vs-bd	control-escapes	canonicalize	622	635	1.02×	+0	+0

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
rs:dragonbox-vs-bd	deep	canonicalize	30490	30020	1.02×	+0	+0
rs:dragonbox-vs-bd	deep-64	canonicalize	11204	11134	1.01×	+0	+0
rs:dragonbox-vs-bd	escaped-key-order	canonicalize	338	832	2.46×	+33	+408
rs:dragonbox-vs-bd	large	canonicalize	5737358	19079970	3.33×	+926334	+12054624
rs:dragonbox-vs-bd	long-string	canonicalize	17345	17722	1.02×	+0	+0
rs:dragonbox-vs-bd	medium	canonicalize	185833	531911	2.86×	+24337	+314864
rs:dragonbox-vs-bd	mixed-prod	canonicalize	67412	123188	1.83×	+3941	+49904
rs:dragonbox-vs-bd	nested-mixed	canonicalize	1033	1981	1.92×	+66	+816
rs:dragonbox-vs-bd	number-heavy	canonicalize	1433	11822	8.25×	+594	+26080
rs:dragonbox-vs-bd	numeric-boundary	canonicalize	966	9141	9.46×	+446	+24168
rs:dragonbox-vs-bd	rfc-key-sorting	canonicalize	1372	1340	1.02×	+0	+0
rs:dragonbox-vs-bd	small	canonicalize	680	1144	1.68×	+33	+408
rs:dragonbox-vs-bd	surrogate-pair	canonicalize	339	321	1.06×	+0	+0
rs:dragonbox-vs-bd	unicode	canonicalize	1601	1578	1.01×	+0	+0
rs:dragonbox-vs-bd	verify-whitespace	canonicalize	411	1182	2.87×	+50	+616
rs:dragonbox-vs-bd	canonical/array-2048	verify	1346174	3357314	2.49×	+141429	+1814032
rs:dragonbox-vs-bd	canonical/array-256	verify	167878	380560	2.27×	+14906	+189584
rs:dragonbox-vs-bd	canonical/canonical-minimal	verify	205	433	2.11×	+16	+200
rs:dragonbox-vs-bd	canonical/control-escapes	verify	622	608	1.02×	+0	+0
rs:dragonbox-vs-bd	canonical/deep	verify	30257	29275	1.03×	+0	+0
rs:dragonbox-vs-bd	canonical/deep-64	verify	11374	10823	1.05×	+0	+0
rs:dragonbox-vs-bd	canonical/escaped-key-order	verify	334	830	2.48×	+33	+408
rs:dragonbox-vs-bd	canonical/large	verify	5869913	19532718	3.33×	+926334	+12054624
rs:dragonbox-vs-bd	canonical/long-string	verify	17474	17308	1.01×	+0	+0
rs:dragonbox-vs-bd	canonical/medium	verify	180672	538538	2.98×	+24337	+314864
rs:dragonbox-vs-bd	canonical/mixed-prod	verify	67153	121970	1.82×	+3941	+49904
rs:dragonbox-vs-bd	canonical/nested-mixed	verify	1011	1985	1.96×	+66	+816
rs:dragonbox-vs-bd	canonical/number-heavy	verify	1419	11718	8.26×	+594	+26080
rs:dragonbox-vs-bd	canonical/numeric-boundary	verify	962	9039	9.39×	+446	+24168
rs:dragonbox-vs-bd	canonical/rfc-key-sorting	verify	1410	1309	1.08×	+0	+0
rs:dragonbox-vs-bd	canonical/small	verify	682	1147	1.68×	+33	+408
rs:dragonbox-vs-bd	canonical/surrogate-pair	verify	339	312	1.09×	+0	+0
rs:dragonbox-vs-bd	canonical/unicode	verify	1634	1546	1.06×	+0	+0
rs:dragonbox-vs-bd	canonical/verify-whitespace	verify	405	1148	2.84×	+50	+616
schubfach:go-vs-rs	array-2048	canonicalize	2531402	1513396	1.67×	-12253	-3191528
schubfach:go-vs-rs	array-256	canonicalize	305209	168411	1.81×	-1509	-387568
schubfach:go-vs-rs	canonical-minimal	canonicalize	354	210	1.68×	+0	-105
schubfach:go-vs-rs	control-escapes	canonicalize	1292	646	2.00×	-8	-1495
schubfach:go-vs-rs	deep	canonicalize	36558	28846	1.27×	-141	-3696
schubfach:go-vs-rs	deep-64	canonicalize	16189	11636	1.39×	-65	-1704
schubfach:go-vs-rs	escaped-key-order	canonicalize	716	326	2.20×	-2	-707
schubfach:go-vs-rs	large	canonicalize	11851487	5816696	2.04×	-57301	-13736557
schubfach:go-vs-rs	long-string	canonicalize	76188	16928	4.50×	-4	-2765
schubfach:go-vs-rs	medium	canonicalize	339803	182570	1.86×	-1758	-396434
schubfach:go-vs-rs	mixed-prod	canonicalize	134635	69235	1.94×	-700	-165818
schubfach:go-vs-rs	nested-mixed	canonicalize	2192	1004	2.18×	-8	-2352

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
schubfach:go-vs-rs	number-heavy	canonicalize	13860	1437	9.65×	-23	-3036
schubfach:go-vs-rs	numeric-boundary	canonicalize	12714	980	12.98×	-14	-2172
schubfach:go-vs-rs	rfc-key-sorting	canonicalize	2988	1370	2.18×	-20	-2569
schubfach:go-vs-rs	small	canonicalize	1240	654	1.89×	-3	-1375
schubfach:go-vs-rs	surrogate-pair	canonicalize	650	317	2.05×	-4	-713
schubfach:go-vs-rs	unicode	canonicalize	1564	1511	1.04×	-6	-1419
schubfach:go-vs-rs	verify-whitespace	canonicalize	914	400	2.29×	-4	-1002
schubfach:go-vs-rs	canonical/array-2048	verify	2475144	1345524	1.84×	-12253	-3191521
schubfach:go-vs-rs	canonical/array-256	verify	317805	168727	1.88×	-1509	-387569
schubfach:go-vs-rs	canonical/canonical-minimal	verify	360	208	1.73×	+0	-105
schubfach:go-vs-rs	canonical/control-escapes	verify	1292	634	2.04×	-8	-1495
schubfach:go-vs-rs	canonical/deep	verify	38126	28599	1.33×	-141	-3696
schubfach:go-vs-rs	canonical/deep-64	verify	16844	11091	1.52×	-65	-1704
schubfach:go-vs-rs	canonical/escaped-key-order	verify	695	329	2.11×	-2	-735
schubfach:go-vs-rs	canonical/large	verify	11434977	5913158	1.93×	-57301	-13737457
schubfach:go-vs-rs	canonical/long-string	verify	77429	17396	4.45×	-4	-2765
schubfach:go-vs-rs	canonical/medium	verify	341094	178455	1.91×	-1758	-396471
schubfach:go-vs-rs	canonical/mixed-prod	verify	133722	66720	2.00×	-700	-165845
schubfach:go-vs-rs	canonical/nested-mixed	verify	2211	1010	2.19×	-8	-2352
schubfach:go-vs-rs	canonical/number-heavy	verify	13970	1424	9.81×	-23	-3040
schubfach:go-vs-rs	canonical/numeric-boundary	verify	12736	956	13.33×	-14	-2172
schubfach:go-vs-rs	canonical/rfc-key-sorting	verify	2852	1329	2.15×	-15	-2575
schubfach:go-vs-rs	canonical/small	verify	1223	665	1.84×	-3	-1378
schubfach:go-vs-rs	canonical/surrogate-pair	verify	665	315	2.11×	-4	-713
schubfach:go-vs-rs	canonical/unicode	verify	1561	1509	1.03×	-6	-1405
schubfach:go-vs-rs	canonical/verify-whitespace	verify	934	392	2.38×	-4	-1009
bd:go-vs-rs	array-2048	canonicalize	4153082	3427870	1.21×	+120966	-1564150
bd:go-vs-rs	array-256	canonicalize	479254	387093	1.24×	+12377	-222961
bd:go-vs-rs	canonical-minimal	canonicalize	529	443	1.19×	+16	+95
bd:go-vs-rs	control-escapes	canonicalize	1310	635	2.06×	-8	-1495
bd:go-vs-rs	deep	canonicalize	36491	30020	1.22×	-141	-3696
bd:go-vs-rs	deep-64	canonicalize	16140	11134	1.45×	-65	-1704
bd:go-vs-rs	escaped-key-order	canonicalize	1138	832	1.37×	+29	-340
bd:go-vs-rs	large	canonicalize	20936873	19079970	1.10×	+836241	-2345268
bd:go-vs-rs	long-string	canonicalize	76724	17722	4.33×	-4	-2765
bd:go-vs-rs	medium	canonicalize	596274	531911	1.12×	+21565	-102700
bd:go-vs-rs	mixed-prod	canonicalize	183236	123188	1.49×	+2986	-122147
bd:go-vs-rs	nested-mixed	canonicalize	3104	1981	1.57×	+54	-1621
bd:go-vs-rs	number-heavy	canonicalize	21718	11822	1.84×	+551	+22214
bd:go-vs-rs	numeric-boundary	canonicalize	19052	9141	2.08×	+418	+21290
bd:go-vs-rs	rfc-key-sorting	canonicalize	3080	1340	2.30×	-20	-2569
bd:go-vs-rs	small	canonicalize	1685	1144	1.47×	+28	-1017
bd:go-vs-rs	surrogate-pair	canonicalize	660	321	2.05×	-4	-713
bd:go-vs-rs	unicode	canonicalize	1561	1578	1.01×	-6	-1419
bd:go-vs-rs	verify-whitespace	canonicalize	1576	1182	1.33×	+42	-468
bd:go-vs-rs	canonical/array-2048	verify	4113295	3357314	1.23×	+120968	-1564086

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
bd:go-vs-rs	canonical/array-256	verify	490966	380560	1.29×	+12375	-223215
bd:go-vs-rs	canonical/canonical-minimal	verify	543	433	1.25×	+16	+95
bd:go-vs-rs	canonical/control-escapes	verify	1322	608	2.18×	-8	-1495
bd:go-vs-rs	canonical/deep	verify	37749	29275	1.29×	-141	-3696
bd:go-vs-rs	canonical/deep-64	verify	16630	10823	1.54×	-65	-1704
bd:go-vs-rs	canonical/escaped-key-order	verify	1120	830	1.35×	+29	-368
bd:go-vs-rs	canonical/large	verify	20458581	19532718	1.05×	+836243	-2345913
bd:go-vs-rs	canonical/long-string	verify	77802	17308	4.50×	-4	-2765
bd:go-vs-rs	canonical/medium	verify	591838	538538	1.10×	+21564	-102978
bd:go-vs-rs	canonical/mixed-prod	verify	179814	121970	1.47×	+2986	-122264
bd:go-vs-rs	canonical/nested-mixed	verify	3111	1985	1.57×	+54	-1622
bd:go-vs-rs	canonical/number-heavy	verify	21748	11718	1.86×	+551	+22206
bd:go-vs-rs	canonical/numeric-boundary	verify	18914	9039	2.09×	+418	+21287
bd:go-vs-rs	canonical/rfc-key-sorting	verify	2873	1309	2.19×	-15	-2575
bd:go-vs-rs	canonical/small	verify	1653	1147	1.44×	+28	-1021
bd:go-vs-rs	canonical/surrogate-pair	verify	668	312	2.14×	-4	-713
bd:go-vs-rs	canonical/unicode	verify	1555	1546	1.01×	-6	-1405
bd:go-vs-rs	canonical/verify-whitespace	verify	1592	1148	1.39×	+42	-475
ryu:go-vs-rs	array-2048	canonicalize	2502576	1402320	1.78×	-12253	-3191523
ryu:go-vs-rs	array-256	canonicalize	302518	172069	1.76×	-1509	-387568
ryu:go-vs-rs	canonical-minimal	canonicalize	344	202	1.70×	+0	-105
ryu:go-vs-rs	control-escapes	canonicalize	1281	668	1.92×	-8	-1495
ryu:go-vs-rs	deep	canonicalize	37235	29262	1.27×	-141	-3696
ryu:go-vs-rs	deep-64	canonicalize	16390	11020	1.49×	-65	-1704
ryu:go-vs-rs	escaped-key-order	canonicalize	711	336	2.11×	-2	-707
ryu:go-vs-rs	large	canonicalize	11785645	5857077	2.01×	-57301	-13736557
ryu:go-vs-rs	long-string	canonicalize	76170	17286	4.41×	-4	-2765
ryu:go-vs-rs	medium	canonicalize	337542	178024	1.90×	-1758	-396434
ryu:go-vs-rs	mixed-prod	canonicalize	135246	70593	1.92×	-700	-165818
ryu:go-vs-rs	nested-mixed	canonicalize	2207	1025	2.15×	-8	-2352
ryu:go-vs-rs	number-heavy	canonicalize	13898	1480	9.39×	-23	-3036
ryu:go-vs-rs	numeric-boundary	canonicalize	12702	1000	12.71×	-14	-2172
ryu:go-vs-rs	rfc-key-sorting	canonicalize	3004	1410	2.13×	-20	-2569
ryu:go-vs-rs	small	canonicalize	1246	696	1.79×	-3	-1375
ryu:go-vs-rs	surrogate-pair	canonicalize	652	318	2.05×	-4	-713
ryu:go-vs-rs	unicode	canonicalize	1577	1537	1.03×	-6	-1419
ryu:go-vs-rs	verify-whitespace	canonicalize	912	408	2.24×	-4	-1002
ryu:go-vs-rs	canonical/array-2048	verify	2487780	1369349	1.82×	-12253	-3191520
ryu:go-vs-rs	canonical/array-256	verify	315702	170020	1.86×	-1509	-387569
ryu:go-vs-rs	canonical/canonical-minimal	verify	359	214	1.68×	+0	-105
ryu:go-vs-rs	canonical/control-escapes	verify	1286	644	2.00×	-8	-1495
ryu:go-vs-rs	canonical/deep	verify	38082	30662	1.24×	-141	-3696
ryu:go-vs-rs	canonical/deep-64	verify	16854	11060	1.52×	-65	-1704
ryu:go-vs-rs	canonical/escaped-key-order	verify	692	335	2.07×	-2	-735
ryu:go-vs-rs	canonical/large	verify	11401357	5927302	1.92×	-57301	-13737448
ryu:go-vs-rs	canonical/long-string	verify	78084	17542	4.45×	-4	-2765

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
ryu:go-vs-rs	canonical/medium	verify	343581	182743	1.88×	-1758	-396471
ryu:go-vs-rs	canonical/mixed-prod	verify	134634	67564	1.99×	-700	-165845
ryu:go-vs-rs	canonical/nested-mixed	verify	2236	1012	2.21×	-8	-2352
ryu:go-vs-rs	canonical/number-heavy	verify	14045	1454	9.66×	-23	-3040
ryu:go-vs-rs	canonical/numeric-boundary	verify	12855	978	13.15×	-14	-2172
ryu:go-vs-rs	canonical/rfc-key-sorting	verify	2873	1394	2.06×	-15	-2575
ryu:go-vs-rs	canonical/small	verify	1233	717	1.72×	-3	-1378
ryu:go-vs-rs	canonical/surrogate-pair	verify	666	306	2.18×	-4	-713
ryu:go-vs-rs	canonical/unicode	verify	1576	1508	1.05×	-6	-1405
ryu:go-vs-rs	canonical/verify-whitespace	verify	939	397	2.36×	-4	-1009
dragonbox:go-vs-rs	array-2048	canonicalize	2530127	1337112	1.89×	-12253	-3191524
dragonbox:go-vs-rs	array-256	canonicalize	302413	168169	1.80×	-1509	-387568
dragonbox:go-vs-rs	canonical-minimal	canonicalize	345	208	1.66×	+0	-105
dragonbox:go-vs-rs	control-escapes	canonicalize	1288	622	2.07×	-8	-1495
dragonbox:go-vs-rs	deep	canonicalize	37345	30490	1.22×	-141	-3696
dragonbox:go-vs-rs	deep-64	canonicalize	16493	11204	1.47×	-65	-1704
dragonbox:go-vs-rs	escaped-key-order	canonicalize	719	338	2.13×	-2	-707
dragonbox:go-vs-rs	large	canonicalize	11787210	5737358	2.05×	-57301	-13736544
dragonbox:go-vs-rs	long-string	canonicalize	76427	17345	4.41×	-4	-2765
dragonbox:go-vs-rs	medium	canonicalize	336597	185833	1.81×	-1758	-396434
dragonbox:go-vs-rs	mixed-prod	canonicalize	134935	67412	2.00×	-700	-165818
dragonbox:go-vs-rs	nested-mixed	canonicalize	2196	1033	2.12×	-8	-2352
dragonbox:go-vs-rs	number-heavy	canonicalize	13771	1433	9.61×	-23	-3036
dragonbox:go-vs-rs	numeric-boundary	canonicalize	12666	966	13.11×	-14	-2172
dragonbox:go-vs-rs	rfc-key-sorting	canonicalize	2996	1372	2.18×	-20	-2569
dragonbox:go-vs-rs	small	canonicalize	1242	680	1.83×	-3	-1375
dragonbox:go-vs-rs	surrogate-pair	canonicalize	659	339	1.94×	-4	-713
dragonbox:go-vs-rs	unicode	canonicalize	1569	1601	1.02×	-6	-1419
dragonbox:go-vs-rs	verify-whitespace	canonicalize	915	411	2.22×	-4	-1002
dragonbox:go-vs-rs	canonical/array-2048	verify	2508195	1346174	1.86×	-12253	-3191520
dragonbox:go-vs-rs	canonical/array-256	verify	315659	167878	1.88×	-1509	-387570
dragonbox:go-vs-rs	canonical/canonical-minimal	verify	358	205	1.75×	+0	-105
dragonbox:go-vs-rs	canonical/control-escapes	verify	1293	622	2.08×	-8	-1495
dragonbox:go-vs-rs	canonical/deep	verify	38406	30257	1.27×	-141	-3696
dragonbox:go-vs-rs	canonical/deep-64	verify	16918	11374	1.49×	-65	-1704
dragonbox:go-vs-rs	canonical/escaped-key-order	verify	694	334	2.08×	-2	-735
dragonbox:go-vs-rs	canonical/large	verify	11524340	5869913	1.96×	-57300	-13737442
dragonbox:go-vs-rs	canonical/long-string	verify	77534	17474	4.44×	-4	-2765
dragonbox:go-vs-rs	canonical/medium	verify	342029	180672	1.89×	-1758	-396472
dragonbox:go-vs-rs	canonical/mixed-prod	verify	134529	67153	2.00×	-700	-165845
dragonbox:go-vs-rs	canonical/nested-mixed	verify	2224	1011	2.20×	-8	-2352
dragonbox:go-vs-rs	canonical/number-heavy	verify	13960	1419	9.84×	-23	-3040
dragonbox:go-vs-rs	canonical/numeric-boundary	verify	12803	962	13.31×	-14	-2172
dragonbox:go-vs-rs	canonical/rfc-key-sorting	verify	2880	1410	2.04×	-15	-2575
dragonbox:go-vs-rs	canonical/small	verify	1228	682	1.80×	-3	-1378
dragonbox:go-vs-rs	canonical/surrogate-pair	verify	664	339	1.96×	-4	-713

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Pair	Workload	Mode	ns/op		Speedup	Δallocs/op	ΔB/op
dragonbox:go-vs-rs	canonical/unicode	verify	1573	1634	1.04×	-6	-1405
dragonbox:go-vs-rs	canonical/verify-whitespace	verify	946	405	2.34×	-4	-1009

B/op = bytes allocated per operation. allocs/op = heap allocations per operation.
Δ columns show Impl B − Impl A for each pair.

A.2 Full CLI Benchmarks

Table 9: CLI benchmark results (end-to-end, $n = 15$).

Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
go:schubfach-vs-bd	array-2048	canonicalize	5.714	7.144	1.25×	schubfach
go:schubfach-vs-bd	array-2048	verify	5.609	7.334	1.31×	schubfach
go:schubfach-vs-bd	array-256	canonicalize	2.008	2.326	1.16×	schubfach
go:schubfach-vs-bd	array-256	verify	2.038	2.321	1.14×	schubfach
go:schubfach-vs-bd	canonical-minimal	canonicalize	1.496	1.529	1.02×	schubfach
go:schubfach-vs-bd	canonical-minimal	verify	1.569	1.775	1.13×	schubfach
go:schubfach-vs-bd	control-escapes	canonicalize	1.465	1.603	1.09×	schubfach
go:schubfach-vs-bd	control-escapes	verify	1.569	1.582	1.01×	schubfach
go:schubfach-vs-bd	deep	canonicalize	1.489	1.749	1.17×	schubfach
go:schubfach-vs-bd	deep	verify	1.663	1.777	1.07×	schubfach
go:schubfach-vs-bd	deep-64	canonicalize	1.521	1.739	1.14×	schubfach
go:schubfach-vs-bd	deep-64	verify	1.610	1.588	1.01×	json-canon
go:schubfach-vs-bd	escaped-key-order	canonicalize	1.525	1.795	1.18×	schubfach
go:schubfach-vs-bd	escaped-key-order	verify	1.558	1.467	1.06×	json-canon
go:schubfach-vs-bd	large	canonicalize	16.338	25.285	1.55×	schubfach
go:schubfach-vs-bd	large	verify	16.748	24.851	1.48×	schubfach
go:schubfach-vs-bd	long-string	canonicalize	1.591	1.811	1.14×	schubfach
go:schubfach-vs-bd	long-string	verify	1.573	1.792	1.14×	schubfach
go:schubfach-vs-bd	medium	canonicalize	2.158	2.631	1.22×	schubfach
go:schubfach-vs-bd	medium	verify	2.088	2.350	1.13×	schubfach
go:schubfach-vs-bd	mixed-prod	canonicalize	1.667	1.965	1.18×	schubfach
go:schubfach-vs-bd	mixed-prod	verify	1.879	1.970	1.05×	schubfach
go:schubfach-vs-bd	nested-mixed	canonicalize	1.631	1.671	1.02×	schubfach
go:schubfach-vs-bd	nested-mixed	verify	1.393	1.663	1.19×	schubfach
go:schubfach-vs-bd	number-heavy	canonicalize	1.493	1.653	1.11×	schubfach
go:schubfach-vs-bd	number-heavy	verify	1.437	1.499	1.04×	schubfach
go:schubfach-vs-bd	numeric-boundary	canonicalize	1.489	1.737	1.17×	schubfach
go:schubfach-vs-bd	numeric-boundary	verify	1.572	1.608	1.02×	schubfach
go:schubfach-vs-bd	rfc-key-sorting	canonicalize	1.463	1.564	1.07×	schubfach
go:schubfach-vs-bd	rfc-key-sorting	verify	1.466	1.597	1.09×	schubfach
go:schubfach-vs-bd	small	canonicalize	1.452	1.567	1.08×	schubfach

Continued on next page

Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
go:schubfach-vs-bd	small	verify	1.423	1.739	1.22×	schubfach
go:schubfach-vs-bd	surrogate-pair	canonicalize	1.479	1.556	1.05×	schubfach
go:schubfach-vs-bd	surrogate-pair	verify	1.518	1.651	1.09×	schubfach
go:schubfach-vs-bd	unicode	canonicalize	1.482	1.625	1.10×	schubfach
go:schubfach-vs-bd	unicode	verify	1.506	1.594	1.06×	schubfach
go:schubfach-vs-bd	verify-whitespace	canonicalize	1.545	1.643	1.06×	schubfach
go:schubfach-vs-bd	verify-whitespace	verify	1.640	1.677	1.02×	schubfach
go:schubfach-vs-ryu	array-2048	canonicalize	5.714	5.440	1.05×	ryu
go:schubfach-vs-ryu	array-2048	verify	5.609	5.590	1.00×	ryu
go:schubfach-vs-ryu	array-256	canonicalize	2.008	2.085	1.04×	schubfach
go:schubfach-vs-ryu	array-256	verify	2.038	2.111	1.04×	schubfach
go:schubfach-vs-ryu	canonical-minimal	canonicalize	1.496	1.379	1.08×	ryu
go:schubfach-vs-ryu	canonical-minimal	verify	1.569	1.625	1.04×	schubfach
go:schubfach-vs-ryu	control-escapes	canonicalize	1.465	1.381	1.06×	ryu
go:schubfach-vs-ryu	control-escapes	verify	1.569	1.498	1.05×	ryu
go:schubfach-vs-ryu	deep	canonicalize	1.489	1.644	1.10×	schubfach
go:schubfach-vs-ryu	deep	verify	1.663	1.574	1.06×	ryu
go:schubfach-vs-ryu	deep-64	canonicalize	1.521	1.608	1.06×	schubfach
go:schubfach-vs-ryu	deep-64	verify	1.610	1.534	1.05×	ryu
go:schubfach-vs-ryu	escaped-key-order	canonicalize	1.525	1.631	1.07×	schubfach
go:schubfach-vs-ryu	escaped-key-order	verify	1.558	1.452	1.07×	ryu
go:schubfach-vs-ryu	large	canonicalize	16.338	16.015	1.02×	ryu
go:schubfach-vs-ryu	large	verify	16.748	16.076	1.04×	ryu
go:schubfach-vs-ryu	long-string	canonicalize	1.591	1.587	1.00×	ryu
go:schubfach-vs-ryu	long-string	verify	1.573	1.792	1.14×	schubfach
go:schubfach-vs-ryu	medium	canonicalize	2.158	1.980	1.09×	ryu
go:schubfach-vs-ryu	medium	verify	2.088	2.213	1.06×	schubfach
go:schubfach-vs-ryu	mixed-prod	canonicalize	1.667	1.775	1.06×	schubfach
go:schubfach-vs-ryu	mixed-prod	verify	1.879	1.765	1.06×	ryu
go:schubfach-vs-ryu	nested-mixed	canonicalize	1.631	1.471	1.11×	ryu
go:schubfach-vs-ryu	nested-mixed	verify	1.393	1.615	1.16×	schubfach
go:schubfach-vs-ryu	number-heavy	canonicalize	1.493	1.553	1.04×	schubfach
go:schubfach-vs-ryu	number-heavy	verify	1.437	1.564	1.09×	schubfach
go:schubfach-vs-ryu	numeric-boundary	canonicalize	1.489	1.599	1.07×	schubfach
go:schubfach-vs-ryu	numeric-boundary	verify	1.572	1.477	1.06×	ryu
go:schubfach-vs-ryu	rfc-key-sorting	canonicalize	1.463	1.488	1.02×	schubfach
go:schubfach-vs-ryu	rfc-key-sorting	verify	1.466	1.512	1.03×	schubfach
go:schubfach-vs-ryu	small	canonicalize	1.452	1.519	1.05×	schubfach
go:schubfach-vs-ryu	small	verify	1.423	1.453	1.02×	schubfach
go:schubfach-vs-ryu	surrogate-pair	canonicalize	1.479	1.445	1.02×	ryu
go:schubfach-vs-ryu	surrogate-pair	verify	1.518	1.601	1.05×	schubfach
go:schubfach-vs-ryu	unicode	canonicalize	1.482	1.502	1.01×	schubfach
go:schubfach-vs-ryu	unicode	verify	1.506	1.440	1.05×	ryu
go:schubfach-vs-ryu	verify-whitespace	canonicalize	1.545	1.566	1.01×	schubfach
go:schubfach-vs-ryu	verify-whitespace	verify	1.640	1.430	1.15×	ryu
go:schubfach-vs-dragonbox	array-2048	canonicalize	5.714	5.363	1.07×	dragonbox

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
go:schubfach-vs-dragonbox	array-2048	verify	5.609	5.828	1.04×	schubfach
go:schubfach-vs-dragonbox	array-256	canonicalize	2.008	1.843	1.09×	dragonbox
go:schubfach-vs-dragonbox	array-256	verify	2.038	1.860	1.10×	dragonbox
go:schubfach-vs-dragonbox	canonical-minimal	canonicalize	1.496	1.455	1.03×	dragonbox
go:schubfach-vs-dragonbox	canonical-minimal	verify	1.569	1.608	1.02×	schubfach
go:schubfach-vs-dragonbox	control-escapes	canonicalize	1.465	1.476	1.01×	schubfach
go:schubfach-vs-dragonbox	control-escapes	verify	1.569	1.417	1.11×	dragonbox
go:schubfach-vs-dragonbox	deep	canonicalize	1.489	1.712	1.15×	schubfach
go:schubfach-vs-dragonbox	deep	verify	1.663	1.613	1.03×	dragonbox
go:schubfach-vs-dragonbox	deep-64	canonicalize	1.521	1.616	1.06×	schubfach
go:schubfach-vs-dragonbox	deep-64	verify	1.610	1.643	1.02×	schubfach
go:schubfach-vs-dragonbox	escaped-key-order	canonicalize	1.525	1.483	1.03×	dragonbox
go:schubfach-vs-dragonbox	escaped-key-order	verify	1.558	1.429	1.09×	dragonbox
go:schubfach-vs-dragonbox	large	canonicalize	16.338	15.905	1.03×	dragonbox
go:schubfach-vs-dragonbox	large	verify	16.748	16.074	1.04×	dragonbox
go:schubfach-vs-dragonbox	long-string	canonicalize	1.591	1.737	1.09×	schubfach
go:schubfach-vs-dragonbox	long-string	verify	1.573	1.749	1.11×	schubfach
go:schubfach-vs-dragonbox	medium	canonicalize	2.158	2.021	1.07×	dragonbox
go:schubfach-vs-dragonbox	medium	verify	2.088	1.888	1.11×	dragonbox
go:schubfach-vs-dragonbox	mixed-prod	canonicalize	1.667	1.659	1.00×	dragonbox
go:schubfach-vs-dragonbox	mixed-prod	verify	1.879	1.757	1.07×	dragonbox
go:schubfach-vs-dragonbox	nested-mixed	canonicalize	1.631	1.514	1.08×	dragonbox
go:schubfach-vs-dragonbox	nested-mixed	verify	1.393	1.580	1.13×	schubfach
go:schubfach-vs-dragonbox	number-heavy	canonicalize	1.493	1.578	1.06×	schubfach
go:schubfach-vs-dragonbox	number-heavy	verify	1.437	1.575	1.10×	schubfach
go:schubfach-vs-dragonbox	numeric-boundary	canonicalize	1.489	1.450	1.03×	dragonbox
go:schubfach-vs-dragonbox	numeric-boundary	verify	1.572	1.463	1.07×	dragonbox
go:schubfach-vs-dragonbox	rfc-key-sorting	canonicalize	1.463	1.466	1.00×	schubfach
go:schubfach-vs-dragonbox	rfc-key-sorting	verify	1.466	1.567	1.07×	schubfach
go:schubfach-vs-dragonbox	small	canonicalize	1.452	1.527	1.05×	schubfach
go:schubfach-vs-dragonbox	small	verify	1.423	1.423	1.00×	dragonbox
go:schubfach-vs-dragonbox	surrogate-pair	canonicalize	1.479	1.572	1.06×	schubfach
go:schubfach-vs-dragonbox	surrogate-pair	verify	1.518	1.610	1.06×	schubfach
go:schubfach-vs-dragonbox	unicode	canonicalize	1.482	1.489	1.00×	schubfach
go:schubfach-vs-dragonbox	unicode	verify	1.506	1.899	1.26×	schubfach
go:schubfach-vs-dragonbox	verify-whitespace	canonicalize	1.545	1.494	1.03×	dragonbox
go:schubfach-vs-dragonbox	verify-whitespace	verify	1.640	1.510	1.09×	dragonbox
go:ryu-vs-dragonbox	array-2048	canonicalize	5.440	5.363	1.01×	dragonbox
go:ryu-vs-dragonbox	array-2048	verify	5.590	5.828	1.04×	ryu
go:ryu-vs-dragonbox	array-256	canonicalize	2.085	1.843	1.13×	dragonbox
go:ryu-vs-dragonbox	array-256	verify	2.111	1.860	1.13×	dragonbox
go:ryu-vs-dragonbox	canonical-minimal	canonicalize	1.379	1.455	1.06×	ryu
go:ryu-vs-dragonbox	canonical-minimal	verify	1.625	1.608	1.01×	dragonbox
go:ryu-vs-dragonbox	control-escapes	canonicalize	1.381	1.476	1.07×	ryu
go:ryu-vs-dragonbox	control-escapes	verify	1.498	1.417	1.06×	dragonbox
go:ryu-vs-dragonbox	deep	canonicalize	1.644	1.712	1.04×	ryu

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
go:ryu-vs-dragonbox	deep	verify	1.574	1.613	1.02×	ryu
go:ryu-vs-dragonbox	deep-64	canonicalize	1.608	1.616	1.01×	ryu
go:ryu-vs-dragonbox	deep-64	verify	1.534	1.643	1.07×	ryu
go:ryu-vs-dragonbox	escaped-key-order	canonicalize	1.631	1.483	1.10×	dragonbox
go:ryu-vs-dragonbox	escaped-key-order	verify	1.452	1.429	1.02×	dragonbox
go:ryu-vs-dragonbox	large	canonicalize	16.015	15.905	1.01×	dragonbox
go:ryu-vs-dragonbox	large	verify	16.076	16.074	1.00×	dragonbox
go:ryu-vs-dragonbox	long-string	canonicalize	1.587	1.737	1.09×	ryu
go:ryu-vs-dragonbox	long-string	verify	1.792	1.749	1.02×	dragonbox
go:ryu-vs-dragonbox	medium	canonicalize	1.980	2.021	1.02×	ryu
go:ryu-vs-dragonbox	medium	verify	2.213	1.888	1.17×	dragonbox
go:ryu-vs-dragonbox	mixed-prod	canonicalize	1.775	1.659	1.07×	dragonbox
go:ryu-vs-dragonbox	mixed-prod	verify	1.765	1.757	1.00×	dragonbox
go:ryu-vs-dragonbox	nested-mixed	canonicalize	1.471	1.514	1.03×	ryu
go:ryu-vs-dragonbox	nested-mixed	verify	1.615	1.580	1.02×	dragonbox
go:ryu-vs-dragonbox	number-heavy	canonicalize	1.553	1.578	1.02×	ryu
go:ryu-vs-dragonbox	number-heavy	verify	1.564	1.575	1.01×	ryu
go:ryu-vs-dragonbox	numeric-boundary	canonicalize	1.599	1.450	1.10×	dragonbox
go:ryu-vs-dragonbox	numeric-boundary	verify	1.477	1.463	1.01×	dragonbox
go:ryu-vs-dragonbox	rfc-key-sorting	canonicalize	1.488	1.466	1.01×	dragonbox
go:ryu-vs-dragonbox	rfc-key-sorting	verify	1.512	1.567	1.04×	ryu
go:ryu-vs-dragonbox	small	canonicalize	1.519	1.527	1.01×	ryu
go:ryu-vs-dragonbox	small	verify	1.453	1.423	1.02×	dragonbox
go:ryu-vs-dragonbox	surrogate-pair	canonicalize	1.445	1.572	1.09×	ryu
go:ryu-vs-dragonbox	surrogate-pair	verify	1.601	1.610	1.01×	ryu
go:ryu-vs-dragonbox	unicode	canonicalize	1.502	1.489	1.01×	dragonbox
go:ryu-vs-dragonbox	unicode	verify	1.440	1.899	1.32×	ryu
go:ryu-vs-dragonbox	verify-whitespace	canonicalize	1.566	1.494	1.05×	dragonbox
go:ryu-vs-dragonbox	verify-whitespace	verify	1.430	1.510	1.06×	ryu
go:ryu-vs-bd	array-2048	canonicalize	5.440	7.144	1.31×	ryu
go:ryu-vs-bd	array-2048	verify	5.590	7.334	1.31×	ryu
go:ryu-vs-bd	array-256	canonicalize	2.085	2.326	1.12×	ryu
go:ryu-vs-bd	array-256	verify	2.111	2.321	1.10×	ryu
go:ryu-vs-bd	canonical-minimal	canonicalize	1.379	1.529	1.11×	ryu
go:ryu-vs-bd	canonical-minimal	verify	1.625	1.775	1.09×	ryu
go:ryu-vs-bd	control-escapes	canonicalize	1.381	1.603	1.16×	ryu
go:ryu-vs-bd	control-escapes	verify	1.498	1.582	1.06×	ryu
go:ryu-vs-bd	deep	canonicalize	1.644	1.749	1.06×	ryu
go:ryu-vs-bd	deep	verify	1.574	1.777	1.13×	ryu
go:ryu-vs-bd	deep-64	canonicalize	1.608	1.739	1.08×	ryu
go:ryu-vs-bd	deep-64	verify	1.534	1.588	1.04×	ryu
go:ryu-vs-bd	escaped-key-order	canonicalize	1.631	1.795	1.10×	ryu
go:ryu-vs-bd	escaped-key-order	verify	1.452	1.467	1.01×	ryu
go:ryu-vs-bd	large	canonicalize	16.015	25.285	1.58×	ryu
go:ryu-vs-bd	large	verify	16.076	24.851	1.55×	ryu
go:ryu-vs-bd	long-string	canonicalize	1.587	1.811	1.14×	ryu

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
go:ryu-vs-bd	long-string	verify	1.792	1.792	1.00×	ryu
go:ryu-vs-bd	medium	canonicalize	1.980	2.631	1.33×	ryu
go:ryu-vs-bd	medium	verify	2.213	2.350	1.06×	ryu
go:ryu-vs-bd	mixed-prod	canonicalize	1.775	1.965	1.11×	ryu
go:ryu-vs-bd	mixed-prod	verify	1.765	1.970	1.12×	ryu
go:ryu-vs-bd	nested-mixed	canonicalize	1.471	1.671	1.14×	ryu
go:ryu-vs-bd	nested-mixed	verify	1.615	1.663	1.03×	ryu
go:ryu-vs-bd	number-heavy	canonicalize	1.553	1.653	1.06×	ryu
go:ryu-vs-bd	number-heavy	verify	1.564	1.499	1.04×	json-canon
go:ryu-vs-bd	numeric-boundary	canonicalize	1.599	1.737	1.09×	ryu
go:ryu-vs-bd	numeric-boundary	verify	1.477	1.608	1.09×	ryu
go:ryu-vs-bd	rfc-key-sorting	canonicalize	1.488	1.564	1.05×	ryu
go:ryu-vs-bd	rfc-key-sorting	verify	1.512	1.597	1.06×	ryu
go:ryu-vs-bd	small	canonicalize	1.519	1.567	1.03×	ryu
go:ryu-vs-bd	small	verify	1.453	1.739	1.20×	ryu
go:ryu-vs-bd	surrogate-pair	canonicalize	1.445	1.556	1.08×	ryu
go:ryu-vs-bd	surrogate-pair	verify	1.601	1.651	1.03×	ryu
go:ryu-vs-bd	unicode	canonicalize	1.502	1.625	1.08×	ryu
go:ryu-vs-bd	unicode	verify	1.440	1.594	1.11×	ryu
go:ryu-vs-bd	verify-whitespace	canonicalize	1.566	1.643	1.05×	ryu
go:ryu-vs-bd	verify-whitespace	verify	1.430	1.677	1.17×	ryu
go:dragonbox-vs-bd	array-2048	canonicalize	5.363	7.144	1.33×	dragonbox
go:dragonbox-vs-bd	array-2048	verify	5.828	7.334	1.26×	dragonbox
go:dragonbox-vs-bd	array-256	canonicalize	1.843	2.326	1.26×	dragonbox
go:dragonbox-vs-bd	array-256	verify	1.860	2.321	1.25×	dragonbox
go:dragonbox-vs-bd	canonical-minimal	canonicalize	1.455	1.529	1.05×	dragonbox
go:dragonbox-vs-bd	canonical-minimal	verify	1.608	1.775	1.10×	dragonbox
go:dragonbox-vs-bd	control-escapes	canonicalize	1.476	1.603	1.09×	dragonbox
go:dragonbox-vs-bd	control-escapes	verify	1.417	1.582	1.12×	dragonbox
go:dragonbox-vs-bd	deep	canonicalize	1.712	1.749	1.02×	dragonbox
go:dragonbox-vs-bd	deep	verify	1.613	1.777	1.10×	dragonbox
go:dragonbox-vs-bd	deep-64	canonicalize	1.616	1.739	1.08×	dragonbox
go:dragonbox-vs-bd	deep-64	verify	1.643	1.588	1.03×	json-canon
go:dragonbox-vs-bd	escaped-key-order	canonicalize	1.483	1.795	1.21×	dragonbox
go:dragonbox-vs-bd	escaped-key-order	verify	1.429	1.467	1.03×	dragonbox
go:dragonbox-vs-bd	large	canonicalize	15.905	25.285	1.59×	dragonbox
go:dragonbox-vs-bd	large	verify	16.074	24.851	1.55×	dragonbox
go:dragonbox-vs-bd	long-string	canonicalize	1.737	1.811	1.04×	dragonbox
go:dragonbox-vs-bd	long-string	verify	1.749	1.792	1.02×	dragonbox
go:dragonbox-vs-bd	medium	canonicalize	2.021	2.631	1.30×	dragonbox
go:dragonbox-vs-bd	medium	verify	1.888	2.350	1.24×	dragonbox
go:dragonbox-vs-bd	mixed-prod	canonicalize	1.659	1.965	1.18×	dragonbox
go:dragonbox-vs-bd	mixed-prod	verify	1.757	1.970	1.12×	dragonbox
go:dragonbox-vs-bd	nested-mixed	canonicalize	1.514	1.671	1.10×	dragonbox
go:dragonbox-vs-bd	nested-mixed	verify	1.580	1.663	1.05×	dragonbox
go:dragonbox-vs-bd	number-heavy	canonicalize	1.578	1.653	1.05×	dragonbox

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
go:dragonbox-vs-bd	number-heavy	verify	1.575	1.499	1.05×	json-canon
go:dragonbox-vs-bd	numeric-boundary	canonicalize	1.450	1.737	1.20×	dragonbox
go:dragonbox-vs-bd	numeric-boundary	verify	1.463	1.608	1.10×	dragonbox
go:dragonbox-vs-bd	rfc-key-sorting	canonicalize	1.466	1.564	1.07×	dragonbox
go:dragonbox-vs-bd	rfc-key-sorting	verify	1.567	1.597	1.02×	dragonbox
go:dragonbox-vs-bd	small	canonicalize	1.527	1.567	1.03×	dragonbox
go:dragonbox-vs-bd	small	verify	1.423	1.739	1.22×	dragonbox
go:dragonbox-vs-bd	surrogate-pair	canonicalize	1.572	1.556	1.01×	json-canon
go:dragonbox-vs-bd	surrogate-pair	verify	1.610	1.651	1.02×	dragonbox
go:dragonbox-vs-bd	unicode	canonicalize	1.489	1.625	1.09×	dragonbox
go:dragonbox-vs-bd	unicode	verify	1.899	1.594	1.19×	json-canon
go:dragonbox-vs-bd	verify-whitespace	canonicalize	1.494	1.643	1.10×	dragonbox
go:dragonbox-vs-bd	verify-whitespace	verify	1.510	1.677	1.11×	dragonbox
rs:schubfach-vs-bd	array-2048	canonicalize	2.534	4.039	1.59×	schubfach-rs
rs:schubfach-vs-bd	array-2048	verify	2.615	3.919	1.50×	schubfach-rs
rs:schubfach-vs-bd	array-256	canonicalize	0.821	1.113	1.36×	schubfach-rs
rs:schubfach-vs-bd	array-256	verify	0.824	1.077	1.31×	schubfach-rs
rs:schubfach-vs-bd	canonical-minimal	canonicalize	0.561	0.615	1.10×	schubfach-rs
rs:schubfach-vs-bd	canonical-minimal	verify	0.553	0.574	1.04×	schubfach-rs
rs:schubfach-vs-bd	control-escapes	canonicalize	0.552	0.614	1.11×	schubfach-rs
rs:schubfach-vs-bd	control-escapes	verify	0.556	0.603	1.08×	schubfach-rs
rs:schubfach-vs-bd	deep	canonicalize	0.659	0.708	1.07×	schubfach-rs
rs:schubfach-vs-bd	deep	verify	0.627	0.712	1.14×	schubfach-rs
rs:schubfach-vs-bd	deep-64	canonicalize	0.577	0.629	1.09×	schubfach-rs
rs:schubfach-vs-bd	deep-64	verify	0.592	0.640	1.08×	schubfach-rs
rs:schubfach-vs-bd	escaped-key-order	canonicalize	0.542	0.679	1.25×	schubfach-rs
rs:schubfach-vs-bd	escaped-key-order	verify	0.544	0.723	1.33×	schubfach-rs
rs:schubfach-vs-bd	large	canonicalize	7.925	15.530	1.96×	schubfach-rs
rs:schubfach-vs-bd	large	verify	7.543	15.446	2.05×	schubfach-rs
rs:schubfach-vs-bd	long-string	canonicalize	0.642	0.670	1.04×	schubfach-rs
rs:schubfach-vs-bd	long-string	verify	0.627	0.667	1.06×	schubfach-rs
rs:schubfach-vs-bd	medium	canonicalize	0.872	1.379	1.58×	schubfach-rs
rs:schubfach-vs-bd	medium	verify	0.802	1.315	1.64×	schubfach-rs
rs:schubfach-vs-bd	mixed-prod	canonicalize	0.718	0.872	1.21×	schubfach-rs
rs:schubfach-vs-bd	mixed-prod	verify	0.679	0.840	1.24×	schubfach-rs
rs:schubfach-vs-bd	nested-mixed	canonicalize	0.558	0.656	1.18×	schubfach-rs
rs:schubfach-vs-bd	nested-mixed	verify	0.554	0.718	1.30×	schubfach-rs
rs:schubfach-vs-bd	number-heavy	canonicalize	0.556	0.746	1.34×	schubfach-rs
rs:schubfach-vs-bd	number-heavy	verify	0.571	0.702	1.23×	schubfach-rs
rs:schubfach-vs-bd	numeric-boundary	canonicalize	0.553	0.705	1.28×	schubfach-rs
rs:schubfach-vs-bd	numeric-boundary	verify	0.548	0.688	1.26×	schubfach-rs
rs:schubfach-vs-bd	rfc-key-sorting	canonicalize	0.547	0.611	1.12×	schubfach-rs
rs:schubfach-vs-bd	rfc-key-sorting	verify	0.570	0.582	1.02×	schubfach-rs
rs:schubfach-vs-bd	small	canonicalize	0.547	0.671	1.23×	schubfach-rs
rs:schubfach-vs-bd	small	verify	0.563	0.703	1.25×	schubfach-rs
rs:schubfach-vs-bd	surrogate-pair	canonicalize	0.566	0.640	1.13×	schubfach-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
rs:schubfach-vs-bd	surrogate-pair	verify	0.569	0.621	1.09×	schubfach-rs
rs:schubfach-vs-bd	unicode	canonicalize	0.551	0.642	1.17×	schubfach-rs
rs:schubfach-vs-bd	unicode	verify	0.557	0.630	1.13×	schubfach-rs
rs:schubfach-vs-bd	verify-whitespace	canonicalize	0.582	0.696	1.19×	schubfach-rs
rs:schubfach-vs-bd	verify-whitespace	verify	0.551	0.692	1.26×	schubfach-rs
rs:schubfach-vs-ryu	array-2048	canonicalize	2.534	2.601	1.03×	schubfach-rs
rs:schubfach-vs-ryu	array-2048	verify	2.615	2.289	1.14×	ryu-rs
rs:schubfach-vs-ryu	array-256	canonicalize	0.821	0.837	1.02×	schubfach-rs
rs:schubfach-vs-ryu	array-256	verify	0.824	0.804	1.02×	ryu-rs
rs:schubfach-vs-ryu	canonical-minimal	canonicalize	0.561	0.575	1.03×	schubfach-rs
rs:schubfach-vs-ryu	canonical-minimal	verify	0.553	0.564	1.02×	schubfach-rs
rs:schubfach-vs-ryu	control-escapes	canonicalize	0.552	0.557	1.01×	schubfach-rs
rs:schubfach-vs-ryu	control-escapes	verify	0.556	0.563	1.01×	schubfach-rs
rs:schubfach-vs-ryu	deep	canonicalize	0.659	0.604	1.09×	ryu-rs
rs:schubfach-vs-ryu	deep	verify	0.627	0.626	1.00×	ryu-rs
rs:schubfach-vs-ryu	deep-64	canonicalize	0.577	0.594	1.03×	schubfach-rs
rs:schubfach-vs-ryu	deep-64	verify	0.592	0.578	1.02×	ryu-rs
rs:schubfach-vs-ryu	escaped-key-order	canonicalize	0.542	0.538	1.01×	ryu-rs
rs:schubfach-vs-ryu	escaped-key-order	verify	0.544	0.593	1.09×	schubfach-rs
rs:schubfach-vs-ryu	large	canonicalize	7.925	7.856	1.01×	ryu-rs
rs:schubfach-vs-ryu	large	verify	7.543	6.980	1.08×	ryu-rs
rs:schubfach-vs-ryu	long-string	canonicalize	0.642	0.629	1.02×	ryu-rs
rs:schubfach-vs-ryu	long-string	verify	0.627	0.624	1.01×	ryu-rs
rs:schubfach-vs-ryu	medium	canonicalize	0.872	0.795	1.10×	ryu-rs
rs:schubfach-vs-ryu	medium	verify	0.802	0.849	1.06×	schubfach-rs
rs:schubfach-vs-ryu	mixed-prod	canonicalize	0.718	0.647	1.11×	ryu-rs
rs:schubfach-vs-ryu	mixed-prod	verify	0.679	0.654	1.04×	ryu-rs
rs:schubfach-vs-ryu	nested-mixed	canonicalize	0.558	0.547	1.02×	ryu-rs
rs:schubfach-vs-ryu	nested-mixed	verify	0.554	0.559	1.01×	schubfach-rs
rs:schubfach-vs-ryu	number-heavy	canonicalize	0.556	0.541	1.03×	ryu-rs
rs:schubfach-vs-ryu	number-heavy	verify	0.571	0.552	1.03×	ryu-rs
rs:schubfach-vs-ryu	numeric-boundary	canonicalize	0.553	0.559	1.01×	schubfach-rs
rs:schubfach-vs-ryu	numeric-boundary	verify	0.548	0.542	1.01×	ryu-rs
rs:schubfach-vs-ryu	rfc-key-sorting	canonicalize	0.547	0.594	1.09×	schubfach-rs
rs:schubfach-vs-ryu	rfc-key-sorting	verify	0.570	0.530	1.08×	ryu-rs
rs:schubfach-vs-ryu	small	canonicalize	0.547	0.569	1.04×	schubfach-rs
rs:schubfach-vs-ryu	small	verify	0.563	0.559	1.01×	ryu-rs
rs:schubfach-vs-ryu	surrogate-pair	canonicalize	0.566	0.545	1.04×	ryu-rs
rs:schubfach-vs-ryu	surrogate-pair	verify	0.569	0.553	1.03×	ryu-rs
rs:schubfach-vs-ryu	unicode	canonicalize	0.551	0.576	1.05×	schubfach-rs
rs:schubfach-vs-ryu	unicode	verify	0.557	0.544	1.02×	ryu-rs
rs:schubfach-vs-ryu	verify-whitespace	canonicalize	0.582	0.554	1.05×	ryu-rs
rs:schubfach-vs-ryu	verify-whitespace	verify	0.551	0.554	1.00×	schubfach-rs
rs:schubfach-vs-dragonbox	array-2048	canonicalize	2.534	2.468	1.03×	dragonbox-rs
rs:schubfach-vs-dragonbox	array-2048	verify	2.615	2.341	1.12×	dragonbox-rs
rs:schubfach-vs-dragonbox	array-256	canonicalize	0.821	0.850	1.04×	schubfach-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
rs:schubfach-vs-dragonbox	array-256	verify	0.824	0.810	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	canonical-minimal	canonicalize	0.561	0.540	1.04×	dragonbox-rs
rs:schubfach-vs-dragonbox	canonical-minimal	verify	0.553	0.563	1.02×	schubfach-rs
rs:schubfach-vs-dragonbox	control-escapes	canonicalize	0.552	0.566	1.03×	schubfach-rs
rs:schubfach-vs-dragonbox	control-escapes	verify	0.556	0.563	1.01×	schubfach-rs
rs:schubfach-vs-dragonbox	deep	canonicalize	0.659	0.620	1.06×	dragonbox-rs
rs:schubfach-vs-dragonbox	deep	verify	0.627	0.656	1.05×	schubfach-rs
rs:schubfach-vs-dragonbox	deep-64	canonicalize	0.577	0.587	1.02×	schubfach-rs
rs:schubfach-vs-dragonbox	deep-64	verify	0.592	0.577	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	escaped-key-order	canonicalize	0.542	0.548	1.01×	schubfach-rs
rs:schubfach-vs-dragonbox	escaped-key-order	verify	0.544	0.554	1.02×	schubfach-rs
rs:schubfach-vs-dragonbox	large	canonicalize	7.925	8.086	1.02×	schubfach-rs
rs:schubfach-vs-dragonbox	large	verify	7.543	7.439	1.01×	dragonbox-rs
rs:schubfach-vs-dragonbox	long-string	canonicalize	0.642	0.631	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	long-string	verify	0.627	0.628	1.00×	schubfach-rs
rs:schubfach-vs-dragonbox	medium	canonicalize	0.872	0.829	1.05×	dragonbox-rs
rs:schubfach-vs-dragonbox	medium	verify	0.802	0.840	1.05×	schubfach-rs
rs:schubfach-vs-dragonbox	mixed-prod	canonicalize	0.718	0.682	1.05×	dragonbox-rs
rs:schubfach-vs-dragonbox	mixed-prod	verify	0.679	0.685	1.01×	schubfach-rs
rs:schubfach-vs-dragonbox	nested-mixed	canonicalize	0.558	0.563	1.01×	schubfach-rs
rs:schubfach-vs-dragonbox	nested-mixed	verify	0.554	0.543	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	number-heavy	canonicalize	0.556	0.564	1.01×	schubfach-rs
rs:schubfach-vs-dragonbox	number-heavy	verify	0.571	0.560	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	numeric-boundary	canonicalize	0.553	0.578	1.05×	schubfach-rs
rs:schubfach-vs-dragonbox	numeric-boundary	verify	0.548	0.565	1.03×	schubfach-rs
rs:schubfach-vs-dragonbox	rfc-key-sorting	canonicalize	0.547	0.545	1.00×	dragonbox-rs
rs:schubfach-vs-dragonbox	rfc-key-sorting	verify	0.570	0.558	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	small	canonicalize	0.547	0.537	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	small	verify	0.563	0.574	1.02×	schubfach-rs
rs:schubfach-vs-dragonbox	surrogate-pair	canonicalize	0.566	0.567	1.00×	schubfach-rs
rs:schubfach-vs-dragonbox	surrogate-pair	verify	0.569	0.559	1.02×	dragonbox-rs
rs:schubfach-vs-dragonbox	unicode	canonicalize	0.551	0.557	1.01×	schubfach-rs
rs:schubfach-vs-dragonbox	unicode	verify	0.557	0.571	1.02×	schubfach-rs
rs:schubfach-vs-dragonbox	verify-whitespace	canonicalize	0.582	0.534	1.09×	dragonbox-rs
rs:schubfach-vs-dragonbox	verify-whitespace	verify	0.551	0.562	1.02×	schubfach-rs
rs:ryu-vs-dragonbox	array-2048	canonicalize	2.601	2.468	1.05×	dragonbox-rs
rs:ryu-vs-dragonbox	array-2048	verify	2.289	2.341	1.02×	ryu-rs
rs:ryu-vs-dragonbox	array-256	canonicalize	0.837	0.850	1.02×	ryu-rs
rs:ryu-vs-dragonbox	array-256	verify	0.804	0.810	1.01×	ryu-rs
rs:ryu-vs-dragonbox	canonical-minimal	canonicalize	0.575	0.540	1.06×	dragonbox-rs
rs:ryu-vs-dragonbox	canonical-minimal	verify	0.564	0.563	1.00×	dragonbox-rs
rs:ryu-vs-dragonbox	control-escapes	canonicalize	0.557	0.566	1.02×	ryu-rs
rs:ryu-vs-dragonbox	control-escapes	verify	0.563	0.563	1.00×	dragonbox-rs
rs:ryu-vs-dragonbox	deep	canonicalize	0.604	0.620	1.03×	ryu-rs
rs:ryu-vs-dragonbox	deep	verify	0.626	0.656	1.05×	ryu-rs
rs:ryu-vs-dragonbox	deep-64	canonicalize	0.594	0.587	1.01×	dragonbox-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
rs:ryu-vs-dragonbox	deep-64	verify	0.578	0.577	1.00×	dragonbox-rs
rs:ryu-vs-dragonbox	escaped-key-order	canonicalize	0.538	0.548	1.02×	ryu-rs
rs:ryu-vs-dragonbox	escaped-key-order	verify	0.593	0.554	1.07×	dragonbox-rs
rs:ryu-vs-dragonbox	large	canonicalize	7.856	8.086	1.03×	ryu-rs
rs:ryu-vs-dragonbox	large	verify	6.980	7.439	1.07×	ryu-rs
rs:ryu-vs-dragonbox	long-string	canonicalize	0.629	0.631	1.00×	ryu-rs
rs:ryu-vs-dragonbox	long-string	verify	0.624	0.628	1.01×	ryu-rs
rs:ryu-vs-dragonbox	medium	canonicalize	0.795	0.829	1.04×	ryu-rs
rs:ryu-vs-dragonbox	medium	verify	0.849	0.840	1.01×	dragonbox-rs
rs:ryu-vs-dragonbox	mixed-prod	canonicalize	0.647	0.682	1.05×	ryu-rs
rs:ryu-vs-dragonbox	mixed-prod	verify	0.654	0.685	1.05×	ryu-rs
rs:ryu-vs-dragonbox	nested-mixed	canonicalize	0.547	0.563	1.03×	ryu-rs
rs:ryu-vs-dragonbox	nested-mixed	verify	0.559	0.543	1.03×	dragonbox-rs
rs:ryu-vs-dragonbox	number-heavy	canonicalize	0.541	0.564	1.04×	ryu-rs
rs:ryu-vs-dragonbox	number-heavy	verify	0.552	0.560	1.01×	ryu-rs
rs:ryu-vs-dragonbox	numeric-boundary	canonicalize	0.559	0.578	1.03×	ryu-rs
rs:ryu-vs-dragonbox	numeric-boundary	verify	0.542	0.565	1.04×	ryu-rs
rs:ryu-vs-dragonbox	rfc-key-sorting	canonicalize	0.594	0.545	1.09×	dragonbox-rs
rs:ryu-vs-dragonbox	rfc-key-sorting	verify	0.530	0.558	1.05×	ryu-rs
rs:ryu-vs-dragonbox	small	canonicalize	0.569	0.537	1.06×	dragonbox-rs
rs:ryu-vs-dragonbox	small	verify	0.559	0.574	1.03×	ryu-rs
rs:ryu-vs-dragonbox	surrogate-pair	canonicalize	0.545	0.567	1.04×	ryu-rs
rs:ryu-vs-dragonbox	surrogate-pair	verify	0.553	0.559	1.01×	ryu-rs
rs:ryu-vs-dragonbox	unicode	canonicalize	0.576	0.557	1.03×	dragonbox-rs
rs:ryu-vs-dragonbox	unicode	verify	0.544	0.571	1.05×	ryu-rs
rs:ryu-vs-dragonbox	verify-whitespace	canonicalize	0.554	0.534	1.04×	dragonbox-rs
rs:ryu-vs-dragonbox	verify-whitespace	verify	0.554	0.562	1.01×	ryu-rs
rs:ryu-vs-bd	array-2048	canonicalize	2.601	4.039	1.55×	ryu-rs
rs:ryu-vs-bd	array-2048	verify	2.289	3.919	1.71×	ryu-rs
rs:ryu-vs-bd	array-256	canonicalize	0.837	1.113	1.33×	ryu-rs
rs:ryu-vs-bd	array-256	verify	0.804	1.077	1.34×	ryu-rs
rs:ryu-vs-bd	canonical-minimal	canonicalize	0.575	0.615	1.07×	ryu-rs
rs:ryu-vs-bd	canonical-minimal	verify	0.564	0.574	1.02×	ryu-rs
rs:ryu-vs-bd	control-escapes	canonicalize	0.557	0.614	1.10×	ryu-rs
rs:ryu-vs-bd	control-escapes	verify	0.563	0.603	1.07×	ryu-rs
rs:ryu-vs-bd	deep	canonicalize	0.604	0.708	1.17×	ryu-rs
rs:ryu-vs-bd	deep	verify	0.626	0.712	1.14×	ryu-rs
rs:ryu-vs-bd	deep-64	canonicalize	0.594	0.629	1.06×	ryu-rs
rs:ryu-vs-bd	deep-64	verify	0.578	0.640	1.11×	ryu-rs
rs:ryu-vs-bd	escaped-key-order	canonicalize	0.538	0.679	1.26×	ryu-rs
rs:ryu-vs-bd	escaped-key-order	verify	0.593	0.723	1.22×	ryu-rs
rs:ryu-vs-bd	large	canonicalize	7.856	15.530	1.98×	ryu-rs
rs:ryu-vs-bd	large	verify	6.980	15.446	2.21×	ryu-rs
rs:ryu-vs-bd	long-string	canonicalize	0.629	0.670	1.07×	ryu-rs
rs:ryu-vs-bd	long-string	verify	0.624	0.667	1.07×	ryu-rs
rs:ryu-vs-bd	medium	canonicalize	0.795	1.379	1.74×	ryu-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
rs:ryu-vs-bd	medium	verify	0.849	1.315	1.55×	ryu-rs
rs:ryu-vs-bd	mixed-prod	canonicalize	0.647	0.872	1.35×	ryu-rs
rs:ryu-vs-bd	mixed-prod	verify	0.654	0.840	1.28×	ryu-rs
rs:ryu-vs-bd	nested-mixed	canonicalize	0.547	0.656	1.20×	ryu-rs
rs:ryu-vs-bd	nested-mixed	verify	0.559	0.718	1.28×	ryu-rs
rs:ryu-vs-bd	number-heavy	canonicalize	0.541	0.746	1.38×	ryu-rs
rs:ryu-vs-bd	number-heavy	verify	0.552	0.702	1.27×	ryu-rs
rs:ryu-vs-bd	numeric-boundary	canonicalize	0.559	0.705	1.26×	ryu-rs
rs:ryu-vs-bd	numeric-boundary	verify	0.542	0.688	1.27×	ryu-rs
rs:ryu-vs-bd	rfc-key-sorting	canonicalize	0.594	0.611	1.03×	ryu-rs
rs:ryu-vs-bd	rfc-key-sorting	verify	0.530	0.582	1.10×	ryu-rs
rs:ryu-vs-bd	small	canonicalize	0.569	0.671	1.18×	ryu-rs
rs:ryu-vs-bd	small	verify	0.559	0.703	1.26×	ryu-rs
rs:ryu-vs-bd	surrogate-pair	canonicalize	0.545	0.640	1.18×	ryu-rs
rs:ryu-vs-bd	surrogate-pair	verify	0.553	0.621	1.12×	ryu-rs
rs:ryu-vs-bd	unicode	canonicalize	0.576	0.642	1.11×	ryu-rs
rs:ryu-vs-bd	unicode	verify	0.544	0.630	1.16×	ryu-rs
rs:ryu-vs-bd	verify-whitespace	canonicalize	0.554	0.696	1.26×	ryu-rs
rs:ryu-vs-bd	verify-whitespace	verify	0.554	0.692	1.25×	ryu-rs
rs:dragonbox-vs-bd	array-2048	canonicalize	2.468	4.039	1.64×	dragonbox-rs
rs:dragonbox-vs-bd	array-2048	verify	2.341	3.919	1.67×	dragonbox-rs
rs:dragonbox-vs-bd	array-256	canonicalize	0.850	1.113	1.31×	dragonbox-rs
rs:dragonbox-vs-bd	array-256	verify	0.810	1.077	1.33×	dragonbox-rs
rs:dragonbox-vs-bd	canonical-minimal	canonicalize	0.540	0.615	1.14×	dragonbox-rs
rs:dragonbox-vs-bd	canonical-minimal	verify	0.563	0.574	1.02×	dragonbox-rs
rs:dragonbox-vs-bd	control-escapes	canonicalize	0.566	0.614	1.08×	dragonbox-rs
rs:dragonbox-vs-bd	control-escapes	verify	0.563	0.603	1.07×	dragonbox-rs
rs:dragonbox-vs-bd	deep	canonicalize	0.620	0.708	1.14×	dragonbox-rs
rs:dragonbox-vs-bd	deep	verify	0.656	0.712	1.09×	dragonbox-rs
rs:dragonbox-vs-bd	deep-64	canonicalize	0.587	0.629	1.07×	dragonbox-rs
rs:dragonbox-vs-bd	deep-64	verify	0.577	0.640	1.11×	dragonbox-rs
rs:dragonbox-vs-bd	escaped-key-order	canonicalize	0.548	0.679	1.24×	dragonbox-rs
rs:dragonbox-vs-bd	escaped-key-order	verify	0.554	0.723	1.31×	dragonbox-rs
rs:dragonbox-vs-bd	large	canonicalize	8.086	15.530	1.92×	dragonbox-rs
rs:dragonbox-vs-bd	large	verify	7.439	15.446	2.08×	dragonbox-rs
rs:dragonbox-vs-bd	long-string	canonicalize	0.631	0.670	1.06×	dragonbox-rs
rs:dragonbox-vs-bd	long-string	verify	0.628	0.667	1.06×	dragonbox-rs
rs:dragonbox-vs-bd	medium	canonicalize	0.829	1.379	1.66×	dragonbox-rs
rs:dragonbox-vs-bd	medium	verify	0.840	1.315	1.57×	dragonbox-rs
rs:dragonbox-vs-bd	mixed-prod	canonicalize	0.682	0.872	1.28×	dragonbox-rs
rs:dragonbox-vs-bd	mixed-prod	verify	0.685	0.840	1.23×	dragonbox-rs
rs:dragonbox-vs-bd	nested-mixed	canonicalize	0.563	0.656	1.17×	dragonbox-rs
rs:dragonbox-vs-bd	nested-mixed	verify	0.543	0.718	1.32×	dragonbox-rs
rs:dragonbox-vs-bd	number-heavy	canonicalize	0.564	0.746	1.32×	dragonbox-rs
rs:dragonbox-vs-bd	number-heavy	verify	0.560	0.702	1.25×	dragonbox-rs
rs:dragonbox-vs-bd	numeric-boundary	canonicalize	0.578	0.705	1.22×	dragonbox-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
rs:dragonbox-vs-bd	numeric-boundary	verify	0.565	0.688	1.22×	dragonbox-rs
rs:dragonbox-vs-bd	rfc-key-sorting	canonicalize	0.545	0.611	1.12×	dragonbox-rs
rs:dragonbox-vs-bd	rfc-key-sorting	verify	0.558	0.582	1.04×	dragonbox-rs
rs:dragonbox-vs-bd	small	canonicalize	0.537	0.671	1.25×	dragonbox-rs
rs:dragonbox-vs-bd	small	verify	0.574	0.703	1.22×	dragonbox-rs
rs:dragonbox-vs-bd	surrogate-pair	canonicalize	0.567	0.640	1.13×	dragonbox-rs
rs:dragonbox-vs-bd	surrogate-pair	verify	0.559	0.621	1.11×	dragonbox-rs
rs:dragonbox-vs-bd	unicode	canonicalize	0.557	0.642	1.15×	dragonbox-rs
rs:dragonbox-vs-bd	unicode	verify	0.571	0.630	1.10×	dragonbox-rs
rs:dragonbox-vs-bd	verify-whitespace	canonicalize	0.534	0.696	1.30×	dragonbox-rs
rs:dragonbox-vs-bd	verify-whitespace	verify	0.562	0.692	1.23×	dragonbox-rs
schubfach:go-vs-rs	array-2048	canonicalize	5.714	2.534	2.25×	schubfach-rs
schubfach:go-vs-rs	array-2048	verify	5.609	2.615	2.14×	schubfach-rs
schubfach:go-vs-rs	array-256	canonicalize	2.008	0.821	2.45×	schubfach-rs
schubfach:go-vs-rs	array-256	verify	2.038	0.824	2.47×	schubfach-rs
schubfach:go-vs-rs	canonical-minimal	canonicalize	1.496	0.561	2.67×	schubfach-rs
schubfach:go-vs-rs	canonical-minimal	verify	1.569	0.553	2.84×	schubfach-rs
schubfach:go-vs-rs	control-escapes	canonicalize	1.465	0.552	2.65×	schubfach-rs
schubfach:go-vs-rs	control-escapes	verify	1.569	0.556	2.82×	schubfach-rs
schubfach:go-vs-rs	deep	canonicalize	1.489	0.659	2.26×	schubfach-rs
schubfach:go-vs-rs	deep	verify	1.663	0.627	2.65×	schubfach-rs
schubfach:go-vs-rs	deep-64	canonicalize	1.521	0.577	2.64×	schubfach-rs
schubfach:go-vs-rs	deep-64	verify	1.610	0.592	2.72×	schubfach-rs
schubfach:go-vs-rs	escaped-key-order	canonicalize	1.525	0.542	2.82×	schubfach-rs
schubfach:go-vs-rs	escaped-key-order	verify	1.558	0.544	2.86×	schubfach-rs
schubfach:go-vs-rs	large	canonicalize	16.338	7.925	2.06×	schubfach-rs
schubfach:go-vs-rs	large	verify	16.748	7.543	2.22×	schubfach-rs
schubfach:go-vs-rs	long-string	canonicalize	1.591	0.642	2.48×	schubfach-rs
schubfach:go-vs-rs	long-string	verify	1.573	0.627	2.51×	schubfach-rs
schubfach:go-vs-rs	medium	canonicalize	2.158	0.872	2.48×	schubfach-rs
schubfach:go-vs-rs	medium	verify	2.088	0.802	2.60×	schubfach-rs
schubfach:go-vs-rs	mixed-prod	canonicalize	1.667	0.718	2.32×	schubfach-rs
schubfach:go-vs-rs	mixed-prod	verify	1.879	0.679	2.77×	schubfach-rs
schubfach:go-vs-rs	nested-mixed	canonicalize	1.631	0.558	2.92×	schubfach-rs
schubfach:go-vs-rs	nested-mixed	verify	1.393	0.554	2.51×	schubfach-rs
schubfach:go-vs-rs	number-heavy	canonicalize	1.493	0.556	2.68×	schubfach-rs
schubfach:go-vs-rs	number-heavy	verify	1.437	0.571	2.52×	schubfach-rs
schubfach:go-vs-rs	numeric-boundary	canonicalize	1.489	0.553	2.69×	schubfach-rs
schubfach:go-vs-rs	numeric-boundary	verify	1.572	0.548	2.87×	schubfach-rs
schubfach:go-vs-rs	rfc-key-sorting	canonicalize	1.463	0.547	2.67×	schubfach-rs
schubfach:go-vs-rs	rfc-key-sorting	verify	1.466	0.570	2.57×	schubfach-rs
schubfach:go-vs-rs	small	canonicalize	1.452	0.547	2.66×	schubfach-rs
schubfach:go-vs-rs	small	verify	1.423	0.563	2.53×	schubfach-rs
schubfach:go-vs-rs	surrogate-pair	canonicalize	1.479	0.566	2.61×	schubfach-rs
schubfach:go-vs-rs	surrogate-pair	verify	1.518	0.569	2.67×	schubfach-rs
schubfach:go-vs-rs	unicode	canonicalize	1.482	0.551	2.69×	schubfach-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
schubfach:go-vs-rs	unicode	verify	1.506	0.557	2.70×	schubfach-rs
schubfach:go-vs-rs	verify-whitespace	canonicalize	1.545	0.582	2.65×	schubfach-rs
schubfach:go-vs-rs	verify-whitespace	verify	1.640	0.551	2.97×	schubfach-rs
bd:go-vs-rs	array-2048	canonicalize	7.144	4.039	1.77×	json-canon-rs
bd:go-vs-rs	array-2048	verify	7.334	3.919	1.87×	json-canon-rs
bd:go-vs-rs	array-256	canonicalize	2.326	1.113	2.09×	json-canon-rs
bd:go-vs-rs	array-256	verify	2.321	1.077	2.15×	json-canon-rs
bd:go-vs-rs	canonical-minimal	canonicalize	1.529	0.615	2.48×	json-canon-rs
bd:go-vs-rs	canonical-minimal	verify	1.775	0.574	3.09×	json-canon-rs
bd:go-vs-rs	control-escapes	canonicalize	1.603	0.614	2.61×	json-canon-rs
bd:go-vs-rs	control-escapes	verify	1.582	0.603	2.63×	json-canon-rs
bd:go-vs-rs	deep	canonicalize	1.749	0.708	2.47×	json-canon-rs
bd:go-vs-rs	deep	verify	1.777	0.712	2.50×	json-canon-rs
bd:go-vs-rs	deep-64	canonicalize	1.739	0.629	2.76×	json-canon-rs
bd:go-vs-rs	deep-64	verify	1.588	0.640	2.48×	json-canon-rs
bd:go-vs-rs	escaped-key-order	canonicalize	1.795	0.679	2.64×	json-canon-rs
bd:go-vs-rs	escaped-key-order	verify	1.467	0.723	2.03×	json-canon-rs
bd:go-vs-rs	large	canonicalize	25.285	15.530	1.63×	json-canon-rs
bd:go-vs-rs	large	verify	24.851	15.446	1.61×	json-canon-rs
bd:go-vs-rs	long-string	canonicalize	1.811	0.670	2.70×	json-canon-rs
bd:go-vs-rs	long-string	verify	1.792	0.667	2.69×	json-canon-rs
bd:go-vs-rs	medium	canonicalize	2.631	1.379	1.91×	json-canon-rs
bd:go-vs-rs	medium	verify	2.350	1.315	1.79×	json-canon-rs
bd:go-vs-rs	mixed-prod	canonicalize	1.965	0.872	2.25×	json-canon-rs
bd:go-vs-rs	mixed-prod	verify	1.970	0.840	2.34×	json-canon-rs
bd:go-vs-rs	nested-mixed	canonicalize	1.671	0.656	2.55×	json-canon-rs
bd:go-vs-rs	nested-mixed	verify	1.663	0.718	2.32×	json-canon-rs
bd:go-vs-rs	number-heavy	canonicalize	1.653	0.746	2.22×	json-canon-rs
bd:go-vs-rs	number-heavy	verify	1.499	0.702	2.13×	json-canon-rs
bd:go-vs-rs	numeric-boundary	canonicalize	1.737	0.705	2.46×	json-canon-rs
bd:go-vs-rs	numeric-boundary	verify	1.608	0.688	2.34×	json-canon-rs
bd:go-vs-rs	rfc-key-sorting	canonicalize	1.564	0.611	2.56×	json-canon-rs
bd:go-vs-rs	rfc-key-sorting	verify	1.597	0.582	2.74×	json-canon-rs
bd:go-vs-rs	small	canonicalize	1.567	0.671	2.33×	json-canon-rs
bd:go-vs-rs	small	verify	1.739	0.703	2.48×	json-canon-rs
bd:go-vs-rs	surrogate-pair	canonicalize	1.556	0.640	2.43×	json-canon-rs
bd:go-vs-rs	surrogate-pair	verify	1.651	0.621	2.66×	json-canon-rs
bd:go-vs-rs	unicode	canonicalize	1.625	0.642	2.53×	json-canon-rs
bd:go-vs-rs	unicode	verify	1.594	0.630	2.53×	json-canon-rs
bd:go-vs-rs	verify-whitespace	canonicalize	1.643	0.696	2.36×	json-canon-rs
bd:go-vs-rs	verify-whitespace	verify	1.677	0.692	2.42×	json-canon-rs
ryu:go-vs-rs	array-2048	canonicalize	5.440	2.601	2.09×	ryu-rs
ryu:go-vs-rs	array-2048	verify	5.590	2.289	2.44×	ryu-rs
ryu:go-vs-rs	array-256	canonicalize	2.085	0.837	2.49×	ryu-rs
ryu:go-vs-rs	array-256	verify	2.111	0.804	2.63×	ryu-rs
ryu:go-vs-rs	canonical-minimal	canonicalize	1.379	0.575	2.40×	ryu-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
ryu:go-vs-rs	canonical-minimal	verify	1.625	0.564	2.88×	ryu-rs
ryu:go-vs-rs	control-escapes	canonicalize	1.381	0.557	2.48×	ryu-rs
ryu:go-vs-rs	control-escapes	verify	1.498	0.563	2.66×	ryu-rs
ryu:go-vs-rs	deep	canonicalize	1.644	0.604	2.72×	ryu-rs
ryu:go-vs-rs	deep	verify	1.574	0.626	2.52×	ryu-rs
ryu:go-vs-rs	deep-64	canonicalize	1.608	0.594	2.70×	ryu-rs
ryu:go-vs-rs	deep-64	verify	1.534	0.578	2.65×	ryu-rs
ryu:go-vs-rs	escaped-key-order	canonicalize	1.631	0.538	3.03×	ryu-rs
ryu:go-vs-rs	escaped-key-order	verify	1.452	0.593	2.45×	ryu-rs
ryu:go-vs-rs	large	canonicalize	16.015	7.856	2.04×	ryu-rs
ryu:go-vs-rs	large	verify	16.076	6.980	2.30×	ryu-rs
ryu:go-vs-rs	long-string	canonicalize	1.587	0.629	2.52×	ryu-rs
ryu:go-vs-rs	long-string	verify	1.792	0.624	2.87×	ryu-rs
ryu:go-vs-rs	medium	canonicalize	1.980	0.795	2.49×	ryu-rs
ryu:go-vs-rs	medium	verify	2.213	0.849	2.61×	ryu-rs
ryu:go-vs-rs	mixed-prod	canonicalize	1.775	0.647	2.74×	ryu-rs
ryu:go-vs-rs	mixed-prod	verify	1.765	0.654	2.70×	ryu-rs
ryu:go-vs-rs	nested-mixed	canonicalize	1.471	0.547	2.69×	ryu-rs
ryu:go-vs-rs	nested-mixed	verify	1.615	0.559	2.89×	ryu-rs
ryu:go-vs-rs	number-heavy	canonicalize	1.553	0.541	2.87×	ryu-rs
ryu:go-vs-rs	number-heavy	verify	1.564	0.552	2.83×	ryu-rs
ryu:go-vs-rs	numeric-boundary	canonicalize	1.599	0.559	2.86×	ryu-rs
ryu:go-vs-rs	numeric-boundary	verify	1.477	0.542	2.73×	ryu-rs
ryu:go-vs-rs	rfc-key-sorting	canonicalize	1.488	0.594	2.50×	ryu-rs
ryu:go-vs-rs	rfc-key-sorting	verify	1.512	0.530	2.85×	ryu-rs
ryu:go-vs-rs	small	canonicalize	1.519	0.569	2.67×	ryu-rs
ryu:go-vs-rs	small	verify	1.453	0.559	2.60×	ryu-rs
ryu:go-vs-rs	surrogate-pair	canonicalize	1.445	0.545	2.65×	ryu-rs
ryu:go-vs-rs	surrogate-pair	verify	1.601	0.553	2.90×	ryu-rs
ryu:go-vs-rs	unicode	canonicalize	1.502	0.576	2.61×	ryu-rs
ryu:go-vs-rs	unicode	verify	1.440	0.544	2.65×	ryu-rs
ryu:go-vs-rs	verify-whitespace	canonicalize	1.566	0.554	2.83×	ryu-rs
ryu:go-vs-rs	verify-whitespace	verify	1.430	0.554	2.58×	ryu-rs
dragonbox:go-vs-rs	array-2048	canonicalize	5.363	2.468	2.17×	dragonbox-rs
dragonbox:go-vs-rs	array-2048	verify	5.828	2.341	2.49×	dragonbox-rs
dragonbox:go-vs-rs	array-256	canonicalize	1.843	0.850	2.17×	dragonbox-rs
dragonbox:go-vs-rs	array-256	verify	1.860	0.810	2.30×	dragonbox-rs
dragonbox:go-vs-rs	canonical-minimal	canonicalize	1.455	0.540	2.69×	dragonbox-rs
dragonbox:go-vs-rs	canonical-minimal	verify	1.608	0.563	2.85×	dragonbox-rs
dragonbox:go-vs-rs	control-escapes	canonicalize	1.476	0.566	2.60×	dragonbox-rs
dragonbox:go-vs-rs	control-escapes	verify	1.417	0.563	2.52×	dragonbox-rs
dragonbox:go-vs-rs	deep	canonicalize	1.712	0.620	2.76×	dragonbox-rs
dragonbox:go-vs-rs	deep	verify	1.613	0.656	2.46×	dragonbox-rs
dragonbox:go-vs-rs	deep-64	canonicalize	1.616	0.587	2.76×	dragonbox-rs
dragonbox:go-vs-rs	deep-64	verify	1.643	0.577	2.85×	dragonbox-rs
dragonbox:go-vs-rs	escaped-key-order	canonicalize	1.483	0.548	2.71×	dragonbox-rs

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Pair	Workload	Mode	Impl A (ms)	Impl B (ms)	Speedup	Winner
dragonbox:go-vs-rs	escaped-key-order	verify	1.429	0.554	2.58×	dragonbox-rs
dragonbox:go-vs-rs	large	canonicalize	15.905	8.086	1.97×	dragonbox-rs
dragonbox:go-vs-rs	large	verify	16.074	7.439	2.16×	dragonbox-rs
dragonbox:go-vs-rs	long-string	canonicalize	1.737	0.631	2.75×	dragonbox-rs
dragonbox:go-vs-rs	long-string	verify	1.749	0.628	2.78×	dragonbox-rs
dragonbox:go-vs-rs	medium	canonicalize	2.021	0.829	2.44×	dragonbox-rs
dragonbox:go-vs-rs	medium	verify	1.888	0.840	2.25×	dragonbox-rs
dragonbox:go-vs-rs	mixed-prod	canonicalize	1.659	0.682	2.43×	dragonbox-rs
dragonbox:go-vs-rs	mixed-prod	verify	1.757	0.685	2.56×	dragonbox-rs
dragonbox:go-vs-rs	nested-mixed	canonicalize	1.514	0.563	2.69×	dragonbox-rs
dragonbox:go-vs-rs	nested-mixed	verify	1.580	0.543	2.91×	dragonbox-rs
dragonbox:go-vs-rs	number-heavy	canonicalize	1.578	0.564	2.80×	dragonbox-rs
dragonbox:go-vs-rs	number-heavy	verify	1.575	0.560	2.81×	dragonbox-rs
dragonbox:go-vs-rs	numeric-boundary	canonicalize	1.450	0.578	2.51×	dragonbox-rs
dragonbox:go-vs-rs	numeric-boundary	verify	1.463	0.565	2.59×	dragonbox-rs
dragonbox:go-vs-rs	rfc-key-sorting	canonicalize	1.466	0.545	2.69×	dragonbox-rs
dragonbox:go-vs-rs	rfc-key-sorting	verify	1.567	0.558	2.81×	dragonbox-rs
dragonbox:go-vs-rs	small	canonicalize	1.527	0.537	2.84×	dragonbox-rs
dragonbox:go-vs-rs	small	verify	1.423	0.574	2.48×	dragonbox-rs
dragonbox:go-vs-rs	surrogate-pair	canonicalize	1.572	0.567	2.77×	dragonbox-rs
dragonbox:go-vs-rs	surrogate-pair	verify	1.610	0.559	2.88×	dragonbox-rs
dragonbox:go-vs-rs	unicode	canonicalize	1.489	0.557	2.67×	dragonbox-rs
dragonbox:go-vs-rs	unicode	verify	1.899	0.571	3.33×	dragonbox-rs
dragonbox:go-vs-rs	verify-whitespace	canonicalize	1.494	0.534	2.80×	dragonbox-rs
dragonbox:go-vs-rs	verify-whitespace	verify	1.510	0.562	2.69×	dragonbox-rs

Speedup = slower/faster mean wall time. $n = 15$ repetitions, 3 warmup runs discarded.

A.3 Full Statistical Comparisons

Table 10: Statistical comparisons with Benjamini-Hochberg corrected p-values.

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}
go:schubfach-vs-bd	e2e	canonicalize	array-2048	schubfach	json-canon	schubfach*	1.25×	[1.183, 1.331]	0.0002	0.0004
go:schubfach-vs-ryu	e2e	canonicalize	array-2048	schubfach	ryu	ryu	1.05×	[0.985, 1.108]	0.1200	0.1896
go:schubfach-vs-dragonbox	e2e	canonicalize	array-2048	schubfach	dragonbox	dragonbox	1.07×	[0.993, 1.139]	0.0886	0.1451
go:ryu-vs-dragonbox	e2e	canonicalize	array-2048	ryu	dragonbox	dragonbox	1.01×	[0.952, 1.081]	0.6629	0.7459
go:ryu-vs-bd	e2e	canonicalize	array-2048	ryu	json-canon	ryu*	1.31×	[1.243, 1.384]	0.0002	0.0004
go:dragonbox-vs-bd	e2e	canonicalize	array-2048	dragonbox	json-canon	dragonbox*	1.33×	[1.258, 1.422]	0.0002	0.0004
rs:schubfach-vs-bd	e2e	canonicalize	array-2048	schubfach-rs	json-canon-rs	schubfach-rs*	1.59×	[1.436, 1.802]	0.0002	0.0004
rs:schubfach-vs-ryu	e2e	canonicalize	array-2048	schubfach-rs	ryu-rs	schubfach-rs	1.03×	[0.914, 1.148]	0.6731	0.7528
rs:schubfach-vs-dragonbox	e2e	canonicalize	array-2048	schubfach-rs	dragonbox-rs	dragonbox-rs	1.03×	[0.909, 1.153]	0.6665	0.7478

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:ryu-vs-dragonbox	e2e	canonicalize	array-2048	ryu-rs	dragonbox-rs	dragonbox-rs	1.05×	[0.925, 1.187]	0.4471	0.5568	0
rs:ryu-vs-bd	e2e	canonicalize	array-2048	ryu-rs	json-canon-rs	ryu-rs*	1.55×	[1.388, 1.771]	0.0002	0.0004	-2
rs:dragonbox-vs-bd	e2e	canonicalize	array-2048	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.64×	[1.449, 1.855]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	canonicalize	array-2048	schubfach	schubfach-rs	schubfach-rs*	2.25×	[2.060, 2.450]	0.0002	0.0004	6
bd:go-vs-rs	e2e	canonicalize	array-2048	json-canon	json-canon-rs	json-canon-rs*	1.77×	[1.591, 1.916]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	array-2048	ryu	ryu-rs	ryu-rs*	2.09×	[1.913, 2.316]	0.0002	0.0004	6
dragonbox:go-vs-rs	e2e	canonicalize	array-2048	dragonbox	dragonbox-rs	dragonbox-rs*	2.17×	[1.947, 2.396]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	canonicalize	array-256	schubfach	json-canon	schubfach	1.16×	[1.024, 1.320]	0.0366	0.0650	-0
go:schubfach-vs-ryu	e2e	canonicalize	array-256	schubfach	ryu	schubfach	1.04×	[0.924, 1.173]	0.5405	0.6387	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	array-256	schubfach	dragonbox	dragonbox	1.09×	[0.950, 1.236]	0.2196	0.3119	0
go:ryu-vs-dragonbox	e2e	canonicalize	array-256	ryu	dragonbox	dragonbox	1.13×	[0.991, 1.275]	0.0762	0.1269	0
go:ryu-vs-bd	e2e	canonicalize	array-256	ryu	json-canon	ryu	1.12×	[0.991, 1.267]	0.0996	0.1597	-0
go:dragonbox-vs-bd	e2e	canonicalize	array-256	dragonbox	json-canon	dragonbox*	1.26×	[1.093, 1.440]	0.0040	0.0079	-1
rs:schubfach-vs-bd	e2e	canonicalize	array-256	schubfach-rs	json-canon-rs	schubfach-rs*	1.36×	[1.234, 1.484]	0.0002	0.0004	-2
rs:schubfach-vs-ryu	e2e	canonicalize	array-256	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.943, 1.096]	0.6153	0.7077	-0
rs:schubfach-vs-dragonbox	e2e	canonicalize	array-256	schubfach-rs	dragonbox-rs	schubfach-rs	1.04×	[0.956, 1.120]	0.3907	0.4991	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	array-256	ryu-rs	dragonbox-rs	ryu-rs	1.02×	[0.951, 1.088]	0.6533	0.7394	-0
rs:ryu-vs-bd	e2e	canonicalize	array-256	ryu-rs	json-canon-rs	ryu-rs*	1.33×	[1.222, 1.443]	0.0002	0.0004	-2
rs:dragonbox-vs-bd	e2e	canonicalize	array-256	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.31×	[1.201, 1.423]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	canonicalize	array-256	schubfach	schubfach-rs	schubfach-rs*	2.45×	[2.190, 2.709]	0.0002	0.0004	4
bd:go-vs-rs	e2e	canonicalize	array-256	json-canon	json-canon-rs	json-canon-rs*	2.09×	[1.876, 2.353]	0.0002	0.0004	3
ryu:go-vs-rs	e2e	canonicalize	array-256	ryu	ryu-rs	ryu-rs*	2.49×	[2.277, 2.729]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	canonicalize	array-256	dragonbox	dragonbox-rs	dragonbox-rs*	2.17×	[1.959, 2.430]	0.0002	0.0004	3
go:schubfach-vs-bd	e2e	canonicalize	canonical-minimal	schubfach	json-canon	schubfach	1.02×	[0.901, 1.171]	0.7495	0.8143	-0
go:schubfach-vs-ryu	e2e	canonicalize	canonical-minimal	schubfach	ryu	ryu	1.08×	[0.937, 1.251]	0.2647	0.3652	0
go:schubfach-vs-dragonbox	e2e	canonicalize	canonical-minimal	schubfach	dragonbox	dragonbox	1.03×	[0.910, 1.158]	0.6647	0.7472	0
go:ryu-vs-dragonbox	e2e	canonicalize	canonical-minimal	ryu	dragonbox	ryu	1.06×	[0.919, 1.211]	0.4737	0.5777	-0
go:ryu-vs-bd	e2e	canonicalize	canonical-minimal	ryu	json-canon	ryu	1.11×	[0.959, 1.288]	0.1990	0.2873	-0
go:dragonbox-vs-bd	e2e	canonicalize	canonical-minimal	dragonbox	json-canon	dragonbox	1.05×	[0.920, 1.193]	0.4673	0.5743	-0
rs:schubfach-vs-bd	e2e	canonicalize	canonical-minimal	schubfach-rs	json-canon-rs	schubfach-rs*	1.10×	[1.044, 1.146]	0.0002	0.0004	-1
rs:schubfach-vs-ryu	e2e	canonicalize	canonical-minimal	schubfach-rs	ryu-rs	schubfach-rs	1.03×	[0.962, 1.096]	0.4939	0.5956	-0
rs:schubfach-vs-dragonbox	e2e	canonicalize	canonical-minimal	schubfach-rs	dragonbox-rs	dragonbox-rs	1.04×	[0.980, 1.119]	0.2813	0.3804	0
rs:ryu-vs-dragonbox	e2e	canonicalize	canonical-minimal	ryu-rs	dragonbox-rs	dragonbox-rs	1.06×	[0.988, 1.163]	0.1534	0.2343	0
rs:ryu-vs-bd	e2e	canonicalize	canonical-minimal	ryu-rs	json-canon-rs	ryu-rs	1.07×	[0.999, 1.138]	0.0474	0.0818	-0
rs:dragonbox-vs-bd	e2e	canonicalize	canonical-minimal	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.14×	[1.073, 1.230]	0.0002	0.0004	-1
schubfach:go-vs-rs	e2e	canonicalize	canonical-minimal	schubfach	schubfach-rs	schubfach-rs*	2.67×	[2.385, 2.885]	0.0002	0.0004	4
bd:go-vs-rs	e2e	canonicalize	canonical-minimal	json-canon	json-canon-rs	json-canon-rs*	2.48×	[2.239, 2.750]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	canonical-minimal	ryu	ryu-rs	ryu-rs*	2.40×	[2.097, 2.694]	0.0002	0.0004	3
dragonbox:go-vs-rs	e2e	canonicalize	canonical-minimal	dragonbox	dragonbox-rs	dragonbox-rs*	2.69×	[2.439, 2.992]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	canonicalize	control-escapes	schubfach	json-canon	schubfach	1.09×	[0.968, 1.232]	0.1678	0.2516	-0
go:schubfach-vs-ryu	e2e	canonicalize	control-escapes	schubfach	ryu	ryu	1.06×	[0.933, 1.199]	0.3803	0.4877	0
go:schubfach-vs-dragonbox	e2e	canonicalize	control-escapes	schubfach	dragonbox	schubfach	1.01×	[0.889, 1.135]	0.9060	0.9342	-0
go:ryu-vs-dragonbox	e2e	canonicalize	control-escapes	ryu	dragonbox	ryu	1.07×	[0.933, 1.205]	0.3201	0.4214	-0
go:ryu-vs-bd	e2e	canonicalize	control-escapes	ryu	json-canon	ryu	1.16×	[1.018, 1.319]	0.0392	0.0692	-0
go:dragonbox-vs-bd	e2e	canonicalize	control-escapes	dragonbox	json-canon	dragonbox	1.09×	[0.961, 1.238]	0.2164	0.3086	-0
rs:schubfach-vs-bd	e2e	canonicalize	control-escapes	schubfach-rs	json-canon-rs	schubfach-rs*	1.11×	[1.046, 1.186]	0.0038	0.0075	-1

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}
rs:schubfach-vs-ryu	e2e	canonicalize	control-escapes	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.946, 1.063]	0.8016	0.8538
rs:schubfach-vs-dragonbox	e2e	canonicalize	control-escapes	schubfach-rs	dragonbox-rs	schubfach-rs	1.03×	[0.950, 1.092]	0.4979	0.5998
rs:ryu-vs-dragonbox	e2e	canonicalize	control-escapes	ryu-rs	dragonbox-rs	ryu-rs	1.02×	[0.938, 1.095]	0.6675	0.7478
rs:ryu-vs-bd	e2e	canonicalize	control-escapes	ryu-rs	json-canon-rs	ryu-rs*	1.10×	[1.027, 1.186]	0.0130	0.0245
rs:dragonbox-vs-bd	e2e	canonicalize	control-escapes	dragonbox-rs	json-canon-rs	dragonbox-rs	1.08×	[1.001, 1.183]	0.0616	0.1050
schubfach:go-vs-rs	e2e	canonicalize	control-escapes	schubfach	schubfach-rs	schubfach-rs*	2.65×	[2.396, 2.890]	0.0002	0.0004
bd:go-vs-rs	e2e	canonicalize	control-escapes	json-canon	json-canon-rs	json-canon-rs*	2.61×	[2.327, 2.872]	0.0002	0.0004
ryu:go-vs-rs	e2e	canonicalize	control-escapes	ryu	ryu-rs	ryu-rs*	2.48×	[2.240, 2.774]	0.0002	0.0004
dragonbox:go-vs-rs	e2e	canonicalize	control-escapes	dragonbox	dragonbox-rs	dragonbox-rs*	2.60×	[2.330, 2.886]	0.0002	0.0004
go:schubfach-vs-bd	e2e	canonicalize	deep-64	schubfach	json-canon	schubfach	1.14×	[0.988, 1.321]	0.0814	0.1347
go:schubfach-vs-ryu	e2e	canonicalize	deep-64	schubfach	ryu	schubfach	1.06×	[0.912, 1.228]	0.4779	0.5807
go:schubfach-vs-dragonbox	e2e	canonicalize	deep-64	schubfach	dragonbox	schubfach	1.06×	[0.929, 1.212]	0.3929	0.5004
go:ryu-vs-dragonbox	e2e	canonicalize	deep-64	ryu	dragonbox	ryu	1.01×	[0.893, 1.129]	0.9354	0.9569
go:ryu-vs-bd	e2e	canonicalize	deep-64	ryu	json-canon	ryu	1.08×	[0.946, 1.214]	0.2486	0.3485
go:dragonbox-vs-bd	e2e	canonicalize	deep-64	dragonbox	json-canon	dragonbox	1.08×	[0.954, 1.197]	0.2162	0.3086
rs:schubfach-vs-bd	e2e	canonicalize	deep-64	schubfach-rs	json-canon-rs	schubfach-rs	1.09×	[0.977, 1.229]	0.1444	0.2232
rs:schubfach-vs-ryu	e2e	canonicalize	deep-64	schubfach-rs	ryu-rs	schubfach-rs	1.03×	[0.929, 1.154]	0.6057	0.6985
rs:schubfach-vs-dragonbox	e2e	canonicalize	deep-64	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.904, 1.153]	0.8100	0.8613
rs:ryu-vs-dragonbox	e2e	canonicalize	deep-64	ryu-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.927, 1.104]	0.7812	0.8389
rs:ryu-vs-bd	e2e	canonicalize	deep-64	ryu-rs	json-canon-rs	ryu-rs	1.06×	[0.974, 1.130]	0.1554	0.2368
rs:dragonbox-vs-bd	e2e	canonicalize	deep-64	dragonbox-rs	json-canon-rs	dragonbox-rs	1.07×	[0.971, 1.175]	0.1812	0.2658
schubfach:go-vs-rs	e2e	canonicalize	deep-64	schubfach	schubfach-rs	schubfach-rs*	2.64×	[2.275, 3.052]	0.0002	0.0004
bd:go-vs-rs	e2e	canonicalize	deep-64	json-canon	json-canon-rs	json-canon-rs*	2.76×	[2.468, 3.036]	0.0002	0.0004
ryu:go-vs-rs	e2e	canonicalize	deep-64	ryu	ryu-rs	ryu-rs*	2.70×	[2.436, 2.991]	0.0002	0.0004
dragonbox:go-vs-rs	e2e	canonicalize	deep-64	dragonbox	dragonbox-rs	dragonbox-rs*	2.76×	[2.468, 3.038]	0.0002	0.0004
go:schubfach-vs-bd	e2e	canonicalize	deep	schubfach	json-canon	schubfach	1.17×	[1.026, 1.336]	0.0326	0.0586
go:schubfach-vs-ryu	e2e	canonicalize	deep	schubfach	ryu	schubfach	1.10×	[0.971, 1.238]	0.1194	0.1888
go:schubfach-vs-dragonbox	e2e	canonicalize	deep	schubfach	dragonbox	schubfach*	1.15×	[1.022, 1.273]	0.0190	0.0353
go:ryu-vs-dragonbox	e2e	canonicalize	deep	ryu	dragonbox	ryu	1.04×	[0.933, 1.154]	0.4619	0.5698
go:ryu-vs-bd	e2e	canonicalize	deep	ryu	json-canon	ryu	1.06×	[0.936, 1.201]	0.3487	0.4531
go:dragonbox-vs-bd	e2e	canonicalize	deep	dragonbox	json-canon	dragonbox	1.02×	[0.907, 1.146]	0.7333	0.7994
rs:schubfach-vs-bd	e2e	canonicalize	deep	schubfach-rs	json-canon-rs	schubfach-rs*	1.07×	[1.011, 1.121]	0.0124	0.0235
rs:schubfach-vs-ryu	e2e	canonicalize	deep	schubfach-rs	ryu-rs	ryu-rs*	1.09×	[1.025, 1.180]	0.0184	0.0342
rs:schubfach-vs-dragonbox	e2e	canonicalize	deep	schubfach-rs	dragonbox-rs	dragonbox-rs	1.06×	[1.012, 1.144]	0.0494	0.0849
rs:ryu-vs-dragonbox	e2e	canonicalize	deep	ryu-rs	dragonbox-rs	ryu-rs	1.03×	[0.949, 1.110]	0.5207	0.6215
rs:ryu-vs-bd	e2e	canonicalize	deep	ryu-rs	json-canon-rs	ryu-rs*	1.17×	[1.087, 1.259]	0.0002	0.0004
rs:dragonbox-vs-bd	e2e	canonicalize	deep	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.14×	[1.077, 1.222]	0.0004	0.0008
schubfach:go-vs-rs	e2e	canonicalize	deep	schubfach	schubfach-rs	schubfach-rs*	2.26×	[2.058, 2.487]	0.0002	0.0004
bd:go-vs-rs	e2e	canonicalize	deep	json-canon	json-canon-rs	json-canon-rs*	2.47×	[2.217, 2.723]	0.0002	0.0004
ryu:go-vs-rs	e2e	canonicalize	deep	ryu	ryu-rs	ryu-rs*	2.72×	[2.461, 3.006]	0.0002	0.0004
dragonbox:go-vs-rs	e2e	canonicalize	deep	dragonbox	dragonbox-rs	dragonbox-rs*	2.76×	[2.528, 2.968]	0.0002	0.0004
go:schubfach-vs-bd	e2e	canonicalize	escaped-key-order	schubfach	json-canon	schubfach	1.18×	[0.996, 1.716]	0.2511	0.3506
go:schubfach-vs-ryu	e2e	canonicalize	escaped-key-order	schubfach	ryu	schubfach	1.07×	[0.965, 1.205]	0.2691	0.3705
go:schubfach-vs-dragonbox	e2e	canonicalize	escaped-key-order	schubfach	dragonbox	dragonbox	1.03×	[0.898, 1.173]	0.6953	0.7670
go:ryu-vs-dragonbox	e2e	canonicalize	escaped-key-order	ryu	dragonbox	dragonbox	1.10×	[0.969, 1.260]	0.1730	0.2564
go:ryu-vs-bd	e2e	canonicalize	escaped-key-order	ryu	json-canon	ryu	1.10×	[0.930, 1.564]	0.6523	0.7394

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:dragonbox-vs-bd	e2e	canonicalize	escaped-key-order	dragonbox	json-canon	dragonbox	1.21×	[1.007, 1.710]	0.1700	0.2534	-0
rs:schubfach-vs-bd	e2e	canonicalize	escaped-key-order	schubfach-rs	json-canon-rs	schubfach-rs*	1.25×	[1.160, 1.366]	0.0002	0.0004	-1
rs:schubfach-vs-ryu	e2e	canonicalize	escaped-key-order	schubfach-rs	ryu-rs	ryu-rs	1.01×	[0.922, 1.103]	0.8738	0.9098	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	escaped-key-order	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.944, 1.104]	0.7840	0.8394	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	escaped-key-order	ryu-rs	dragonbox-rs	ryu-rs	1.02×	[0.945, 1.109]	0.6605	0.7443	-0
rs:ryu-vs-bd	e2e	canonicalize	escaped-key-order	ryu-rs	json-canon-rs	ryu-rs*	1.26×	[1.165, 1.376]	0.0002	0.0004	-1
rs:dragonbox-vs-bd	e2e	canonicalize	escaped-key-order	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.24×	[1.158, 1.324]	0.0004	0.0008	-2
schubfach:go-vs-rs	e2e	canonicalize	escaped-key-order	schubfach	schubfach-rs	schubfach-rs*	2.82×	[2.537, 3.108]	0.0002	0.0004	5
bd:go-vs-rs	e2e	canonicalize	escaped-key-order	json-canon	json-canon-rs	json-canon-rs*	2.64×	[2.256, 3.747]	0.0002	0.0004	1
ryu:go-vs-rs	e2e	canonicalize	escaped-key-order	ryu	ryu-rs	ryu-rs*	3.03×	[2.744, 3.369]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	canonicalize	escaped-key-order	dragonbox	dragonbox-rs	dragonbox-rs*	2.71×	[2.406, 3.021]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	canonicalize	large	schubfach	json-canon	schubfach*	1.55×	[1.500, 1.594]	0.0002	0.0004	-9
go:schubfach-vs-ryu	e2e	canonicalize	large	schubfach	ryu	ryu	1.02×	[0.990, 1.053]	0.2320	0.3280	0
go:schubfach-vs-dragonbox	e2e	canonicalize	large	schubfach	dragonbox	dragonbox	1.03×	[0.995, 1.059]	0.1206	0.1903	0
go:ryu-vs-dragonbox	e2e	canonicalize	large	ryu	dragonbox	dragonbox	1.01×	[0.970, 1.040]	0.6911	0.7643	0
go:ryu-vs-bd	e2e	canonicalize	large	ryu	json-canon	ryu*	1.58×	[1.528, 1.632]	0.0002	0.0004	-9
go:dragonbox-vs-bd	e2e	canonicalize	large	dragonbox	json-canon	dragonbox*	1.59×	[1.536, 1.642]	0.0002	0.0004	-9
rs:schubfach-vs-bd	e2e	canonicalize	large	schubfach-rs	json-canon-rs	schubfach-rs*	1.96×	[1.842, 2.072]	0.0002	0.0004	-9
rs:schubfach-vs-ryu	e2e	canonicalize	large	schubfach-rs	ryu-rs	ryu-rs	1.01×	[0.931, 1.088]	0.8348	0.8790	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	large	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.935, 1.151]	0.7315	0.7987	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	large	ryu-rs	dragonbox-rs	ryu-rs	1.03×	[0.946, 1.152]	0.6053	0.6985	-0
rs:ryu-vs-bd	e2e	canonicalize	large	ryu-rs	json-canon-rs	ryu-rs*	1.98×	[1.847, 2.079]	0.0002	0.0004	-10
rs:dragonbox-vs-bd	e2e	canonicalize	large	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.92×	[1.718, 2.052]	0.0002	0.0004	-7
schubfach:go-vs-rs	e2e	canonicalize	large	schubfach	schubfach-rs	schubfach-rs*	2.06×	[1.938, 2.177]	0.0002	0.0004	10
bd:go-vs-rs	e2e	canonicalize	large	json-canon	json-canon-rs	json-canon-rs*	1.63×	[1.578, 1.676]	0.0002	0.0004	10
ryu:go-vs-rs	e2e	canonicalize	large	ryu	ryu-rs	ryu-rs*	2.04×	[1.899, 2.145]	0.0002	0.0004	9
dragonbox:go-vs-rs	e2e	canonicalize	large	dragonbox	dragonbox-rs	dragonbox-rs*	1.97×	[1.755, 2.102]	0.0002	0.0004	7
go:schubfach-vs-bd	e2e	canonicalize	long-string	schubfach	json-canon	schubfach	1.14×	[0.979, 1.340]	0.1378	0.2135	-0
go:schubfach-vs-ryu	e2e	canonicalize	long-string	schubfach	ryu	ryu	1.00×	[0.861, 1.176]	0.9738	0.9795	0
go:schubfach-vs-dragonbox	e2e	canonicalize	long-string	schubfach	dragonbox	schubfach	1.09×	[0.966, 1.244]	0.1940	0.2814	-0
go:ryu-vs-dragonbox	e2e	canonicalize	long-string	ryu	dragonbox	ryu	1.09×	[0.981, 1.246]	0.1620	0.2451	-0
go:ryu-vs-bd	e2e	canonicalize	long-string	ryu	json-canon	ryu	1.14×	[0.993, 1.340]	0.1136	0.1808	-0
go:dragonbox-vs-bd	e2e	canonicalize	long-string	dragonbox	json-canon	dragonbox	1.04×	[0.919, 1.198]	0.5565	0.6534	-0
rs:schubfach-vs-bd	e2e	canonicalize	long-string	schubfach-rs	json-canon-rs	schubfach-rs	1.04×	[0.966, 1.108]	0.2450	0.3449	-0
rs:schubfach-vs-ryu	e2e	canonicalize	long-string	schubfach-rs	ryu-rs	ryu-rs	1.02×	[0.984, 1.077]	0.4059	0.5129	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	long-string	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.974, 1.064]	0.4993	0.6005	0
rs:ryu-vs-dragonbox	e2e	canonicalize	long-string	ryu-rs	dragonbox-rs	ryu-rs	1.00×	[0.967, 1.059]	0.8624	0.9014	-0
rs:ryu-vs-bd	e2e	canonicalize	long-string	ryu-rs	json-canon-rs	ryu-rs	1.07×	[0.993, 1.136]	0.0892	0.1459	-0
rs:dragonbox-vs-bd	e2e	canonicalize	long-string	dragonbox-rs	json-canon-rs	dragonbox-rs	1.06×	[0.982, 1.126]	0.1218	0.1917	-0
schubfach:go-vs-rs	e2e	canonicalize	long-string	schubfach	schubfach-rs	schubfach-rs*	2.48×	[2.207, 2.771]	0.0002	0.0004	3
bd:go-vs-rs	e2e	canonicalize	long-string	json-canon	json-canon-rs	json-canon-rs*	2.70×	[2.392, 3.085]	0.0002	0.0004	3
ryu:go-vs-rs	e2e	canonicalize	long-string	ryu	ryu-rs	ryu-rs*	2.52×	[2.249, 2.779]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	canonicalize	long-string	dragonbox	dragonbox-rs	dragonbox-rs*	2.75×	[2.548, 2.926]	0.0002	0.0004	7
go:schubfach-vs-bd	e2e	canonicalize	medium	schubfach	json-canon	schubfach*	1.22×	[1.070, 1.376]	0.0058	0.0113	-1
go:schubfach-vs-ryu	e2e	canonicalize	medium	schubfach	ryu	ryu	1.09×	[0.956, 1.220]	0.1744	0.2582	0
go:schubfach-vs-dragonbox	e2e	canonicalize	medium	schubfach	dragonbox	dragonbox	1.07×	[0.942, 1.202]	0.3135	0.4153	0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:ryu-vs-dragonbox	e2e	canonicalize	medium	ryu	dragonbox	ryu	1.02×	[0.888, 1.158]	0.7720	0.8311	-0
go:ryu-vs-bd	e2e	canonicalize	medium	ryu	json-canon	ryu*	1.33×	[1.159, 1.503]	0.0008	0.0017	-1
go:dragonbox-vs-bd	e2e	canonicalize	medium	dragonbox	json-canon	dragonbox*	1.30×	[1.127, 1.473]	0.0020	0.0041	-1
rs:schubfach-vs-bd	e2e	canonicalize	medium	schubfach-rs	json-canon-rs	schubfach-rs*	1.58×	[1.471, 1.711]	0.0002	0.0004	-4
rs:schubfach-vs-ryu	e2e	canonicalize	medium	schubfach-rs	ryu-rs	ryu-rs	1.10×	[1.001, 1.204]	0.0702	0.1178	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	medium	schubfach-rs	dragonbox-rs	dragonbox-rs	1.05×	[0.963, 1.144]	0.2699	0.3708	0
rs:ryu-vs-dragonbox	e2e	canonicalize	medium	ryu-rs	dragonbox-rs	ryu-rs	1.04×	[0.950, 1.154]	0.4163	0.5235	-0
rs:ryu-vs-bd	e2e	canonicalize	medium	ryu-rs	json-canon-rs	ryu-rs*	1.74×	[1.589, 1.902]	0.0002	0.0004	-4
rs:dragonbox-vs-bd	e2e	canonicalize	medium	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.66×	[1.538, 1.798]	0.0002	0.0004	-4
schubfach:go-vs-rs	e2e	canonicalize	medium	schubfach	schubfach-rs	schubfach-rs*	2.48×	[2.240, 2.732]	0.0002	0.0004	5
bd:go-vs-rs	e2e	canonicalize	medium	json-canon	json-canon-rs	json-canon-rs*	1.91×	[1.700, 2.098]	0.0002	0.0004	3
ryu:go-vs-rs	e2e	canonicalize	medium	ryu	ryu-rs	ryu-rs*	2.49×	[2.228, 2.820]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	canonicalize	medium	dragonbox	dragonbox-rs	dragonbox-rs*	2.44×	[2.172, 2.727]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	canonicalize	mixed-prod	schubfach	json-canon	schubfach*	1.18×	[1.041, 1.340]	0.0182	0.0339	-0
go:schubfach-vs-ryu	e2e	canonicalize	mixed-prod	schubfach	ryu	schubfach	1.06×	[0.946, 1.192]	0.3195	0.4214	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	mixed-prod	schubfach	dragonbox	dragonbox	1.00×	[0.878, 1.154]	0.9408	0.9601	0
go:ryu-vs-dragonbox	e2e	canonicalize	mixed-prod	ryu	dragonbox	dragonbox	1.07×	[0.945, 1.210]	0.3197	0.4214	0
go:ryu-vs-bd	e2e	canonicalize	mixed-prod	ryu	json-canon	ryu	1.11×	[0.988, 1.249]	0.1158	0.1838	-0
go:dragonbox-vs-bd	e2e	canonicalize	mixed-prod	dragonbox	json-canon	dragonbox*	1.18×	[1.042, 1.359]	0.0208	0.0384	-0
rs:schubfach-vs-bd	e2e	canonicalize	mixed-prod	schubfach-rs	json-canon-rs	schubfach-rs*	1.21×	[1.115, 1.310]	0.0004	0.0008	-1
rs:schubfach-vs-ryu	e2e	canonicalize	mixed-prod	schubfach-rs	ryu-rs	ryu-rs*	1.11×	[1.027, 1.209]	0.0176	0.0329	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	mixed-prod	schubfach-rs	dragonbox-rs	dragonbox-rs	1.05×	[0.986, 1.125]	0.1590	0.2412	0
rs:ryu-vs-dragonbox	e2e	canonicalize	mixed-prod	ryu-rs	dragonbox-rs	ryu-rs	1.05×	[0.985, 1.144]	0.1704	0.2534	-0
rs:ryu-vs-bd	e2e	canonicalize	mixed-prod	ryu-rs	json-canon-rs	ryu-rs*	1.35×	[1.239, 1.468]	0.0002	0.0004	-2
rs:dragonbox-vs-bd	e2e	canonicalize	mixed-prod	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.28×	[1.180, 1.365]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	canonicalize	mixed-prod	schubfach	schubfach-rs	schubfach-rs*	2.32×	[2.107, 2.585]	0.0002	0.0004	4
bd:go-vs-rs	e2e	canonicalize	mixed-prod	json-canon	json-canon-rs	json-canon-rs*	2.25×	[2.038, 2.516]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	mixed-prod	ryu	ryu-rs	ryu-rs*	2.74×	[2.501, 3.045]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	canonicalize	mixed-prod	dragonbox	dragonbox-rs	dragonbox-rs*	2.43×	[2.183, 2.705]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	canonicalize	nested-mixed	schubfach	json-canon	schubfach	1.02×	[0.914, 1.148]	0.6707	0.7507	-0
go:schubfach-vs-ryu	e2e	canonicalize	nested-mixed	schubfach	ryu	ryu	1.11×	[1.005, 1.246]	0.0778	0.1290	0
go:schubfach-vs-dragonbox	e2e	canonicalize	nested-mixed	schubfach	dragonbox	dragonbox	1.08×	[0.965, 1.203]	0.1984	0.2871	0
go:ryu-vs-dragonbox	e2e	canonicalize	nested-mixed	ryu	dragonbox	ryu	1.03×	[0.918, 1.172]	0.6609	0.7443	-0
go:ryu-vs-bd	e2e	canonicalize	nested-mixed	ryu	json-canon	ryu	1.14×	[1.007, 1.288]	0.0684	0.1154	-0
go:dragonbox-vs-bd	e2e	canonicalize	nested-mixed	dragonbox	json-canon	dragonbox	1.10×	[0.968, 1.245]	0.1476	0.2273	-0
rs:schubfach-vs-bd	e2e	canonicalize	nested-mixed	schubfach-rs	json-canon-rs	schubfach-rs*	1.18×	[1.070, 1.295]	0.0030	0.0061	-1
rs:schubfach-vs-ryu	e2e	canonicalize	nested-mixed	schubfach-rs	ryu-rs	ryu-rs	1.02×	[0.924, 1.130]	0.6853	0.7614	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	nested-mixed	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.953, 1.094]	0.8548	0.8949	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	nested-mixed	ryu-rs	dragonbox-rs	ryu-rs	1.03×	[0.964, 1.118]	0.4629	0.5705	-0
rs:ryu-vs-bd	e2e	canonicalize	nested-mixed	ryu-rs	json-canon-rs	ryu-rs*	1.20×	[1.086, 1.326]	0.0018	0.0037	-1
rs:dragonbox-vs-bd	e2e	canonicalize	nested-mixed	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.17×	[1.071, 1.233]	0.0006	0.0013	-1
schubfach:go-vs-rs	e2e	canonicalize	nested-mixed	schubfach	schubfach-rs	schubfach-rs*	2.92×	[2.668, 3.209]	0.0002	0.0004	6
bd:go-vs-rs	e2e	canonicalize	nested-mixed	json-canon	json-canon-rs	json-canon-rs*	2.55×	[2.257, 2.836]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	nested-mixed	ryu	ryu-rs	ryu-rs*	2.69×	[2.385, 2.993]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	canonicalize	nested-mixed	dragonbox	dragonbox-rs	dragonbox-rs*	2.69×	[2.437, 2.924]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	canonicalize	number-heavy	schubfach	json-canon	schubfach	1.11×	[0.998, 1.237]	0.0934	0.1512	-0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:schubfach-vs-ryu	e2e	canonicalize	number-heavy	schubfach	ryu	schubfach	1.04×	[0.919, 1.160]	0.5237	0.6229	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	number-heavy	schubfach	dragonbox	schubfach	1.06×	[0.949, 1.175]	0.3169	0.4193	-0
go:ryu-vs-dragonbox	e2e	canonicalize	number-heavy	ryu	dragonbox	ryu	1.02×	[0.910, 1.147]	0.7826	0.8391	-0
go:ryu-vs-bd	e2e	canonicalize	number-heavy	ryu	json-canon	ryu	1.06×	[0.951, 1.208]	0.3211	0.4223	-0
go:dragonbox-vs-bd	e2e	canonicalize	number-heavy	dragonbox	json-canon	dragonbox	1.05×	[0.939, 1.167]	0.4101	0.5172	-0
rs:schubfach-vs-bd	e2e	canonicalize	number-heavy	schubfach-rs	json-canon-rs	schubfach-rs*	1.34×	[1.242, 1.449]	0.0002	0.0004	-2
rs:schubfach-vs-ryu	e2e	canonicalize	number-heavy	schubfach-rs	ryu-rs	ryu-rs	1.03×	[0.946, 1.101]	0.4927	0.5953	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	number-heavy	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.957, 1.094]	0.6999	0.7701	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	number-heavy	ryu-rs	dragonbox-rs	ryu-rs	1.04×	[0.977, 1.122]	0.2567	0.3557	-0
rs:ryu-vs-bd	e2e	canonicalize	number-heavy	ryu-rs	json-canon-rs	ryu-rs*	1.38×	[1.273, 1.487]	0.0002	0.0004	-2
rs:dragonbox-vs-bd	e2e	canonicalize	number-heavy	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.32×	[1.233, 1.411]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	canonicalize	number-heavy	schubfach	schubfach-rs	schubfach-rs*	2.68×	[2.463, 2.956]	0.0002	0.0004	5
bd:go-vs-rs	e2e	canonicalize	number-heavy	json-canon	json-canon-rs	json-canon-rs*	2.22×	[2.018, 2.448]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	number-heavy	ryu	ryu-rs	ryu-rs*	2.87×	[2.562, 3.166]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	canonicalize	number-heavy	dragonbox	dragonbox-rs	dragonbox-rs*	2.80×	[2.567, 3.041]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	canonicalize	numeric-boundary	schubfach	json-canon	schubfach*	1.17×	[1.028, 1.316]	0.0262	0.0479	-0
go:schubfach-vs-ryu	e2e	canonicalize	numeric-boundary	schubfach	ryu	schubfach	1.07×	[0.957, 1.206]	0.2368	0.3345	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	numeric-boundary	schubfach	dragonbox	dragonbox	1.03×	[0.896, 1.184]	0.7199	0.7887	0
go:ryu-vs-dragonbox	e2e	canonicalize	numeric-boundary	ryu	dragonbox	dragonbox	1.10×	[0.962, 1.265]	0.1706	0.2534	0
go:ryu-vs-bd	e2e	canonicalize	numeric-boundary	ryu	json-canon	ryu	1.09×	[0.967, 1.225]	0.1854	0.2707	-0
go:dragonbox-vs-bd	e2e	canonicalize	numeric-boundary	dragonbox	json-canon	dragonbox*	1.20×	[1.039, 1.382]	0.0226	0.0417	-0
rs:schubfach-vs-bd	e2e	canonicalize	numeric-boundary	schubfach-rs	json-canon-rs	schubfach-rs*	1.28×	[1.187, 1.381]	0.0002	0.0004	-2
rs:schubfach-vs-ryu	e2e	canonicalize	numeric-boundary	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.913, 1.117]	0.8406	0.8837	-0
rs:schubfach-vs-dragonbox	e2e	canonicalize	numeric-boundary	schubfach-rs	dragonbox-rs	schubfach-rs	1.05×	[0.971, 1.140]	0.2741	0.3749	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	numeric-boundary	ryu-rs	dragonbox-rs	ryu-rs	1.03×	[0.948, 1.141]	0.4933	0.5955	-0
rs:ryu-vs-bd	e2e	canonicalize	numeric-boundary	ryu-rs	json-canon-rs	ryu-rs*	1.26×	[1.157, 1.386]	0.0002	0.0004	-1
rs:dragonbox-vs-bd	e2e	canonicalize	numeric-boundary	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.22×	[1.140, 1.291]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	canonicalize	numeric-boundary	schubfach	schubfach-rs	schubfach-rs*	2.69×	[2.427, 2.978]	0.0002	0.0004	5
bd:go-vs-rs	e2e	canonicalize	numeric-boundary	json-canon	json-canon-rs	json-canon-rs*	2.46×	[2.229, 2.706]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	numeric-boundary	ryu	ryu-rs	ryu-rs*	2.86×	[2.554, 3.191]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	canonicalize	numeric-boundary	dragonbox	dragonbox-rs	dragonbox-rs*	2.51×	[2.206, 2.803]	0.0002	0.0004	3
go:schubfach-vs-bd	e2e	canonicalize	rfc-key-sorting	schubfach	json-canon	schubfach	1.07×	[0.957, 1.208]	0.2789	0.3784	-0
go:schubfach-vs-ryu	e2e	canonicalize	rfc-key-sorting	schubfach	ryu	schubfach	1.02×	[0.900, 1.147]	0.7886	0.8427	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	rfc-key-sorting	schubfach	dragonbox	schubfach	1.00×	[0.889, 1.128]	0.9718	0.9790	-0
go:ryu-vs-dragonbox	e2e	canonicalize	rfc-key-sorting	ryu	dragonbox	dragonbox	1.01×	[0.895, 1.135]	0.8158	0.8660	0
go:ryu-vs-bd	e2e	canonicalize	rfc-key-sorting	ryu	json-canon	ryu	1.05×	[0.943, 1.191]	0.4143	0.5215	-0
go:dragonbox-vs-bd	e2e	canonicalize	rfc-key-sorting	dragonbox	json-canon	dragonbox	1.07×	[0.954, 1.197]	0.2879	0.3873	-0
rs:schubfach-vs-bd	e2e	canonicalize	rfc-key-sorting	schubfach-rs	json-canon-rs	schubfach-rs*	1.12×	[1.037, 1.194]	0.0040	0.0079	-1
rs:schubfach-vs-ryu	e2e	canonicalize	rfc-key-sorting	schubfach-rs	ryu-rs	schubfach-rs	1.09×	[1.011, 1.164]	0.0320	0.0576	-0
rs:schubfach-vs-dragonbox	e2e	canonicalize	rfc-key-sorting	schubfach-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.925, 1.111]	0.9384	0.9592	0
rs:ryu-vs-dragonbox	e2e	canonicalize	rfc-key-sorting	ryu-rs	dragonbox-rs	dragonbox-rs	1.09×	[1.002, 1.197]	0.0774	0.1285	0
rs:ryu-vs-bd	e2e	canonicalize	rfc-key-sorting	ryu-rs	json-canon-rs	ryu-rs	1.03×	[0.955, 1.102]	0.4767	0.5807	-0
rs:dragonbox-vs-bd	e2e	canonicalize	rfc-key-sorting	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.12×	[1.030, 1.238]	0.0206	0.0381	-0
schubfach:go-vs-rs	e2e	canonicalize	rfc-key-sorting	schubfach	schubfach-rs	schubfach-rs*	2.67×	[2.430, 2.951]	0.0002	0.0004	5
bd:go-vs-rs	e2e	canonicalize	rfc-key-sorting	json-canon	json-canon-rs	json-canon-rs*	2.56×	[2.334, 2.813]	0.0002	0.0004	5
ryu:go-vs-rs	e2e	canonicalize	rfc-key-sorting	ryu	ryu-rs	ryu-rs*	2.50×	[2.246, 2.745]	0.0002	0.0004	4

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
dragonbox:go-vs-rs	e2e	canonicalize	rfc-key-sorting	dragonbox	dragonbox-rs	dragonbox-rs*	2.69×	[2.424, 3.011]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	canonicalize	small	schubfach	json-canon	schubfach	1.08×	[0.955, 1.218]	0.2412	0.3399	-0
go:schubfach-vs-ryu	e2e	canonicalize	small	schubfach	ryu	schubfach	1.05×	[0.930, 1.180]	0.4723	0.5766	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	small	schubfach	dragonbox	schubfach	1.05×	[0.932, 1.175]	0.4057	0.5129	-0
go:ryu-vs-dragonbox	e2e	canonicalize	small	ryu	dragonbox	ryu	1.01×	[0.902, 1.116]	0.9336	0.9558	-0
go:ryu-vs-bd	e2e	canonicalize	small	ryu	json-canon	ryu	1.03×	[0.921, 1.161]	0.6071	0.6994	-0
go:dragonbox-vs-bd	e2e	canonicalize	small	dragonbox	json-canon	dragonbox	1.03×	[0.924, 1.151]	0.6673	0.7478	-0
rs:schubfach-vs-bd	e2e	canonicalize	small	schubfach-rs	json-canon-rs	schubfach-rs*	1.23×	[1.125, 1.336]	0.0008	0.0017	-1
rs:schubfach-vs-ryu	e2e	canonicalize	small	schubfach-rs	ryu-rs	schubfach-rs	1.04×	[0.978, 1.130]	0.2751	0.3752	-0
rs:schubfach-vs-dragonbox	e2e	canonicalize	small	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.914, 1.120]	0.7399	0.8052	0
rs:ryu-vs-dragonbox	e2e	canonicalize	small	ryu-rs	dragonbox-rs	dragonbox-rs	1.06×	[0.961, 1.149]	0.2154	0.3079	0
rs:ryu-vs-bd	e2e	canonicalize	small	ryu-rs	json-canon-rs	ryu-rs*	1.18×	[1.092, 1.253]	0.0004	0.0008	-1
rs:dragonbox-vs-bd	e2e	canonicalize	small	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.25×	[1.119, 1.370]	0.0006	0.0013	-1
schubfach:go-vs-rs	e2e	canonicalize	small	schubfach	schubfach-rs	schubfach-rs*	2.66×	[2.383, 2.958]	0.0002	0.0004	4
bd:go-vs-rs	e2e	canonicalize	small	json-canon	json-canon-rs	json-canon-rs*	2.33×	[2.107, 2.579]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	small	ryu	ryu-rs	ryu-rs*	2.67×	[2.437, 2.880]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	canonicalize	small	dragonbox	dragonbox-rs	dragonbox-rs*	2.84×	[2.522, 3.142]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	canonicalize	surrogate-pair	schubfach	json-canon	schubfach	1.05×	[0.954, 1.177]	0.3617	0.4665	-0
go:schubfach-vs-ryu	e2e	canonicalize	surrogate-pair	schubfach	ryu	ryu	1.02×	[0.900, 1.166]	0.7331	0.7994	0
go:schubfach-vs-dragonbox	e2e	canonicalize	surrogate-pair	schubfach	dragonbox	schubfach	1.06×	[0.965, 1.193]	0.2603	0.3599	-0
go:ryu-vs-dragonbox	e2e	canonicalize	surrogate-pair	ryu	dragonbox	ryu	1.09×	[0.974, 1.227]	0.1764	0.2602	-0
go:ryu-vs-bd	e2e	canonicalize	surrogate-pair	ryu	json-canon	ryu	1.08×	[0.954, 1.216]	0.2408	0.3397	-0
go:dragonbox-vs-bd	e2e	canonicalize	surrogate-pair	dragonbox	json-canon	json-canon	1.01×	[0.926, 1.099]	0.8198	0.8696	0
rs:schubfach-vs-bd	e2e	canonicalize	surrogate-pair	schubfach-rs	json-canon-rs	schubfach-rs*	1.13×	[1.061, 1.206]	0.0002	0.0004	-1
rs:schubfach-vs-ryu	e2e	canonicalize	surrogate-pair	schubfach-rs	ryu-rs	ryu-rs	1.04×	[0.973, 1.102]	0.2563	0.3555	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	surrogate-pair	schubfach-rs	dragonbox-rs	schubfach-rs	1.00×	[0.930, 1.076]	0.9566	0.9707	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	surrogate-pair	ryu-rs	dragonbox-rs	ryu-rs	1.04×	[0.967, 1.121]	0.3229	0.4242	-0
rs:ryu-vs-bd	e2e	canonicalize	surrogate-pair	ryu-rs	json-canon-rs	ryu-rs*	1.18×	[1.102, 1.251]	0.0002	0.0004	-1
rs:dragonbox-vs-bd	e2e	canonicalize	surrogate-pair	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.13×	[1.046, 1.216]	0.0030	0.0061	-1
schubfach:go-vs-rs	e2e	canonicalize	surrogate-pair	schubfach	schubfach-rs	schubfach-rs*	2.61×	[2.345, 2.848]	0.0002	0.0004	4
bd:go-vs-rs	e2e	canonicalize	surrogate-pair	json-canon	json-canon-rs	json-canon-rs*	2.43×	[2.239, 2.620]	0.0002	0.0004	6
ryu:go-vs-rs	e2e	canonicalize	surrogate-pair	ryu	ryu-rs	ryu-rs*	2.65×	[2.358, 2.943]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	canonicalize	surrogate-pair	dragonbox	dragonbox-rs	dragonbox-rs*	2.77×	[2.540, 2.997]	0.0002	0.0004	7
go:schubfach-vs-bd	e2e	canonicalize	unicode	schubfach	json-canon	schubfach	1.10×	[0.983, 1.219]	0.1160	0.1839	-0
go:schubfach-vs-ryu	e2e	canonicalize	unicode	schubfach	ryu	schubfach	1.01×	[0.898, 1.132]	0.8220	0.8707	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	unicode	schubfach	dragonbox	schubfach	1.00×	[0.899, 1.118]	0.9328	0.9557	-0
go:ryu-vs-dragonbox	e2e	canonicalize	unicode	ryu	dragonbox	dragonbox	1.01×	[0.898, 1.132]	0.8834	0.9167	0
go:ryu-vs-bd	e2e	canonicalize	unicode	ryu	json-canon	ryu	1.08×	[0.967, 1.216]	0.1944	0.2816	-0
go:dragonbox-vs-bd	e2e	canonicalize	unicode	dragonbox	json-canon	dragonbox	1.09×	[0.982, 1.215]	0.1424	0.2204	-0
rs:schubfach-vs-bd	e2e	canonicalize	unicode	schubfach-rs	json-canon-rs	schubfach-rs*	1.17×	[1.093, 1.254]	0.0004	0.0008	-1
rs:schubfach-vs-ryu	e2e	canonicalize	unicode	schubfach-rs	ryu-rs	schubfach-rs	1.05×	[0.995, 1.106]	0.1232	0.1930	-0
rs:schubfach-vs-dragonbox	e2e	canonicalize	unicode	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.928, 1.106]	0.8300	0.8772	-0
rs:ryu-vs-dragonbox	e2e	canonicalize	unicode	ryu-rs	dragonbox-rs	dragonbox-rs	1.03×	[0.949, 1.131]	0.4577	0.5662	0
rs:ryu-vs-bd	e2e	canonicalize	unicode	ryu-rs	json-canon-rs	ryu-rs*	1.11×	[1.043, 1.194]	0.0054	0.0106	-1
rs:dragonbox-vs-bd	e2e	canonicalize	unicode	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.15×	[1.051, 1.271]	0.0078	0.0151	-1
schubfach:go-vs-rs	e2e	canonicalize	unicode	schubfach	schubfach-rs	schubfach-rs*	2.69×	[2.465, 2.924]	0.0002	0.0004	5

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
bd:go-vs-rs	e2e	canonicalize	unicode	json-canon	json-canon-rs	json-canon-rs*	2.53×	[2.287, 2.767]	0.0002	0.0004	5
ryu:go-vs-rs	e2e	canonicalize	unicode	ryu	ryu-rs	ryu-rs*	2.61×	[2.353, 2.827]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	canonicalize	unicode	dragonbox	dragonbox-rs	dragonbox-rs*	2.67×	[2.371, 2.964]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	canonicalize	verify-whitespace	schubfach	json-canon	schubfach	1.06×	[0.939, 1.210]	0.3623	0.4665	-0
go:schubfach-vs-ryu	e2e	canonicalize	verify-whitespace	schubfach	ryu	schubfach	1.01×	[0.897, 1.138]	0.8316	0.8778	-0
go:schubfach-vs-dragonbox	e2e	canonicalize	verify-whitespace	schubfach	dragonbox	dragonbox	1.03×	[0.918, 1.163]	0.5927	0.6878	0
go:ryu-vs-dragonbox	e2e	canonicalize	verify-whitespace	ryu	dragonbox	dragonbox	1.05×	[0.937, 1.169]	0.4461	0.5567	0
go:ryu-vs-bd	e2e	canonicalize	verify-whitespace	ryu	json-canon	ryu	1.05×	[0.928, 1.195]	0.4697	0.5756	-0
go:dragonbox-vs-bd	e2e	canonicalize	verify-whitespace	dragonbox	json-canon	dragonbox	1.10×	[0.972, 1.247]	0.1706	0.2534	-0
rs:schubfach-vs-bd	e2e	canonicalize	verify-whitespace	schubfach-rs	json-canon-rs	schubfach-rs*	1.19×	[1.101, 1.298]	0.0004	0.0008	-1
rs:schubfach-vs-ryu	e2e	canonicalize	verify-whitespace	schubfach-rs	ryu-rs	ryu-rs	1.05×	[0.966, 1.154]	0.2953	0.3956	0
rs:schubfach-vs-dragonbox	e2e	canonicalize	verify-whitespace	schubfach-rs	dragonbox-rs	dragonbox-rs	1.09×	[1.016, 1.182]	0.0330	0.0591	0
rs:ryu-vs-dragonbox	e2e	canonicalize	verify-whitespace	ryu-rs	dragonbox-rs	dragonbox-rs	1.04×	[0.949, 1.130]	0.4173	0.5243	0
rs:ryu-vs-bd	e2e	canonicalize	verify-whitespace	ryu-rs	json-canon-rs	ryu-rs*	1.26×	[1.150, 1.392]	0.0002	0.0004	-1
rs:dragonbox-vs-bd	e2e	canonicalize	verify-whitespace	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.30×	[1.209, 1.423]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	canonicalize	verify-whitespace	schubfach	schubfach-rs	schubfach-rs*	2.65×	[2.366, 2.911]	0.0002	0.0004	4
bd:go-vs-rs	e2e	canonicalize	verify-whitespace	json-canon	json-canon-rs	json-canon-rs*	2.36×	[2.091, 2.636]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	canonicalize	verify-whitespace	ryu	ryu-rs	ryu-rs*	2.83×	[2.528, 3.131]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	canonicalize	verify-whitespace	dragonbox	dragonbox-rs	dragonbox-rs*	2.80×	[2.523, 3.059]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	verify	array-2048	schubfach	json-canon	schubfach*	1.31×	[1.221, 1.398]	0.0002	0.0004	-2
go:schubfach-vs-ryu	e2e	verify	array-2048	schubfach	ryu	ryu	1.00×	[0.922, 1.084]	0.9448	0.9627	0
go:schubfach-vs-dragonbox	e2e	verify	array-2048	schubfach	dragonbox	schubfach	1.04×	[0.969, 1.111]	0.2991	0.3995	-0
go:ryu-vs-dragonbox	e2e	verify	array-2048	ryu	dragonbox	ryu	1.04×	[0.960, 1.110]	0.2885	0.3877	-0
go:ryu-vs-bd	e2e	verify	array-2048	ryu	json-canon	ryu*	1.31×	[1.215, 1.404]	0.0002	0.0004	-2
go:dragonbox-vs-bd	e2e	verify	array-2048	dragonbox	json-canon	dragonbox*	1.26×	[1.190, 1.338]	0.0002	0.0004	-2
rs:schubfach-vs-bd	e2e	verify	array-2048	schubfach-rs	json-canon-rs	schubfach-rs*	1.50×	[1.347, 1.713]	0.0002	0.0004	-2
rs:schubfach-vs-ryu	e2e	verify	array-2048	schubfach-rs	ryu-rs	ryu-rs	1.14×	[1.001, 1.271]	0.0420	0.0733	0
rs:schubfach-vs-dragonbox	e2e	verify	array-2048	schubfach-rs	dragonbox-rs	dragonbox-rs	1.12×	[0.972, 1.263]	0.1220	0.1918	0
rs:ryu-vs-dragonbox	e2e	verify	array-2048	ryu-rs	dragonbox-rs	ryu-rs	1.02×	[0.908, 1.155]	0.7147	0.7839	-0
rs:ryu-vs-bd	e2e	verify	array-2048	ryu-rs	json-canon-rs	ryu-rs*	1.71×	[1.556, 1.921]	0.0002	0.0004	-3
rs:dragonbox-vs-bd	e2e	verify	array-2048	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.67×	[1.489, 1.903]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	verify	array-2048	schubfach	schubfach-rs	schubfach-rs*	2.14×	[1.946, 2.412]	0.0002	0.0004	5
bd:go-vs-rs	e2e	verify	array-2048	json-canon	json-canon-rs	json-canon-rs*	1.87×	[1.679, 2.011]	0.0002	0.0004	5
ryu:go-vs-rs	e2e	verify	array-2048	ryu	ryu-rs	ryu-rs*	2.44×	[2.221, 2.682]	0.0002	0.0004	6
dragonbox:go-vs-rs	e2e	verify	array-2048	dragonbox	dragonbox-rs	dragonbox-rs*	2.49×	[2.230, 2.741]	0.0002	0.0004	7
go:schubfach-vs-bd	e2e	verify	array-256	schubfach	json-canon	schubfach	1.14×	[1.013, 1.281]	0.0420	0.0733	-0
go:schubfach-vs-ryu	e2e	verify	array-256	schubfach	ryu	schubfach	1.04×	[0.927, 1.151]	0.5491	0.6476	-0
go:schubfach-vs-dragonbox	e2e	verify	array-256	schubfach	dragonbox	dragonbox	1.10×	[0.955, 1.229]	0.1780	0.2620	0
go:ryu-vs-dragonbox	e2e	verify	array-256	ryu	dragonbox	dragonbox	1.13×	[0.992, 1.258]	0.0414	0.0725	0
go:ryu-vs-bd	e2e	verify	array-256	ryu	json-canon	ryu	1.10×	[0.994, 1.228]	0.0956	0.1542	-0
go:dragonbox-vs-bd	e2e	verify	array-256	dragonbox	json-canon	dragonbox*	1.25×	[1.092, 1.395]	0.0018	0.0037	-1
rs:schubfach-vs-bd	e2e	verify	array-256	schubfach-rs	json-canon-rs	schubfach-rs*	1.31×	[1.193, 1.447]	0.0002	0.0004	-1
rs:schubfach-vs-ryu	e2e	verify	array-256	schubfach-rs	ryu-rs	ryu-rs	1.02×	[0.943, 1.101]	0.5641	0.6606	0
rs:schubfach-vs-dragonbox	e2e	verify	array-256	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.936, 1.101]	0.7005	0.7701	0
rs:ryu-vs-dragonbox	e2e	verify	array-256	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.928, 1.088]	0.8744	0.9098	-0
rs:ryu-vs-bd	e2e	verify	array-256	ryu-rs	json-canon-rs	ryu-rs*	1.34×	[1.219, 1.471]	0.0002	0.0004	-2

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:dragonbox-vs-bd	e2e	verify	array-256	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.33×	[1.209, 1.468]	0.0004	0.0008	-1
schubfach:go-vs-rs	e2e	verify	array-256	schubfach	schubfach-rs	schubfach-rs*	2.47×	[2.247, 2.756]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	array-256	json-canon	json-canon-rs	json-canon-rs*	2.15×	[1.939, 2.410]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	array-256	ryu	ryu-rs	ryu-rs*	2.63×	[2.413, 2.876]	0.0002	0.0004	6
dragonbox:go-vs-rs	e2e	verify	array-256	dragonbox	dragonbox-rs	dragonbox-rs*	2.30×	[2.092, 2.612]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	verify	canonical-minimal	schubfach	json-canon	schubfach*	1.13×	[1.041, 1.257]	0.0178	0.0332	-0
go:schubfach-vs-ryu	e2e	verify	canonical-minimal	schubfach	ryu	schubfach	1.04×	[0.935, 1.154]	0.5345	0.6334	-0
go:schubfach-vs-dragonbox	e2e	verify	canonical-minimal	schubfach	dragonbox	schubfach	1.02×	[0.938, 1.137]	0.6221	0.7109	-0
go:ryu-vs-dragonbox	e2e	verify	canonical-minimal	ryu	dragonbox	dragonbox	1.01×	[0.911, 1.110]	0.8378	0.8814	0
go:ryu-vs-bd	e2e	verify	canonical-minimal	ryu	json-canon	ryu	1.09×	[0.991, 1.205]	0.0924	0.1500	-0
go:dragonbox-vs-bd	e2e	verify	canonical-minimal	dragonbox	json-canon	dragonbox	1.10×	[1.012, 1.203]	0.0362	0.0644	-0
rs:schubfach-vs-bd	e2e	verify	canonical-minimal	schubfach-rs	json-canon-rs	schubfach-rs	1.04×	[0.942, 1.128]	0.4535	0.5621	-0
rs:schubfach-vs-ryu	e2e	verify	canonical-minimal	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.944, 1.103]	0.6367	0.7252	-0
rs:schubfach-vs-dragonbox	e2e	verify	canonical-minimal	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.953, 1.093]	0.6177	0.7084	-0
rs:ryu-vs-dragonbox	e2e	verify	canonical-minimal	ryu-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.940, 1.070]	0.9550	0.9707	0
rs:ryu-vs-bd	e2e	verify	canonical-minimal	ryu-rs	json-canon-rs	ryu-rs	1.02×	[0.925, 1.105]	0.7273	0.7962	-0
rs:dragonbox-vs-bd	e2e	verify	canonical-minimal	dragonbox-rs	json-canon-rs	dragonbox-rs	1.02×	[0.937, 1.096]	0.6547	0.7399	-0
schubfach:go-vs-rs	e2e	verify	canonical-minimal	schubfach	schubfach-rs	schubfach-rs*	2.84×	[2.555, 3.073]	0.0002	0.0004	6
bd:go-vs-rs	e2e	verify	canonical-minimal	json-canon	json-canon-rs	json-canon-rs*	3.09×	[2.826, 3.398]	0.0002	0.0004	7
ryu:go-vs-rs	e2e	verify	canonical-minimal	ryu	ryu-rs	ryu-rs*	2.88×	[2.604, 3.144]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	verify	canonical-minimal	dragonbox	dragonbox-rs	dragonbox-rs*	2.85×	[2.638, 3.051]	0.0002	0.0004	7
go:schubfach-vs-bd	e2e	verify	control-escapes	schubfach	json-canon	schubfach	1.01×	[0.907, 1.141]	0.8986	0.9280	-0
go:schubfach-vs-ryu	e2e	verify	control-escapes	schubfach	ryu	ryu	1.05×	[0.918, 1.167]	0.4593	0.5677	0
go:schubfach-vs-dragonbox	e2e	verify	control-escapes	schubfach	dragonbox	dragonbox	1.11×	[0.967, 1.254]	0.1494	0.2298	0
go:ryu-vs-dragonbox	e2e	verify	control-escapes	ryu	dragonbox	dragonbox	1.06×	[0.938, 1.206]	0.3957	0.5033	0
go:ryu-vs-bd	e2e	verify	control-escapes	ryu	json-canon	ryu	1.06×	[0.952, 1.182]	0.3385	0.4411	-0
go:dragonbox-vs-bd	e2e	verify	control-escapes	dragonbox	json-canon	dragonbox	1.12×	[0.990, 1.272]	0.0904	0.1475	-0
rs:schubfach-vs-bd	e2e	verify	control-escapes	schubfach-rs	json-canon-rs	schubfach-rs	1.08×	[1.006, 1.184]	0.0724	0.1210	-0
rs:schubfach-vs-ryu	e2e	verify	control-escapes	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.949, 1.103]	0.7646	0.8266	-0
rs:schubfach-vs-dragonbox	e2e	verify	control-escapes	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.943, 1.113]	0.7962	0.8487	-0
rs:ryu-vs-dragonbox	e2e	verify	control-escapes	ryu-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.927, 1.065]	0.9674	0.9772	0
rs:ryu-vs-bd	e2e	verify	control-escapes	ryu-rs	json-canon-rs	ryu-rs	1.07×	[0.996, 1.141]	0.0686	0.1156	-0
rs:dragonbox-vs-bd	e2e	verify	control-escapes	dragonbox-rs	json-canon-rs	dragonbox-rs	1.07×	[0.985, 1.148]	0.1058	0.1690	-0
schubfach:go-vs-rs	e2e	verify	control-escapes	schubfach	schubfach-rs	schubfach-rs*	2.82×	[2.513, 3.115]	0.0002	0.0004	5
bd:go-vs-rs	e2e	verify	control-escapes	json-canon	json-canon-rs	json-canon-rs*	2.63×	[2.399, 2.872]	0.0002	0.0004	5
ryu:go-vs-rs	e2e	verify	control-escapes	ryu	ryu-rs	ryu-rs*	2.66×	[2.423, 2.901]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	verify	control-escapes	dragonbox	dragonbox-rs	dragonbox-rs*	2.52×	[2.230, 2.787]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	verify	deep-64	schubfach	json-canon	json-canon	1.01×	[0.872, 1.163]	0.8512	0.8933	0
go:schubfach-vs-ryu	e2e	verify	deep-64	schubfach	ryu	ryu	1.05×	[0.907, 1.208]	0.5305	0.6298	0
go:schubfach-vs-dragonbox	e2e	verify	deep-64	schubfach	dragonbox	schubfach	1.02×	[0.894, 1.173]	0.7730	0.8315	-0
go:ryu-vs-dragonbox	e2e	verify	deep-64	ryu	dragonbox	ryu	1.07×	[0.952, 1.220]	0.2935	0.3936	-0
go:ryu-vs-bd	e2e	verify	deep-64	ryu	json-canon	ryu	1.04×	[0.907, 1.188]	0.6197	0.7096	-0
go:dragonbox-vs-bd	e2e	verify	deep-64	dragonbox	json-canon	json-canon	1.03×	[0.913, 1.171]	0.5925	0.6878	0
rs:schubfach-vs-bd	e2e	verify	deep-64	schubfach-rs	json-canon-rs	schubfach-rs*	1.08×	[1.025, 1.137]	0.0074	0.0143	-1
rs:schubfach-vs-ryu	e2e	verify	deep-64	schubfach-rs	ryu-rs	ryu-rs	1.02×	[0.972, 1.086]	0.4283	0.5360	0
rs:schubfach-vs-dragonbox	e2e	verify	deep-64	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.969, 1.110]	0.4825	0.5846	0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:ryu-vs-dragonbox	e2e	verify	deep-64	ryu-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.935, 1.089]	0.9926	0.9949	0
rs:ryu-vs-bd	e2e	verify	deep-64	ryu-rs	json-canon-rs	ryu-rs*	1.11×	[1.040, 1.183]	0.0046	0.0090	-1
rs:dragonbox-vs-bd	e2e	verify	deep-64	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.11×	[1.038, 1.204]	0.0100	0.0191	-0
schubfach:go-vs-rs	e2e	verify	deep-64	schubfach	schubfach-rs	schubfach-rs*	2.72×	[2.414, 3.018]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	deep-64	json-canon	json-canon-rs	json-canon-rs*	2.48×	[2.208, 2.736]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	deep-64	ryu	ryu-rs	ryu-rs*	2.65×	[2.368, 2.928]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	verify	deep-64	dragonbox	dragonbox-rs	dragonbox-rs*	2.85×	[2.573, 3.130]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	verify	deep	schubfach	json-canon	schubfach	1.07×	[0.960, 1.203]	0.2857	0.3852	-0
go:schubfach-vs-ryu	e2e	verify	deep	schubfach	ryu	ryu	1.06×	[0.937, 1.185]	0.3843	0.4924	0
go:schubfach-vs-dragonbox	e2e	verify	deep	schubfach	dragonbox	dragonbox	1.03×	[0.913, 1.154]	0.6225	0.7109	0
go:ryu-vs-dragonbox	e2e	verify	deep	ryu	dragonbox	ryu	1.02×	[0.900, 1.173]	0.7357	0.8013	-0
go:ryu-vs-bd	e2e	verify	deep	ryu	json-canon	ryu	1.13×	[0.992, 1.285]	0.0848	0.1398	-0
go:dragonbox-vs-bd	e2e	verify	deep	dragonbox	json-canon	dragonbox	1.10×	[0.970, 1.250]	0.1680	0.2516	-0
rs:schubfach-vs-bd	e2e	verify	deep	schubfach-rs	json-canon-rs	schubfach-rs*	1.14×	[1.064, 1.240]	0.0034	0.0068	-1
rs:schubfach-vs-ryu	e2e	verify	deep	schubfach-rs	ryu-rs	ryu-rs	1.00×	[0.916, 1.065]	0.9572	0.9707	0
rs:schubfach-vs-dragonbox	e2e	verify	deep	schubfach-rs	dragonbox-rs	schubfach-rs	1.05×	[0.992, 1.147]	0.2456	0.3454	-0
rs:ryu-vs-dragonbox	e2e	verify	deep	ryu-rs	dragonbox-rs	ryu-rs	1.05×	[1.003, 1.115]	0.0982	0.1576	-0
rs:ryu-vs-bd	e2e	verify	deep	ryu-rs	json-canon-rs	ryu-rs*	1.14×	[1.069, 1.208]	0.0006	0.0013	-1
rs:dragonbox-vs-bd	e2e	verify	deep	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.09×	[1.015, 1.139]	0.0156	0.0292	-0
schubfach:go-vs-rs	e2e	verify	deep	schubfach	schubfach-rs	schubfach-rs*	2.65×	[2.411, 2.921]	0.0002	0.0004	5
bd:go-vs-rs	e2e	verify	deep	json-canon	json-canon-rs	json-canon-rs*	2.50×	[2.259, 2.757]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	deep	ryu	ryu-rs	ryu-rs*	2.52×	[2.249, 2.769]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	verify	deep	dragonbox	dragonbox-rs	dragonbox-rs*	2.46×	[2.234, 2.715]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	verify	escaped-key-order	schubfach	json-canon	json-canon	1.06×	[0.939, 1.190]	0.3371	0.4398	0
go:schubfach-vs-ryu	e2e	verify	escaped-key-order	schubfach	ryu	ryu	1.07×	[0.965, 1.189]	0.2138	0.3060	0
go:schubfach-vs-dragonbox	e2e	verify	escaped-key-order	schubfach	dragonbox	dragonbox	1.09×	[0.972, 1.200]	0.1372	0.2128	0
go:ryu-vs-dragonbox	e2e	verify	escaped-key-order	ryu	dragonbox	dragonbox	1.02×	[0.897, 1.142]	0.8098	0.8613	0
go:ryu-vs-bd	e2e	verify	escaped-key-order	ryu	json-canon	ryu	1.01×	[0.890, 1.152]	0.8746	0.9098	-0
go:dragonbox-vs-bd	e2e	verify	escaped-key-order	dragonbox	json-canon	dragonbox	1.03×	[0.905, 1.171]	0.6969	0.7681	-0
rs:schubfach-vs-bd	e2e	verify	escaped-key-order	schubfach-rs	json-canon-rs	schubfach-rs*	1.33×	[1.272, 1.444]	0.0002	0.0004	-3
rs:schubfach-vs-ryu	e2e	verify	escaped-key-order	schubfach-rs	ryu-rs	schubfach-rs	1.09×	[0.991, 1.317]	0.2482	0.3485	-0
rs:schubfach-vs-dragonbox	e2e	verify	escaped-key-order	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.950, 1.105]	0.6669	0.7478	-0
rs:ryu-vs-dragonbox	e2e	verify	escaped-key-order	ryu-rs	dragonbox-rs	dragonbox-rs	1.07×	[0.973, 1.283]	0.3989	0.5063	0
rs:ryu-vs-bd	e2e	verify	escaped-key-order	ryu-rs	json-canon-rs	ryu-rs*	1.22×	[1.016, 1.329]	0.0008	0.0017	-1
rs:dragonbox-vs-bd	e2e	verify	escaped-key-order	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.31×	[1.239, 1.394]	0.0002	0.0004	-3
schubfach:go-vs-rs	e2e	verify	escaped-key-order	schubfach	schubfach-rs	schubfach-rs*	2.86×	[2.631, 3.116]	0.0002	0.0004	6
bd:go-vs-rs	e2e	verify	escaped-key-order	json-canon	json-canon-rs	json-canon-rs*	2.03×	[1.838, 2.254]	0.0002	0.0004	3
ryu:go-vs-rs	e2e	verify	escaped-key-order	ryu	ryu-rs	ryu-rs*	2.45×	[2.022, 2.740]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	verify	escaped-key-order	dragonbox	dragonbox-rs	dragonbox-rs*	2.58×	[2.347, 2.870]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	verify	large	schubfach	json-canon	schubfach*	1.48×	[1.433, 1.519]	0.0002	0.0004	-10
go:schubfach-vs-ryu	e2e	verify	large	schubfach	ryu	ryu	1.04×	[0.999, 1.084]	0.0774	0.1285	0
go:schubfach-vs-dragonbox	e2e	verify	large	schubfach	dragonbox	dragonbox	1.04×	[1.000, 1.088]	0.0692	0.1164	0
go:ryu-vs-dragonbox	e2e	verify	large	ryu	dragonbox	dragonbox	1.00×	[0.957, 1.050]	0.9954	0.9969	0
go:ryu-vs-bd	e2e	verify	large	ryu	json-canon	ryu*	1.55×	[1.487, 1.593]	0.0002	0.0004	-10
go:dragonbox-vs-bd	e2e	verify	large	dragonbox	json-canon	dragonbox*	1.55×	[1.492, 1.601]	0.0002	0.0004	-10
rs:schubfach-vs-bd	e2e	verify	large	schubfach-rs	json-canon-rs	schubfach-rs*	2.05×	[1.893, 2.171]	0.0002	0.0004	-9

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:schubfach-vs-ryu	e2e	verify	large	schubfach-rs	ryu-rs	ryu-rs	1.08×	[0.995, 1.177]	0.0924	0.1500	0
rs:schubfach-vs-dragonbox	e2e	verify	large	schubfach-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.929, 1.102]	0.7612	0.8250	0
rs:ryu-vs-dragonbox	e2e	verify	large	ryu-rs	dragonbox-rs	ryu-rs	1.07×	[0.988, 1.156]	0.1530	0.2340	-0
rs:ryu-vs-bd	e2e	verify	large	ryu-rs	json-canon-rs	ryu-rs*	2.21×	[2.044, 2.309]	0.0002	0.0004	-11
rs:dragonbox-vs-bd	e2e	verify	large	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.08×	[1.910, 2.178]	0.0002	0.0004	-10
schubfach:go-vs-rs	e2e	verify	large	schubfach	schubfach-rs	schubfach-rs*	2.22×	[2.046, 2.357]	0.0002	0.0004	9
bd:go-vs-rs	e2e	verify	large	json-canon	json-canon-rs	json-canon-rs*	1.61×	[1.566, 1.645]	0.0002	0.0004	14
ryu:go-vs-rs	e2e	verify	large	ryu	ryu-rs	ryu-rs*	2.30×	[2.124, 2.425]	0.0002	0.0004	9
dragonbox:go-vs-rs	e2e	verify	large	dragonbox	dragonbox-rs	dragonbox-rs*	2.16×	[1.984, 2.281]	0.0002	0.0004	8
go:schubfach-vs-bd	e2e	verify	long-string	schubfach	json-canon	schubfach	1.14×	[0.987, 1.303]	0.0874	0.1433	-0
go:schubfach-vs-ryu	e2e	verify	long-string	schubfach	ryu	schubfach	1.14×	[0.994, 1.290]	0.0660	0.1116	-0
go:schubfach-vs-dragonbox	e2e	verify	long-string	schubfach	dragonbox	schubfach	1.11×	[0.964, 1.264]	0.1350	0.2097	-0
go:ryu-vs-dragonbox	e2e	verify	long-string	ryu	dragonbox	dragonbox	1.02×	[0.924, 1.131]	0.6537	0.7394	0
go:ryu-vs-bd	e2e	verify	long-string	ryu	json-canon	ryu	1.00×	[0.894, 1.118]	0.9994	0.9994	-0
go:dragonbox-vs-bd	e2e	verify	long-string	dragonbox	json-canon	dragonbox	1.02×	[0.920, 1.154]	0.6821	0.7595	-0
rs:schubfach-vs-bd	e2e	verify	long-string	schubfach-rs	json-canon-rs	schubfach-rs	1.06×	[0.948, 1.163]	0.2539	0.3532	-0
rs:schubfach-vs-ryu	e2e	verify	long-string	schubfach-rs	ryu-rs	ryu-rs	1.01×	[0.926, 1.132]	0.9234	0.9498	0
rs:schubfach-vs-dragonbox	e2e	verify	long-string	schubfach-rs	dragonbox-rs	schubfach-rs	1.00×	[0.897, 1.085]	0.9652	0.9758	-0
rs:ryu-vs-dragonbox	e2e	verify	long-string	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.940, 1.090]	0.8538	0.8946	-0
rs:ryu-vs-bd	e2e	verify	long-string	ryu-rs	json-canon-rs	ryu-rs	1.07×	[0.986, 1.165]	0.1336	0.2082	-0
rs:dragonbox-vs-bd	e2e	verify	long-string	dragonbox-rs	json-canon-rs	dragonbox-rs	1.06×	[0.986, 1.150]	0.1642	0.2476	-0
schubfach:go-vs-rs	e2e	verify	long-string	schubfach	schubfach-rs	schubfach-rs*	2.51×	[2.175, 2.876]	0.0002	0.0004	3
bd:go-vs-rs	e2e	verify	long-string	json-canon	json-canon-rs	json-canon-rs*	2.69×	[2.416, 2.992]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	long-string	ryu	ryu-rs	ryu-rs*	2.87×	[2.612, 3.128]	0.0002	0.0004	6
dragonbox:go-vs-rs	e2e	verify	long-string	dragonbox	dragonbox-rs	dragonbox-rs*	2.78×	[2.538, 3.023]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	verify	medium	schubfach	json-canon	schubfach	1.13×	[0.995, 1.270]	0.0718	0.1202	-0
go:schubfach-vs-ryu	e2e	verify	medium	schubfach	ryu	schubfach	1.06×	[0.937, 1.185]	0.3345	0.4373	-0
go:schubfach-vs-dragonbox	e2e	verify	medium	schubfach	dragonbox	dragonbox	1.11×	[0.974, 1.243]	0.1230	0.1929	0
go:ryu-vs-dragonbox	e2e	verify	medium	ryu	dragonbox	dragonbox*	1.17×	[1.044, 1.303]	0.0128	0.0242	0
go:ryu-vs-bd	e2e	verify	medium	ryu	json-canon	ryu	1.06×	[0.953, 1.198]	0.3201	0.4214	-0
go:dragonbox-vs-bd	e2e	verify	medium	dragonbox	json-canon	dragonbox*	1.24×	[1.110, 1.395]	0.0024	0.0049	-1
rs:schubfach-vs-bd	e2e	verify	medium	schubfach-rs	json-canon-rs	schubfach-rs*	1.64×	[1.475, 1.814]	0.0002	0.0004	-3
rs:schubfach-vs-ryu	e2e	verify	medium	schubfach-rs	ryu-rs	schubfach-rs	1.06×	[0.956, 1.173]	0.3025	0.4034	-0
rs:schubfach-vs-dragonbox	e2e	verify	medium	schubfach-rs	dragonbox-rs	schubfach-rs	1.05×	[0.945, 1.152]	0.3661	0.4709	-0
rs:ryu-vs-dragonbox	e2e	verify	medium	ryu-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.928, 1.106]	0.8328	0.8783	0
rs:ryu-vs-bd	e2e	verify	medium	ryu-rs	json-canon-rs	ryu-rs*	1.55×	[1.403, 1.704]	0.0002	0.0004	-3
rs:dragonbox-vs-bd	e2e	verify	medium	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.57×	[1.434, 1.715]	0.0002	0.0004	-3
schubfach:go-vs-rs	e2e	verify	medium	schubfach	schubfach-rs	schubfach-rs*	2.60×	[2.317, 2.928]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	medium	json-canon	json-canon-rs	json-canon-rs*	1.79×	[1.615, 1.995]	0.0002	0.0004	3
ryu:go-vs-rs	e2e	verify	medium	ryu	ryu-rs	ryu-rs*	2.61×	[2.338, 2.889]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	verify	medium	dragonbox	dragonbox-rs	dragonbox-rs*	2.25×	[2.059, 2.514]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	verify	mixed-prod	schubfach	json-canon	schubfach	1.05×	[0.936, 1.182]	0.4447	0.5555	-0
go:schubfach-vs-ryu	e2e	verify	mixed-prod	schubfach	ryu	ryu	1.06×	[0.932, 1.209]	0.3607	0.4659	0
go:schubfach-vs-dragonbox	e2e	verify	mixed-prod	schubfach	dragonbox	dragonbox	1.07×	[0.947, 1.211]	0.3125	0.4146	0
go:ryu-vs-dragonbox	e2e	verify	mixed-prod	ryu	dragonbox	dragonbox	1.00×	[0.886, 1.133]	0.9472	0.9644	0
go:ryu-vs-bd	e2e	verify	mixed-prod	ryu	json-canon	ryu	1.12×	[1.000, 1.264]	0.0894	0.1460	-0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:dragonbox-vs-bd	e2e	verify	mixed-prod	dragonbox	json-canon	dragonbox	1.12×	[1.004, 1.252]	0.0660	0.1116	-0
rs:schubfach-vs-bd	e2e	verify	mixed-prod	schubfach-rs	json-canon-rs	schubfach-rs*	1.24×	[1.125, 1.353]	0.0002	0.0004	-1
rs:schubfach-vs-ryu	e2e	verify	mixed-prod	schubfach-rs	ryu-rs	ryu-rs	1.04×	[0.959, 1.126]	0.3913	0.4994	0
rs:schubfach-vs-dragonbox	e2e	verify	mixed-prod	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.928, 1.090]	0.8304	0.8772	-0
rs:ryu-vs-dragonbox	e2e	verify	mixed-prod	ryu-rs	dragonbox-rs	ryu-rs	1.05×	[0.963, 1.130]	0.2805	0.3797	-0
rs:ryu-vs-bd	e2e	verify	mixed-prod	ryu-rs	json-canon-rs	ryu-rs*	1.28×	[1.166, 1.408]	0.0004	0.0008	-1
rs:dragonbox-vs-bd	e2e	verify	mixed-prod	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.23×	[1.114, 1.341]	0.0004	0.0008	-1
schubfach:go-vs-rs	e2e	verify	mixed-prod	schubfach	schubfach-rs	schubfach-rs*	2.77×	[2.479, 3.062]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	mixed-prod	json-canon	json-canon-rs	json-canon-rs*	2.34×	[2.131, 2.621]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	mixed-prod	ryu	ryu-rs	ryu-rs*	2.70×	[2.411, 2.991]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	verify	mixed-prod	dragonbox	dragonbox-rs	dragonbox-rs*	2.56×	[2.311, 2.832]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	verify	nested-mixed	schubfach	json-canon	schubfach	1.19×	[1.022, 1.383]	0.0364	0.0647	-0
go:schubfach-vs-ryu	e2e	verify	nested-mixed	schubfach	ryu	schubfach	1.16×	[1.008, 1.316]	0.0472	0.0815	-0
go:schubfach-vs-dragonbox	e2e	verify	nested-mixed	schubfach	dragonbox	schubfach	1.13×	[0.990, 1.293]	0.0872	0.1432	-0
go:ryu-vs-dragonbox	e2e	verify	nested-mixed	ryu	dragonbox	dragonbox	1.02×	[0.911, 1.144]	0.7291	0.7975	0
go:ryu-vs-bd	e2e	verify	nested-mixed	ryu	json-canon	ryu	1.03×	[0.902, 1.181]	0.6859	0.7614	-0
go:dragonbox-vs-bd	e2e	verify	nested-mixed	dragonbox	json-canon	dragonbox	1.05×	[0.925, 1.205]	0.4775	0.5807	-0
rs:schubfach-vs-bd	e2e	verify	nested-mixed	schubfach-rs	json-canon-rs	schubfach-rs*	1.30×	[1.202, 1.408]	0.0002	0.0004	-2
rs:schubfach-vs-ryu	e2e	verify	nested-mixed	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.933, 1.095]	0.8426	0.8850	-0
rs:schubfach-vs-dragonbox	e2e	verify	nested-mixed	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.929, 1.136]	0.7067	0.7762	0
rs:ryu-vs-dragonbox	e2e	verify	nested-mixed	ryu-rs	dragonbox-rs	dragonbox-rs	1.03×	[0.950, 1.133]	0.5581	0.6541	0
rs:ryu-vs-bd	e2e	verify	nested-mixed	ryu-rs	json-canon-rs	ryu-rs*	1.28×	[1.217, 1.371]	0.0002	0.0004	-2
rs:dragonbox-vs-bd	e2e	verify	nested-mixed	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.32×	[1.229, 1.464]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	verify	nested-mixed	schubfach	schubfach-rs	schubfach-rs*	2.51×	[2.220, 2.856]	0.0002	0.0004	3
bd:go-vs-rs	e2e	verify	nested-mixed	json-canon	json-canon-rs	json-canon-rs*	2.32×	[2.057, 2.588]	0.0002	0.0004	3
ryu:go-vs-rs	e2e	verify	nested-mixed	ryu	ryu-rs	ryu-rs*	2.89×	[2.613, 3.150]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	verify	nested-mixed	dragonbox	dragonbox-rs	dragonbox-rs*	2.91×	[2.624, 3.257]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	verify	number-heavy	schubfach	json-canon	schubfach	1.04×	[0.922, 1.209]	0.5581	0.6541	-0
go:schubfach-vs-ryu	e2e	verify	number-heavy	schubfach	ryu	schubfach	1.09×	[0.951, 1.221]	0.2060	0.2961	-0
go:schubfach-vs-dragonbox	e2e	verify	number-heavy	schubfach	dragonbox	schubfach	1.10×	[0.966, 1.234]	0.1688	0.2522	-0
go:ryu-vs-dragonbox	e2e	verify	number-heavy	ryu	dragonbox	ryu	1.01×	[0.903, 1.134]	0.9060	0.9342	-0
go:ryu-vs-bd	e2e	verify	number-heavy	ryu	json-canon	json-canon	1.04×	[0.904, 1.174]	0.5345	0.6334	0
go:dragonbox-vs-bd	e2e	verify	number-heavy	dragonbox	json-canon	json-canon	1.05×	[0.918, 1.188]	0.4637	0.5709	0
rs:schubfach-vs-bd	e2e	verify	number-heavy	schubfach-rs	json-canon-rs	schubfach-rs*	1.23×	[1.143, 1.320]	0.0004	0.0008	-1
rs:schubfach-vs-ryu	e2e	verify	number-heavy	schubfach-rs	ryu-rs	ryu-rs	1.03×	[0.960, 1.121]	0.4081	0.5152	0
rs:schubfach-vs-dragonbox	e2e	verify	number-heavy	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.930, 1.117]	0.6861	0.7614	0
rs:ryu-vs-dragonbox	e2e	verify	number-heavy	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.924, 1.122]	0.7942	0.8473	-0
rs:ryu-vs-bd	e2e	verify	number-heavy	ryu-rs	json-canon-rs	ryu-rs*	1.27×	[1.182, 1.378]	0.0002	0.0004	-2
rs:dragonbox-vs-bd	e2e	verify	number-heavy	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.25×	[1.133, 1.373]	0.0004	0.0008	-1
schubfach:go-vs-rs	e2e	verify	number-heavy	schubfach	schubfach-rs	schubfach-rs*	2.52×	[2.263, 2.791]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	number-heavy	json-canon	json-canon-rs	json-canon-rs*	2.13×	[1.918, 2.403]	0.0002	0.0004	3
ryu:go-vs-rs	e2e	verify	number-heavy	ryu	ryu-rs	ryu-rs*	2.83×	[2.542, 3.116]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	verify	number-heavy	dragonbox	dragonbox-rs	dragonbox-rs*	2.81×	[2.490, 3.134]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	verify	numeric-boundary	schubfach	json-canon	schubfach	1.02×	[0.924, 1.127]	0.6593	0.7438	-0
go:schubfach-vs-ryu	e2e	verify	numeric-boundary	schubfach	ryu	ryu	1.06×	[0.955, 1.188]	0.2773	0.3774	0
go:schubfach-vs-dragonbox	e2e	verify	numeric-boundary	schubfach	dragonbox	dragonbox	1.07×	[0.941, 1.203]	0.2601	0.3599	0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:ryu-vs-dragonbox	e2e	verify	numeric-boundary	ryu	dragonbox	dragonbox	1.01×	[0.883, 1.145]	0.8910	0.9216	0
go:ryu-vs-bd	e2e	verify	numeric-boundary	ryu	json-canon	ryu	1.09×	[0.973, 1.216]	0.1640	0.2476	-0
go:dragonbox-vs-bd	e2e	verify	numeric-boundary	dragonbox	json-canon	dragonbox	1.10×	[0.960, 1.240]	0.1720	0.2552	-0
rs:schubfach-vs-bd	e2e	verify	numeric-boundary	schubfach-rs	json-canon-rs	schubfach-rs*	1.26×	[1.163, 1.352]	0.0002	0.0004	-1
rs:schubfach-vs-ryu	e2e	verify	numeric-boundary	schubfach-rs	ryu-rs	ryu-rs	1.01×	[0.936, 1.094]	0.7792	0.8375	0
rs:schubfach-vs-dragonbox	e2e	verify	numeric-boundary	schubfach-rs	dragonbox-rs	schubfach-rs	1.03×	[0.971, 1.102]	0.3593	0.4645	-0
rs:ryu-vs-dragonbox	e2e	verify	numeric-boundary	ryu-rs	dragonbox-rs	ryu-rs	1.04×	[0.971, 1.131]	0.3027	0.4034	-0
rs:ryu-vs-bd	e2e	verify	numeric-boundary	ryu-rs	json-canon-rs	ryu-rs*	1.27×	[1.164, 1.392]	0.0004	0.0008	-1
rs:dragonbox-vs-bd	e2e	verify	numeric-boundary	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.22×	[1.125, 1.308]	0.0002	0.0004	-1
schubfach:go-vs-rs	e2e	verify	numeric-boundary	schubfach	schubfach-rs	schubfach-rs*	2.87×	[2.653, 3.133]	0.0002	0.0004	6
bd:go-vs-rs	e2e	verify	numeric-boundary	json-canon	json-canon-rs	json-canon-rs*	2.34×	[2.116, 2.562]	0.0002	0.0004	5
ryu:go-vs-rs	e2e	verify	numeric-boundary	ryu	ryu-rs	ryu-rs*	2.73×	[2.454, 3.027]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	verify	numeric-boundary	dragonbox	dragonbox-rs	dragonbox-rs*	2.59×	[2.326, 2.915]	0.0002	0.0004	4
go:schubfach-vs-bd	e2e	verify	rfc-key-sorting	schubfach	json-canon	schubfach	1.09×	[0.948, 1.258]	0.2699	0.3708	-0
go:schubfach-vs-ryu	e2e	verify	rfc-key-sorting	schubfach	ryu	schubfach	1.03×	[0.889, 1.190]	0.6987	0.7694	-0
go:schubfach-vs-dragonbox	e2e	verify	rfc-key-sorting	schubfach	dragonbox	schubfach	1.07×	[0.931, 1.222]	0.3701	0.4751	-0
go:ryu-vs-dragonbox	e2e	verify	rfc-key-sorting	ryu	dragonbox	ryu	1.04×	[0.903, 1.190]	0.6169	0.7083	-0
go:ryu-vs-bd	e2e	verify	rfc-key-sorting	ryu	json-canon	ryu	1.06×	[0.912, 1.218]	0.4619	0.5698	-0
go:dragonbox-vs-bd	e2e	verify	rfc-key-sorting	dragonbox	json-canon	dragonbox	1.02×	[0.898, 1.161]	0.7852	0.8398	-0
rs:schubfach-vs-bd	e2e	verify	rfc-key-sorting	schubfach-rs	json-canon-rs	schubfach-rs	1.02×	[0.947, 1.100]	0.5989	0.6944	-0
rs:schubfach-vs-ryu	e2e	verify	rfc-key-sorting	schubfach-rs	ryu-rs	ryu-rs	1.08×	[0.987, 1.169]	0.0982	0.1576	0
rs:schubfach-vs-dragonbox	e2e	verify	rfc-key-sorting	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.953, 1.072]	0.4801	0.5822	0
rs:ryu-vs-dragonbox	e2e	verify	rfc-key-sorting	ryu-rs	dragonbox-rs	ryu-rs	1.05×	[0.978, 1.137]	0.2054	0.2956	-0
rs:ryu-vs-bd	e2e	verify	rfc-key-sorting	ryu-rs	json-canon-rs	ryu-rs	1.10×	[1.005, 1.204]	0.0494	0.0849	-0
rs:dragonbox-vs-bd	e2e	verify	rfc-key-sorting	dragonbox-rs	json-canon-rs	dragonbox-rs	1.04×	[0.975, 1.107]	0.2080	0.2987	-0
schubfach:go-vs-rs	e2e	verify	rfc-key-sorting	schubfach	schubfach-rs	schubfach-rs*	2.57×	[2.271, 2.890]	0.0002	0.0004	3
bd:go-vs-rs	e2e	verify	rfc-key-sorting	json-canon	json-canon-rs	json-canon-rs*	2.74×	[2.450, 3.064]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	rfc-key-sorting	ryu	ryu-rs	ryu-rs*	2.85×	[2.516, 3.219]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	verify	rfc-key-sorting	dragonbox	dragonbox-rs	dragonbox-rs*	2.81×	[2.540, 3.052]	0.0002	0.0004	5
go:schubfach-vs-bd	e2e	verify	small	schubfach	json-canon	schubfach*	1.22×	[1.069, 1.407]	0.0122	0.0231	-1
go:schubfach-vs-ryu	e2e	verify	small	schubfach	ryu	schubfach	1.02×	[0.896, 1.172]	0.7626	0.8258	-0
go:schubfach-vs-dragonbox	e2e	verify	small	schubfach	dragonbox	dragonbox	1.00×	[0.862, 1.154]	0.9988	0.9994	0
go:ryu-vs-dragonbox	e2e	verify	small	ryu	dragonbox	dragonbox	1.02×	[0.893, 1.158]	0.7674	0.8282	0
go:ryu-vs-bd	e2e	verify	small	ryu	json-canon	ryu*	1.20×	[1.062, 1.361]	0.0114	0.0217	-0
go:dragonbox-vs-bd	e2e	verify	small	dragonbox	json-canon	dragonbox*	1.22×	[1.063, 1.404]	0.0110	0.0210	-1
rs:schubfach-vs-bd	e2e	verify	small	schubfach-rs	json-canon-rs	schubfach-rs*	1.25×	[1.184, 1.331]	0.0002	0.0004	-2
rs:schubfach-vs-ryu	e2e	verify	small	schubfach-rs	ryu-rs	ryu-rs	1.01×	[0.953, 1.083]	0.8572	0.8967	0
rs:schubfach-vs-dragonbox	e2e	verify	small	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.975, 1.081]	0.4721	0.5766	-0
rs:ryu-vs-dragonbox	e2e	verify	small	ryu-rs	dragonbox-rs	ryu-rs	1.03×	[0.967, 1.109]	0.4487	0.5578	-0
rs:ryu-vs-bd	e2e	verify	small	ryu-rs	json-canon-rs	ryu-rs*	1.26×	[1.180, 1.366]	0.0002	0.0004	-2
rs:dragonbox-vs-bd	e2e	verify	small	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.22×	[1.151, 1.301]	0.0002	0.0004	-2
schubfach:go-vs-rs	e2e	verify	small	schubfach	schubfach-rs	schubfach-rs*	2.53×	[2.254, 2.797]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	small	json-canon	json-canon-rs	json-canon-rs*	2.48×	[2.224, 2.741]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	small	ryu	ryu-rs	ryu-rs*	2.60×	[2.371, 2.891]	0.0002	0.0004	5
dragonbox:go-vs-rs	e2e	verify	small	dragonbox	dragonbox-rs	dragonbox-rs*	2.48×	[2.217, 2.766]	0.0002	0.0004	3
go:schubfach-vs-bd	e2e	verify	surrogate-pair	schubfach	json-canon	schubfach	1.09×	[0.948, 1.240]	0.2515	0.3508	-0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:schubfach-vs-ryu	e2e	verify	surrogate-pair	schubfach	ryu	schubfach	1.05×	[0.945, 1.179]	0.3573	0.4624	-0
go:schubfach-vs-dragonbox	e2e	verify	surrogate-pair	schubfach	dragonbox	schubfach	1.06×	[0.960, 1.185]	0.2779	0.3778	-0
go:ryu-vs-dragonbox	e2e	verify	surrogate-pair	ryu	dragonbox	ryu	1.01×	[0.933, 1.099]	0.8854	0.9173	-0
go:ryu-vs-bd	e2e	verify	surrogate-pair	ryu	json-canon	ryu	1.03×	[0.919, 1.161]	0.6271	0.7156	-0
go:dragonbox-vs-bd	e2e	verify	surrogate-pair	dragonbox	json-canon	dragonbox	1.02×	[0.911, 1.145]	0.6883	0.7623	-0
rs:schubfach-vs-bd	e2e	verify	surrogate-pair	schubfach-rs	json-canon-rs	schubfach-rs*	1.09×	[1.023, 1.152]	0.0088	0.0169	-1
rs:schubfach-vs-ryu	e2e	verify	surrogate-pair	schubfach-rs	ryu-rs	ryu-rs	1.03×	[0.946, 1.124]	0.5363	0.6349	0
rs:schubfach-vs-dragonbox	e2e	verify	surrogate-pair	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.934, 1.089]	0.6805	0.7584	0
rs:ryu-vs-dragonbox	e2e	verify	surrogate-pair	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.911, 1.130]	0.8296	0.8772	-0
rs:ryu-vs-bd	e2e	verify	surrogate-pair	ryu-rs	json-canon-rs	ryu-rs*	1.12×	[1.021, 1.237]	0.0264	0.0481	-0
rs:dragonbox-vs-bd	e2e	verify	surrogate-pair	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.11×	[1.012, 1.197]	0.0194	0.0360	-0
schubfach:go-vs-rs	e2e	verify	surrogate-pair	schubfach	schubfach-rs	schubfach-rs*	2.67×	[2.398, 2.925]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	surrogate-pair	json-canon	json-canon-rs	json-canon-rs*	2.66×	[2.367, 2.947]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	surrogate-pair	ryu	ryu-rs	ryu-rs*	2.90×	[2.585, 3.188]	0.0002	0.0004	6
dragonbox:go-vs-rs	e2e	verify	surrogate-pair	dragonbox	dragonbox-rs	dragonbox-rs*	2.88×	[2.610, 3.115]	0.0002	0.0004	7
go:schubfach-vs-bd	e2e	verify	unicode	schubfach	json-canon	schubfach	1.06×	[0.958, 1.179]	0.3065	0.4079	-0
go:schubfach-vs-ryu	e2e	verify	unicode	schubfach	ryu	ryu	1.05×	[0.930, 1.169]	0.4685	0.5746	0
go:schubfach-vs-dragonbox	e2e	verify	unicode	schubfach	dragonbox	schubfach	1.26×	[1.055, 1.899]	0.0468	0.0811	-0
go:ryu-vs-dragonbox	e2e	verify	unicode	ryu	dragonbox	ryu*	1.32×	[1.101, 1.990]	0.0056	0.0109	-0
go:ryu-vs-bd	e2e	verify	unicode	ryu	json-canon	ryu	1.11×	[0.995, 1.243]	0.0928	0.1505	-0
go:dragonbox-vs-bd	e2e	verify	unicode	dragonbox	json-canon	json-canon	1.19×	[0.999, 1.760]	0.2737	0.3748	0
rs:schubfach-vs-bd	e2e	verify	unicode	schubfach-rs	json-canon-rs	schubfach-rs*	1.13×	[1.033, 1.229]	0.0132	0.0248	-0
rs:schubfach-vs-ryu	e2e	verify	unicode	schubfach-rs	ryu-rs	ryu-rs	1.02×	[0.964, 1.097]	0.4895	0.5925	0
rs:schubfach-vs-dragonbox	e2e	verify	unicode	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.960, 1.083]	0.4477	0.5571	-0
rs:ryu-vs-dragonbox	e2e	verify	unicode	ryu-rs	dragonbox-rs	ryu-rs	1.05×	[0.982, 1.115]	0.1670	0.2509	-0
rs:ryu-vs-bd	e2e	verify	unicode	ryu-rs	json-canon-rs	ryu-rs*	1.16×	[1.057, 1.260]	0.0040	0.0079	-1
rs:dragonbox-vs-bd	e2e	verify	unicode	dragonbox-rs	json-canon-rs	dragonbox-rs	1.10×	[1.013, 1.196]	0.0380	0.0673	-0
schubfach:go-vs-rs	e2e	verify	unicode	schubfach	schubfach-rs	schubfach-rs*	2.70×	[2.459, 2.937]	0.0002	0.0004	5
bd:go-vs-rs	e2e	verify	unicode	json-canon	json-canon-rs	json-canon-rs*	2.53×	[2.295, 2.805]	0.0002	0.0004	5
ryu:go-vs-rs	e2e	verify	unicode	ryu	ryu-rs	ryu-rs*	2.65×	[2.400, 2.915]	0.0002	0.0004	4
dragonbox:go-vs-rs	e2e	verify	unicode	dragonbox	dragonbox-rs	dragonbox-rs*	3.33×	[2.818, 5.147]	0.0002	0.0004	1
go:schubfach-vs-bd	e2e	verify	verify-whitespace	schubfach	json-canon	schubfach	1.02×	[0.891, 1.180]	0.7598	0.8242	-0
go:schubfach-vs-ryu	e2e	verify	verify-whitespace	schubfach	ryu	ryu	1.15×	[0.988, 1.314]	0.0918	0.1494	0
go:schubfach-vs-dragonbox	e2e	verify	verify-whitespace	schubfach	dragonbox	dragonbox	1.09×	[0.941, 1.246]	0.2831	0.3825	0
go:ryu-vs-dragonbox	e2e	verify	verify-whitespace	ryu	dragonbox	ryu	1.06×	[0.919, 1.199]	0.4279	0.5360	-0
go:ryu-vs-bd	e2e	verify	verify-whitespace	ryu	json-canon	ryu	1.17×	[1.028, 1.340]	0.0370	0.0656	-0
go:dragonbox-vs-bd	e2e	verify	verify-whitespace	dragonbox	json-canon	dragonbox	1.11×	[0.986, 1.261]	0.1340	0.2086	-0
rs:schubfach-vs-bd	e2e	verify	verify-whitespace	schubfach-rs	json-canon-rs	schubfach-rs*	1.26×	[1.161, 1.361]	0.0004	0.0008	-1
rs:schubfach-vs-ryu	e2e	verify	verify-whitespace	schubfach-rs	ryu-rs	schubfach-rs	1.00×	[0.910, 1.090]	0.9404	0.9601	-0
rs:schubfach-vs-dragonbox	e2e	verify	verify-whitespace	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.940, 1.107]	0.6777	0.7559	-0
rs:ryu-vs-dragonbox	e2e	verify	verify-whitespace	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.932, 1.121]	0.7680	0.8282	-0
rs:ryu-vs-bd	e2e	verify	verify-whitespace	ryu-rs	json-canon-rs	ryu-rs*	1.25×	[1.146, 1.379]	0.0002	0.0004	-1
rs:dragonbox-vs-bd	e2e	verify	verify-whitespace	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.23×	[1.137, 1.341]	0.0002	0.0004	-1
schubfach:go-vs-rs	e2e	verify	verify-whitespace	schubfach	schubfach-rs	schubfach-rs*	2.97×	[2.612, 3.340]	0.0002	0.0004	4
bd:go-vs-rs	e2e	verify	verify-whitespace	json-canon	json-canon-rs	json-canon-rs*	2.42×	[2.172, 2.692]	0.0002	0.0004	4
ryu:go-vs-rs	e2e	verify	verify-whitespace	ryu	ryu-rs	ryu-rs*	2.58×	[2.287, 2.927]	0.0002	0.0004	4

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
dragonbox:go-vs-rs	e2e	verify	verify-whitespace	dragonbox	dragonbox-rs	dragonbox-rs*	2.69×	[2.382, 2.968]	0.0002	0.0004	4
go:schubfach-vs-bd	api	canonicalize	array-2048	schubfach	json-canon	schubfach*	1.64×	[1.631, 1.648]	0.0002	0.0004	-81
go:schubfach-vs-ryu	api	canonicalize	array-2048	schubfach	ryu	ryu*	1.01×	[1.007, 1.018]	0.0006	0.0013	1
go:schubfach-vs-dragonbox	api	canonicalize	array-2048	schubfach	dragonbox	dragonbox	1.00×	[0.996, 1.007]	0.8756	0.9101	0
go:ryu-vs-dragonbox	api	canonicalize	array-2048	ryu	dragonbox	ryu*	1.01×	[1.006, 1.016]	0.0022	0.0045	-1
go:ryu-vs-bd	api	canonicalize	array-2048	ryu	json-canon	ryu*	1.66×	[1.652, 1.668]	0.0002	0.0004	-87
go:dragonbox-vs-bd	api	canonicalize	array-2048	dragonbox	json-canon	dragonbox*	1.64×	[1.635, 1.650]	0.0002	0.0004	-85
rs:schubfach-vs-bd	api	canonicalize	array-2048	schubfach-rs	json-canon-rs	schubfach-rs*	2.27×	[1.910, 2.545]	0.0002	0.0004	-7
rs:schubfach-vs-ryu	api	canonicalize	array-2048	schubfach-rs	ryu-rs	ryu-rs	1.08×	[0.958, 1.310]	0.3667	0.4712	0
rs:schubfach-vs-dragonbox	api	canonicalize	array-2048	schubfach-rs	dragonbox-rs	dragonbox-rs*	1.13×	[1.008, 1.340]	0.0040	0.0079	0
rs:ryu-vs-dragonbox	api	canonicalize	array-2048	ryu-rs	dragonbox-rs	dragonbox-rs*	1.05×	[1.006, 1.174]	0.0060	0.0117	0
rs:ryu-vs-bd	api	canonicalize	array-2048	ryu-rs	json-canon-rs	ryu-rs*	2.44×	[2.117, 2.548]	0.0002	0.0004	-16
rs:dragonbox-vs-bd	api	canonicalize	array-2048	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.56×	[2.551, 2.576]	0.0002	0.0004	-130
schubfach:go-vs-rs	api	canonicalize	array-2048	schubfach	schubfach-rs	schubfach-rs*	1.67×	[1.418, 1.880]	0.0002	0.0004	4
bd:go-vs-rs	api	canonicalize	array-2048	json-canon	json-canon-rs	json-canon-rs*	1.21×	[1.206, 1.218]	0.0002	0.0004	33
ryu:go-vs-rs	api	canonicalize	array-2048	ryu	ryu-rs	ryu-rs*	1.78×	[1.546, 1.860]	0.0002	0.0004	9
dragonbox:go-vs-rs	api	canonicalize	array-2048	dragonbox	dragonbox-rs	dragonbox-rs*	1.89×	[1.883, 1.901]	0.0002	0.0004	98
go:schubfach-vs-bd	api	canonicalize	array-256	schubfach	json-canon	schubfach*	1.57×	[1.558, 1.581]	0.0002	0.0004	-54
go:schubfach-vs-ryu	api	canonicalize	array-256	schubfach	ryu	ryu*	1.01×	[1.002, 1.016]	0.0268	0.0488	1
go:schubfach-vs-dragonbox	api	canonicalize	array-256	schubfach	dragonbox	dragonbox*	1.01×	[1.003, 1.017]	0.0142	0.0267	1
go:ryu-vs-dragonbox	api	canonicalize	array-256	ryu	dragonbox	dragonbox	1.00×	[0.996, 1.004]	0.8696	0.9075	0
go:ryu-vs-bd	api	canonicalize	array-256	ryu	json-canon	ryu*	1.58×	[1.576, 1.592]	0.0002	0.0004	-69
go:dragonbox-vs-bd	api	canonicalize	array-256	dragonbox	json-canon	dragonbox*	1.58×	[1.577, 1.592]	0.0002	0.0004	-72
rs:schubfach-vs-bd	api	canonicalize	array-256	schubfach-rs	json-canon-rs	schubfach-rs*	2.30×	[2.252, 2.353]	0.0002	0.0004	-22
rs:schubfach-vs-ryu	api	canonicalize	array-256	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[1.001, 1.083]	0.2306	0.3268	-0
rs:schubfach-vs-dragonbox	api	canonicalize	array-256	schubfach-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.994, 1.014]	0.8124	0.8631	0
rs:ryu-vs-dragonbox	api	canonicalize	array-256	ryu-rs	dragonbox-rs	dragonbox-rs	1.02×	[1.004, 1.078]	0.0394	0.0694	0
rs:ryu-vs-bd	api	canonicalize	array-256	ryu-rs	json-canon-rs	ryu-rs*	2.25×	[2.136, 2.314]	0.0002	0.0004	-18
rs:dragonbox-vs-bd	api	canonicalize	array-256	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.30×	[2.260, 2.356]	0.0002	0.0004	-22
schubfach:go-vs-rs	api	canonicalize	array-256	schubfach	schubfach-rs	schubfach-rs*	1.81×	[1.789, 1.829]	0.0002	0.0004	47
bd:go-vs-rs	api	canonicalize	array-256	json-canon	json-canon-rs	json-canon-rs*	1.24×	[1.209, 1.261]	0.0002	0.0004	9
ryu:go-vs-rs	api	canonicalize	array-256	ryu	ryu-rs	ryu-rs*	1.76×	[1.646, 1.792]	0.0002	0.0004	19
dragonbox:go-vs-rs	api	canonicalize	array-256	dragonbox	dragonbox-rs	dragonbox-rs*	1.80×	[1.789, 1.804]	0.0002	0.0004	121
go:schubfach-vs-bd	api	canonicalize	canonical-minimal	schubfach	json-canon	schubfach*	1.49×	[1.482, 1.510]	0.0002	0.0004	-38
go:schubfach-vs-ryu	api	canonicalize	canonical-minimal	schubfach	ryu	ryu*	1.03×	[1.018, 1.038]	0.0002	0.0004	2
go:schubfach-vs-dragonbox	api	canonicalize	canonical-minimal	schubfach	dragonbox	dragonbox*	1.02×	[1.014, 1.032]	0.0006	0.0013	2
go:ryu-vs-dragonbox	api	canonicalize	canonical-minimal	ryu	dragonbox	ryu	1.00×	[0.999, 1.011]	0.1680	0.2516	-0
go:ryu-vs-bd	api	canonicalize	canonical-minimal	ryu	json-canon	ryu*	1.54×	[1.527, 1.549]	0.0002	0.0004	-51
go:dragonbox-vs-bd	api	canonicalize	canonical-minimal	dragonbox	json-canon	dragonbox*	1.53×	[1.522, 1.539]	0.0002	0.0004	-56
rs:schubfach-vs-bd	api	canonicalize	canonical-minimal	schubfach-rs	json-canon-rs	schubfach-rs*	2.10×	[2.023, 2.191]	0.0002	0.0004	-14
rs:schubfach-vs-ryu	api	canonicalize	canonical-minimal	schubfach-rs	ryu-rs	ryu-rs	1.04×	[0.989, 1.079]	0.1028	0.1646	0
rs:schubfach-vs-dragonbox	api	canonicalize	canonical-minimal	schubfach-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.942, 1.061]	0.6945	0.7670	0
rs:ryu-vs-dragonbox	api	canonicalize	canonical-minimal	ryu-rs	dragonbox-rs	ryu-rs	1.03×	[0.978, 1.102]	0.4983	0.5998	-0
rs:ryu-vs-bd	api	canonicalize	canonical-minimal	ryu-rs	json-canon-rs	ryu-rs*	2.19×	[2.085, 2.259]	0.0002	0.0004	-15
rs:dragonbox-vs-bd	api	canonicalize	canonical-minimal	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.13×	[1.981, 2.226]	0.0002	0.0004	-12
schubfach:go-vs-rs	api	canonicalize	canonical-minimal	schubfach	schubfach-rs	schubfach-rs*	1.68×	[1.624, 1.729]	0.0002	0.0004	16

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
bd:go-vs-rs	api	canonicalize	canonical-minimal	json-canon	json-canon-rs	json-canon-rs*	1.19×	[1.158, 1.220]	0.0002	0.0004	6
ryu:go-vs-rs	api	canonicalize	canonical-minimal	ryu	ryu-rs	ryu-rs*	1.70×	[1.612, 1.730]	0.0002	0.0004	18
dragonbox:go-vs-rs	api	canonicalize	canonical-minimal	dragonbox	dragonbox-rs	dragonbox-rs*	1.66×	[1.541, 1.722]	0.0002	0.0004	10
go:schubfach-vs-bd	api	canonicalize	control-escapes	schubfach	json-canon	schubfach*	1.01×	[1.009, 1.019]	0.0004	0.0008	-2
go:schubfach-vs-ryu	api	canonicalize	control-escapes	schubfach	ryu	ryu	1.01×	[1.001, 1.016]	0.0504	0.0865	0
go:schubfach-vs-dragonbox	api	canonicalize	control-escapes	schubfach	dragonbox	dragonbox	1.00×	[0.996, 1.012]	0.4271	0.5355	0
go:ryu-vs-dragonbox	api	canonicalize	control-escapes	ryu	dragonbox	ryu	1.00×	[0.996, 1.013]	0.2959	0.3960	-0
go:ryu-vs-bd	api	canonicalize	control-escapes	ryu	json-canon	ryu*	1.02×	[1.016, 1.029]	0.0004	0.0008	-2
go:dragonbox-vs-bd	api	canonicalize	control-escapes	dragonbox	json-canon	dragonbox*	1.02×	[1.011, 1.026]	0.0002	0.0004	-2
rs:schubfach-vs-bd	api	canonicalize	control-escapes	schubfach-rs	json-canon-rs	json-canon-rs	1.02×	[0.981, 1.052]	0.3429	0.4464	0
rs:schubfach-vs-ryu	api	canonicalize	control-escapes	schubfach-rs	ryu-rs	schubfach-rs	1.03×	[0.989, 1.097]	0.2747	0.3750	-0
rs:schubfach-vs-dragonbox	api	canonicalize	control-escapes	schubfach-rs	dragonbox-rs	dragonbox-rs	1.04×	[0.995, 1.071]	0.0524	0.0898	0
rs:ryu-vs-dragonbox	api	canonicalize	control-escapes	ryu-rs	dragonbox-rs	dragonbox-rs	1.08×	[1.017, 1.147]	0.0356	0.0635	1
rs:ryu-vs-bd	api	canonicalize	control-escapes	ryu-rs	json-canon-rs	json-canon-rs	1.05×	[0.998, 1.126]	0.1336	0.2082	0
rs:dragonbox-vs-bd	api	canonicalize	control-escapes	dragonbox-rs	json-canon-rs	dragonbox-rs	1.02×	[0.974, 1.066]	0.3929	0.5004	-0
schubfach:go-vs-rs	api	canonicalize	control-escapes	schubfach	schubfach-rs	schubfach-rs*	2.00×	[1.971, 2.028]	0.0002	0.0004	49
bd:go-vs-rs	api	canonicalize	control-escapes	json-canon	json-canon-rs	json-canon-rs*	2.06×	[1.996, 2.126]	0.0002	0.0004	27
ryu:go-vs-rs	api	canonicalize	control-escapes	ryu	ryu-rs	ryu-rs*	1.92×	[1.799, 1.998]	0.0002	0.0004	14
dragonbox:go-vs-rs	api	canonicalize	control-escapes	dragonbox	dragonbox-rs	dragonbox-rs*	2.07×	[1.985, 2.127]	0.0002	0.0004	25
go:schubfach-vs-bd	api	canonicalize	deep-64	schubfach	json-canon	json-canon	1.00×	[0.999, 1.007]	0.1824	0.2669	0
go:schubfach-vs-ryu	api	canonicalize	deep-64	schubfach	ryu	schubfach*	1.01×	[1.008, 1.016]	0.0002	0.0004	-2
go:schubfach-vs-dragonbox	api	canonicalize	deep-64	schubfach	dragonbox	schubfach*	1.02×	[1.013, 1.025]	0.0004	0.0008	-2
go:ryu-vs-dragonbox	api	canonicalize	deep-64	ryu	dragonbox	ryu	1.01×	[1.000, 1.012]	0.0646	0.1097	-0
go:ryu-vs-bd	api	canonicalize	deep-64	ryu	json-canon	json-canon*	1.02×	[1.011, 1.020]	0.0002	0.0004	3
go:dragonbox-vs-bd	api	canonicalize	deep-64	dragonbox	json-canon	json-canon*	1.02×	[1.016, 1.028]	0.0002	0.0004	3
rs:schubfach-vs-bd	api	canonicalize	deep-64	schubfach-rs	json-canon-rs	json-canon-rs	1.05×	[1.004, 1.154]	0.1570	0.2387	0
rs:schubfach-vs-ryu	api	canonicalize	deep-64	schubfach-rs	ryu-rs	ryu-rs*	1.06×	[1.021, 1.186]	0.0002	0.0004	0
rs:schubfach-vs-dragonbox	api	canonicalize	deep-64	schubfach-rs	dragonbox-rs	dragonbox-rs	1.04×	[1.003, 1.139]	0.0918	0.1494	0
rs:ryu-vs-dragonbox	api	canonicalize	deep-64	ryu-rs	dragonbox-rs	ryu-rs*	1.02×	[1.011, 1.023]	0.0002	0.0004	-2
rs:ryu-vs-bd	api	canonicalize	deep-64	ryu-rs	json-canon-rs	ryu-rs	1.01×	[0.994, 1.036]	0.3901	0.4988	-0
rs:dragonbox-vs-bd	api	canonicalize	deep-64	dragonbox-rs	json-canon-rs	json-canon-rs	1.01×	[0.981, 1.024]	0.5913	0.6877	0
schubfach:go-vs-rs	api	canonicalize	deep-64	schubfach	schubfach-rs	schubfach-rs*	1.39×	[1.241, 1.439]	0.0002	0.0004	5
bd:go-vs-rs	api	canonicalize	deep-64	json-canon	json-canon-rs	json-canon-rs*	1.45×	[1.415, 1.474]	0.0002	0.0004	18
ryu:go-vs-rs	api	canonicalize	deep-64	ryu	ryu-rs	ryu-rs*	1.49×	[1.481, 1.493]	0.0002	0.0004	76
dragonbox:go-vs-rs	api	canonicalize	deep-64	dragonbox	dragonbox-rs	dragonbox-rs*	1.47×	[1.461, 1.483]	0.0002	0.0004	43
go:schubfach-vs-bd	api	canonicalize	deep	schubfach	json-canon	json-canon	1.00×	[0.996, 1.008]	0.6075	0.6994	0
go:schubfach-vs-ryu	api	canonicalize	deep	schubfach	ryu	schubfach*	1.02×	[1.011, 1.025]	0.0004	0.0008	-2
go:schubfach-vs-dragonbox	api	canonicalize	deep	schubfach	dragonbox	schubfach*	1.02×	[1.014, 1.028]	0.0004	0.0008	-2
go:ryu-vs-dragonbox	api	canonicalize	deep	ryu	dragonbox	ryu	1.00×	[0.998, 1.009]	0.3521	0.4566	-0
go:ryu-vs-bd	api	canonicalize	deep	ryu	json-canon	json-canon*	1.02×	[1.015, 1.025]	0.0002	0.0004	3
go:dragonbox-vs-bd	api	canonicalize	deep	dragonbox	json-canon	json-canon*	1.02×	[1.019, 1.029]	0.0002	0.0004	3
rs:schubfach-vs-bd	api	canonicalize	deep	schubfach-rs	json-canon-rs	schubfach-rs*	1.04×	[1.026, 1.054]	0.0004	0.0008	-2
rs:schubfach-vs-ryu	api	canonicalize	deep	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.993, 1.059]	0.5063	0.6077	-0
rs:schubfach-vs-dragonbox	api	canonicalize	deep	schubfach-rs	dragonbox-rs	schubfach-rs*	1.06×	[1.044, 1.069]	0.0002	0.0004	-3
rs:ryu-vs-dragonbox	api	canonicalize	deep	ryu-rs	dragonbox-rs	ryu-rs*	1.04×	[0.996, 1.063]	0.0130	0.0245	-1
rs:ryu-vs-bd	api	canonicalize	deep	ryu-rs	json-canon-rs	ryu-rs	1.03×	[0.979, 1.048]	0.1194	0.1888	-0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:dragonbox-vs-bd	api	canonicalize	deep	dragonbox-rs	json-canon-rs	json-canon-rs	1.02×	[1.003, 1.028]	0.0316	0.0571	1
schubfach:go-vs-rs	api	canonicalize	deep	schubfach	schubfach-rs	schubfach-rs*	1.27×	[1.252, 1.281]	0.0002	0.0004	18
bd:go-vs-rs	api	canonicalize	deep	json-canon	json-canon-rs	json-canon-rs*	1.22×	[1.203, 1.228]	0.0002	0.0004	16
ryu:go-vs-rs	api	canonicalize	deep	ryu	ryu-rs	ryu-rs*	1.27×	[1.215, 1.298]	0.0002	0.0004	7
dragonbox:go-vs-rs	api	canonicalize	deep	dragonbox	dragonbox-rs	dragonbox-rs*	1.22×	[1.215, 1.235]	0.0002	0.0004	21
go:schubfach-vs-bd	api	canonicalize	escaped-key-order	schubfach	json-canon	schubfach*	1.59×	[1.571, 1.609]	0.0002	0.0004	-33
go:schubfach-vs-ryu	api	canonicalize	escaped-key-order	schubfach	ryu	ryu	1.01×	[0.998, 1.019]	0.2178	0.3096	0
go:schubfach-vs-dragonbox	api	canonicalize	escaped-key-order	schubfach	dragonbox	schubfach	1.00×	[0.990, 1.033]	0.7473	0.8126	-0
go:ryu-vs-dragonbox	api	canonicalize	escaped-key-order	ryu	dragonbox	ryu	1.01×	[1.000, 1.040]	0.2006	0.2893	-0
go:ryu-vs-bd	api	canonicalize	escaped-key-order	ryu	json-canon	ryu*	1.60×	[1.591, 1.617]	0.0002	0.0004	-41
go:dragonbox-vs-bd	api	canonicalize	escaped-key-order	dragonbox	json-canon	dragonbox*	1.58×	[1.541, 1.605]	0.0002	0.0004	-24
rs:schubfach-vs-bd	api	canonicalize	escaped-key-order	schubfach-rs	json-canon-rs	schubfach-rs*	2.55×	[2.505, 2.601]	0.0002	0.0004	-38
rs:schubfach-vs-ryu	api	canonicalize	escaped-key-order	schubfach-rs	ryu-rs	schubfach-rs*	1.03×	[1.008, 1.055]	0.0248	0.0455	-1
rs:schubfach-vs-dragonbox	api	canonicalize	escaped-key-order	schubfach-rs	dragonbox-rs	schubfach-rs*	1.04×	[1.013, 1.064]	0.0118	0.0224	-1
rs:ryu-vs-dragonbox	api	canonicalize	escaped-key-order	ryu-rs	dragonbox-rs	ryu-rs	1.00×	[0.979, 1.033]	0.7680	0.8282	-0
rs:ryu-vs-bd	api	canonicalize	escaped-key-order	ryu-rs	json-canon-rs	ryu-rs*	2.47×	[2.421, 2.527]	0.0002	0.0004	-35
rs:dragonbox-vs-bd	api	canonicalize	escaped-key-order	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.46×	[2.401, 2.516]	0.0002	0.0004	-34
schubfach:go-vs-rs	api	canonicalize	escaped-key-order	schubfach	schubfach-rs	schubfach-rs*	2.20×	[2.162, 2.234]	0.0002	0.0004	40
bd:go-vs-rs	api	canonicalize	escaped-key-order	json-canon	json-canon-rs	json-canon-rs*	1.37×	[1.351, 1.390]	0.0002	0.0004	19
ryu:go-vs-rs	api	canonicalize	escaped-key-order	ryu	ryu-rs	ryu-rs*	2.11×	[2.076, 2.154]	0.0002	0.0004	46
dragonbox:go-vs-rs	api	canonicalize	escaped-key-order	dragonbox	dragonbox-rs	dragonbox-rs*	2.13×	[2.080, 2.189]	0.0002	0.0004	23
go:schubfach-vs-bd	api	canonicalize	large	schubfach	json-canon	schubfach*	1.77×	[1.761, 1.774]	0.0002	0.0004	-130
go:schubfach-vs-ryu	api	canonicalize	large	schubfach	ryu	ryu	1.01×	[1.000, 1.009]	0.0404	0.0709	1
go:schubfach-vs-dragonbox	api	canonicalize	large	schubfach	dragonbox	dragonbox	1.01×	[1.000, 1.011]	0.0954	0.1541	0
go:ryu-vs-dragonbox	api	canonicalize	large	ryu	dragonbox	ryu	1.00×	[0.993, 1.006]	0.9638	0.9751	-0
go:ryu-vs-bd	api	canonicalize	large	ryu	json-canon	ryu*	1.78×	[1.767, 1.783]	0.0002	0.0004	-122
go:dragonbox-vs-bd	api	canonicalize	large	dragonbox	json-canon	dragonbox*	1.78×	[1.767, 1.786]	0.0002	0.0004	-101
rs:schubfach-vs-bd	api	canonicalize	large	schubfach-rs	json-canon-rs	schubfach-rs*	3.28×	[3.181, 3.344]	0.0002	0.0004	-48
rs:schubfach-vs-ryu	api	canonicalize	large	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.974, 1.048]	0.6511	0.7394	-0
rs:schubfach-vs-dragonbox	api	canonicalize	large	schubfach-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.985, 1.042]	0.3257	0.4271	0
rs:ryu-vs-dragonbox	api	canonicalize	large	ryu-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.990, 1.062]	0.2484	0.3485	0
rs:ryu-vs-bd	api	canonicalize	large	ryu-rs	json-canon-rs	ryu-rs*	3.26×	[3.135, 3.338]	0.0002	0.0004	-43
rs:dragonbox-vs-bd	api	canonicalize	large	dragonbox-rs	json-canon-rs	dragonbox-rs*	3.33×	[3.228, 3.373]	0.0002	0.0004	-52
schubfach:go-vs-rs	api	canonicalize	large	schubfach	schubfach-rs	schubfach-rs*	2.04×	[1.979, 2.074]	0.0002	0.0004	37
bd:go-vs-rs	api	canonicalize	large	json-canon	json-canon-rs	json-canon-rs*	1.10×	[1.087, 1.108]	0.0002	0.0004	8
ryu:go-vs-rs	api	canonicalize	large	ryu	ryu-rs	ryu-rs*	2.01×	[1.937, 2.059]	0.0002	0.0004	27
dragonbox:go-vs-rs	api	canonicalize	large	dragonbox	dragonbox-rs	dragonbox-rs*	2.05×	[1.993, 2.079]	0.0002	0.0004	43
go:schubfach-vs-bd	api	canonicalize	long-string	schubfach	json-canon	schubfach	1.01×	[0.996, 1.021]	0.2975	0.3977	-0
go:schubfach-vs-ryu	api	canonicalize	long-string	schubfach	ryu	ryu	1.00×	[0.987, 1.014]	0.9742	0.9795	0
go:schubfach-vs-dragonbox	api	canonicalize	long-string	schubfach	dragonbox	schubfach	1.00×	[0.992, 1.015]	0.6451	0.7335	-0
go:ryu-vs-dragonbox	api	canonicalize	long-string	ryu	dragonbox	ryu	1.00×	[0.991, 1.017]	0.6589	0.7438	-0
go:ryu-vs-bd	api	canonicalize	long-string	ryu	json-canon	ryu	1.01×	[0.995, 1.022]	0.3197	0.4214	-0
go:dragonbox-vs-bd	api	canonicalize	long-string	dragonbox	json-canon	dragonbox	1.00×	[0.993, 1.016]	0.5297	0.6294	-0
rs:schubfach-vs-bd	api	canonicalize	long-string	schubfach-rs	json-canon-rs	schubfach-rs*	1.05×	[1.039, 1.051]	0.0002	0.0004	-6
rs:schubfach-vs-ryu	api	canonicalize	long-string	schubfach-rs	ryu-rs	schubfach-rs*	1.02×	[1.020, 1.022]	0.0002	0.0004	-13
rs:schubfach-vs-dragonbox	api	canonicalize	long-string	schubfach-rs	dragonbox-rs	schubfach-rs*	1.02×	[1.023, 1.026]	0.0002	0.0004	-15

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:ryu-vs-dragonbox	api	canonicalize	long-string	ryu-rs	dragonbox-rs	ryu-rs*	1.00×	[1.002, 1.005]	0.0004	0.0008	-2
rs:ryu-vs-bd	api	canonicalize	long-string	ryu-rs	json-canon-rs	ryu-rs*	1.03×	[1.018, 1.030]	0.0002	0.0004	-3
rs:dragonbox-vs-bd	api	canonicalize	long-string	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.02×	[1.014, 1.026]	0.0002	0.0004	-3
schubfach:go-vs-rs	api	canonicalize	long-string	schubfach	schubfach-rs	schubfach-rs*	4.50×	[4.456, 4.536]	0.0002	0.0004	74
bd:go-vs-rs	api	canonicalize	long-string	json-canon	json-canon-rs	json-canon-rs*	4.33×	[4.293, 4.378]	0.0002	0.0004	77
ryu:go-vs-rs	api	canonicalize	long-string	ryu	ryu-rs	ryu-rs*	4.41×	[4.356, 4.448]	0.0002	0.0004	62
dragonbox:go-vs-rs	api	canonicalize	long-string	dragonbox	dragonbox-rs	dragonbox-rs*	4.41×	[4.372, 4.445]	0.0002	0.0004	78
go:schubfach-vs-bd	api	canonicalize	medium	schubfach	json-canon	schubfach*	1.75×	[1.743, 1.766]	0.0002	0.0004	-75
go:schubfach-vs-ryu	api	canonicalize	medium	schubfach	ryu	ryu	1.01×	[1.000, 1.013]	0.0642	0.1093	0
go:schubfach-vs-dragonbox	api	canonicalize	medium	schubfach	dragonbox	dragonbox*	1.01×	[1.003, 1.016]	0.0148	0.0277	1
go:ryu-vs-dragonbox	api	canonicalize	medium	ryu	dragonbox	dragonbox	1.00×	[0.999, 1.006]	0.1766	0.2602	0
go:ryu-vs-bd	api	canonicalize	medium	ryu	json-canon	ryu*	1.77×	[1.759, 1.773]	0.0002	0.0004	-101
go:dragonbox-vs-bd	api	canonicalize	medium	dragonbox	json-canon	dragonbox*	1.77×	[1.763, 1.779]	0.0002	0.0004	-95
rs:schubfach-vs-bd	api	canonicalize	medium	schubfach-rs	json-canon-rs	schubfach-rs*	2.91×	[2.484, 3.068]	0.0002	0.0004	-17
rs:schubfach-vs-ryu	api	canonicalize	medium	schubfach-rs	ryu-rs	ryu-rs	1.03×	[0.974, 1.176]	0.9924	0.9949	0
rs:schubfach-vs-dragonbox	api	canonicalize	medium	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.916, 1.178]	0.5131	0.6142	-0
rs:ryu-vs-dragonbox	api	canonicalize	medium	ryu-rs	dragonbox-rs	ryu-rs	1.04×	[0.986, 1.216]	0.9884	0.9922	-0
rs:ryu-vs-bd	api	canonicalize	medium	ryu-rs	json-canon-rs	ryu-rs*	2.99×	[2.943, 3.020]	0.0002	0.0004	-57
rs:dragonbox-vs-bd	api	canonicalize	medium	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.86×	[2.447, 3.032]	0.0002	0.0004	-15
schubfach:go-vs-rs	api	canonicalize	medium	schubfach	schubfach-rs	schubfach-rs*	1.86×	[1.577, 1.959]	0.0002	0.0004	8
bd:go-vs-rs	api	canonicalize	medium	json-canon	json-canon-rs	json-canon-rs*	1.12×	[1.110, 1.132]	0.0002	0.0004	10
ryu:go-vs-rs	api	canonicalize	medium	ryu	ryu-rs	ryu-rs*	1.90×	[1.867, 1.908]	0.0002	0.0004	76
dragonbox:go-vs-rs	api	canonicalize	medium	dragonbox	dragonbox-rs	dragonbox-rs*	1.81×	[1.497, 1.915]	0.0002	0.0004	6
go:schubfach-vs-bd	api	canonicalize	mixed-prod	schubfach	json-canon	schubfach*	1.36×	[1.351, 1.370]	0.0002	0.0004	-39
go:schubfach-vs-ryu	api	canonicalize	mixed-prod	schubfach	ryu	schubfach	1.00×	[0.997, 1.011]	0.2725	0.3739	-0
go:schubfach-vs-dragonbox	api	canonicalize	mixed-prod	schubfach	dragonbox	schubfach	1.00×	[0.995, 1.008]	0.5385	0.6369	-0
go:ryu-vs-dragonbox	api	canonicalize	mixed-prod	ryu	dragonbox	dragonbox	1.00×	[0.996, 1.009]	0.5061	0.6077	0
go:ryu-vs-bd	api	canonicalize	mixed-prod	ryu	json-canon	ryu*	1.35×	[1.347, 1.364]	0.0002	0.0004	-42
go:dragonbox-vs-bd	api	canonicalize	mixed-prod	dragonbox	json-canon	dragonbox*	1.36×	[1.351, 1.366]	0.0002	0.0004	-46
rs:schubfach-vs-bd	api	canonicalize	mixed-prod	schubfach-rs	json-canon-rs	schubfach-rs*	1.78×	[1.738, 1.802]	0.0002	0.0004	-33
rs:schubfach-vs-ryu	api	canonicalize	mixed-prod	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.997, 1.041]	0.1238	0.1937	-0
rs:schubfach-vs-dragonbox	api	canonicalize	mixed-prod	schubfach-rs	dragonbox-rs	dragonbox-rs*	1.03×	[1.013, 1.051]	0.0048	0.0094	1
rs:ryu-vs-dragonbox	api	canonicalize	mixed-prod	ryu-rs	dragonbox-rs	dragonbox-rs*	1.05×	[1.032, 1.068]	0.0002	0.0004	2
rs:ryu-vs-bd	api	canonicalize	mixed-prod	ryu-rs	json-canon-rs	ryu-rs*	1.75×	[1.710, 1.768]	0.0002	0.0004	-32
rs:dragonbox-vs-bd	api	canonicalize	mixed-prod	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.83×	[1.807, 1.845]	0.0002	0.0004	-47
schubfach:go-vs-rs	api	canonicalize	mixed-prod	schubfach	schubfach-rs	schubfach-rs*	1.94×	[1.900, 1.970]	0.0002	0.0004	41
bd:go-vs-rs	api	canonicalize	mixed-prod	json-canon	json-canon-rs	json-canon-rs*	1.49×	[1.476, 1.498]	0.0002	0.0004	47
ryu:go-vs-rs	api	canonicalize	mixed-prod	ryu	ryu-rs	ryu-rs*	1.92×	[1.877, 1.941]	0.0002	0.0004	42
dragonbox:go-vs-rs	api	canonicalize	mixed-prod	dragonbox	dragonbox-rs	dragonbox-rs*	2.00×	[1.980, 2.018]	0.0002	0.0004	73
go:schubfach-vs-bd	api	canonicalize	nested-mixed	schubfach	json-canon	schubfach*	1.42×	[1.408, 1.424]	0.0002	0.0004	-48
go:schubfach-vs-ryu	api	canonicalize	nested-mixed	schubfach	ryu	schubfach	1.01×	[1.002, 1.013]	0.0328	0.0589	-1
go:schubfach-vs-dragonbox	api	canonicalize	nested-mixed	schubfach	dragonbox	schubfach	1.00×	[0.996, 1.007]	0.5915	0.6877	-0
go:ryu-vs-dragonbox	api	canonicalize	nested-mixed	ryu	dragonbox	dragonbox	1.01×	[1.000, 1.012]	0.1224	0.1922	0
go:ryu-vs-bd	api	canonicalize	nested-mixed	ryu	json-canon	ryu*	1.41×	[1.398, 1.414]	0.0002	0.0004	-46
go:dragonbox-vs-bd	api	canonicalize	nested-mixed	dragonbox	json-canon	dragonbox*	1.41×	[1.405, 1.423]	0.0002	0.0004	-46
rs:schubfach-vs-bd	api	canonicalize	nested-mixed	schubfach-rs	json-canon-rs	schubfach-rs*	1.97×	[1.923, 2.010]	0.0002	0.0004	-33

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:schubfach-vs-ryu	api	canonicalize	nested-mixed	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.988, 1.052]	0.2519	0.3509	-0
rs:schubfach-vs-dragonbox	api	canonicalize	nested-mixed	schubfach-rs	dragonbox-rs	schubfach-rs	1.03×	[0.999, 1.052]	0.0466	0.0808	-0
rs:ryu-vs-dragonbox	api	canonicalize	nested-mixed	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.979, 1.035]	0.5805	0.6773	-0
rs:ryu-vs-bd	api	canonicalize	nested-mixed	ryu-rs	json-canon-rs	ryu-rs*	1.93×	[1.880, 1.977]	0.0002	0.0004	-28
rs:dragonbox-vs-bd	api	canonicalize	nested-mixed	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.92×	[1.889, 1.950]	0.0002	0.0004	-39
schubfach:go-vs-rs	api	canonicalize	nested-mixed	schubfach	schubfach-rs	schubfach-rs*	2.18×	[2.128, 2.221]	0.0002	0.0004	44
bd:go-vs-rs	api	canonicalize	nested-mixed	json-canon	json-canon-rs	json-canon-rs*	1.57×	[1.556, 1.580]	0.0002	0.0004	49
ryu:go-vs-rs	api	canonicalize	nested-mixed	ryu	ryu-rs	ryu-rs*	2.15×	[2.097, 2.202]	0.0002	0.0004	37
dragonbox:go-vs-rs	api	canonicalize	nested-mixed	dragonbox	dragonbox-rs	dragonbox-rs*	2.12×	[2.096, 2.160]	0.0002	0.0004	53
go:schubfach-vs-bd	api	canonicalize	number-heavy	schubfach	json-canon	schubfach*	1.57×	[1.552, 1.579]	0.0002	0.0004	-43
go:schubfach-vs-ryu	api	canonicalize	number-heavy	schubfach	ryu	schubfach	1.00×	[0.993, 1.011]	0.6015	0.6962	-0
go:schubfach-vs-dragonbox	api	canonicalize	number-heavy	schubfach	dragonbox	dragonbox	1.01×	[0.999, 1.016]	0.1844	0.2696	0
go:ryu-vs-dragonbox	api	canonicalize	number-heavy	ryu	dragonbox	dragonbox	1.01×	[1.002, 1.017]	0.0472	0.0815	0
go:ryu-vs-bd	api	canonicalize	number-heavy	ryu	json-canon	ryu*	1.56×	[1.550, 1.575]	0.0002	0.0004	-45
go:dragonbox-vs-bd	api	canonicalize	number-heavy	dragonbox	json-canon	dragonbox*	1.58×	[1.565, 1.588]	0.0002	0.0004	-48
rs:schubfach-vs-bd	api	canonicalize	number-heavy	schubfach-rs	json-canon-rs	schubfach-rs*	8.23×	[8.106, 8.340]	0.0004	0.0008	-106
rs:schubfach-vs-ryu	api	canonicalize	number-heavy	schubfach-rs	ryu-rs	schubfach-rs	1.03×	[1.004, 1.054]	0.0338	0.0604	-1
rs:schubfach-vs-dragonbox	api	canonicalize	number-heavy	schubfach-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.986, 1.021]	0.7640	0.8266	0
rs:ryu-vs-dragonbox	api	canonicalize	number-heavy	ryu-rs	dragonbox-rs	dragonbox-rs*	1.03×	[1.009, 1.059]	0.0234	0.0430	1
rs:ryu-vs-bd	api	canonicalize	number-heavy	ryu-rs	json-canon-rs	ryu-rs*	7.99×	[7.815, 8.161]	0.0002	0.0004	-101
rs:dragonbox-vs-bd	api	canonicalize	number-heavy	dragonbox-rs	json-canon-rs	dragonbox-rs*	8.25×	[8.129, 8.371]	0.0002	0.0004	-106
schubfach:go-vs-rs	api	canonicalize	number-heavy	schubfach	schubfach-rs	schubfach-rs*	9.65×	[9.505, 9.784]	0.0002	0.0004	106
bd:go-vs-rs	api	canonicalize	number-heavy	json-canon	json-canon-rs	json-canon-rs*	1.84×	[1.821, 1.852]	0.0002	0.0004	58
ryu:go-vs-rs	api	canonicalize	number-heavy	ryu	ryu-rs	ryu-rs*	9.39×	[9.190, 9.605]	0.0002	0.0004	117
dragonbox:go-vs-rs	api	canonicalize	number-heavy	dragonbox	dragonbox-rs	dragonbox-rs*	9.61×	[9.472, 9.743]	0.0002	0.0004	142
go:schubfach-vs-bd	api	canonicalize	numeric-boundary	schubfach	json-canon	schubfach*	1.50×	[1.487, 1.510]	0.0002	0.0004	-41
go:schubfach-vs-ryu	api	canonicalize	numeric-boundary	schubfach	ryu	ryu	1.00×	[0.991, 1.009]	0.8524	0.8939	0
go:schubfach-vs-dragonbox	api	canonicalize	numeric-boundary	schubfach	dragonbox	dragonbox	1.00×	[0.998, 1.010]	0.2561	0.3555	0
go:ryu-vs-dragonbox	api	canonicalize	numeric-boundary	ryu	dragonbox	dragonbox	1.00×	[0.996, 1.013]	0.5421	0.6400	0
go:ryu-vs-bd	api	canonicalize	numeric-boundary	ryu	json-canon	ryu*	1.50×	[1.485, 1.513]	0.0002	0.0004	-36
go:dragonbox-vs-bd	api	canonicalize	numeric-boundary	dragonbox	json-canon	dragonbox*	1.50×	[1.495, 1.515]	0.0002	0.0004	-43
rs:schubfach-vs-bd	api	canonicalize	numeric-boundary	schubfach-rs	json-canon-rs	schubfach-rs*	9.33×	[9.163, 9.542]	0.0002	0.0004	-178
rs:schubfach-vs-ryu	api	canonicalize	numeric-boundary	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.992, 1.058]	0.2767	0.3770	-0
rs:schubfach-vs-dragonbox	api	canonicalize	numeric-boundary	schubfach-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.982, 1.038]	0.3353	0.4379	0
rs:ryu-vs-dragonbox	api	canonicalize	numeric-boundary	ryu-rs	dragonbox-rs	dragonbox-rs	1.03×	[1.004, 1.070]	0.0556	0.0949	0
rs:ryu-vs-bd	api	canonicalize	numeric-boundary	ryu-rs	json-canon-rs	ryu-rs*	9.15×	[8.861, 9.348]	0.0002	0.0004	-161
rs:dragonbox-vs-bd	api	canonicalize	numeric-boundary	dragonbox-rs	json-canon-rs	dragonbox-rs*	9.46×	[9.239, 9.614]	0.0002	0.0004	-179
schubfach:go-vs-rs	api	canonicalize	numeric-boundary	schubfach	schubfach-rs	schubfach-rs*	12.98×	[12.744, 13.263]	0.0002	0.0004	154
bd:go-vs-rs	api	canonicalize	numeric-boundary	json-canon	json-canon-rs	json-canon-rs*	2.08×	[2.071, 2.099]	0.0002	0.0004	70
ryu:go-vs-rs	api	canonicalize	numeric-boundary	ryu	ryu-rs	ryu-rs*	12.71×	[12.314, 13.008]	0.0002	0.0004	100
dragonbox:go-vs-rs	api	canonicalize	numeric-boundary	dragonbox	dragonbox-rs	dragonbox-rs*	13.11×	[12.813, 13.320]	0.0002	0.0004	199
go:schubfach-vs-bd	api	canonicalize	rfc-key-sorting	schubfach	json-canon	schubfach*	1.03×	[1.022, 1.044]	0.0002	0.0004	-2
go:schubfach-vs-ryu	api	canonicalize	rfc-key-sorting	schubfach	ryu	schubfach	1.01×	[0.998, 1.012]	0.1656	0.2491	-0
go:schubfach-vs-dragonbox	api	canonicalize	rfc-key-sorting	schubfach	dragonbox	schubfach	1.00×	[0.997, 1.008]	0.3959	0.5033	-0
go:ryu-vs-dragonbox	api	canonicalize	rfc-key-sorting	ryu	dragonbox	dragonbox	1.00×	[0.995, 1.010]	0.5119	0.6133	0
go:ryu-vs-bd	api	canonicalize	rfc-key-sorting	ryu	json-canon	ryu*	1.03×	[1.015, 1.039]	0.0002	0.0004	-1

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:dragonbox-vs-bd	api	canonicalize	rfc-key-sorting	dragonbox	json-canon	dragonbox*	1.03×	[1.019, 1.042]	0.0002	0.0004	-2
rs:schubfach-vs-bd	api	canonicalize	rfc-key-sorting	schubfach-rs	json-canon-rs	json-canon-rs	1.02×	[0.985, 1.060]	0.2727	0.3739	0
rs:schubfach-vs-ryu	api	canonicalize	rfc-key-sorting	schubfach-rs	ryu-rs	schubfach-rs	1.03×	[0.990, 1.071]	0.1932	0.2805	-0
rs:schubfach-vs-dragonbox	api	canonicalize	rfc-key-sorting	schubfach-rs	dragonbox-rs	schubfach-rs	1.00×	[0.964, 1.044]	0.9438	0.9624	-0
rs:ryu-vs-dragonbox	api	canonicalize	rfc-key-sorting	ryu-rs	dragonbox-rs	dragonbox-rs	1.03×	[0.997, 1.059]	0.1088	0.1736	0
rs:ryu-vs-bd	api	canonicalize	rfc-key-sorting	ryu-rs	json-canon-rs	json-canon-rs*	1.05×	[1.025, 1.083]	0.0032	0.0064	1
rs:dragonbox-vs-bd	api	canonicalize	rfc-key-sorting	dragonbox-rs	json-canon-rs	json-canon-rs	1.02×	[0.999, 1.055]	0.1342	0.2087	0
schubfach:go-vs-rs	api	canonicalize	rfc-key-sorting	schubfach	schubfach-rs	schubfach-rs*	2.18×	[2.113, 2.255]	0.0002	0.0004	29
bd:go-vs-rs	api	canonicalize	rfc-key-sorting	json-canon	json-canon-rs	json-canon-rs*	2.30×	[2.254, 2.348]	0.0002	0.0004	38
ryu:go-vs-rs	api	canonicalize	rfc-key-sorting	ryu	ryu-rs	ryu-rs*	2.13×	[2.079, 2.175]	0.0002	0.0004	38
dragonbox:go-vs-rs	api	canonicalize	rfc-key-sorting	dragonbox	dragonbox-rs	dragonbox-rs*	2.18×	[2.130, 2.226]	0.0002	0.0004	42
go:schubfach-vs-bd	api	canonicalize	small	schubfach	json-canon	schubfach*	1.36×	[1.354, 1.365]	0.0002	0.0004	-58
go:schubfach-vs-ryu	api	canonicalize	small	schubfach	ryu	schubfach	1.01×	[1.000, 1.012]	0.1032	0.1650	-0
go:schubfach-vs-dragonbox	api	canonicalize	small	schubfach	dragonbox	schubfach	1.00×	[0.997, 1.009]	0.5211	0.6215	-0
go:ryu-vs-dragonbox	api	canonicalize	small	ryu	dragonbox	dragonbox	1.00×	[0.995, 1.011]	0.4711	0.5766	0
go:ryu-vs-bd	api	canonicalize	small	ryu	json-canon	ryu*	1.35×	[1.343, 1.360]	0.0002	0.0004	-41
go:dragonbox-vs-bd	api	canonicalize	small	dragonbox	json-canon	dragonbox*	1.36×	[1.346, 1.365]	0.0002	0.0004	-39
rs:schubfach-vs-bd	api	canonicalize	small	schubfach-rs	json-canon-rs	schubfach-rs*	1.75×	[1.675, 1.807]	0.0002	0.0004	-13
rs:schubfach-vs-ryu	api	canonicalize	small	schubfach-rs	ryu-rs	schubfach-rs*	1.06×	[1.020, 1.107]	0.0102	0.0195	-1
rs:schubfach-vs-dragonbox	api	canonicalize	small	schubfach-rs	dragonbox-rs	schubfach-rs	1.04×	[0.983, 1.118]	0.3127	0.4146	-0
rs:ryu-vs-dragonbox	api	canonicalize	small	ryu-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.953, 1.079]	0.4721	0.5766	0
rs:ryu-vs-bd	api	canonicalize	small	ryu-rs	json-canon-rs	ryu-rs*	1.64×	[1.588, 1.696]	0.0002	0.0004	-12
rs:dragonbox-vs-bd	api	canonicalize	small	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.68×	[1.560, 1.761]	0.0002	0.0004	-8
schubfach:go-vs-rs	api	canonicalize	small	schubfach	schubfach-rs	schubfach-rs*	1.89×	[1.821, 1.945]	0.0002	0.0004	23
bd:go-vs-rs	api	canonicalize	small	json-canon	json-canon-rs	json-canon-rs*	1.47×	[1.445, 1.504]	0.0002	0.0004	19
ryu:go-vs-rs	api	canonicalize	small	ryu	ryu-rs	ryu-rs*	1.79×	[1.735, 1.835]	0.0002	0.0004	22
dragonbox:go-vs-rs	api	canonicalize	small	dragonbox	dragonbox-rs	dragonbox-rs*	1.83×	[1.687, 1.905]	0.0002	0.0004	12
go:schubfach-vs-bd	api	canonicalize	surrogate-pair	schubfach	json-canon	schubfach*	1.02×	[1.011, 1.021]	0.0002	0.0004	-2
go:schubfach-vs-ryu	api	canonicalize	surrogate-pair	schubfach	ryu	schubfach	1.00×	[0.998, 1.005]	0.3067	0.4079	-0
go:schubfach-vs-dragonbox	api	canonicalize	surrogate-pair	schubfach	dragonbox	schubfach	1.01×	[1.002, 1.040]	0.1610	0.2439	-0
go:ryu-vs-dragonbox	api	canonicalize	surrogate-pair	ryu	dragonbox	ryu	1.01×	[1.000, 1.038]	0.2783	0.3779	-0
go:ryu-vs-bd	api	canonicalize	surrogate-pair	ryu	json-canon	ryu*	1.01×	[1.009, 1.019]	0.0002	0.0004	-2
go:dragonbox-vs-bd	api	canonicalize	surrogate-pair	dragonbox	json-canon	dragonbox	1.00×	[0.977, 1.014]	0.9568	0.9707	-0
rs:schubfach-vs-bd	api	canonicalize	surrogate-pair	schubfach-rs	json-canon-rs	schubfach-rs	1.01×	[0.933, 1.073]	0.7303	0.7981	-0
rs:schubfach-vs-ryu	api	canonicalize	surrogate-pair	schubfach-rs	ryu-rs	schubfach-rs	1.00×	[0.924, 1.056]	0.9508	0.9673	-0
rs:schubfach-vs-dragonbox	api	canonicalize	surrogate-pair	schubfach-rs	dragonbox-rs	schubfach-rs	1.07×	[0.985, 1.158]	0.1510	0.2315	-0
rs:ryu-vs-dragonbox	api	canonicalize	surrogate-pair	ryu-rs	dragonbox-rs	ryu-rs	1.07×	[1.007, 1.157]	0.0968	0.1560	-0
rs:ryu-vs-bd	api	canonicalize	surrogate-pair	ryu-rs	json-canon-rs	ryu-rs	1.01×	[0.962, 1.062]	0.6887	0.7623	-0
rs:dragonbox-vs-bd	api	canonicalize	surrogate-pair	dragonbox-rs	json-canon-rs	json-canon-rs	1.06×	[0.990, 1.148]	0.1908	0.2777	0
schubfach:go-vs-rs	api	canonicalize	surrogate-pair	schubfach	schubfach-rs	schubfach-rs*	2.05×	[1.873, 2.129]	0.0002	0.0004	15
bd:go-vs-rs	api	canonicalize	surrogate-pair	json-canon	json-canon-rs	json-canon-rs*	2.05×	[1.964, 2.126]	0.0002	0.0004	22
ryu:go-vs-rs	api	canonicalize	surrogate-pair	ryu	ryu-rs	ryu-rs*	2.05×	[1.973, 2.109]	0.0002	0.0004	26
dragonbox:go-vs-rs	api	canonicalize	surrogate-pair	dragonbox	dragonbox-rs	dragonbox-rs*	1.94×	[1.801, 2.048]	0.0002	0.0004	11
go:schubfach-vs-bd	api	canonicalize	unicode	schubfach	json-canon	json-canon	1.00×	[0.998, 1.006]	0.4655	0.5726	0
go:schubfach-vs-ryu	api	canonicalize	unicode	schubfach	ryu	schubfach*	1.01×	[1.003, 1.015]	0.0096	0.0184	-1
go:schubfach-vs-dragonbox	api	canonicalize	unicode	schubfach	dragonbox	schubfach	1.00×	[0.999, 1.008]	0.1476	0.2273	-0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:ryu-vs-dragonbox	api	canonicalize	unicode	ryu	dragonbox	dragonbox	1.00×	[0.999, 1.012]	0.1764	0.2602	0
go:ryu-vs-bd	api	canonicalize	unicode	ryu	json-canon	json-canon*	1.01×	[1.004, 1.017]	0.0036	0.0072	1
go:dragonbox-vs-bd	api	canonicalize	unicode	dragonbox	json-canon	json-canon	1.01×	[1.000, 1.010]	0.0710	0.1190	0
rs:schubfach-vs-bd	api	canonicalize	unicode	schubfach-rs	json-canon-rs	schubfach-rs*	1.04×	[1.021, 1.076]	0.0040	0.0079	-1
rs:schubfach-vs-ryu	api	canonicalize	unicode	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.996, 1.045]	0.2110	0.3023	-0
rs:schubfach-vs-dragonbox	api	canonicalize	unicode	schubfach-rs	dragonbox-rs	schubfach-rs*	1.06×	[1.040, 1.079]	0.0004	0.0008	-2
rs:ryu-vs-dragonbox	api	canonicalize	unicode	ryu-rs	dragonbox-rs	ryu-rs*	1.04×	[1.017, 1.063]	0.0022	0.0045	-1
rs:ryu-vs-bd	api	canonicalize	unicode	ryu-rs	json-canon-rs	ryu-rs	1.03×	[0.998, 1.058]	0.1096	0.1746	-0
rs:dragonbox-vs-bd	api	canonicalize	unicode	dragonbox-rs	json-canon-rs	json-canon-rs	1.01×	[0.984, 1.035]	0.2931	0.3935	0
schubfach:go-vs-rs	api	canonicalize	unicode	schubfach	schubfach-rs	schubfach-rs*	1.04×	[1.020, 1.049]	0.0006	0.0013	2
bd:go-vs-rs	api	canonicalize	unicode	json-canon	json-canon-rs	json-canon	1.01×	[0.993, 1.039]	0.4135	0.5210	-0
ryu:go-vs-rs	api	canonicalize	unicode	ryu	ryu-rs	ryu-rs*	1.03×	[1.002, 1.044]	0.0256	0.0469	1
dragonbox:go-vs-rs	api	canonicalize	unicode	dragonbox	dragonbox-rs	dragonbox*	1.02×	[1.010, 1.033]	0.0026	0.0053	-1
go:schubfach-vs-bd	api	canonicalize	verify-whitespace	schubfach	json-canon	schubfach*	1.73×	[1.716, 1.734]	0.0002	0.0004	-76
go:schubfach-vs-ryu	api	canonicalize	verify-whitespace	schubfach	ryu	ryu	1.00×	[0.998, 1.006]	0.4207	0.5280	0
go:schubfach-vs-dragonbox	api	canonicalize	verify-whitespace	schubfach	dragonbox	schubfach	1.00×	[0.995, 1.005]	0.9304	0.9548	-0
go:ryu-vs-dragonbox	api	canonicalize	verify-whitespace	ryu	dragonbox	ryu	1.00×	[0.997, 1.007]	0.3991	0.5063	-0
go:ryu-vs-bd	api	canonicalize	verify-whitespace	ryu	json-canon	ryu*	1.73×	[1.720, 1.737]	0.0002	0.0004	-78
go:dragonbox-vs-bd	api	canonicalize	verify-whitespace	dragonbox	json-canon	dragonbox*	1.72×	[1.715, 1.734]	0.0002	0.0004	-74
rs:schubfach-vs-bd	api	canonicalize	verify-whitespace	schubfach-rs	json-canon-rs	schubfach-rs*	2.96×	[2.866, 3.039]	0.0002	0.0004	-30
rs:schubfach-vs-ryu	api	canonicalize	verify-whitespace	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.985, 1.051]	0.2625	0.3626	-0
rs:schubfach-vs-dragonbox	api	canonicalize	verify-whitespace	schubfach-rs	dragonbox-rs	schubfach-rs	1.03×	[0.995, 1.058]	0.0980	0.1576	-0
rs:ryu-vs-dragonbox	api	canonicalize	verify-whitespace	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.982, 1.036]	0.5683	0.6649	-0
rs:ryu-vs-bd	api	canonicalize	verify-whitespace	ryu-rs	json-canon-rs	ryu-rs*	2.90×	[2.830, 2.980]	0.0002	0.0004	-31
rs:dragonbox-vs-bd	api	canonicalize	verify-whitespace	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.87×	[2.813, 2.947]	0.0002	0.0004	-32
schubfach:go-vs-rs	api	canonicalize	verify-whitespace	schubfach	schubfach-rs	schubfach-rs*	2.28×	[2.222, 2.337]	0.0002	0.0004	41
bd:go-vs-rs	api	canonicalize	verify-whitespace	json-canon	json-canon-rs	json-canon-rs*	1.33×	[1.307, 1.352]	0.0002	0.0004	16
ryu:go-vs-rs	api	canonicalize	verify-whitespace	ryu	ryu-rs	ryu-rs*	2.24×	[2.190, 2.283]	0.0002	0.0004	47
dragonbox:go-vs-rs	api	canonicalize	verify-whitespace	dragonbox	dragonbox-rs	dragonbox-rs*	2.22×	[2.185, 2.263]	0.0002	0.0004	52
go:schubfach-vs-bd	api	verify	canonical/array-2048	schubfach	json-canon	schubfach*	1.66×	[1.652, 1.670]	0.0002	0.0004	-62
go:schubfach-vs-ryu	api	verify	canonical/array-2048	schubfach	ryu	schubfach	1.01×	[1.000, 1.009]	0.0360	0.0642	-1
go:schubfach-vs-dragonbox	api	verify	canonical/array-2048	schubfach	dragonbox	schubfach*	1.01×	[1.011, 1.017]	0.0002	0.0004	-3
go:ryu-vs-dragonbox	api	verify	canonical/array-2048	ryu	dragonbox	ryu*	1.01×	[1.004, 1.014]	0.0036	0.0072	-1
go:ryu-vs-bd	api	verify	canonical/array-2048	ryu	json-canon	ryu*	1.65×	[1.643, 1.664]	0.0002	0.0004	-57
go:dragonbox-vs-bd	api	verify	canonical/array-2048	dragonbox	json-canon	dragonbox*	1.64×	[1.630, 1.649]	0.0002	0.0004	-58
rs:schubfach-vs-bd	api	verify	canonical/array-2048	schubfach-rs	json-canon-rs	schubfach-rs*	2.50×	[2.472, 2.517]	0.0002	0.0004	-85
rs:schubfach-vs-ryu	api	verify	canonical/array-2048	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[1.001, 1.060]	0.1818	0.2664	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/array-2048	schubfach-rs	dragonbox-rs	schubfach-rs	1.00×	[0.991, 1.009]	0.9152	0.9429	-0
rs:ryu-vs-dragonbox	api	verify	canonical/array-2048	ryu-rs	dragonbox-rs	dragonbox-rs	1.02×	[1.001, 1.063]	0.1582	0.2402	0
rs:ryu-vs-bd	api	verify	canonical/array-2048	ryu-rs	json-canon-rs	ryu-rs*	2.45×	[2.349, 2.491]	0.0002	0.0004	-44
rs:dragonbox-vs-bd	api	verify	canonical/array-2048	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.49×	[2.478, 2.512]	0.0002	0.0004	-91
schubfach:go-vs-rs	api	verify	canonical/array-2048	schubfach	schubfach-rs	schubfach-rs*	1.84×	[1.824, 1.851]	0.0002	0.0004	88
bd:go-vs-rs	api	verify	canonical/array-2048	json-canon	json-canon-rs	json-canon-rs*	1.23×	[1.215, 1.233]	0.0002	0.0004	22
ryu:go-vs-rs	api	verify	canonical/array-2048	ryu	ryu-rs	ryu-rs*	1.82×	[1.728, 1.845]	0.0002	0.0004	27
dragonbox:go-vs-rs	api	verify	canonical/array-2048	dragonbox	dragonbox-rs	dragonbox-rs*	1.86×	[1.853, 1.874]	0.0002	0.0004	98
go:schubfach-vs-bd	api	verify	canonical/array-256	schubfach	json-canon	schubfach*	1.54×	[1.534, 1.554]	0.0002	0.0004	-57

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:schubfach-vs-ryu	api	verify	canonical/array-256	schubfach	ryu	ryu	1.01×	[1.001, 1.013]	0.0432	0.0750	0
go:schubfach-vs-dragonbox	api	verify	canonical/array-256	schubfach	dragonbox	dragonbox	1.01×	[1.001, 1.014]	0.0674	0.1138	0
go:ryu-vs-dragonbox	api	verify	canonical/array-256	ryu	dragonbox	dragonbox	1.00×	[0.995, 1.006]	0.9630	0.9751	0
go:ryu-vs-bd	api	verify	canonical/array-256	ryu	json-canon	ryu*	1.56×	[1.548, 1.564]	0.0002	0.0004	-65
go:dragonbox-vs-bd	api	verify	canonical/array-256	dragonbox	json-canon	dragonbox*	1.56×	[1.547, 1.565]	0.0002	0.0004	-60
rs:schubfach-vs-bd	api	verify	canonical/array-256	schubfach-rs	json-canon-rs	schubfach-rs*	2.26×	[2.237, 2.274]	0.0002	0.0004	-71
rs:schubfach-vs-ryu	api	verify	canonical/array-256	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.987, 1.025]	0.4521	0.5609	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/array-256	schubfach-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.998, 1.012]	0.1986	0.2871	0
rs:ryu-vs-dragonbox	api	verify	canonical/array-256	ryu-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.993, 1.030]	0.2304	0.3268	0
rs:ryu-vs-bd	api	verify	canonical/array-256	ryu-rs	json-canon-rs	ryu-rs*	2.24×	[2.199, 2.283]	0.0002	0.0004	-46
rs:dragonbox-vs-bd	api	verify	canonical/array-256	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.27×	[2.250, 2.283]	0.0002	0.0004	-74
schubfach:go-vs-rs	api	verify	canonical/array-256	schubfach	schubfach-rs	schubfach-rs*	1.88×	[1.869, 1.898]	0.0002	0.0004	67
bd:go-vs-rs	api	verify	canonical/array-256	json-canon	json-canon-rs	json-canon-rs*	1.29×	[1.280, 1.299]	0.0002	0.0004	30
ryu:go-vs-rs	api	verify	canonical/array-256	ryu	ryu-rs	ryu-rs*	1.86×	[1.826, 1.894]	0.0002	0.0004	38
dragonbox:go-vs-rs	api	verify	canonical/array-256	dragonbox	dragonbox-rs	dragonbox-rs*	1.88×	[1.868, 1.891]	0.0002	0.0004	80
go:schubfach-vs-bd	api	verify	canonical/canonical-minimal	schubfach	json-canon	schubfach*	1.51×	[1.504, 1.518]	0.0002	0.0004	-70
go:schubfach-vs-ryu	api	verify	canonical/canonical-minimal	schubfach	ryu	ryu	1.00×	[0.996, 1.008]	0.6033	0.6976	0
go:schubfach-vs-dragonbox	api	verify	canonical/canonical-minimal	schubfach	dragonbox	dragonbox	1.00×	[0.999, 1.009]	0.1696	0.2531	0
go:ryu-vs-dragonbox	api	verify	canonical/canonical-minimal	ryu	dragonbox	dragonbox	1.00×	[0.995, 1.010]	0.5559	0.6533	0
go:ryu-vs-bd	api	verify	canonical/canonical-minimal	ryu	json-canon	ryu*	1.51×	[1.503, 1.524]	0.0002	0.0004	-54
go:dragonbox-vs-bd	api	verify	canonical/canonical-minimal	dragonbox	json-canon	dragonbox*	1.52×	[1.509, 1.526]	0.0002	0.0004	-61
rs:schubfach-vs-bd	api	verify	canonical/canonical-minimal	schubfach-rs	json-canon-rs	schubfach-rs*	2.08×	[2.021, 2.147]	0.0002	0.0004	-18
rs:schubfach-vs-ryu	api	verify	canonical/canonical-minimal	schubfach-rs	ryu-rs	schubfach-rs	1.03×	[0.988, 1.080]	0.2541	0.3532	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/canonical-minimal	schubfach-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.979, 1.045]	0.4795	0.5821	0
rs:ryu-vs-dragonbox	api	verify	canonical/canonical-minimal	ryu-rs	dragonbox-rs	dragonbox-rs	1.04×	[1.000, 1.093]	0.1158	0.1838	0
rs:ryu-vs-bd	api	verify	canonical/canonical-minimal	ryu-rs	json-canon-rs	ryu-rs*	2.03×	[1.932, 2.107]	0.0002	0.0004	-14
rs:dragonbox-vs-bd	api	verify	canonical/canonical-minimal	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.11×	[2.043, 2.176]	0.0002	0.0004	-17
schubfach:go-vs-rs	api	verify	canonical/canonical-minimal	schubfach	schubfach-rs	schubfach-rs*	1.73×	[1.678, 1.756]	0.0002	0.0004	27
bd:go-vs-rs	api	verify	canonical/canonical-minimal	json-canon	json-canon-rs	json-canon-rs*	1.25×	[1.221, 1.276]	0.0002	0.0004	9
ryu:go-vs-rs	api	verify	canonical/canonical-minimal	ryu	ryu-rs	ryu-rs*	1.68×	[1.596, 1.732]	0.0002	0.0004	14
dragonbox:go-vs-rs	api	verify	canonical/canonical-minimal	dragonbox	dragonbox-rs	dragonbox-rs*	1.74×	[1.690, 1.778]	0.0002	0.0004	24
go:schubfach-vs-bd	api	verify	canonical/control-escapes	schubfach	json-canon	schubfach*	1.02×	[1.017, 1.029]	0.0002	0.0004	-3
go:schubfach-vs-ryu	api	verify	canonical/control-escapes	schubfach	ryu	ryu	1.00×	[0.999, 1.011]	0.1552	0.2368	0
go:schubfach-vs-dragonbox	api	verify	canonical/control-escapes	schubfach	dragonbox	schubfach	1.00×	[0.994, 1.007]	0.9200	0.9471	-0
go:ryu-vs-dragonbox	api	verify	canonical/control-escapes	ryu	dragonbox	ryu	1.01×	[1.001, 1.010]	0.0404	0.0709	-0
go:ryu-vs-bd	api	verify	canonical/control-escapes	ryu	json-canon	ryu*	1.03×	[1.025, 1.032]	0.0002	0.0004	-7
go:dragonbox-vs-bd	api	verify	canonical/control-escapes	dragonbox	json-canon	dragonbox*	1.02×	[1.018, 1.027]	0.0002	0.0004	-4
rs:schubfach-vs-bd	api	verify	canonical/control-escapes	schubfach-rs	json-canon-rs	json-canon-rs	1.04×	[0.998, 1.083]	0.0784	0.1299	0
rs:schubfach-vs-ryu	api	verify	canonical/control-escapes	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.972, 1.076]	0.5995	0.6945	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/control-escapes	schubfach-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.988, 1.053]	0.2851	0.3847	0
rs:ryu-vs-dragonbox	api	verify	canonical/control-escapes	ryu-rs	dragonbox-rs	dragonbox-rs	1.04×	[1.003, 1.093]	0.1210	0.1907	0
rs:ryu-vs-bd	api	verify	canonical/control-escapes	ryu-rs	json-canon-rs	json-canon-rs	1.06×	[1.014, 1.117]	0.0334	0.0598	0
rs:dragonbox-vs-bd	api	verify	canonical/control-escapes	dragonbox-rs	json-canon-rs	json-canon-rs	1.02×	[0.981, 1.049]	0.1886	0.2748	0
schubfach:go-vs-rs	api	verify	canonical/control-escapes	schubfach	schubfach-rs	schubfach-rs*	2.04×	[1.976, 2.102]	0.0002	0.0004	27
bd:go-vs-rs	api	verify	canonical/control-escapes	json-canon	json-canon-rs	json-canon-rs*	2.18×	[2.096, 2.228]	0.0002	0.0004	33
ryu:go-vs-rs	api	verify	canonical/control-escapes	ryu	ryu-rs	ryu-rs*	2.00×	[1.882, 2.054]	0.0002	0.0004	21

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
dragonbox:go-vs-rs	api	verify	canonical/control-escapes	dragonbox	dragonbox-rs	dragonbox-rs*	2.08×	[2.051, 2.105]	0.0002	0.0004	62
go:schubfach-vs-bd	api	verify	canonical/deep-64	schubfach	json-canon	json-canon*	1.01×	[1.009, 1.017]	0.0002	0.0004	2
go:schubfach-vs-ryu	api	verify	canonical/deep-64	schubfach	ryu	schubfach	1.00×	[0.996, 1.006]	0.8342	0.8790	-0
go:schubfach-vs-dragonbox	api	verify	canonical/deep-64	schubfach	dragonbox	schubfach	1.00×	[1.001, 1.008]	0.0318	0.0573	-1
go:ryu-vs-dragonbox	api	verify	canonical/deep-64	ryu	dragonbox	ryu	1.00×	[0.998, 1.008]	0.1760	0.2602	-0
go:ryu-vs-bd	api	verify	canonical/deep-64	ryu	json-canon	json-canon*	1.01×	[1.009, 1.019]	0.0004	0.0008	2
go:dragonbox-vs-bd	api	verify	canonical/deep-64	dragonbox	json-canon	json-canon*	1.02×	[1.014, 1.021]	0.0002	0.0004	3
rs:schubfach-vs-bd	api	verify	canonical/deep-64	schubfach-rs	json-canon-rs	json-canon-rs*	1.02×	[1.019, 1.031]	0.0002	0.0004	3
rs:schubfach-vs-ryu	api	verify	canonical/deep-64	schubfach-rs	ryu-rs	ryu-rs	1.00×	[0.996, 1.010]	0.4519	0.5609	0
rs:schubfach-vs-dragonbox	api	verify	canonical/deep-64	schubfach-rs	dragonbox-rs	schubfach-rs*	1.03×	[1.010, 1.067]	0.0022	0.0045	-0
rs:ryu-vs-dragonbox	api	verify	canonical/deep-64	ryu-rs	dragonbox-rs	ryu-rs*	1.03×	[1.013, 1.071]	0.0010	0.0021	-0
rs:ryu-vs-bd	api	verify	canonical/deep-64	ryu-rs	json-canon-rs	json-canon-rs*	1.02×	[1.015, 1.029]	0.0002	0.0004	2
rs:dragonbox-vs-bd	api	verify	canonical/deep-64	dragonbox-rs	json-canon-rs	json-canon-rs*	1.05×	[1.035, 1.094]	0.0002	0.0004	1
schubfach:go-vs-rs	api	verify	canonical/deep-64	schubfach	schubfach-rs	schubfach-rs*	1.52×	[1.510, 1.525]	0.0004	0.0008	74
bd:go-vs-rs	api	verify	canonical/deep-64	json-canon	json-canon-rs	json-canon-rs*	1.54×	[1.528, 1.544]	0.0002	0.0004	74
ryu:go-vs-rs	api	verify	canonical/deep-64	ryu	ryu-rs	ryu-rs*	1.52×	[1.513, 1.535]	0.0002	0.0004	51
dragonbox:go-vs-rs	api	verify	canonical/deep-64	dragonbox	dragonbox-rs	dragonbox-rs*	1.49×	[1.428, 1.509]	0.0002	0.0004	17
go:schubfach-vs-bd	api	verify	canonical/deep	schubfach	json-canon	json-canon*	1.01×	[1.005, 1.017]	0.0014	0.0029	1
go:schubfach-vs-ryu	api	verify	canonical/deep	schubfach	ryu	ryu	1.00×	[0.995, 1.008]	0.7580	0.8229	0
go:schubfach-vs-dragonbox	api	verify	canonical/deep	schubfach	dragonbox	schubfach	1.01×	[1.000, 1.015]	0.0950	0.1536	-0
go:ryu-vs-dragonbox	api	verify	canonical/deep	ryu	dragonbox	ryu	1.01×	[1.000, 1.017]	0.0760	0.1267	-0
go:ryu-vs-bd	api	verify	canonical/deep	ryu	json-canon	json-canon*	1.01×	[1.002, 1.015]	0.0200	0.0370	1
go:dragonbox-vs-bd	api	verify	canonical/deep	dragonbox	json-canon	json-canon*	1.02×	[1.010, 1.025]	0.0008	0.0017	1
rs:schubfach-vs-bd	api	verify	canonical/deep	schubfach-rs	json-canon-rs	schubfach-rs*	1.02×	[1.014, 1.033]	0.0004	0.0008	-2
rs:schubfach-vs-ryu	api	verify	canonical/deep	schubfach-rs	ryu-rs	schubfach-rs*	1.07×	[1.061, 1.083]	0.0004	0.0008	-5
rs:schubfach-vs-dragonbox	api	verify	canonical/deep	schubfach-rs	dragonbox-rs	schubfach-rs*	1.06×	[1.047, 1.068]	0.0002	0.0004	-4
rs:ryu-vs-dragonbox	api	verify	canonical/deep	ryu-rs	dragonbox-rs	dragonbox-rs*	1.01×	[1.006, 1.022]	0.0038	0.0075	1
rs:ryu-vs-bd	api	verify	canonical/deep	ryu-rs	json-canon-rs	json-canon-rs*	1.05×	[1.041, 1.056]	0.0002	0.0004	5
rs:dragonbox-vs-bd	api	verify	canonical/deep	dragonbox-rs	json-canon-rs	json-canon-rs*	1.03×	[1.027, 1.040]	0.0004	0.0008	4
schubfach:go-vs-rs	api	verify	canonical/deep	schubfach	schubfach-rs	schubfach-rs*	1.33×	[1.321, 1.346]	0.0002	0.0004	26
bd:go-vs-rs	api	verify	canonical/deep	json-canon	json-canon-rs	json-canon-rs*	1.29×	[1.284, 1.297]	0.0002	0.0004	40
ryu:go-vs-rs	api	verify	canonical/deep	ryu	ryu-rs	ryu-rs*	1.24×	[1.232, 1.252]	0.0002	0.0004	22
dragonbox:go-vs-rs	api	verify	canonical/deep	dragonbox	dragonbox-rs	dragonbox-rs*	1.27×	[1.260, 1.280]	0.0002	0.0004	23
go:schubfach-vs-bd	api	verify	canonical/escaped-key-order	schubfach	json-canon	schubfach*	1.61×	[1.605, 1.622]	0.0002	0.0004	-90
go:schubfach-vs-ryu	api	verify	canonical/escaped-key-order	schubfach	ryu	ryu	1.00×	[0.998, 1.010]	0.1648	0.2482	0
go:schubfach-vs-dragonbox	api	verify	canonical/escaped-key-order	schubfach	dragonbox	dragonbox	1.00×	[0.996, 1.008]	0.6399	0.7282	0
go:ryu-vs-dragonbox	api	verify	canonical/escaped-key-order	ryu	dragonbox	ryu	1.00×	[0.997, 1.008]	0.3105	0.4126	-0
go:ryu-vs-bd	api	verify	canonical/escaped-key-order	ryu	json-canon	ryu*	1.62×	[1.613, 1.627]	0.0002	0.0004	-108
go:dragonbox-vs-bd	api	verify	canonical/escaped-key-order	dragonbox	json-canon	dragonbox*	1.62×	[1.609, 1.624]	0.0002	0.0004	-100
rs:schubfach-vs-bd	api	verify	canonical/escaped-key-order	schubfach-rs	json-canon-rs	schubfach-rs*	2.53×	[2.335, 2.622]	0.0002	0.0004	-19
rs:schubfach-vs-ryu	api	verify	canonical/escaped-key-order	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.943, 1.058]	0.6165	0.7083	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/escaped-key-order	schubfach-rs	dragonbox-rs	schubfach-rs	1.02×	[0.937, 1.055]	0.6181	0.7084	-0
rs:ryu-vs-dragonbox	api	verify	canonical/escaped-key-order	ryu-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.968, 1.032]	0.9696	0.9787	0
rs:ryu-vs-bd	api	verify	canonical/escaped-key-order	ryu-rs	json-canon-rs	ryu-rs*	2.48×	[2.407, 2.552]	0.0002	0.0004	-24
rs:dragonbox-vs-bd	api	verify	canonical/escaped-key-order	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.48×	[2.414, 2.553]	0.0002	0.0004	-25
schubfach:go-vs-rs	api	verify	canonical/escaped-key-order	schubfach	schubfach-rs	schubfach-rs*	2.11×	[1.947, 2.180]	0.0002	0.0004	18

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
bd:go-vs-rs	api	verify	canonical/escaped-key-order	json-canon	json-canon-rs	json-canon-rs*	1.35×	[1.320, 1.369]	0.0002	0.0004	16
ryu:go-vs-rs	api	verify	canonical/escaped-key-order	ryu	ryu-rs	ryu-rs*	2.07×	[2.009, 2.109]	0.0002	0.0004	36
dragonbox:go-vs-rs	api	verify	canonical/escaped-key-order	dragonbox	dragonbox-rs	dragonbox-rs*	2.07×	[2.022, 2.115]	0.0002	0.0004	38
go:schubfach-vs-bd	api	verify	canonical/large	schubfach	json-canon	schubfach*	1.79×	[1.779, 1.799]	0.0002	0.0004	-79
go:schubfach-vs-ryu	api	verify	canonical/large	schubfach	ryu	ryu	1.00×	[0.998, 1.008]	0.2797	0.3790	0
go:schubfach-vs-dragonbox	api	verify	canonical/large	schubfach	dragonbox	schubfach	1.01×	[1.001, 1.014]	0.0430	0.0748	-0
go:ryu-vs-dragonbox	api	verify	canonical/large	ryu	dragonbox	ryu*	1.01×	[1.004, 1.016]	0.0038	0.0075	-1
go:ryu-vs-bd	api	verify	canonical/large	ryu	json-canon	ryu*	1.79×	[1.785, 1.802]	0.0002	0.0004	-86
go:dragonbox-vs-bd	api	verify	canonical/large	dragonbox	json-canon	dragonbox*	1.78×	[1.765, 1.787]	0.0002	0.0004	-73
rs:schubfach-vs-bd	api	verify	canonical/large	schubfach-rs	json-canon-rs	schubfach-rs*	3.30×	[3.219, 3.357]	0.0002	0.0004	-66
rs:schubfach-vs-ryu	api	verify	canonical/large	schubfach-rs	ryu-rs	schubfach-rs	1.00×	[0.977, 1.042]	0.8978	0.9279	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/large	schubfach-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.977, 1.032]	0.6949	0.7670	0
rs:ryu-vs-dragonbox	api	verify	canonical/large	ryu-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.983, 1.047]	0.5075	0.6086	0
rs:ryu-vs-bd	api	verify	canonical/large	ryu-rs	json-canon-rs	ryu-rs*	3.30×	[3.156, 3.349]	0.0002	0.0004	-58
rs:dragonbox-vs-bd	api	verify	canonical/large	dragonbox-rs	json-canon-rs	dragonbox-rs*	3.33×	[3.223, 3.372]	0.0002	0.0004	-66
schubfach:go-vs-rs	api	verify	canonical/large	schubfach	schubfach-rs	schubfach-rs*	1.93×	[1.885, 1.964]	0.0002	0.0004	37
bd:go-vs-rs	api	verify	canonical/large	json-canon	json-canon-rs	json-canon-rs*	1.05×	[1.038, 1.055]	0.0002	0.0004	5
ryu:go-vs-rs	api	verify	canonical/large	ryu	ryu-rs	ryu-rs*	1.92×	[1.844, 1.954]	0.0002	0.0004	30
dragonbox:go-vs-rs	api	verify	canonical/large	dragonbox	dragonbox-rs	dragonbox-rs*	1.96×	[1.892, 1.988]	0.0002	0.0004	36
go:schubfach-vs-bd	api	verify	canonical/long-string	schubfach	json-canon	schubfach	1.00×	[0.991, 1.021]	0.5505	0.6476	-0
go:schubfach-vs-ryu	api	verify	canonical/long-string	schubfach	ryu	schubfach	1.01×	[0.998, 1.020]	0.1684	0.2519	-0
go:schubfach-vs-dragonbox	api	verify	canonical/long-string	schubfach	dragonbox	schubfach	1.00×	[0.990, 1.013]	0.8222	0.8707	-0
go:ryu-vs-dragonbox	api	verify	canonical/long-string	ryu	dragonbox	dragonbox	1.01×	[0.998, 1.018]	0.2026	0.2919	0
go:ryu-vs-bd	api	verify	canonical/long-string	ryu	json-canon	json-canon	1.00×	[0.989, 1.017]	0.6283	0.7163	0
go:dragonbox-vs-bd	api	verify	canonical/long-string	dragonbox	json-canon	dragonbox	1.00×	[0.990, 1.018]	0.6531	0.7394	-0
rs:schubfach-vs-bd	api	verify	canonical/long-string	schubfach-rs	json-canon-rs	json-canon-rs	1.01×	[0.992, 1.028]	0.5499	0.6476	0
rs:schubfach-vs-ryu	api	verify	canonical/long-string	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.987, 1.021]	0.3533	0.4577	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/long-string	schubfach-rs	dragonbox-rs	schubfach-rs	1.00×	[0.980, 1.018]	0.6039	0.6977	-0
rs:ryu-vs-dragonbox	api	verify	canonical/long-string	ryu-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.999, 1.015]	0.3619	0.4665	0
rs:ryu-vs-bd	api	verify	canonical/long-string	ryu-rs	json-canon-rs	json-canon-rs*	1.01×	[1.009, 1.022]	0.0002	0.0004	1
rs:dragonbox-vs-bd	api	verify	canonical/long-string	dragonbox-rs	json-canon-rs	json-canon-rs	1.01×	[1.000, 1.017]	0.0646	0.1097	0
schubfach:go-vs-rs	api	verify	canonical/long-string	schubfach	schubfach-rs	schubfach-rs*	4.45×	[4.350, 4.515]	0.0002	0.0004	69
bd:go-vs-rs	api	verify	canonical/long-string	json-canon	json-canon-rs	json-canon-rs*	4.50×	[4.440, 4.560]	0.0002	0.0004	52
ryu:go-vs-rs	api	verify	canonical/long-string	ryu	ryu-rs	ryu-rs*	4.45×	[4.422, 4.483]	0.0002	0.0004	96
dragonbox:go-vs-rs	api	verify	canonical/long-string	dragonbox	dragonbox-rs	dragonbox-rs*	4.44×	[4.400, 4.489]	0.0002	0.0004	86
go:schubfach-vs-bd	api	verify	canonical/medium	schubfach	json-canon	schubfach*	1.74×	[1.727, 1.742]	0.0002	0.0004	-101
go:schubfach-vs-ryu	api	verify	canonical/medium	schubfach	ryu	schubfach	1.01×	[1.001, 1.013]	0.0424	0.0739	-0
go:schubfach-vs-dragonbox	api	verify	canonical/medium	schubfach	dragonbox	schubfach	1.00×	[0.997, 1.008]	0.3855	0.4934	-0
go:ryu-vs-dragonbox	api	verify	canonical/medium	ryu	dragonbox	dragonbox	1.00×	[0.998, 1.011]	0.2490	0.3486	0
go:ryu-vs-bd	api	verify	canonical/medium	ryu	json-canon	ryu*	1.72×	[1.712, 1.732]	0.0002	0.0004	-84
go:dragonbox-vs-bd	api	verify	canonical/medium	dragonbox	json-canon	dragonbox*	1.73×	[1.722, 1.740]	0.0002	0.0004	-92
rs:schubfach-vs-bd	api	verify	canonical/medium	schubfach-rs	json-canon-rs	schubfach-rs*	3.02×	[2.954, 3.088]	0.0002	0.0004	-49
rs:schubfach-vs-ryu	api	verify	canonical/medium	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.990, 1.060]	0.2168	0.3086	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/medium	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.993, 1.033]	0.2743	0.3749	-0
rs:ryu-vs-dragonbox	api	verify	canonical/medium	ryu-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.982, 1.038]	0.4433	0.5542	0
rs:ryu-vs-bd	api	verify	canonical/medium	ryu-rs	json-canon-rs	ryu-rs*	2.95×	[2.869, 3.040]	0.0002	0.0004	-41

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:dragonbox-vs-bd	api	verify	canonical/medium	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.98×	[2.959, 3.024]	0.0002	0.0004	-59
schubfach:go-vs-rs	api	verify	canonical/medium	schubfach	schubfach-rs	schubfach-rs*	1.91×	[1.873, 1.951]	0.0002	0.0004	36
bd:go-vs-rs	api	verify	canonical/medium	json-canon	json-canon-rs	json-canon-rs*	1.10×	[1.082, 1.106]	0.0002	0.0004	8
ryu:go-vs-rs	api	verify	canonical/medium	ryu	ryu-rs	ryu-rs*	1.88×	[1.831, 1.935]	0.0002	0.0004	25
dragonbox:go-vs-rs	api	verify	canonical/medium	dragonbox	dragonbox-rs	dragonbox-rs*	1.89×	[1.881, 1.904]	0.0002	0.0004	78
go:schubfach-vs-bd	api	verify	canonical/mixed-prod	schubfach	json-canon	schubfach*	1.34×	[1.336, 1.353]	0.0002	0.0004	-36
go:schubfach-vs-ryu	api	verify	canonical/mixed-prod	schubfach	ryu	schubfach	1.01×	[1.000, 1.017]	0.1504	0.2311	-0
go:schubfach-vs-dragonbox	api	verify	canonical/mixed-prod	schubfach	dragonbox	schubfach	1.01×	[1.001, 1.012]	0.0556	0.0949	-0
go:ryu-vs-dragonbox	api	verify	canonical/mixed-prod	ryu	dragonbox	dragonbox	1.00×	[0.994, 1.011]	0.8646	0.9030	0
go:ryu-vs-bd	api	verify	canonical/mixed-prod	ryu	json-canon	ryu*	1.34×	[1.322, 1.346]	0.0002	0.0004	-28
go:dragonbox-vs-bd	api	verify	canonical/mixed-prod	dragonbox	json-canon	dragonbox*	1.34×	[1.329, 1.345]	0.0002	0.0004	-37
rs:schubfach-vs-bd	api	verify	canonical/mixed-prod	schubfach-rs	json-canon-rs	schubfach-rs*	1.83×	[1.807, 1.848]	0.0002	0.0004	-46
rs:schubfach-vs-ryu	api	verify	canonical/mixed-prod	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.997, 1.032]	0.2166	0.3086	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/mixed-prod	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.994, 1.028]	0.5185	0.6195	-0
rs:ryu-vs-dragonbox	api	verify	canonical/mixed-prod	ryu-rs	dragonbox-rs	dragonbox-rs	1.01×	[0.986, 1.026]	0.5785	0.6756	0
rs:ryu-vs-bd	api	verify	canonical/mixed-prod	ryu-rs	json-canon-rs	ryu-rs*	1.81×	[1.772, 1.832]	0.0002	0.0004	-35
rs:dragonbox-vs-bd	api	verify	canonical/mixed-prod	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.82×	[1.781, 1.834]	0.0002	0.0004	-40
schubfach:go-vs-rs	api	verify	canonical/mixed-prod	schubfach	schubfach-rs	schubfach-rs*	2.00×	[1.981, 2.023]	0.0002	0.0004	65
bd:go-vs-rs	api	verify	canonical/mixed-prod	json-canon	json-canon-rs	json-canon-rs*	1.47×	[1.463, 1.486]	0.0002	0.0004	41
ryu:go-vs-rs	api	verify	canonical/mixed-prod	ryu	ryu-rs	ryu-rs*	1.99×	[1.954, 2.022]	0.0002	0.0004	39
dragonbox:go-vs-rs	api	verify	canonical/mixed-prod	dragonbox	dragonbox-rs	dragonbox-rs*	2.00×	[1.961, 2.021]	0.0002	0.0004	58
go:schubfach-vs-bd	api	verify	canonical/nested-mixed	schubfach	json-canon	schubfach*	1.41×	[1.398, 1.413]	0.0002	0.0004	-50
go:schubfach-vs-ryu	api	verify	canonical/nested-mixed	schubfach	ryu	schubfach*	1.01×	[1.004, 1.016]	0.0030	0.0061	-1
go:schubfach-vs-dragonbox	api	verify	canonical/nested-mixed	schubfach	dragonbox	schubfach	1.01×	[1.000, 1.010]	0.0658	0.1116	-0
go:ryu-vs-dragonbox	api	verify	canonical/nested-mixed	ryu	dragonbox	dragonbox	1.01×	[1.000, 1.010]	0.0696	0.1169	0
go:ryu-vs-bd	api	verify	canonical/nested-mixed	ryu	json-canon	ryu*	1.39×	[1.384, 1.400]	0.0002	0.0004	-49
go:dragonbox-vs-bd	api	verify	canonical/nested-mixed	dragonbox	json-canon	dragonbox*	1.40×	[1.392, 1.405]	0.0002	0.0004	-54
rs:schubfach-vs-bd	api	verify	canonical/nested-mixed	schubfach-rs	json-canon-rs	schubfach-rs*	1.97×	[1.913, 2.010]	0.0002	0.0004	-26
rs:schubfach-vs-ryu	api	verify	canonical/nested-mixed	schubfach-rs	ryu-rs	schubfach-rs	1.00×	[0.966, 1.039]	0.9322	0.9557	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/nested-mixed	schubfach-rs	dragonbox-rs	schubfach-rs	1.00×	[0.971, 1.026]	0.9282	0.9540	-0
rs:ryu-vs-dragonbox	api	verify	canonical/nested-mixed	ryu-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.971, 1.039]	0.9722	0.9790	0
rs:ryu-vs-bd	api	verify	canonical/nested-mixed	ryu-rs	json-canon-rs	ryu-rs*	1.96×	[1.895, 2.016]	0.0002	0.0004	-22
rs:dragonbox-vs-bd	api	verify	canonical/nested-mixed	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.96×	[1.931, 2.014]	0.0002	0.0004	-29
schubfach:go-vs-rs	api	verify	canonical/nested-mixed	schubfach	schubfach-rs	schubfach-rs*	2.19×	[2.135, 2.234]	0.0002	0.0004	40
bd:go-vs-rs	api	verify	canonical/nested-mixed	json-canon	json-canon-rs	json-canon-rs*	1.57×	[1.548, 1.583]	0.0002	0.0004	38
ryu:go-vs-rs	api	verify	canonical/nested-mixed	ryu	ryu-rs	ryu-rs*	2.21×	[2.141, 2.267]	0.0002	0.0004	33
dragonbox:go-vs-rs	api	verify	canonical/nested-mixed	dragonbox	dragonbox-rs	dragonbox-rs*	2.20×	[2.172, 2.253]	0.0002	0.0004	54
go:schubfach-vs-bd	api	verify	canonical/number-heavy	schubfach	json-canon	schubfach*	1.56×	[1.546, 1.569]	0.0002	0.0004	-55
go:schubfach-vs-ryu	api	verify	canonical/number-heavy	schubfach	ryu	schubfach	1.01×	[0.996, 1.016]	0.3435	0.4468	-0
go:schubfach-vs-dragonbox	api	verify	canonical/number-heavy	schubfach	dragonbox	dragonbox	1.00×	[0.991, 1.008]	0.8806	0.9145	0
go:ryu-vs-dragonbox	api	verify	canonical/number-heavy	ryu	dragonbox	dragonbox	1.01×	[0.997, 1.016]	0.2498	0.3494	0
go:ryu-vs-bd	api	verify	canonical/number-heavy	ryu	json-canon	ryu*	1.55×	[1.534, 1.560]	0.0002	0.0004	-49
go:dragonbox-vs-bd	api	verify	canonical/number-heavy	dragonbox	json-canon	dragonbox*	1.56×	[1.546, 1.566]	0.0002	0.0004	-64
rs:schubfach-vs-bd	api	verify	canonical/number-heavy	schubfach-rs	json-canon-rs	schubfach-rs*	8.23×	[8.115, 8.394]	0.0002	0.0004	-136
rs:schubfach-vs-ryu	api	verify	canonical/number-heavy	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.998, 1.043]	0.0838	0.1383	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/number-heavy	schubfach-rs	dragonbox-rs	dragonbox-rs	1.00×	[0.974, 1.027]	0.7910	0.8446	0

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:ryu-vs-dragonbox	api	verify	canonical/number-heavy	ryu-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.997, 1.048]	0.0816	0.1348	0
rs:ryu-vs-bd	api	verify	canonical/number-heavy	ryu-rs	json-canon-rs	ryu-rs*	8.06×	[7.965, 8.225]	0.0002	0.0004	-137
rs:dragonbox-vs-bd	api	verify	canonical/number-heavy	dragonbox-rs	json-canon-rs	dragonbox-rs*	8.26×	[8.069, 8.414]	0.0002	0.0004	-131
schubfach:go-vs-rs	api	verify	canonical/number-heavy	schubfach	schubfach-rs	schubfach-rs*	9.81×	[9.666, 10.010]	0.0002	0.0004	110
bd:go-vs-rs	api	verify	canonical/number-heavy	json-canon	json-canon-rs	json-canon-rs*	1.86×	[1.844, 1.867]	0.0002	0.0004	89
ryu:go-vs-rs	api	verify	canonical/number-heavy	ryu	ryu-rs	ryu-rs*	9.66×	[9.531, 9.858]	0.0002	0.0004	94
dragonbox:go-vs-rs	api	verify	canonical/number-heavy	dragonbox	dragonbox-rs	dragonbox-rs*	9.84×	[9.601, 10.022]	0.0002	0.0004	137
go:schubfach-vs-bd	api	verify	canonical/numeric-boundary	schubfach	json-canon	schubfach*	1.49×	[1.475, 1.495]	0.0002	0.0004	-45
go:schubfach-vs-ryu	api	verify	canonical/numeric-boundary	schubfach	ryu	schubfach	1.01×	[1.000, 1.019]	0.0856	0.1409	-0
go:schubfach-vs-dragonbox	api	verify	canonical/numeric-boundary	schubfach	dragonbox	schubfach	1.01×	[0.998, 1.013]	0.1872	0.2730	-0
go:ryu-vs-dragonbox	api	verify	canonical/numeric-boundary	ryu	dragonbox	dragonbox	1.00×	[0.994, 1.014]	0.4471	0.5568	0
go:ryu-vs-bd	api	verify	canonical/numeric-boundary	ryu	json-canon	ryu*	1.47×	[1.457, 1.485]	0.0002	0.0004	-35
go:dragonbox-vs-bd	api	verify	canonical/numeric-boundary	dragonbox	json-canon	dragonbox*	1.48×	[1.465, 1.488]	0.0002	0.0004	-41
rs:schubfach-vs-bd	api	verify	canonical/numeric-boundary	schubfach-rs	json-canon-rs	schubfach-rs*	9.46×	[9.328, 9.773]	0.0002	0.0004	-154
rs:schubfach-vs-ryu	api	verify	canonical/numeric-boundary	schubfach-rs	ryu-rs	schubfach-rs	1.02×	[0.990, 1.051]	0.1784	0.2623	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/numeric-boundary	schubfach-rs	dragonbox-rs	schubfach-rs	1.01×	[0.988, 1.037]	0.6219	0.7109	-0
rs:ryu-vs-dragonbox	api	verify	canonical/numeric-boundary	ryu-rs	dragonbox-rs	dragonbox-rs	1.02×	[0.979, 1.036]	0.2879	0.3873	0
rs:ryu-vs-bd	api	verify	canonical/numeric-boundary	ryu-rs	json-canon-rs	ryu-rs*	9.25×	[9.107, 9.587]	0.0002	0.0004	-149
rs:dragonbox-vs-bd	api	verify	canonical/numeric-boundary	dragonbox-rs	json-canon-rs	dragonbox-rs*	9.39×	[9.261, 9.521]	0.0002	0.0004	-163
schubfach:go-vs-rs	api	verify	canonical/numeric-boundary	schubfach	schubfach-rs	schubfach-rs*	13.33×	[13.142, 13.737]	0.0002	0.0004	162
bd:go-vs-rs	api	verify	canonical/numeric-boundary	json-canon	json-canon-rs	json-canon-rs*	2.09×	[2.077, 2.106]	0.0002	0.0004	78
ryu:go-vs-rs	api	verify	canonical/numeric-boundary	ryu	ryu-rs	ryu-rs*	13.15×	[12.934, 13.603]	0.0002	0.0004	93
dragonbox:go-vs-rs	api	verify	canonical/numeric-boundary	dragonbox	dragonbox-rs	dragonbox-rs*	13.31×	[13.112, 13.488]	0.0002	0.0004	133
go:schubfach-vs-bd	api	verify	canonical/rfc-key-sorting	schubfach	json-canon	schubfach*	1.01×	[1.002, 1.012]	0.0230	0.0423	-1
go:schubfach-vs-ryu	api	verify	canonical/rfc-key-sorting	schubfach	ryu	schubfach*	1.01×	[1.002, 1.012]	0.0166	0.0310	-1
go:schubfach-vs-dragonbox	api	verify	canonical/rfc-key-sorting	schubfach	dragonbox	schubfach*	1.01×	[1.003, 1.015]	0.0096	0.0184	-1
go:ryu-vs-dragonbox	api	verify	canonical/rfc-key-sorting	ryu	dragonbox	ryu	1.00×	[0.996, 1.008]	0.4919	0.5949	-0
go:ryu-vs-bd	api	verify	canonical/rfc-key-sorting	ryu	json-canon	json-canon	1.00×	[0.995, 1.005]	0.9706	0.9789	0
go:dragonbox-vs-bd	api	verify	canonical/rfc-key-sorting	dragonbox	json-canon	json-canon	1.00×	[0.996, 1.008]	0.4775	0.5807	0
rs:schubfach-vs-bd	api	verify	canonical/rfc-key-sorting	schubfach-rs	json-canon-rs	json-canon-rs	1.01×	[0.988, 1.043]	0.3265	0.4277	0
rs:schubfach-vs-ryu	api	verify	canonical/rfc-key-sorting	schubfach-rs	ryu-rs	schubfach-rs*	1.05×	[1.022, 1.076]	0.0036	0.0072	-1
rs:schubfach-vs-dragonbox	api	verify	canonical/rfc-key-sorting	schubfach-rs	dragonbox-rs	schubfach-rs*	1.06×	[1.025, 1.107]	0.0110	0.0210	-1
rs:ryu-vs-dragonbox	api	verify	canonical/rfc-key-sorting	ryu-rs	dragonbox-rs	ryu-rs	1.01×	[0.977, 1.051]	0.5831	0.6791	-0
rs:ryu-vs-bd	api	verify	canonical/rfc-key-sorting	ryu-rs	json-canon-rs	json-canon-rs*	1.06×	[1.039, 1.090]	0.0002	0.0004	2
rs:dragonbox-vs-bd	api	verify	canonical/rfc-key-sorting	dragonbox-rs	json-canon-rs	json-canon-rs*	1.08×	[1.038, 1.121]	0.0024	0.0049	1
schubfach:go-vs-rs	api	verify	canonical/rfc-key-sorting	schubfach	schubfach-rs	schubfach-rs*	2.15×	[2.101, 2.188]	0.0002	0.0004	44
bd:go-vs-rs	api	verify	canonical/rfc-key-sorting	json-canon	json-canon-rs	json-canon-rs*	2.19×	[2.153, 2.236]	0.0002	0.0004	49
ryu:go-vs-rs	api	verify	canonical/rfc-key-sorting	ryu	ryu-rs	ryu-rs*	2.06×	[2.028, 2.096]	0.0002	0.0004	50
dragonbox:go-vs-rs	api	verify	canonical/rfc-key-sorting	dragonbox	dragonbox-rs	dragonbox-rs*	2.04×	[1.967, 2.107]	0.0002	0.0004	24
go:schubfach-vs-bd	api	verify	canonical/small	schubfach	json-canon	schubfach*	1.35×	[1.344, 1.360]	0.0002	0.0004	-43
go:schubfach-vs-ryu	api	verify	canonical/small	schubfach	ryu	schubfach	1.01×	[1.003, 1.016]	0.0278	0.0504	-1
go:schubfach-vs-dragonbox	api	verify	canonical/small	schubfach	dragonbox	schubfach	1.00×	[0.998, 1.010]	0.1806	0.2652	-0
go:ryu-vs-dragonbox	api	verify	canonical/small	ryu	dragonbox	dragonbox	1.00×	[0.999, 1.011]	0.2092	0.3001	0
go:ryu-vs-bd	api	verify	canonical/small	ryu	json-canon	ryu*	1.34×	[1.333, 1.348]	0.0002	0.0004	-44
go:dragonbox-vs-bd	api	verify	canonical/small	dragonbox	json-canon	dragonbox*	1.35×	[1.340, 1.353]	0.0002	0.0004	-49
rs:schubfach-vs-bd	api	verify	canonical/small	schubfach-rs	json-canon-rs	schubfach-rs*	1.73×	[1.654, 1.795]	0.0002	0.0004	-10

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
rs:schubfach-vs-ryu	api	verify	canonical/small	schubfach-rs	ryu-rs	schubfach-rs*	1.08×	[1.033, 1.118]	0.0046	0.0090	-1
rs:schubfach-vs-dragonbox	api	verify	canonical/small	schubfach-rs	dragonbox-rs	schubfach-rs	1.03×	[0.985, 1.069]	0.2511	0.3506	-0
rs:ryu-vs-dragonbox	api	verify	canonical/small	ryu-rs	dragonbox-rs	dragonbox-rs*	1.05×	[1.008, 1.088]	0.0264	0.0481	1
rs:ryu-vs-bd	api	verify	canonical/small	ryu-rs	json-canon-rs	ryu-rs*	1.60×	[1.545, 1.668]	0.0002	0.0004	-9
rs:dragonbox-vs-bd	api	verify	canonical/small	dragonbox-rs	json-canon-rs	dragonbox-rs*	1.68×	[1.611, 1.747]	0.0002	0.0004	-10
schubfach:go-vs-rs	api	verify	canonical/small	schubfach	schubfach-rs	schubfach-rs*	1.84×	[1.780, 1.890]	0.0002	0.0004	22
bd:go-vs-rs	api	verify	canonical/small	json-canon	json-canon-rs	json-canon-rs*	1.44×	[1.399, 1.485]	0.0002	0.0004	12
ryu:go-vs-rs	api	verify	canonical/small	ryu	ryu-rs	ryu-rs*	1.72×	[1.681, 1.769]	0.0002	0.0004	22
dragonbox:go-vs-rs	api	verify	canonical/small	dragonbox	dragonbox-rs	dragonbox-rs*	1.80×	[1.746, 1.846]	0.0002	0.0004	23
go:schubfach-vs-bd	api	verify	canonical/surrogate-pair	schubfach	json-canon	schubfach	1.01×	[1.000, 1.009]	0.0408	0.0715	-0
go:schubfach-vs-ryu	api	verify	canonical/surrogate-pair	schubfach	ryu	schubfach	1.00×	[0.995, 1.006]	0.6767	0.7555	-0
go:schubfach-vs-dragonbox	api	verify	canonical/surrogate-pair	schubfach	dragonbox	dragonbox	1.00×	[0.996, 1.010]	0.7842	0.8394	0
go:ryu-vs-dragonbox	api	verify	canonical/surrogate-pair	ryu	dragonbox	dragonbox	1.00×	[0.996, 1.010]	0.5503	0.6476	0
go:ryu-vs-bd	api	verify	canonical/surrogate-pair	ryu	json-canon	ryu	1.00×	[0.999, 1.009]	0.1556	0.2368	-0
go:dragonbox-vs-bd	api	verify	canonical/surrogate-pair	dragonbox	json-canon	dragonbox	1.01×	[1.002, 1.014]	0.0404	0.0709	-0
rs:schubfach-vs-bd	api	verify	canonical/surrogate-pair	schubfach-rs	json-canon-rs	json-canon-rs	1.01×	[0.970, 1.062]	0.6843	0.7613	0
rs:schubfach-vs-ryu	api	verify	canonical/surrogate-pair	schubfach-rs	ryu-rs	ryu-rs	1.03×	[0.993, 1.081]	0.2312	0.3273	0
rs:schubfach-vs-dragonbox	api	verify	canonical/surrogate-pair	schubfach-rs	dragonbox-rs	schubfach-rs	1.08×	[1.016, 1.137]	0.0306	0.0554	-1
rs:ryu-vs-dragonbox	api	verify	canonical/surrogate-pair	ryu-rs	dragonbox-rs	ryu-rs*	1.11×	[1.062, 1.165]	0.0006	0.0013	-1
rs:ryu-vs-bd	api	verify	canonical/surrogate-pair	ryu-rs	json-canon-rs	ryu-rs	1.02×	[0.994, 1.056]	0.2685	0.3701	-0
rs:dragonbox-vs-bd	api	verify	canonical/surrogate-pair	dragonbox-rs	json-canon-rs	json-canon-rs*	1.09×	[1.035, 1.146]	0.0078	0.0151	1
schubfach:go-vs-rs	api	verify	canonical/surrogate-pair	schubfach	schubfach-rs	schubfach-rs*	2.11×	[2.013, 2.177]	0.0002	0.0004	23
bd:go-vs-rs	api	verify	canonical/surrogate-pair	json-canon	json-canon-rs	json-canon-rs*	2.14×	[2.067, 2.178]	0.0002	0.0004	39
ryu:go-vs-rs	api	verify	canonical/surrogate-pair	ryu	ryu-rs	ryu-rs*	2.18×	[2.128, 2.211]	0.0002	0.0004	47
dragonbox:go-vs-rs	api	verify	canonical/surrogate-pair	dragonbox	dragonbox-rs	dragonbox-rs*	1.96×	[1.860, 2.035]	0.0002	0.0004	17
go:schubfach-vs-bd	api	verify	canonical/unicode	schubfach	json-canon	json-canon	1.00×	[0.999, 1.009]	0.1526	0.2336	0
go:schubfach-vs-ryu	api	verify	canonical/unicode	schubfach	ryu	schubfach	1.01×	[1.001, 1.016]	0.0292	0.0529	-1
go:schubfach-vs-dragonbox	api	verify	canonical/unicode	schubfach	dragonbox	schubfach	1.01×	[1.001, 1.014]	0.0390	0.0689	-1
go:ryu-vs-dragonbox	api	verify	canonical/unicode	ryu	dragonbox	dragonbox	1.00×	[0.993, 1.009]	0.6873	0.7621	0
go:ryu-vs-bd	api	verify	canonical/unicode	ryu	json-canon	json-canon*	1.01×	[1.005, 1.020]	0.0034	0.0068	1
go:dragonbox-vs-bd	api	verify	canonical/unicode	dragonbox	json-canon	json-canon*	1.01×	[1.006, 1.018]	0.0020	0.0041	1
rs:schubfach-vs-bd	api	verify	canonical/unicode	schubfach-rs	json-canon-rs	schubfach-rs*	1.02×	[1.010, 1.045]	0.0098	0.0188	-1
rs:schubfach-vs-ryu	api	verify	canonical/unicode	schubfach-rs	ryu-rs	ryu-rs	1.00×	[0.990, 1.014]	0.9304	0.9548	0
rs:schubfach-vs-dragonbox	api	verify	canonical/unicode	schubfach-rs	dragonbox-rs	schubfach-rs*	1.08×	[1.066, 1.102]	0.0002	0.0004	-3
rs:ryu-vs-dragonbox	api	verify	canonical/unicode	ryu-rs	dragonbox-rs	ryu-rs*	1.08×	[1.067, 1.103]	0.0002	0.0004	-3
rs:ryu-vs-bd	api	verify	canonical/unicode	ryu-rs	json-canon-rs	ryu-rs*	1.03×	[1.011, 1.048]	0.0082	0.0158	-1
rs:dragonbox-vs-bd	api	verify	canonical/unicode	dragonbox-rs	json-canon-rs	json-canon-rs*	1.06×	[1.033, 1.077]	0.0002	0.0004	2
schubfach:go-vs-rs	api	verify	canonical/unicode	schubfach	schubfach-rs	schubfach-rs*	1.03×	[1.023, 1.043]	0.0004	0.0008	2
bd:go-vs-rs	api	verify	canonical/unicode	json-canon	json-canon-rs	json-canon-rs	1.01×	[0.986, 1.018]	0.5167	0.6179	0
ryu:go-vs-rs	api	verify	canonical/unicode	ryu	ryu-rs	ryu-rs*	1.05×	[1.034, 1.056]	0.0004	0.0008	3
dragonbox:go-vs-rs	api	verify	canonical/unicode	dragonbox	dragonbox-rs	dragonbox*	1.04×	[1.024, 1.056]	0.0002	0.0004	-2
go:schubfach-vs-bd	api	verify	canonical/verify-whitespace	schubfach	json-canon	schubfach*	1.71×	[1.698, 1.717]	0.0004	0.0008	-72
go:schubfach-vs-ryu	api	verify	canonical/verify-whitespace	schubfach	ryu	schubfach	1.01×	[1.001, 1.011]	0.0546	0.0934	-0
go:schubfach-vs-dragonbox	api	verify	canonical/verify-whitespace	schubfach	dragonbox	schubfach*	1.01×	[1.007, 1.019]	0.0004	0.0008	-1
go:ryu-vs-dragonbox	api	verify	canonical/verify-whitespace	ryu	dragonbox	ryu	1.01×	[1.001, 1.013]	0.0318	0.0573	-1
go:ryu-vs-bd	api	verify	canonical/verify-whitespace	ryu	json-canon	ryu*	1.70×	[1.687, 1.707]	0.0002	0.0004	-69

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:dragonbox-vs-bd	api	verify	canonical/verify-whitespace	dragonbox	json-canon	dragonbox*	1.68×	[1.674, 1.696]	0.0002	0.0004	-65
rs:schubfach-vs-bd	api	verify	canonical/verify-whitespace	schubfach-rs	json-canon-rs	schubfach-rs*	2.93×	[2.801, 3.004]	0.0002	0.0004	-33
rs:schubfach-vs-ryu	api	verify	canonical/verify-whitespace	schubfach-rs	ryu-rs	schubfach-rs	1.01×	[0.965, 1.046]	0.5223	0.6222	-0
rs:schubfach-vs-dragonbox	api	verify	canonical/verify-whitespace	schubfach-rs	dragonbox-rs	schubfach-rs	1.03×	[0.982, 1.060]	0.0860	0.1414	-0
rs:ryu-vs-dragonbox	api	verify	canonical/verify-whitespace	ryu-rs	dragonbox-rs	ryu-rs	1.02×	[0.992, 1.044]	0.1928	0.2803	-0
rs:ryu-vs-bd	api	verify	canonical/verify-whitespace	ryu-rs	json-canon-rs	ryu-rs*	2.89×	[2.818, 2.965]	0.0002	0.0004	-37
rs:dragonbox-vs-bd	api	verify	canonical/verify-whitespace	dragonbox-rs	json-canon-rs	dragonbox-rs*	2.84×	[2.788, 2.902]	0.0002	0.0004	-39
schubfach:go-vs-rs	api	verify	canonical/verify-whitespace	schubfach	schubfach-rs	schubfach-rs*	2.38×	[2.273, 2.434]	0.0002	0.0004	35
bd:go-vs-rs	api	verify	canonical/verify-whitespace	json-canon	json-canon-rs	json-canon-rs*	1.39×	[1.367, 1.404]	0.0002	0.0004	23
ryu:go-vs-rs	api	verify	canonical/verify-whitespace	ryu	ryu-rs	ryu-rs*	2.36×	[2.309, 2.413]	0.0002	0.0004	48
dragonbox:go-vs-rs	api	verify	canonical/verify-whitespace	dragonbox	dragonbox-rs	dragonbox-rs*	2.33×	[2.306, 2.388]	0.0002	0.0004	59
go:schubfach-vs-bd	api	verify	noncanonical/control-escapes	schubfach	json-canon	schubfach*	1.03×	[1.028, 1.042]	0.0002	0.0004	-3
go:schubfach-vs-ryu	api	verify	noncanonical/control-escapes	schubfach	ryu	schubfach*	1.01×	[1.006, 1.021]	0.0032	0.0064	-1
go:schubfach-vs-dragonbox	api	verify	noncanonical/control-escapes	schubfach	dragonbox	schubfach*	1.01×	[1.005, 1.015]	0.0016	0.0033	-1
go:ryu-vs-dragonbox	api	verify	noncanonical/control-escapes	ryu	dragonbox	dragonbox	1.00×	[0.997, 1.010]	0.3505	0.4550	0
go:ryu-vs-bd	api	verify	noncanonical/control-escapes	ryu	json-canon	ryu*	1.02×	[1.012, 1.030]	0.0004	0.0008	-1
go:dragonbox-vs-bd	api	verify	noncanonical/control-escapes	dragonbox	json-canon	dragonbox*	1.02×	[1.018, 1.031]	0.0002	0.0004	-2
go:schubfach-vs-bd	api	verify	noncanonical/escaped-key-order	schubfach	json-canon	schubfach*	1.62×	[1.612, 1.627]	0.0002	0.0004	-84
go:schubfach-vs-ryu	api	verify	noncanonical/escaped-key-order	schubfach	ryu	schubfach*	1.01×	[1.009, 1.020]	0.0002	0.0004	-2
go:schubfach-vs-dragonbox	api	verify	noncanonical/escaped-key-order	schubfach	dragonbox	schubfach*	1.01×	[1.002, 1.018]	0.0274	0.0498	-1
go:ryu-vs-dragonbox	api	verify	noncanonical/escaped-key-order	ryu	dragonbox	dragonbox	1.00×	[0.996, 1.013]	0.4033	0.5107	0
go:ryu-vs-bd	api	verify	noncanonical/escaped-key-order	ryu	json-canon	ryu*	1.60×	[1.589, 1.605]	0.0002	0.0004	-78
go:dragonbox-vs-bd	api	verify	noncanonical/escaped-key-order	dragonbox	json-canon	dragonbox*	1.60×	[1.590, 1.617]	0.0002	0.0004	-56
go:schubfach-vs-bd	api	verify	noncanonical/large	schubfach	json-canon	schubfach*	1.77×	[1.762, 1.774]	0.0002	0.0004	-141
go:schubfach-vs-ryu	api	verify	noncanonical/large	schubfach	ryu	schubfach	1.01×	[1.000, 1.011]	0.0734	0.1225	-0
go:schubfach-vs-dragonbox	api	verify	noncanonical/large	schubfach	dragonbox	schubfach*	1.01×	[1.005, 1.016]	0.0032	0.0064	-1
go:ryu-vs-dragonbox	api	verify	noncanonical/large	ryu	dragonbox	ryu	1.01×	[0.998, 1.011]	0.1454	0.2245	-0
go:ryu-vs-bd	api	verify	noncanonical/large	ryu	json-canon	ryu*	1.76×	[1.750, 1.766]	0.0002	0.0004	-117
go:dragonbox-vs-bd	api	verify	noncanonical/large	dragonbox	json-canon	dragonbox*	1.75×	[1.742, 1.759]	0.0002	0.0004	-113
go:schubfach-vs-bd	api	verify	noncanonical/medium	schubfach	json-canon	schubfach*	1.73×	[1.721, 1.739]	0.0002	0.0004	-115
go:schubfach-vs-ryu	api	verify	noncanonical/medium	schubfach	ryu	ryu	1.00×	[0.995, 1.007]	0.8732	0.9098	0
go:schubfach-vs-dragonbox	api	verify	noncanonical/medium	schubfach	dragonbox	dragonbox	1.00×	[0.995, 1.007]	0.9774	0.9820	0
go:ryu-vs-dragonbox	api	verify	noncanonical/medium	ryu	dragonbox	ryu	1.00×	[0.994, 1.005]	0.8852	0.9173	-0
go:ryu-vs-bd	api	verify	noncanonical/medium	ryu	json-canon	ryu*	1.73×	[1.725, 1.738]	0.0002	0.0004	-144
go:dragonbox-vs-bd	api	verify	noncanonical/medium	dragonbox	json-canon	dragonbox*	1.73×	[1.726, 1.741]	0.0002	0.0004	-147
go:schubfach-vs-bd	api	verify	noncanonical/mixed-prod	schubfach	json-canon	schubfach*	1.34×	[1.336, 1.352]	0.0002	0.0004	-41
go:schubfach-vs-ryu	api	verify	noncanonical/mixed-prod	schubfach	ryu	schubfach*	1.01×	[1.005, 1.020]	0.0070	0.0136	-1
go:schubfach-vs-dragonbox	api	verify	noncanonical/mixed-prod	schubfach	dragonbox	schubfach*	1.01×	[1.004, 1.019]	0.0114	0.0217	-1
go:ryu-vs-dragonbox	api	verify	noncanonical/mixed-prod	ryu	dragonbox	ryu	1.00×	[0.990, 1.008]	0.8900	0.9214	-0
go:ryu-vs-bd	api	verify	noncanonical/mixed-prod	ryu	json-canon	ryu*	1.33×	[1.317, 1.337]	0.0002	0.0004	-32
go:dragonbox-vs-bd	api	verify	noncanonical/mixed-prod	dragonbox	json-canon	dragonbox*	1.33×	[1.318, 1.339]	0.0004	0.0008	-31
go:schubfach-vs-bd	api	verify	noncanonical/nested-mixed	schubfach	json-canon	schubfach*	1.41×	[1.399, 1.412]	0.0002	0.0004	-59
go:schubfach-vs-ryu	api	verify	noncanonical/nested-mixed	schubfach	ryu	schubfach*	1.01×	[1.005, 1.014]	0.0028	0.0057	-1
go:schubfach-vs-dragonbox	api	verify	noncanonical/nested-mixed	schubfach	dragonbox	schubfach*	1.01×	[1.003, 1.011]	0.0062	0.0121	-1
go:ryu-vs-dragonbox	api	verify	noncanonical/nested-mixed	ryu	dragonbox	dragonbox	1.00×	[0.999, 1.007]	0.3251	0.4267	0
go:ryu-vs-bd	api	verify	noncanonical/nested-mixed	ryu	json-canon	ryu*	1.39×	[1.386, 1.399]	0.0002	0.0004	-58

Pair	Track	Mode	Workload	Impl A	Impl B	Winner	Speedup	CI_{95}	p	p_{adj}	
go:dragonbox-vs-bd	api	verify	noncanonical/nested-mixed	dragonbox	json-canon	dragonbox*	1.40×	[1.391, 1.402]	0.0002	0.0004	-65
go:schubfach-vs-bd	api	verify	noncanonical/number-heavy	schubfach	json-canon	schubfach*	1.56×	[1.542, 1.573]	0.0004	0.0008	-38
go:schubfach-vs-ryu	api	verify	noncanonical/number-heavy	schubfach	ryu	schubfach	1.00×	[0.992, 1.010]	0.5825	0.6790	-0
go:schubfach-vs-dragonbox	api	verify	noncanonical/number-heavy	schubfach	dragonbox	schubfach	1.00×	[0.991, 1.009]	0.7149	0.7839	-0
go:ryu-vs-dragonbox	api	verify	noncanonical/number-heavy	ryu	dragonbox	dragonbox	1.00×	[0.996, 1.007]	0.7822	0.8391	0
go:ryu-vs-bd	api	verify	noncanonical/number-heavy	ryu	json-canon	ryu*	1.56×	[1.545, 1.566]	0.0002	0.0004	-47
go:dragonbox-vs-bd	api	verify	noncanonical/number-heavy	dragonbox	json-canon	dragonbox*	1.56×	[1.547, 1.568]	0.0002	0.0004	-46
go:schubfach-vs-bd	api	verify	noncanonical/numeric-boundary	schubfach	json-canon	schubfach*	1.50×	[1.485, 1.505]	0.0002	0.0004	-44
go:schubfach-vs-ryu	api	verify	noncanonical/numeric-boundary	schubfach	ryu	schubfach	1.00×	[0.995, 1.007]	0.6537	0.7394	-0
go:schubfach-vs-dragonbox	api	verify	noncanonical/numeric-boundary	schubfach	dragonbox	schubfach	1.00×	[0.993, 1.007]	0.9622	0.9750	-0
go:ryu-vs-dragonbox	api	verify	noncanonical/numeric-boundary	ryu	dragonbox	dragonbox	1.00×	[0.994, 1.009]	0.7706	0.8303	0
go:ryu-vs-bd	api	verify	noncanonical/numeric-boundary	ryu	json-canon	ryu*	1.49×	[1.481, 1.503]	0.0002	0.0004	-42
go:dragonbox-vs-bd	api	verify	noncanonical/numeric-boundary	dragonbox	json-canon	dragonbox*	1.49×	[1.483, 1.508]	0.0002	0.0004	-39
go:schubfach-vs-bd	api	verify	noncanonical/rfc-key-sorting	schubfach	json-canon	schubfach*	1.02×	[1.017, 1.027]	0.0002	0.0004	-3
go:schubfach-vs-ryu	api	verify	noncanonical/rfc-key-sorting	schubfach	ryu	schubfach*	1.02×	[1.011, 1.022]	0.0002	0.0004	-2
go:schubfach-vs-dragonbox	api	verify	noncanonical/rfc-key-sorting	schubfach	dragonbox	schubfach*	1.02×	[1.015, 1.022]	0.0002	0.0004	-4
go:ryu-vs-dragonbox	api	verify	noncanonical/rfc-key-sorting	ryu	dragonbox	ryu	1.00×	[0.997, 1.009]	0.4681	0.5746	-0
go:ryu-vs-bd	api	verify	noncanonical/rfc-key-sorting	ryu	json-canon	ryu	1.00×	[0.998, 1.012]	0.3341	0.4372	-0
go:dragonbox-vs-bd	api	verify	noncanonical/rfc-key-sorting	dragonbox	json-canon	dragonbox	1.00×	[0.997, 1.009]	0.6753	0.7546	-0
go:schubfach-vs-bd	api	verify	noncanonical/small	schubfach	json-canon	schubfach*	1.35×	[1.344, 1.357]	0.0002	0.0004	-47
go:schubfach-vs-ryu	api	verify	noncanonical/small	schubfach	ryu	schubfach*	1.02×	[1.012, 1.024]	0.0002	0.0004	-2
go:schubfach-vs-dragonbox	api	verify	noncanonical/small	schubfach	dragonbox	schubfach*	1.02×	[1.014, 1.025]	0.0002	0.0004	-3
go:ryu-vs-dragonbox	api	verify	noncanonical/small	ryu	dragonbox	ryu	1.00×	[0.996, 1.007]	0.4573	0.5662	-0
go:ryu-vs-bd	api	verify	noncanonical/small	ryu	json-canon	ryu*	1.33×	[1.320, 1.334]	0.0002	0.0004	-43
go:dragonbox-vs-bd	api	verify	noncanonical/small	dragonbox	json-canon	dragonbox*	1.32×	[1.318, 1.331]	0.0002	0.0004	-45
go:schubfach-vs-bd	api	verify	noncanonical/surrogate-pair	schubfach	json-canon	schubfach*	1.01×	[1.004, 1.015]	0.0066	0.0128	-1
go:schubfach-vs-ryu	api	verify	noncanonical/surrogate-pair	schubfach	ryu	schubfach	1.00×	[0.999, 1.010]	0.1506	0.2311	-0
go:schubfach-vs-dragonbox	api	verify	noncanonical/surrogate-pair	schubfach	dragonbox	schubfach	1.01×	[1.001, 1.012]	0.0478	0.0823	-0
go:ryu-vs-dragonbox	api	verify	noncanonical/surrogate-pair	ryu	dragonbox	ryu	1.00×	[0.996, 1.008]	0.5705	0.6669	-0
go:ryu-vs-bd	api	verify	noncanonical/surrogate-pair	ryu	json-canon	ryu	1.00×	[0.999, 1.011]	0.1634	0.2470	-0
go:dragonbox-vs-bd	api	verify	noncanonical/surrogate-pair	dragonbox	json-canon	dragonbox	1.00×	[0.997, 1.009]	0.3995	0.5063	-0
go:schubfach-vs-bd	api	verify	noncanonical/unicode	schubfach	json-canon	schubfach	1.00×	[0.997, 1.006]	0.5227	0.6222	-0
go:schubfach-vs-ryu	api	verify	noncanonical/unicode	schubfach	ryu	schubfach*	1.01×	[1.005, 1.014]	0.0016	0.0033	-1
go:schubfach-vs-dragonbox	api	verify	noncanonical/unicode	schubfach	dragonbox	schubfach*	1.02×	[1.010, 1.020]	0.0004	0.0008	-2
go:ryu-vs-dragonbox	api	verify	noncanonical/unicode	ryu	dragonbox	ryu*	1.01×	[1.001, 1.010]	0.0260	0.0476	-1
go:ryu-vs-bd	api	verify	noncanonical/unicode	ryu	json-canon	json-canon*	1.01×	[1.004, 1.011]	0.0004	0.0008	1
go:dragonbox-vs-bd	api	verify	noncanonical/unicode	dragonbox	json-canon	json-canon*	1.01×	[1.009, 1.018]	0.0002	0.0004	2
go:schubfach-vs-bd	api	verify	noncanonical/verify-whitespace	schubfach	json-canon	schubfach*	1.72×	[1.706, 1.726]	0.0002	0.0004	-78
go:schubfach-vs-ryu	api	verify	noncanonical/verify-whitespace	schubfach	ryu	schubfach	1.01×	[0.999, 1.011]	0.1280	0.2000	-0
go:schubfach-vs-dragonbox	api	verify	noncanonical/verify-whitespace	schubfach	dragonbox	schubfach*	1.01×	[1.005, 1.018]	0.0020	0.0041	-1
go:ryu-vs-dragonbox	api	verify	noncanonical/verify-whitespace	ryu	dragonbox	ryu	1.01×	[1.001, 1.012]	0.0426	0.0742	-1
go:ryu-vs-bd	api	verify	noncanonical/verify-whitespace	ryu	json-canon	ryu*	1.71×	[1.699, 1.717]	0.0002	0.0004	-83
go:dragonbox-vs-bd	api	verify	noncanonical/verify-whitespace	dragonbox	json-canon	dragonbox*	1.70×	[1.687, 1.706]	0.0002	0.0004	-80

*Significant after Benjamini-Hochberg FDR correction at $q = 0.05$.
d = Cohen's d: $(\bar{x}_A - \bar{x}_B)/s_p$; negative = impl A faster. Track: e2e = end-to-end CLI.
Noise = CV × mean of faster impl. MDE = minimum detectable effect at 80% power (z-approx).
Rust allocs/op and B/op measured via counting global allocator (§8).

B Reproducibility Checklist

This appendix documents the information required to reproduce the benchmark results reported in this paper.

Software Versions

- Go 1.25.6 (`go version go1.25.6 linux/amd64`)
- Rust 1.94.0 (`rustc 1.94.0`)
- Node.js v23.3.0 (V8 12.9.202.28-node.11) — oracle generation
- SpiderMonkey (JavaScript-C115.10.0) — oracle validation
- JavaScriptCore (WebKitGTK 2.50.4-0ubuntu0.24.04.1) — oracle validation
- Linux 6.8.0-101-generic (Ubuntu)

Hardware Specification

- CPU: Intel Core i7-12700K (8 P-cores at 3.6–5.0 GHz, 4 E-cores at 2.7–3.8 GHz)
- RAM: 64 GB DDR4
- Architecture: x86-64
- CPU governor: `performance`

Oracle Vector Checksums (SHA-256)

- `golden_vectors.csv` (54,445 vectors):
  `593bdecbe0dccbc182bc3baf570b716887db25739fc61b7808764ecb966d5636`
- `golden_stress_vectors.csv` (231,917 vectors):
  `287d21ac87e5665550f1baf86038302a0afc67a74a020dffb872f1a93b26d410`

Reproduction Commands

```
git clone <repository-url> && cd jcs-bench-lab
go run ./cmd/lab setup
go run ./cmd/lab conformance
go run ./cmd/lab bench-api -count 10
go run ./cmd/lab bench-cli -repeats 15
go run ./cmd/lab fuzz -cases 2000
go run ./cmd/lab stats -resamples 5000
go run ./cmd/lab gate
go run scripts/generate-paper-tables.go
```

Build Configuration

- Go implementations: default `go build` flags
- Rust implementations: `opt-level=3`, `lto=true`, `codegen-units=1`, `strip=true`
- Rust API benchmarks use a counting global allocator (`std::alloc::GlobalAlloc`) to measure per-operation allocation bytes and counts

Data Availability

All raw benchmark data, statistical reports, oracle vectors, and the benchmark infrastructure are archived at Zenodo.[3]

[3]Zenodo DOI to be inserted for camera-ready version.